MEAMS: A REAL-TIME TRACKING AND FORECASTING ASSET MANAGEMENT SYSTEM

An IT Capstone Project Presented to the Faculty of

The College of Computing Studies

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In Partial Fulfillment for the Degree in Bachelor of

Science in Information Technology with

Specialization in Data Science

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ABSTRACT

The Maintenance and Engineering Asset Management System (MEAMS) was developed to modernize the existing manual processes of the Maintenance and Engineering Division (MED) of Universidad de Manila. The system integrates real-time tracking, forecasting, and analytical capabilities to enhance asset visibility, operational efficiency, and decision-making. It employs Quick Response (QR) code labeling for easy asset identification, a digital stock card for monitoring supply usage, and a repair history module that records repair activities and estimates equipment life for cost-benefit analysis. A forecasting feature powered by time-series analytics predicts future supply and equipment trends, while a threshold-based alert system identifies understocking and asset deterioration. The platform is designed with a user-friendly interface built using React.js, Node.js, and MongoDB, ensuring data accuracy and secure storage. MEAMS aligns with ISO 25010 software quality standards—specifically Security, Functional Suitability, Performance Efficiency, Usability, Maintainability, and Reliability. The implementation of MEAMS demonstrated how automation and forecast insights can significantly improve institutional maintenance management, reduce operational delays, and promote transparency within university operations.

Keywords: Asset Management, Forecasting Analytics, QR Code Tracking, Repair Monitoring, Real-Time System

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Chapter I

INTRODUCTION

Project Context

The digital revolution fundamentally transformed how organizations lived, worked, and interacted. Yet for many organizations—especially governments—the shift to modern asset management systems had been slow and uneven. Inefficient tracking, outdated processes, and data trapped in disconnected systems wasted billions in lost productivity and resources. The Philippine Government Asset Management Plan (PGAMP) highlighted that the Asset Management Information System (AMIS) provided the ability to understand assets and make the best decisions to optimize asset lifecycle costs and performance. With the rise of digital transformation, organizations increasingly adopted automated systems to track, maintain, and utilize assets efficiently.

However, many institutions still relied on outdated manual processes, which led to mismanagement, financial losses, and operational inefficiencies. Recent studies emphasized the need for agile and technology-driven asset management solutions. In the Philippines, where public sector accountability and resource optimization were pressing concerns, a sufficient Asset Management System (AMS) was not just a technological upgrade, it was a necessity for sustainable development (PGAMP, 2022-2023, p. 34). This study aimed to address these challenges by developing an optimized AMS that enhanced accuracy, security, and operational efficiency.

This research focused on developing an AMS for the Maintenance and Engineering Division (MED) of Universidad de Manila (UDM) to modernize its outdated manual tracking

processes. At the time of the study, the MED relied on paper-based records for asset transactions, including purchases and repairs, which led to an unreliable inventory system. While historical data existed, the lack of a centralized digital platform resulted in inefficiencies, discrepancies in assets, and difficulties in monitoring asset conditions issues that aligned with the challenges highlighted in the PGAMP regarding poor asset visibility. Studies, such as those by findings in the International Journal of Innovation, Management, and Technology (IJIMT), confirmed that manual systems increased errors and hindered accountability. By implementing an automated AMS, this study aimed to optimize MED's asset tracking, ensure real-time updates, and enhance compliance with modern asset management standards, ultimately supporting better resource utilization and institutional transparency.

The persistent issue in MED's asset system highlighted an urgent need for digitization. Without real-time tracking, staff struggled with inaccurate stock records, which led to preventable delays in maintenance work. Additionally, the lack of a centralized repair history made it difficult to predict equipment failures or schedule servicing and left the useful life of the assets unidentified. These challenges not only wasted resources but also disrupted operations and productivity. By implementing a digital asset management system, MED could eliminate manual errors, reduce material waste, and ensure faster response times for repairs. This solution would ultimately lower costs, extend equipment lifespan, and create a more efficient and reliable repair process for the entire MED.

Purpose and Description of the Project

The proponents developed the MEAMS for the MED of UDM to replace the existing manual procedures with a digital solution. The system was created to address recurring

issues such as stock discrepancies, delayed repair services, and asset deterioration. Through the integration of features like Quick Response (QR) code-based labeling, digital stock cards, repair history tracking, and estimated equipment life monitoring, the MEAMS aims to ensure accurate inventory tracking, immediate repair action, and efficient utilization of resources. Its overall purpose is to enhance the operational efficiency of the MED, reduce operational costs, and align asset management practices with modern technological standards.

Objectives of the Study

General Objective

The study is conducted to design and develop “MEAMS: A Real-Time Tracking and Forecasting Asset Management System” that optimizes inventory processes, tracks supply usage, monitors repair history, and improves the overall management of assets. The system aims to successfully reduce discrepancies, minimize wastage, and promote efficient maintenance operations within the MED.

Specific Objectives

The following were the study's specific objectives:

1. To develop a system that tracks stock records using an updated and digitized stock card for real-time inventory monitoring.
2. To create a feature that provides repair history and estimates the useful life of equipment for cost-benefit analysis.
3. To implement forecasting and data analytics features that display stock levels and equipment status trends to optimize decision-making.

4. To integrate QR code label generation for classifying and organizing items on stockroom shelves.

5. To evaluate the MEAMS based on its compliance with ISO 25010 standards, focusing on the following criteria:

    5.1. Security

    5.2. Functional Suitability

    5.3. Performance Efficiency

    5.4. Usability

    5.5. Maintainability

    5.6 Reliability

Conceptual Framework

The Figure 1-1 below demonstrates the input, process, and output diagram required for developing MEAMS for the MED. The Input phase involves gathering knowledge through interviews and brainstorming, which helps in understanding MED requirements and system functionalities. Additionally, it incorporates a comprehensive set of software tools—such as React.js, Chart.js, Python, Node.js, and MongoDB—chosen for their ability to support efficient, scalable, and real-time application development. In the Process phase, the system follows the SCRUM Agile Model, a step-by-step development approach consisting of product backlog creation, sprint planning meeting, sprint backlog, daily stand-up, sprint review, and retrospective. This cyclical process ensures continuous improvement and adaptability based on stakeholder feedback. The Output will be the MEAMS itself, a robust and responsive asset management system that will enhance operational efficiency by enabling real-time tracking and insights for supplies and

equipment. A feedback mechanism connects the output back to the input, allowing the system to evolve based on real-life performance and user experiences.

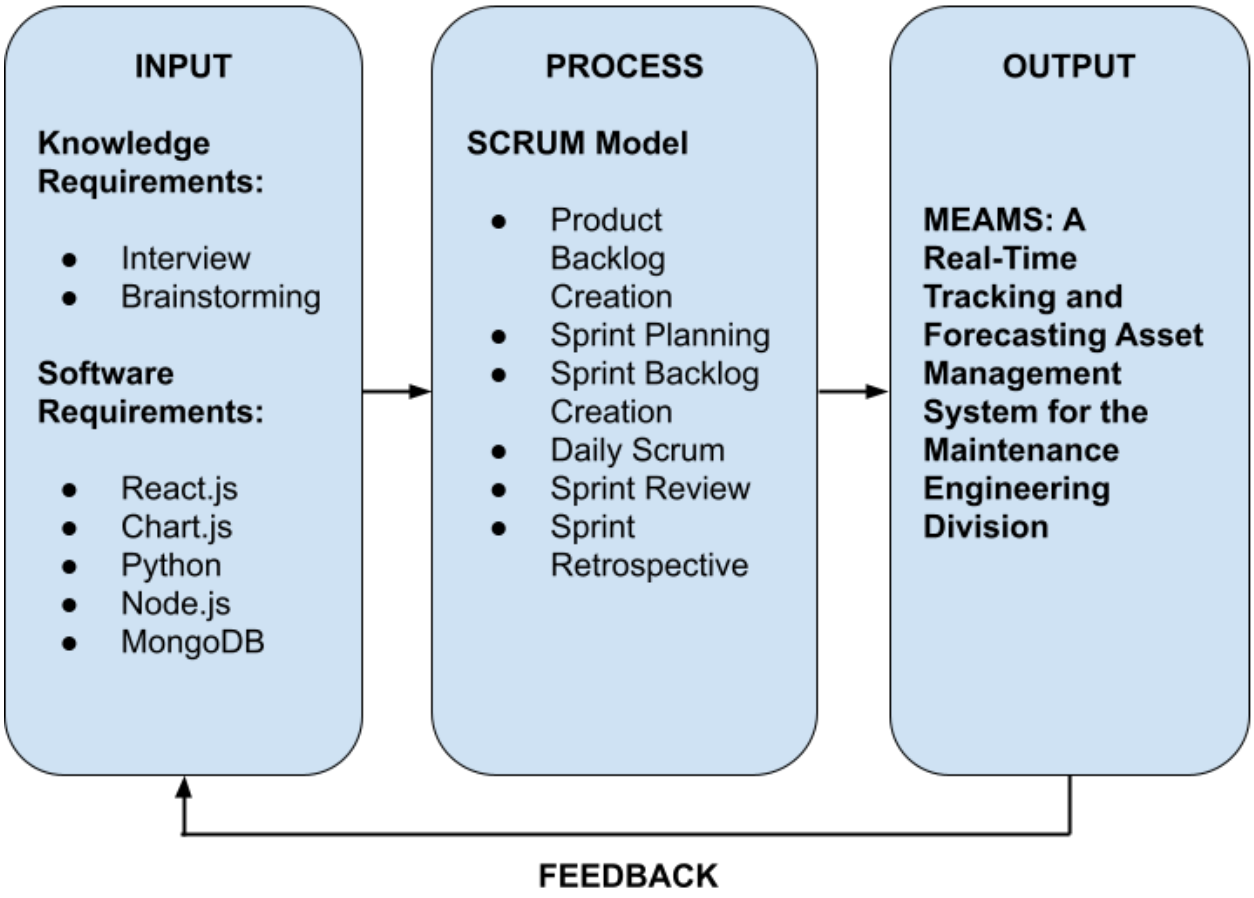


Figure 1-1. Input-Process-Output Diagram

Scope and Limitation

This project, which started in March 2025, is entitled “MEAMS: A Real-Time Tracking and Forecasting Asset Management System” and aims to enhance decision-making, resource allocation, and operational efficiency within the MED of UDM. The system focuses on providing real-time data and analytics on both supplies and equipment

while aligning with existing MED workflows and procedures. Its key features include a secure cloud database that stores migrated asset records, QR code integration for mobile access to updated stock cards and repair history, and an interactive real-time dashboard that visually categorizes stock levels, tracks equipment lifespan, and provides alerts for critical stock levels and equipment for replacement and beyond their useful life. To ensure smooth adoption, staff training on the new digital interface will also be conducted. The target implementation for the system within the MED is set for November 2025, with the goal of building a secure, efficient, and user-friendly platform for asset management.

Despite its potential to modernize asset management for the MED of UDM, the implementation of MEAMS also carries certain limitations. First, the system is designed solely for the MED and does not integrate with other university departments. Second, real-time synchronization depends entirely on stable internet connectivity, and interruptions in connection delay updates until service is restored. Third, the accuracy of the system's dashboards relies heavily on the quality of input data; errors or incomplete information from legacy data migration required manual corrections. Fourth, operational accuracy is vulnerable to human error, as asset movements must be recorded consistently and correctly to prevent discrepancies. Lastly, since the system still depends on manual data entry without automated validation, typographical errors or missing entries may compromise data integrity.

Significance of the Study

This study was considered significant as it provided several benefits to the following:

Administrators benefited from the development of the system as it improved the overall operational efficiency within the MED. It provided them with real-time data on equipment and maintenance needs, which enabled them to make more informed decisions.

The **Maintenance and Engineering Division** overcome its challenges through the system, which offered a more organized, accurate, and efficient way of managing the division's assets. It allowed the division to monitor machines, equipment, and tools in real time, reducing stock discrepancies and supporting better decision-making for resource allocation. As a result, the MED avoided issues related to overstocking and understocking, which improved material management and ensured quicker response times for repairs. This may have contributed to enhancing the overall service delivery within the university.

Staff benefited from the system by gaining access to real-time information about inventory and equipment. With features such as QR code labels, repair history tracking, and updated stock cards, the MED staff could plan and perform tasks more efficiently, which reduced delays and enabled faster responses to repair requests. This improved their ability to perform duties effectively and minimized errors caused by outdated inventory control.

Future researchers will find this study a valuable reference in the areas of asset management, inventory systems, and smart solutions in educational institutions. It may serve as a foundation for further exploration of digital asset management systems and may provide insights into addressing challenges associated with manual processes.

Technical Definition of Terms

Agile Methodology. A software development and project management framework that emphasizes iterative progress, collaboration, customer feedback, and flexibility in response to changing requirements

Analysis. The process of examining data or information in detail to identify patterns, trends, or insights, often to support decision-making or problem-solving.

Asset. Any resource with economic value that an individual, company, or country owns or controls with the expectation that it will provide a future benefit.

Asset Management System. A digital platform or tool used to track, manage, and monitor the status, location, usage, and condition of assets within an organization.

Discrepancy. A difference or inconsistency between recorded data and actual values, often referring to inventory or asset records.

Forecasting. The process of estimating or predicting future trends, behaviors, or outcomes, often using historical data and statistical models, especially to anticipate demand or equipment failure.

Intermittent. Occurring at irregular or random intervals; not continuous or steady. Commonly used to describe system faults, signal issues, or equipment behavior that starts and stops unpredictably.

Iterative. A process where the design of a product or application is improved by repeated review and testing.

Quick Response Code (QR) Code. A type of two-dimensional barcode that stores information in the form of black and white squares, easily scanned by digital devices to quickly access the data.

Real-Time Monitoring. The continuous observation and updating of inventory and asset status in the system, provide immediate data for decision-making.

Resource Allocation. The process of assigning available resources (such as materials, tools, and personnel) to tasks or projects in a manner that optimizes efficiency and effectiveness.

Scrum Methodology. A flexible, agile framework used for managing complex projects, particularly in software development. It emphasizes iterative development, collaboration, and continuous improvement.

Stock Card. A document or digital record is used to track the quantity, movement, and status of inventory items, including details on incoming and outgoing stocks.

Useful Life. The period during which an asset remains functional and cost-effective for its intended purpose is often tracked for maintenance and replacement planning.

Chapter II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews relevant literature and studies on asset management systems, analyzing global technological solutions with cloud implementation challenges. The review emphasizes the need for cost-effective, scalable digital solutions that enhance asset utilization while supporting data-driven decision-making in institutional environments.

Related Literature

Modern asset management systems (AMS) require structured frameworks to ensure operational efficiency, as outlined in the ISO 55000:2024 standard (ISO, 2024), which establishes globally recognized principles for lifecycle optimization and data-driven governance. MEAMS system directly implements these principles through its integrated QR-code tracking and predictive analytics capabilities. The system addresses three key ISO requirements: (1) lifecycle management through real-time equipment status updates, (2) risk-based maintenance planning via machine learning-powered forecasting algorithms, and (3) value realization through dashboard visualizations of asset utilization patterns. While maintaining compliance with ISO's strategic framework, the AMS demonstrates how localized SQL database architecture can effectively support advanced asset management functions without requiring extensive cloud infrastructure. This adaptation proves particularly valuable for institutional maintenance operations where reliable, scan-verified data forms the foundation for both regulatory compliance and operational efficiency.

As emphasized by Katina et al. (2024), effective asset management in critical infrastructure requires a governance approach, as demonstrated through Asset Integrity Management (AIM) systems in high-risk industries including nuclear energy, power generation, and hydrocarbon transportation. The authors present insights from the 2024 AIM-CI conference, which brought together industry experts to share concepts, processes, and technologies for optimizing infrastructure reliability and safety. The conference proceedings identified key gaps in current asset management practices, emphasizing the need for integrated governance frameworks to address systemic risks. The findings underscore the value of cross-industry collaboration and advanced methodologies in enhancing asset performance and longevity. These principles are highly relevant to educational infrastructure management, where robust governance and risk mitigation strategies are equally critical for ensuring safe and sustainable facilities.

As highlighted by Preethichandra et al. (2023), smart technologies are transforming asset management through innovations like Structural Health Monitoring (SHM) and IoT-enabled predictive maintenance. Their research demonstrates how sensor networks and non-destructive testing methods (e.g., ultrasound, LiDAR) enable real-time infrastructure monitoring without operational disruptions. The study further highlights AI-driven deterioration prediction and Digital Twin integration as breakthrough approaches for optimizing maintenance schedules and lifecycle management. These data-driven solutions—particularly relevant for large-scale facility management—demonstrate measurable improvements in safety standards, cost efficiency, and asset longevity when applied to critical infrastructure systems

In the work of Torrecampo (2023), the effectiveness of data visualization in AMS was demonstrated through their database monitoring solution. Similar to their approach of presenting historical data in analytical dashboards, the study AMS for UDM's MED implements a comprehensive dashboard interface for tracking maintenance equipment. While their system focused on database performance metrics, our dashboard is specifically designed to display maintenance asset utilization patterns, equipment status updates, and inventory levels, all powered by our localized SQL database with QR-code verification. This adaptation applies Torrecampo's dashboard principles to physical asset management, replacing their web-based database monitoring with our localized system tailored for maintenance operations.

As explained by Fauziah and Mediawati (2024), assets originate from “wealth” in Indonesian, stating that anything of economic value that can be owned by people, businesses, or the government and can be evaluated financially is considered an asset. Effective asset management in government sectors requires systematic optimization through four critical phases: inventory, legal audit, valuation, and supervision. MEAMS system directly implements these principles by transforming its conceptual framework into operational reality. The system's QR-code tracking automates their inventory phase through digital recording of equipment conditions, while the analytical dashboard fulfills their supervision requirement with real-time maintenance asset monitoring. Where the authors emphasize financial valuation, our AMS introduces operational valuation through equipment utilization analytics that predict optimal maintenance and replacement schedules. This digital approach not only achieves Fauziah and Mediawati's optimization

objectives but also specifically adapts them for public sector maintenance operations where physical asset reliability is dominant.

As stated in the National Asset Management Plan (NAMP) 2022–2023, the critical role of AMS in modern governance is highlighted through its importance for efficient and cost-effective management of non-financial assets. The document underscores the need for an integrated Asset Management Information System (AMIS) to support data-driven decision-making, lifecycle management, and risk mitigation. It also stresses the value of open data and public disclosure, advocating for transparency in asset management through digital platforms. The NAMP further identifies the National Asset Registry System (NARS) as a key tool for inventorying and monitoring government assets, demonstrating the shift toward web-based solutions. Additionally, the plan highlights the importance of capability-building and training to ensure effective AMS implementation, aligning with the need for institutional adoption of such systems. These points collectively affirm the relevance of web-based AMS in enhancing accountability, sustainability, and service delivery in today's institutions.

As mandated by COA Circular No. 2020-006 (Commission on Audit, 2020), effective management of Property, Plant, and Equipment (PPE) is critical for public institutions. The circular mandates the digitalization of asset records to enhance transparency, accountability, and operational efficiency. Implementing our AMS for UDM's MED directly aligns with this COA mandate by establishing a compliant digital tracking system for maintenance assets. Specifically, it requires government agencies, including state universities, to maintain updated property cards, conduct regular inventory inspections, and implement digital tracking systems to minimize discrepancies and ensure

accurate financial reporting. By enforcing these measures, the COA aims to strengthen governance and reduce risks associated with manual record-keeping, aligning with modern public sector accountability standards.

As discussed by Kusumojati and Mediawati (2023), higher education institutions often lack integrated systems for asset monitoring, leading to inefficiencies in data tracking and condition assessment. Their Systematic Literature Review (2013–2023) underscores the need for automated solutions—an issue the system addresses through its QR-code-enabled tracking and local SQL database, which ensures real-time asset visibility without requiring web-based infrastructure. While their study advocates for centralized web systems, the study demonstrates how localized digital tools (scannable property cards, offline-capable mobile verification) can similarly resolve traceability issues identified in their research. The AMS' condition-based categorization and maintenance dashboards directly respond to their call for better decision-support tools, although fit for institutional maintenance workflows rather than broad asset types.

In the research of Ahmad, the development of an electronic property inventory system titled "e-AIMSS (Electronic Asset Inventory and Management System in School)" was designed to optimize school resource management and organizational productivity. The study employed the "ISSO" framework (Ignite, Strategic, Systematic, Operationalize) and a waterfall model approach, demonstrating that real-time digital monitoring significantly improved asset custodianship, reliability, and stakeholder engagement—key objectives for AMS in the MED. Findings revealed that the system enhanced transparency in budget allocation and audits, while comparative testing proved its superiority over manual methods. The study's emphasis on process complexity in outsourcing decisions

and the system's ability to provide swift, precise accounting aligns with our goal of ensuring equitable access to school equipment and supplies. Ahmad's work underscores the broader applicability of such systems in educational institutions, reinforcing the potential of the study to transform traditional inventory processes into efficient, digitally driven workflows.

In the findings of Odasco and Saong (2023), the University of Baguio's inventory management system offers relevant considerations for our desktop-based AMS with QR scanning functionality. Their findings regarding barcode implementation challenges (reported by 38.6% of respondents) particularly inform our AMS design, prompting us to implement a robust QR code system with: (1) standardized label placement guidelines, (2) high-contrast QR designs for better scanning reliability, and (3) offline-capable mobile scanning for field verification. While their study identified inventory visibility gaps (46.5% of respondents), our AMS addresses this through a centralized desktop interface that maintains real-time synchronization with mobile scanning data. The system architecture specifically avoids cloud dependencies, instead utilizing local network synchronization between desktop terminals and mobile scanning devices. This approach maintains the cost-effectiveness recommended by Odasco and Saong while providing more reliable asset tracking than their studied implementation.

As pointed out by Campbell et al. (2022), excellence in asset management requires a comprehensive lifecycle approach integrating cost analysis, risk assessment, and reliability-centered maintenance strategies. The system embodies these principles by implementing predictive maintenance capabilities through its analytical dashboard, which processes QR-verified equipment data to forecast maintenance needs and optimize

replacement schedules. The system specifically addresses their emphasized strategies by: (1) applying life cycle costing models to maintenance decisions, (2) incorporating risk-based prioritization of equipment servicing, and (3) enabling predictive maintenance through algorithm-driven failure pattern recognition. Where Campbell et al. highlight the importance of advanced diagnostics, our AMS demonstrates how even localized SQL-based systems can achieve sophisticated predictive capabilities when fed with consistent scan-verified operational data. This implementation proves particularly valuable for public sector maintenance operations where maximizing asset lifespan and minimizing downtime are critical to service delivery

In the research conducted by Ladia, Cuaresma, and Deynata (n.d.), a study titled "Web-Based Abacus Ship Management Asset Tracking and Order Processing with Decision Support System" addressed challenges in manual inventory workflows by developing an automated system for ABACUS Ship Management Ltd., which streamlined order tracking, stock monitoring, and report generation—functionalities critical to AMS for the school's MED. Their findings demonstrated the system's effectiveness in supporting multi-user access, supplier management, and real-time data accuracy, all of which align with our goal of managing school equipment and supplies efficiently. The study also emphasized the system's user-friendly interface and reliability, key features we aim to replicate to reduce manual errors in our institution. While their context focused on maritime logistics, the study's framework for centralized asset tracking and decision support offers valuable insights for educational settings. The researchers' recommendations, such as SMS notifications and mobile accessibility, further suggest scalable enhancements applicable to our system.

In the work of Javier, Parala, and Rifion, a web-based management information system was developed with 2D mapping and predictive analytics to address operational inefficiencies at the Sta. Cruz River Irrigation System. While their context focused on agricultural irrigation, their methodology using the Iterative Waterfall Model and system evaluation through Likert scale analysis demonstrates an effective approach to modernizing legacy systems—a challenge parallel to our school's MED's need to upgrade manual asset tracking. Their incorporation of predictive analytics for forecasting suggests potential applications for our AMS, such as predictive maintenance scheduling for school equipment. The study's emphasis on data centralization and visualization through 2D mapping could inspire similar geospatial tracking features for our school's physical assets. Furthermore, their findings regarding system acceptability among users reinforce the importance of user-centered design in the study's development process.

Related Study/Systems

In the study of Iluore et al. (2020), the researchers focused on an industrial company where they discovered critical issues due to the lack of modern technology for tracking and monitoring movable assets. Specifically, there was no established system to oversee the real-time location, usage, or condition of assets, even though these assets frequently moved to meet customer demands. To address this gap, they developed a Real-Time Equipment Monitoring System that integrates GPS, barcode scanning, and RFID technology. This system provided up-to-date information on the assets' location, operational status, and maintenance history. As a result, the company experienced a 30.8% increase in asset utilization. The benefit analysis in the study showed that the new

system helps maintenance engineers and managers access real-time data and equipment details, improving decision-making and maintenance planning.

As noted in the work of Rijadi and Suakanto (2024), the development of an asset management information system at Telkom University implemented a web-based platform using the Feature-Driven Development (FDD) methodology, PHP with Laravel framework, and MySQL database. The system included building/room administration modules, asset categorization, asset search, and maintenance reporting. Testing showed an 87.5% acceptance rate, and the system improved efficiency in asset tracking and data integrity compared to manual methods. Future enhancements may incorporate RFID for real-time asset monitoring.

As described by Tungcul and Kummer (2021), their study focused on addressing the challenges encountered by the Supply Office staff of Cagayan State University in managing inventory through manual processes. The existing system posed significant difficulties in record-keeping, tracking, report generation, and decision-making, which led to inefficiencies in supply and equipment management. To solve these problems, they developed a web-based system called SEIMTMS (Supply and Equipment Inventory Management and Tracking Management System), which integrated data mining techniques, specifically classification and clustering, to produce decision support reports. These reports aided the supply officer and university administration in forecasting frequently purchased supplies and frequently repaired equipment, allowing for more informed budget allocation and planning. The system was evaluated using ISO/IEC 25010:2011 Software Quality Standards and received a high rating from IT experts,

indicating strong compliance and effectiveness in revolutionizing inventory management processes.

In the research of Odasco and Saong (2023), the University of Baguio's Requisition, Procurement, and Inventory System (UBRPIS) was analyzed to identify the challenges faced in inventory management within higher education institutions. The study revealed that while employees were generally familiar with the processes of the system, such as requesting non-consumable items and conducting physical inventories, there were notable limitations. These included issues with the location of barcodes, the lack of non-consumable items in the system, and inefficiencies in updating and managing inventory records. Despite the system's automation, these challenges hinder the effective utilization of UBRPIS, leading to delays in the inventory management process. To address these issues, the study proposed a two-fold solution: first, to invest in an upgraded inventory management system that better meets the needs of all users, particularly in terms of handling non-consumable items and improving barcode functionality. Secondly, it recommended periodic staff training to ensure users are fully equipped to maximize the system's capabilities. These improvements aim to streamline inventory management processes, reduce errors, and enhance the overall efficiency of the system

As discussed by Wulandari et al. (2022), the study focused on designing a web-based asset management system to address the inefficiencies of the semi-computerized system used at PT. Indonesia Toray Synthetics. The primary challenge was the manual use of Microsoft Excel, which resulted in slow data processing and inaccuracies in tracking incoming and outgoing goods. To address the issue, the study proposed the integration of barcode technology into the asset management process, enabling real-time, accurate

tracking of items. This solution aimed to streamline inventory management, reduce human errors, and improve overall efficiency by providing faster and more reliable data.

In the work of Tong (2023), the focus was on enhancing warehouse inventory stocktaking through intelligent automation. The researchers developed a semi-automated system integrating barcode technology and distributed applications to improve inventory accuracy and monitoring. By leveraging big data analytics, the system provided real-time insights, reducing reliance on manual processes and minimizing errors. The implementation led to improved inventory precision and operational efficiency.

As presented in the study of Tanaman et al. (2023), a web-based inventory management system was developed for a small business enterprise in Pagadian City, Zamboanga del Sur, Philippines, to address the challenges of manual inventory management. The business, comprising four branches and a mobile store, was heavily reliant on paper-based systems, which led to inefficiencies, data inaccuracies, and difficulties in distributing reports. The proposed solution automated these processes through a web platform, electronically recording data, generating reports, and streamlining inventory management. The system was evaluated by both IT experts and end-users, who found it to be highly functional, efficient, compatible, usable, and reliable. The positive evaluation results suggest that the system has the potential to significantly improve business operations by enabling better inventory tracking and data-driven decision-making.

As described by Sharo (2020), the study aimed to address the challenges faced by the maintenance units at the University of Zakho. The university had separate sections

handling different maintenance tasks, such as electricity, computers, and inventories. However, the lack of an integrated management system led to communication issues between the maintenance staff and faculty members. Faculty and staff often did not know where to turn when facing a problem, which led to delays and frustration. By developing a web-based system that integrated all maintenance units, it became easier for faculty and staff to submit maintenance requests. The system automatically routed requests to the relevant maintenance section within the faculty or college, and if the problem could not be solved at that level, the request was forwarded to the main unit at the university presidency. Additionally, the system allowed the main unit to monitor the progress of the requests and conduct statistical analysis of the work being done.

In the research of Parilla et al. (2022), the study conducted on healthcare facilities in Ilocos Norte examined how manual inventory tracking contributed to stock-outs, overstocking, and expired supplies. The absence of digital systems hindered inventory visibility, leading to service interruptions and patient dissatisfaction. To combat this, the study suggested implementing a digital inventory management system integrated with automated alerts and real-time tracking. As a result, inventory accuracy and service delivery in healthcare facilities were notably improved.

As emphasized by Moreno (2023), road and bridge maintenance challenges in Zamboanga Peninsula, Philippines, revealed critical gaps in funding, skilled personnel, and maintenance strategies through DPWH data analysis. The study revealed that inadequate maintenance led to traffic congestion, safety risks, and economic disruptions in the region. Key findings emphasized the need for preventive maintenance approaches rather than reactive repairs to ensure the longevity of infrastructure. The research

recommended increased budget allocations and capacity-building programs for maintenance teams to address these systemic issues. These insights are particularly relevant for developing web-based asset management systems that could automate maintenance tracking and resource allocation. The engineering-focused perspective provides valuable technical considerations for infrastructure management platforms.

As stated in the study of Murniasih and Wiranata (2024), the development of an Asset Inventory Management Information System using the DeLone and McLean Success Model approach addressed challenges in managing over 9,000 devices at an Indonesian company, where manual processes led to inefficiencies and data inaccuracies. They developed a web-based system and evaluated its success using the DeLone and McLean model, measuring six variables: Information Quality, System Quality, Service Quality, Usage, User Satisfaction, and Net Benefits. Testing with Likert-scale questionnaires (5 respondents) showed high performance across all variables, with User Satisfaction (88.89%) and Net Benefits (87.5%) scoring highest. The study concluded that the system improved accuracy and efficiency in asset tracking, though it recommended expanding respondent samples for future research.

In the research conducted by Gong and Hui (2024), an intelligent inventory management system was developed to address the challenges of manual tracking, overstocking, understocking, and discrepancies in modern enterprises. The system integrates automation and real-time data processing using technologies such as Spring Boot and Vue, combined with UML diagrams for system design. It encompasses core functions like warehousing, inventory management, and report data analysis, while also considering non-functional requirements such as ease of use, scalability, and security.

The implementation of this system aims to improve the efficiency and accuracy of inventory management, reduce errors, optimize costs, and enhance the competitiveness of enterprises through automation and digitalization.

Synthesis

The foreign literature establishes that effective asset management revolves around four core principles: creating value, ensuring organizational alignment, demonstrating leadership commitment, and providing assurance through proper controls. These studies reveal how modern digital solutions like IoT networks, artificial intelligence, and blockchain technology are revolutionizing asset management through real-time monitoring capabilities, data-driven decision making, and process automation. Such technological implementations have demonstrated concrete benefits including minimized equipment downtime, extended asset life cycles, and substantial reductions in manual labor requirements.

In contrast, Philippine-based research presents a different reality where many institutions still struggle with legacy systems and paper-based processes. Common challenges include inconsistent data quality, limited technology budgets, and workforce skill gaps that hinder digital transformation efforts. These constraints often result in inefficient resource utilization, compliance risks, and lack of transparency in asset tracking and reporting.

To bridge these issues, scholars recommend adopting practical digital solutions tailored to local constraints. Web-based platforms, QR code systems, and mobile applications emerge as particularly suitable options that balance functionality with

affordability and ease of use. These technologies offer significant advantages over traditional methods by improving data accuracy, streamlining workflows, and strengthening accountability measures. Government sector experiments with cloud-based management systems have already shown promising results in enhancing regulatory compliance and service quality.

The collective findings emphasize how strategic technology adoption can transform asset management practices. Modern systems not only boost operational performance through automation and analytics but also support better strategic planning and risk management. When properly implemented, these solutions enable organizations to maximize asset productivity, control costs, and align physical resources with institutional goals - ultimately creating a foundation for sustainable, evidence-based management practices.

Technological Background

The system utilized a cloud hosted MongoDB database to securely store asset records, repair histories, and stock levels. Using its flexible document-based structure, the system enabled efficient data handling and supported dynamic asset information. Asset details were instantly retrievable via scannable QR codes linked to the database. The system incorporated two major AI-powered and rule-based decision features integrated into its dashboard visualizations. The first feature focused on predictive forecasting, where a SARIMA model was utilized to generate trend forecasts for both overall supply levels and the total quantity of operational equipment. These forecasts were displayed as

interactive line graphs on the system dashboard, allowing administrators to monitor future inventory and equipment conditions based on historical patterns.

The second decision feature involved the threshold-based alert and notification system, which operated independently from the SARIMA model. For supplies, understock alerts were triggered based on predefined thresholds derived from past consumption data, with editable settings that allowed administrators to adjust limits per category or item. For equipment, maintenance alerts appeared only when a repair request was reported, while replacement alerts were generated through Life Cycle Cost (LCC) analysis—specifically when repair costs exceeded replacement value, when frequent repairs occurred, or when an item had surpassed its useful life. These alerts and forecasts together enabled data-driven monitoring and decision-making across MED's operational assets.

Development followed the Scrum framework, with two-to-four-week sprints and cross-functional collaboration that iteratively refined requirements and validated outputs. The frontend, built with React.js and TypeScript, displayed these recommendations through an intuitive dashboard with Chart.js visualizations of asset levels, status, and matrix. The backend combined Python (FastAPI) for machine learning models with Node.js (Express) for API management, while Redis caching ensured responsive performance. Designed primarily for cloud-based network operation, the system accommodated offline scanning with automatic synchronization upon reconnection, though its accuracy remained dependent on consistent asset records and LAN connectivity for real-time updates. This integrated approach enabled data-driven decisions for both asset maintenance and inventory management within MED's operational framework.

Chapter III

METHODOLOGY

This chapter presents the detailed operational framework, feasibility assessments, and techniques utilized in developing MEAMS for the MED. It discusses the methods and processes followed during the system's design and development, including requirement analysis, feasibility studies, system design, development, and testing. Each diagram is elaborated to demonstrate how data flows throughout the system and how users interact with MEAMS in achieving efficient asset tracking and management.

Methodology Used

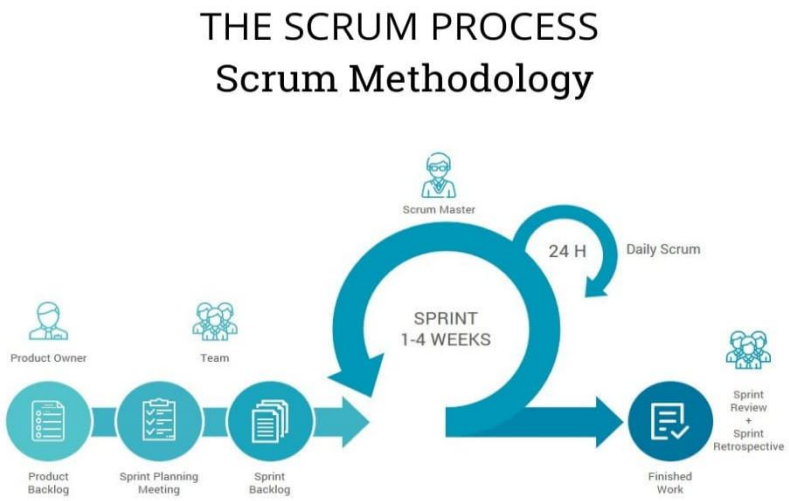


Figure 3-1. SCRUM Methodology

The proponents adopted the Agile methodology, specifically the Scrum framework, for the development of MEAMS. This methodology was chosen because it provided a step-by-step and adaptable method for handling software development and project tasks, focusing on delivering small, useful parts of the system through regular collaboration with

the MED. The Scrum framework was implemented through a set of development phases known as sprints, which lasted between two to four weeks. At the beginning of each sprint, the team conducted a sprint planning meeting to define specific goals and determine which features or tasks from the product backlog would be addressed, such as digitized stock cards, analytical dashboard, repair history, and QR code generation. These tasks were prioritized based on MED's operational needs at the time. During the sprint, the team held daily meetings to monitor progress and maintain consistent coordination among members. At the end of each sprint, the team conducted a sprint review to present the finished tasks or features to the MED and gather their feedback. This review helped the team assess whether the output met the necessary requirements and make any needed adjustments for future sprints. This cyclical approach supported continuous feedback, ensured that the system components were completed on time, and allowed the project to remain adaptable to evolving requirements throughout the development of MEAMS.

Requirement Analysis

The MEAMS aimed to address the significant challenges faced by the MED of UDM in managing their assets. At that time, the processes for recording equipment, supplies, and repair activities were predominantly manual, relying on logbooks and spreadsheets. This manual approach frequently led to delays in information retrieval, instances of missing or incomplete records, and considerable difficulties in accurately tracking and updating inventory levels.

The people involved in these manual operations included various roles within the MED’s staff, such as repair technicians responsible for logging repairs, inventory clerks

who managed stock levels, and administrative personnel who oversaw record-keeping. These business activities occurred regularly within the MED, in various storage areas for supplies and equipment, and even within specific operational units where assets were utilized. The timing of these manual procedures was often event-driven and periodic, but the manual nature consistently slowed down the workflow and introduced potential errors, as information was physically written down, manually updated across different logs or spreadsheets, and communication often relied on physical documents or verbal updates.

To address these inefficiencies, MEAMS provided key functional requirements. It enabled authorized users to track the real-time location and status of equipment and allowed the system to forecast potential equipment repair needs based on historical usage data or predefined lifespan parameters. Furthermore, it offered the capability to forecast supply levels to proactively identify potential overstocking or understocking situations and provided digital stock cards for each asset, offering a centralized view of its key information. The system also maintained a detailed digital repair history for each piece of equipment, including dates, descriptions, and costs, and generated automated alerts for critical events such as low stock levels or equipment nearing its end of life. Finally, MEAMS facilitated the generation of digital request forms for supplies or repair and displayed key asset and supply information through interactive dashboards utilizing graphs and charts for easy visualization and understanding of the current status. The information that MEAMS managed included detailed records of each asset, supply inventory, repair history, user roles and permissions, and system logs.

The system was developed over a span of approximately eight (8) months, from March to November 2025, using the Agile methodology with the Scrum framework.

Instead of following rigid phases, the development was carried out through a series of iterative sprints, each lasting two (2) to four (4) weeks. These sprints allowed the team to focus on prioritized features, gather feedback from the MED, and make continuous improvements throughout the project. At the start of each sprint, the team conducted sprint planning sessions to define specific goals and select tasks from the product backlog, such as real-time tracking, forecasting, dashboard components, and repair history. Regular stand-up meetings were held to monitor progress and ensure close collaboration among members. At the end of each sprint, the team conducted sprint reviews and retrospectives to present completed features, gather feedback from the MED, and identify areas for improvement.

Organizational Structure

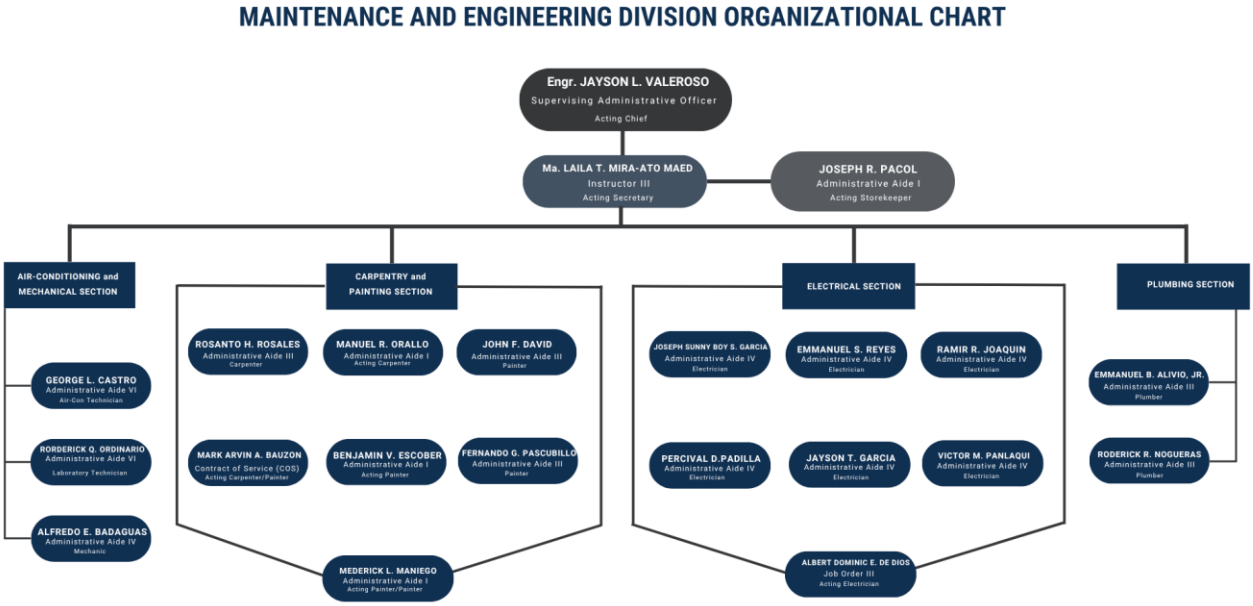


Figure 3-2. MED Organizational Chart

The organizational structure of the MED outlines the hierarchy and distribution of responsibilities essential to maintaining institutional operations. At the top, Engr. Jayson L. Valeroso serves as the Head, supported by Ma. Laila T. Mira-Ato MAEd. as Acting Secretary and Joseph R. Pacol as Acting Storekeeper. The division is composed of four (4) sections: (1) Air-Conditioning and Mechanical, handling HVAC, laboratory, and mechanical systems; (2) Carpentry and Painting, responsible for construction, repairs, and finishing works; (3) Electrical, managing installation, maintenance, and troubleshooting of power systems; and (4) Plumbing, ensuring reliable water supply and sanitation. Together, these sections provide comprehensive maintenance and technical support to keep institutional facilities safe, efficient, and fully operational.

Team Organizational Profile



Figure 3-3. Team Organizational Profile of MEAMS

Our team is organized with specific roles to ensure the project is completed efficiently. Ethyl Jane V. Garcia leads the group as the Project Manager, overseeing the schedule and overall progress. The technical building of the system is handled by Christoper Uriel B. Caole and Crystal Shane E. Piamonte, who serve as Back-end Developers and Researchers to create the main functions and gather necessary data.

Justine S. Dazo manages the visual side as the Front-end Developer and also checks the system for errors as the QA Tester. Finally, Renan Jay S. Moral focuses on Documentation and Research, ensuring the paper is written correctly and all information is accurate.

Requirement Documentation

This section provides an overview of the requirements documentation, highlighting important feasibility figures that assess the benefits and practicality of the MEAMS.

Technical Feasibility

The technical feasibility of MEAMS is supported by lightweight, well-established technologies that run effectively on budget-friendly hardware. The system employs a React.js frontend for fast rendering, while Node.js and Python handle backend operations. MongoDB functions as the primary database, requiring only a modest setup of at least an Intel Core i3 or Ryzen 3 processor, 8GB RAM, and a 256GB SSD. Performance is enhanced through React's virtual DOM, MongoDB memory tuning, scheduled SARIMA processing for forecasting, and QR code generation via cached Python libraries. To prevent memory issues, CSV/XLSX imports are limited to 20MB. Unlike earlier offline designs, MEAMS now requires stable internet connectivity to ensure real-time synchronization of asset tracking, repair history alerts, and dashboard analytics across devices. Development and deployment costs are minimized through the use of free tools such as VS Code and MongoDB Community Edition, while ensuring compatibility with existing university PCs. This design ensures MEAMS remains technically feasible, cost-effective, and scalable for the MED of UDM.

Operational Feasibility

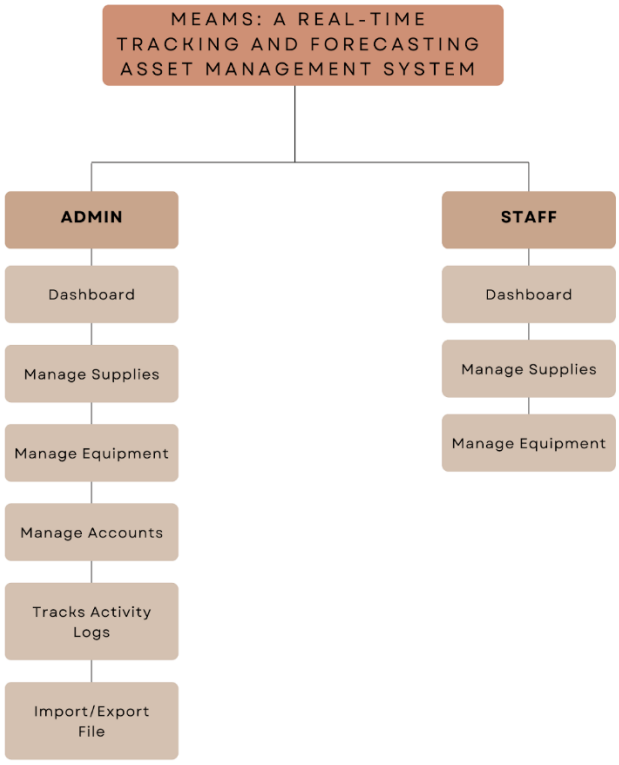


Figure 3-4. Functional Decomposition of MEAMS

The Figure 3-4 illustrates the functional decomposition of MEAMS, showing the functions available to administrators and staff. Both roles share access to the dashboard, supply management, and equipment management, while administrators have additional capabilities such as managing user accounts, tracking activity logs, and importing or exporting files. This structure ensures efficient operations, secure access control, and clear division of responsibilities within the MED of UDM.

Schedule Feasibility

The work breakdown structure for the MEAMS, illustrates the key stages of the project development process. It begins with the Requirements phase, which involves

conducting research, interviews, and problem gathering to define the system's purpose, solutions, and features. The Design phase focuses on planning the system's layout and creating the software design based on the identified requirements. During the Development phase, coding and iteration cycles are carried out to implement the planned features. The Testing phase ensures that all system functions operate correctly by checking features and resolving bugs. The Deployment phase includes evaluating the system with the selected client, while the Review phase involves maintaining and improving the system based on client feedback.

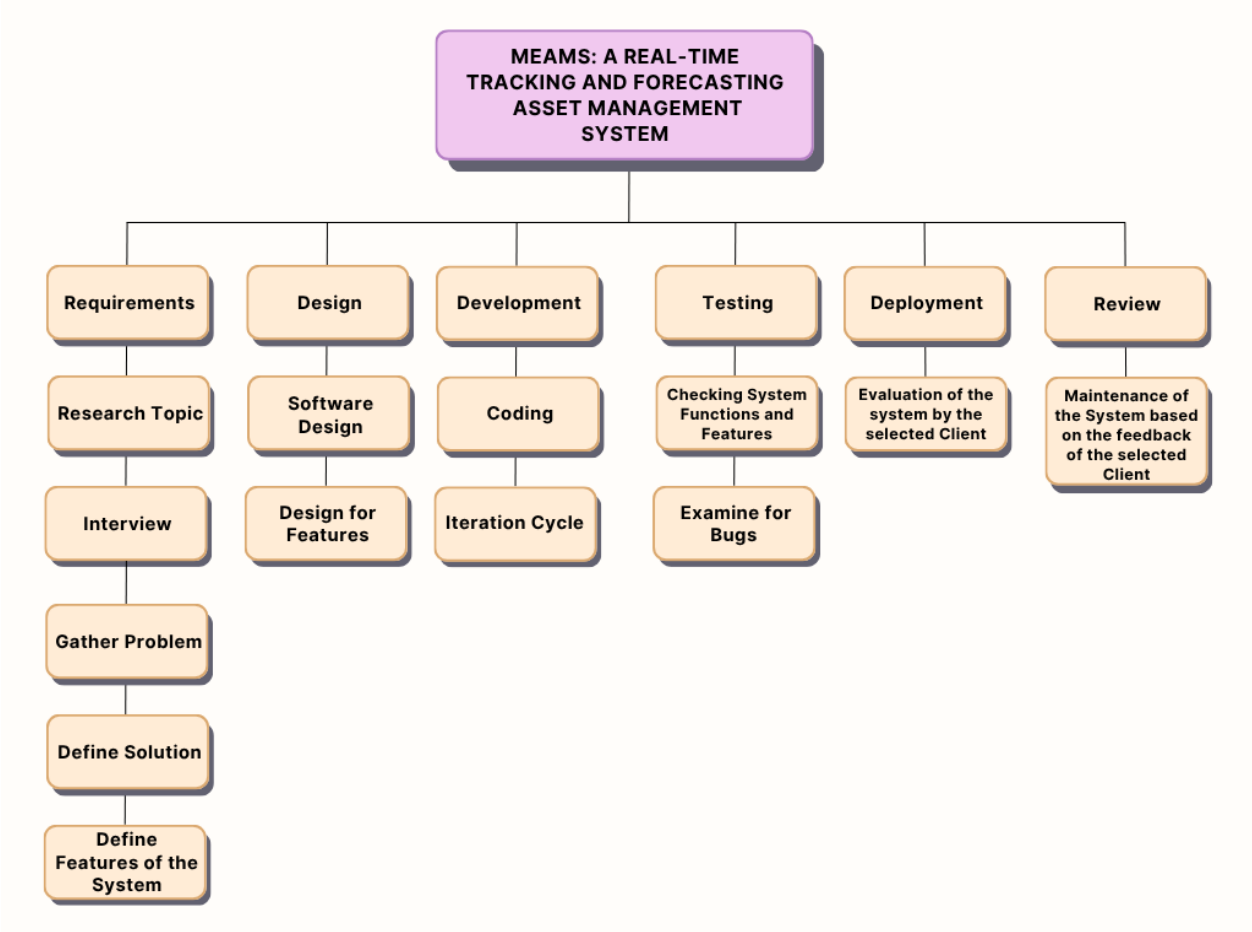


Figure 3-5. Work Breakdown Structure of MEAMS

Gantt Chart

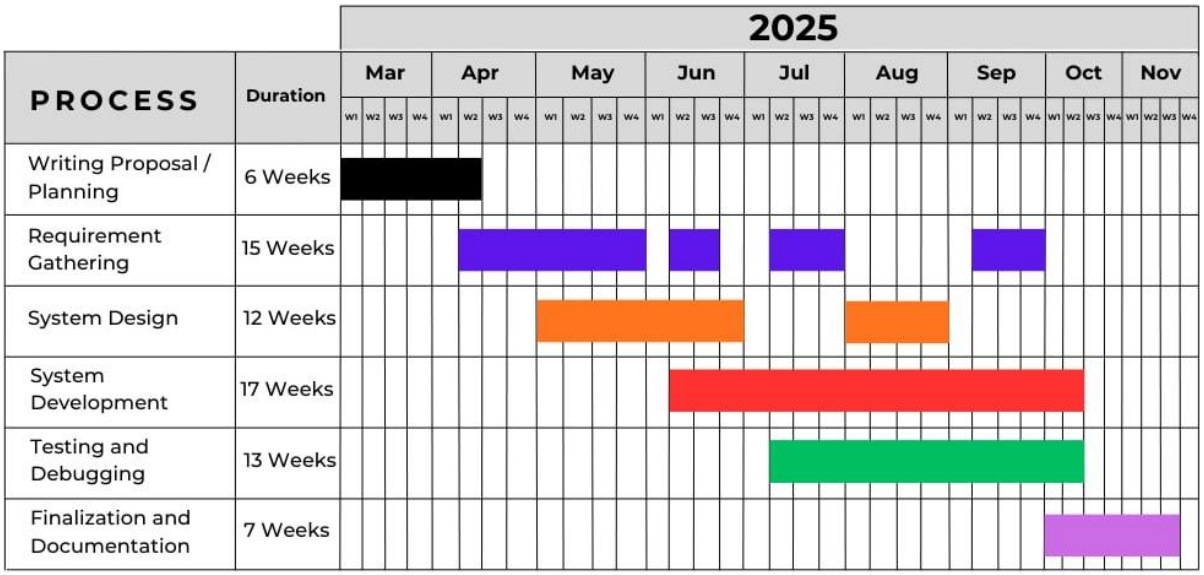


Figure 3-6. Gantt Chart of MEAMS

The project begins with the Planning phase from March to mid-April, where the objectives and scope of MEAMS are established. The Requirement Gathering phase follows from April to September and is emphasized under the Agile methodology. This phase is iterative, as requirements are continuously refined through research, feedback, and collaboration. The System Design and Development phases span from May to October, with Development continuing until the end of October. Testing and Debugging occur alongside development from July to October to ensure functionality and system stability. Finally, the Finalization and Documentation phase starts in mid-October to November, concluding the project with preparation for deployment.

Economic Feasibility

The proposed system has a total projected cost of ₱31,000.00, covering all essential resources for its development and operation. The major expense of ₱30,000.00 is allocated for hardware and devices used in building and testing the system, while

₱1,000.00 is allotted for web hosting to ensure online accessibility. These costs demonstrate the project's economic feasibility, as they are justified by the long-term benefits of enhanced efficiency, automation, and accessibility.

Table 3-1. Economic Feasibility of MEAMS

Cost Benefits of Analysis	Cost in Php	Benefits
Web hosting	Php 1,000.00	Ensures that the system is accessible online anytime, providing users with reliable access to data and services.
Hardware and Devices	Php 30,000.00	Provides the necessary tools for developing, testing, and operating the system efficiently.
Total:	**Php 31,000.00**	

Existing System Process

In Figure 3-7 below the MED supply management workflow starts when MED personnel identify the specific items or supplies required for upcoming repair work. Following this identification, they seek assistance from the storekeeper to locate these necessary materials within the storage area. The storekeeper then undertakes a check of the current inventory to ascertain the actual quantity of the requested items. A crucial decision point follows: if the stock is deemed sufficient to meet the needs, the storekeeper proceeds to issue the requested items or supplies directly to the MED personnel. Then, this transaction is logged, marking the completion of the supply process for those items. However, if the storekeeper finds that the existing stock is insufficient, a different path is initiated. In this scenario, a requisition form is prepared, detailing the needed supplies. This form then requires the signature and approval of the MED Head. Once approved, the

requisition form is forwarded to the Supply and Equipment Office (SEO) for preparation, initiating the procurement process. Eventually, the newly acquired supplies are delivered to the MED. Upon arrival, the storekeeper undertakes a check to verify the delivered items. These supplies are then properly stored in the designated storage room. Finally, regardless of whether the supplies were directly issued from existing stock or newly procured, the transaction is manually recorded.

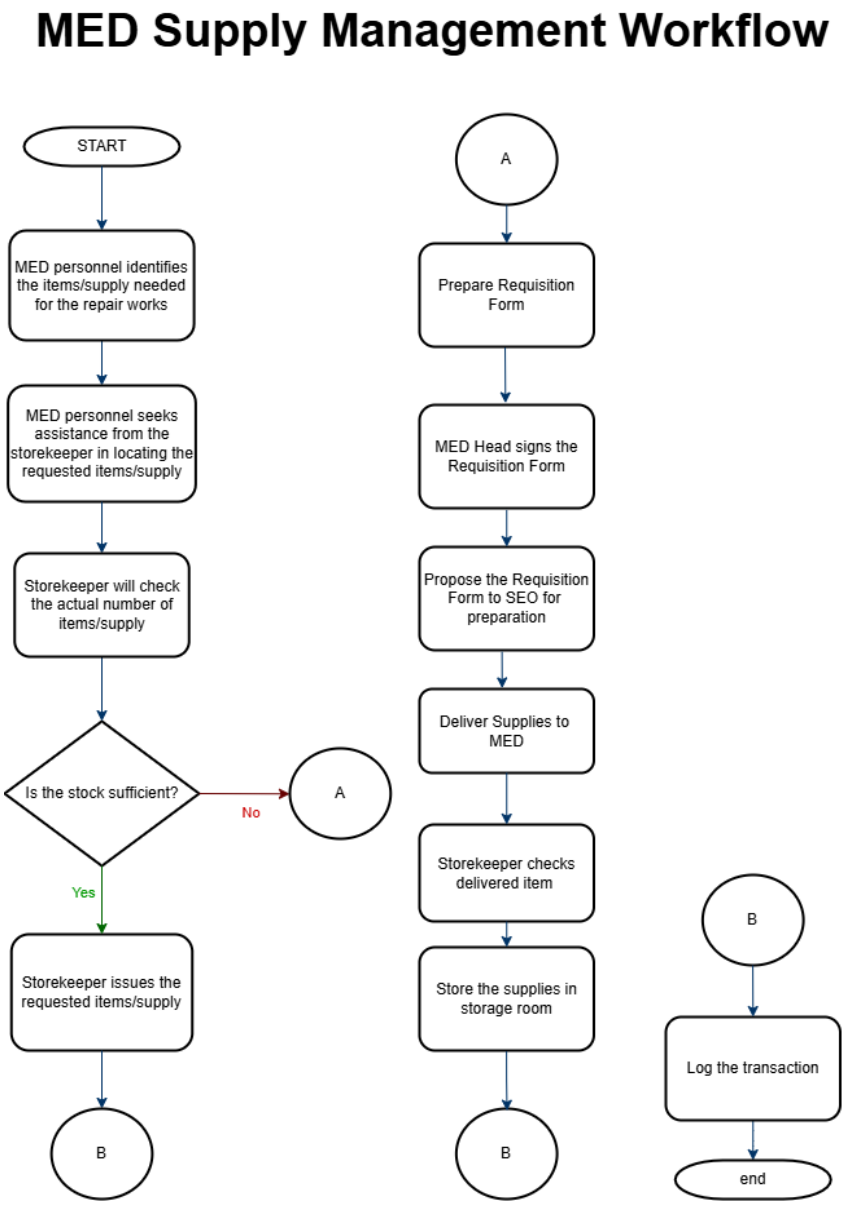


Figure 3-7. MED Existing Supply Management Workflow

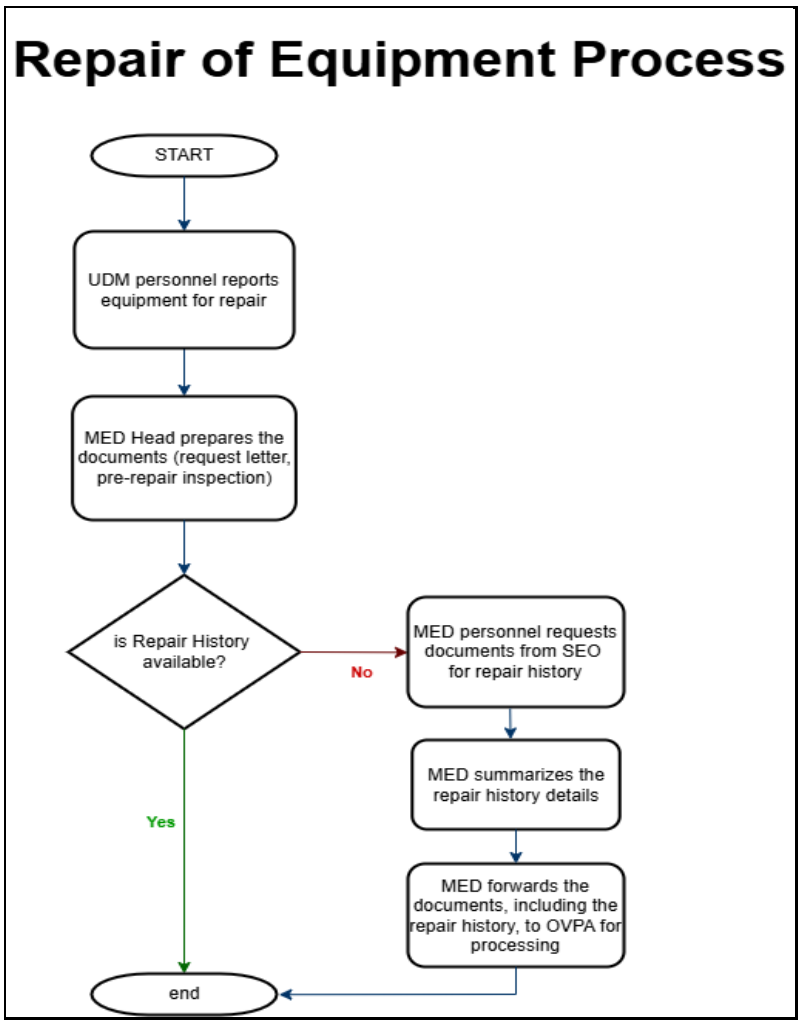


Figure 3-8. MED Existing Repair of Equipment Process

In Figure 3-8 the equipment repair process begins when UDM personnel identify a faulty piece of equipment and report it for repair. Following this report, the MED Head takes charge of preparing the necessary documentation, which typically includes a formal request letter outlining the need for repair and a pre-repair inspection report detailing the equipment's condition. A key step in this initial phase is determining if a repair history exists for the reported equipment. If a repair history is readily available, the process, implying that this information is sufficient to proceed with the repair. However, if no prior

repair history is found, MED personnel must request these records from the SEO. Upon receiving the repair history, the relevant details are summarized by the MED personnel. Finally, the initial documents prepared by the MED Head, along with the summarized repair history, are forwarded to the OVPA for further processing, after which the repair processes.

Design of the Software, Systems, Products, and/or Processes

The goal of the system design phase is to transform the requirements of “MEAMS: A Real-Time Tracking and Forecasting Asset Management System” into clear and accurate specifications that guide development. Following an agile methodology, the design process employed tools such as the Context Diagram, Data Flow Diagram, Use Case Diagram, and Activity Diagram to visualize system flow and break down complex requirements into manageable components. The architecture was structured into three layers: the front end, back end, and data management. The front end, developed with React.js, emphasized usability, consistency, and responsiveness, featuring a minimalist color scheme (gray, black, and white) with an optional light mode theme for user preference, and Chart.js integration for dynamic dashboard visualizations. The back end combined Node.js (Express) for API management and Python (FastAPI) for machine learning tasks, ensuring efficient performance and seamless integration between system components. The cloud-based MongoDB database facilitated efficient and reliable data management, while role-based access control and encryption mechanisms ensured robust security and data integrity. Agile iteration and stakeholder feedback played a key role throughout the design process, enabling flexibility, scalability, and adaptability for

future enhancements. This approach ensured MEAMS delivers a user-friendly, efficient, and secure platform tailored to the operational needs of the MED of UDM.

Context Flow Diagram of the System

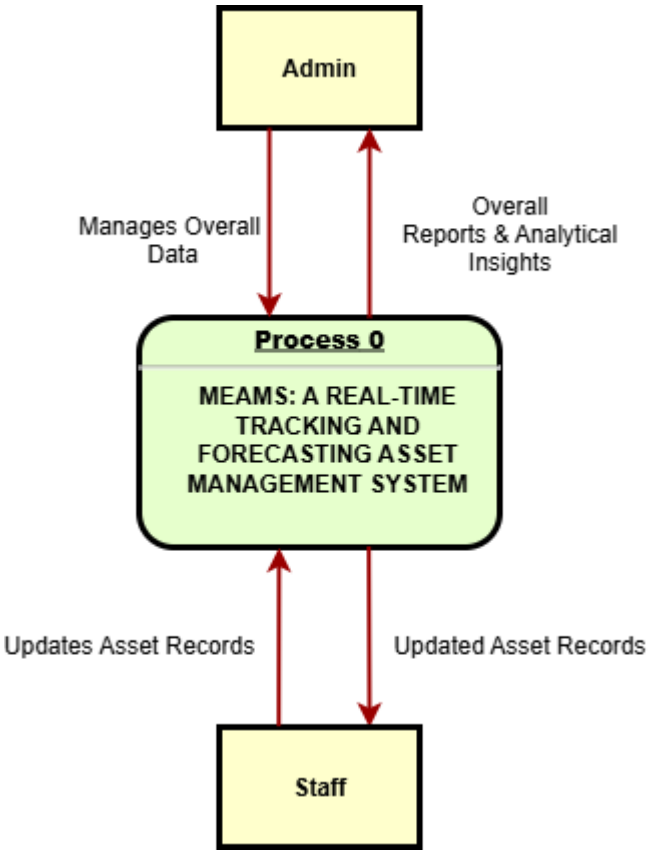


Figure 3-9. Context Flow Diagram of MEAMS

Figure 3-9 illustrates how data flows between the MEAMS: A Real-Time Tracking and Forecasting Asset Management System and its users. The system serves as the central hub for managing and processing asset-related information. The Admin manages overall data, oversees reports, and gains analytical insights for decision-making, while the Staff updates asset records and ensures accurate and timely information within the system. Data moves between users and the system as updates, reports, and records are exchanged. Through this structure, MEAMS ensures efficient coordination between

administrators and staff, promoting organized and transparent asset management operations.

Data Flow Diagram of the System

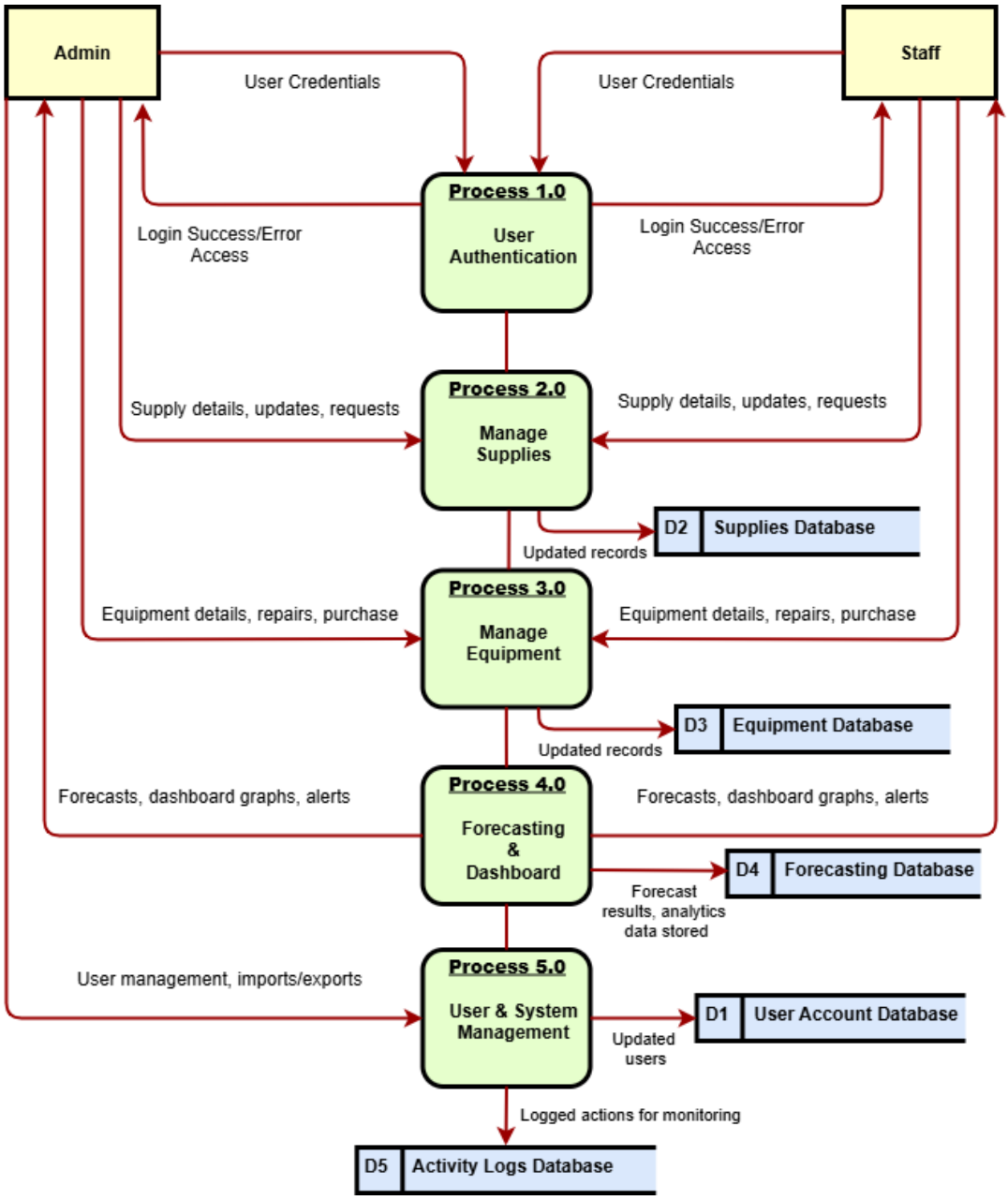


Figure 3-10. Data Flow Diagram of MEAMS

Figure 3-10 illustrates how data moves within the MEAMS and its interactions with users and databases. Admins and staff perform various functions such as logging in, updating supplies and equipment records, and viewing dashboards. The system processes these actions through interconnected modules—User Authentication, Manage Supplies, Manage Equipment, Forecasting & Dashboard, and User & System Management—each linked to dedicated databases. These include user accounts, supplies, equipment, forecasting, and activity logs databases, which securely store and update information in real time. The admin oversees user management, imports or exports data, and monitors activities, while staff handle daily operational tasks. Together, these processes ensure that MEAMS delivers accurate data tracking, seamless asset updates, and reliable forecasting for effective maintenance and engineering management.

Use-Case Diagram

In Figure 3-11 below presents the use case diagram of MEAMS, showing the distribution of functions between administrators and staff. Both administrators and staff share access to the system dashboard, where they can monitor asset information in real time. They also have the ability to manage their own profiles, report system bugs, and perform core operations such as managing supplies and equipment.

Certain privileges, however, are reserved exclusively for administrators. These include deleting supplies and equipment, managing user accounts, monitoring activity logs, and performing import or export of files for data backup and migration. By granting administrators broader control while allowing staff to focus on day-to-day asset operations, the system ensures accountability, accuracy, and efficient task execution.

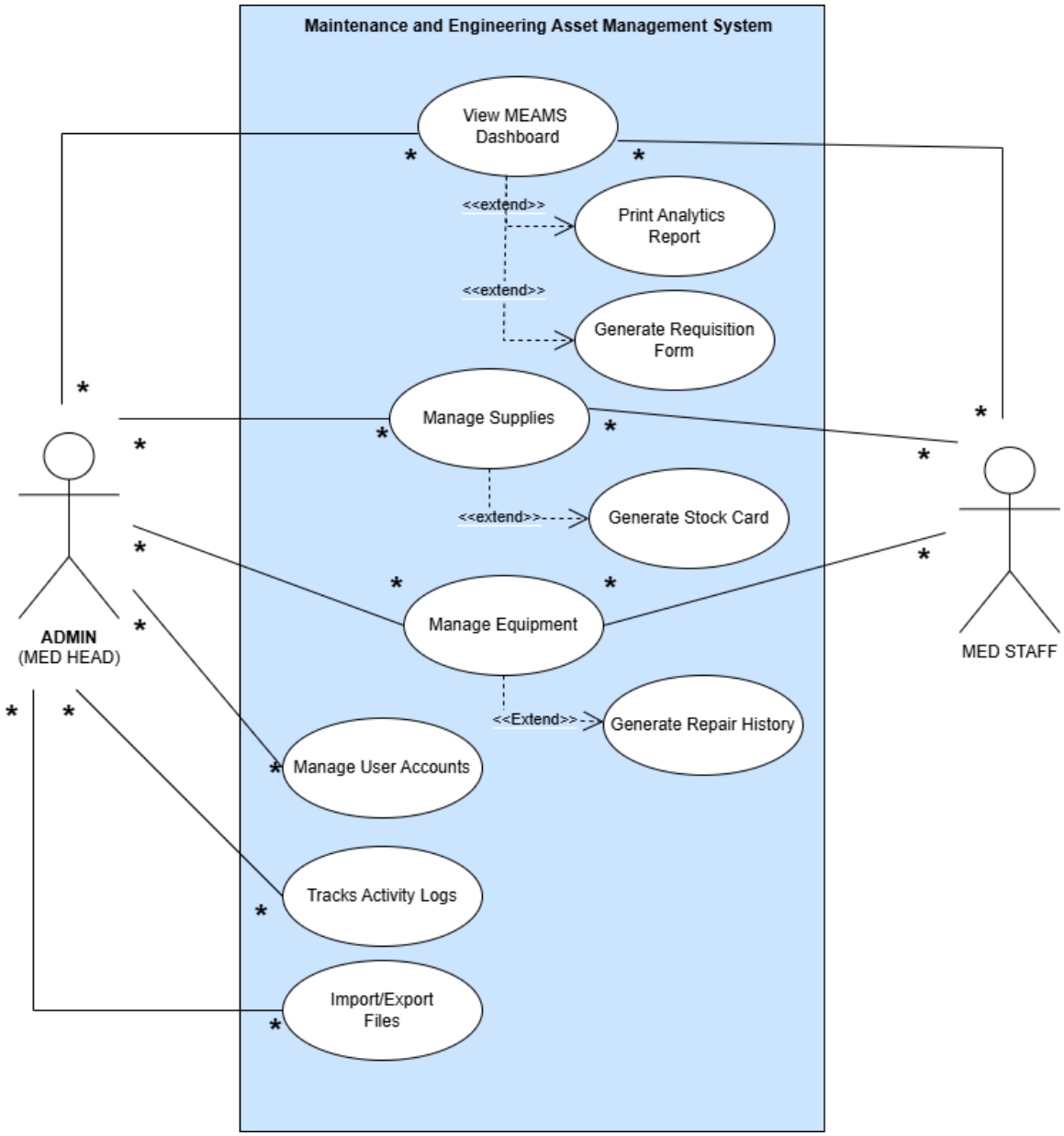


Figure 3-11. Use Case Diagram of MEAMS

Activity Diagram of the System

The activity flow diagram is a behavioral representation that visually captures the sequence of actions and decision points within the MEAMS. It is designed to clearly and systematically model the behavior of specific system processes, making them easier to understand, analyze, and implement. These diagrams illustrate how users interact with

the system—from the beginning of a task to its completion—providing a step-by-step visualization of the workflow involved in various system functions.

Each activity diagram highlights user navigation and system behavior in detail, ensuring that both developers and end users can follow the logic and structure of the system's processes. These diagrams not only assist developers in system design and implementation but also serve as a guide for users in understanding the operational flow and functionalities of the MEAMS platform.

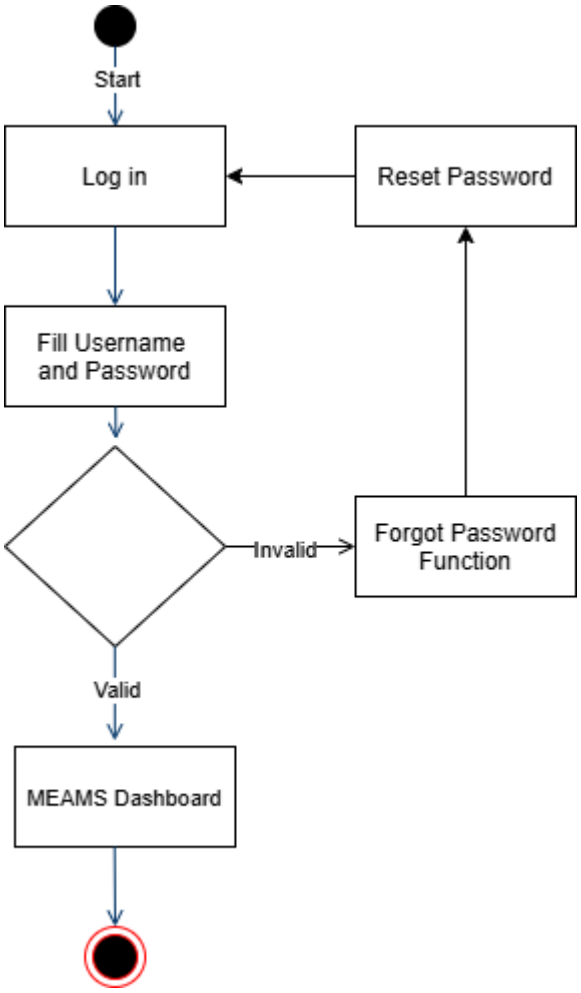


Figure 3-12. Activity Diagram of Login

In Figure 3-12, the login process begins when a user initiates the system. They are then prompted to "Log in," which involves filling in their username and password. The system validates these credentials. If the login information is deemed "Invalid," the process branches to the "Forgot Password Function," allowing the user to recover their account and subsequently "Reset Password." After a successful password reset, the user is directed back to the "Log in" step. If the initial username and password are "Valid," the user gains access to the "MEAMS Dashboard," and the login activity concludes.

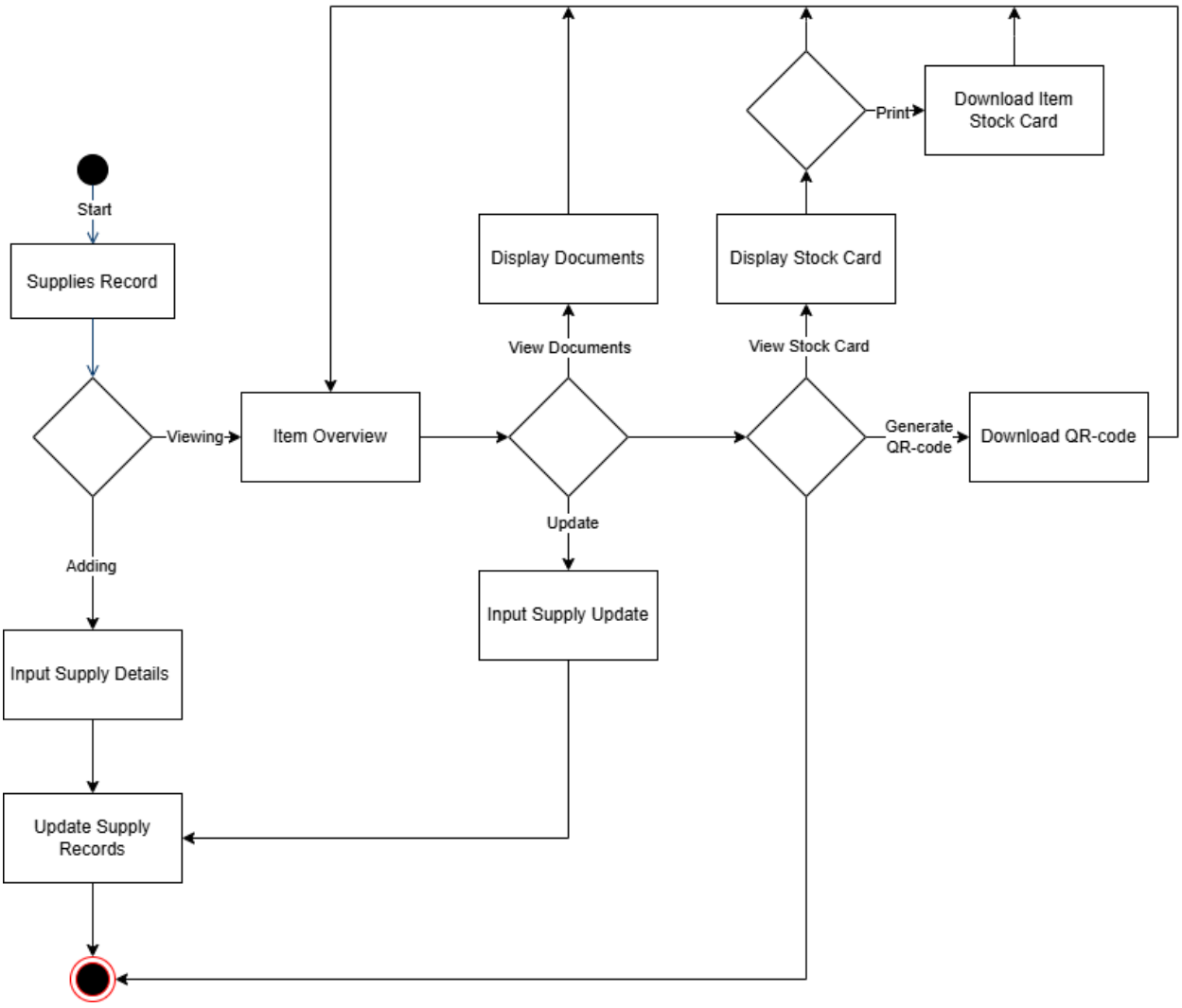


Figure 3-13. Activity Diagram of Supply

The Supplies Record activity commences with user access, offering two primary paths: viewing or adding items. Selecting "Viewing" leads to an "Item Overview" where users can "View Documents" (leading to "Display Documents" and returning to the overview) or "View Stock Card" (leading to "Display Stock Card" with optional actions to "Print" and "Download Item Stock Card" or "Generate QR-code" and "Download QR-code," both returning to the overview). Choosing "Adding" directs the user to "Input Supply Details" followed by "Update Supply Records." Additionally, from the "Item Overview," users can "Update" item details via "Input Supply Update," which then loops back to the "Item Overview." Ultimately, both the "Update Supply Records" and the pathways from the "Item Overview" conclude the activity, outlining a comprehensive flow for managing supply records through viewing, adding, and updating functionalities.

In Figure 3-14 below, the Equipment Record activity initiates upon user access, presenting options for viewing or adding equipment. Choosing "Viewing" leads to an "Item Overview," enabling users to "View Documents" (resulting in "Display Documents" and a return to the overview) or "View Repair History" (leading to "Display Repair History" with optional actions to "Print" and "Download Repair History" or "Generate QR-code" and "Download QR-code," both returning to the overview). Selecting "Adding" directs the user to "Input Equipment Details" followed by "Update Equipment Details." Furthermore, the "Item Overview" allows users to "Record Repair," which involves "Input Repair Details" and subsequently updates the "Item Overview." Ultimately, both the "Update Equipment Details" and the paths from the "Item Overview" conclude the activity, illustrating a

structured process for managing equipment records through viewing, adding, and recording repair functionalities.

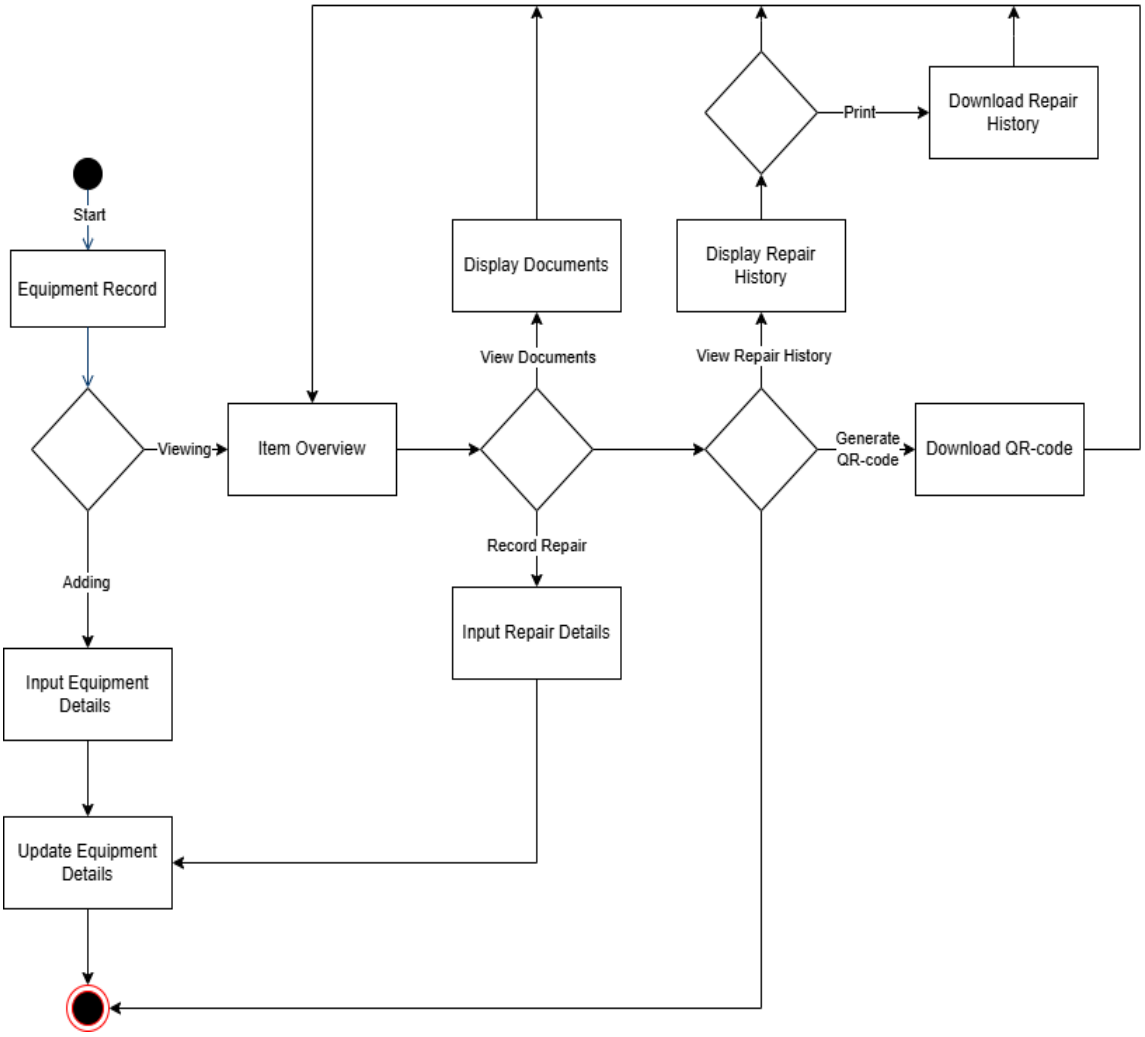


Figure 3-14. Activity Diagram of Equipment

In Figure 3-15 below, upon initiating the "Manage Accounts List," the admin is presented with existing accounts and can choose to "Delete account," which updates the accounts list after deletion. Alternatively, the admin can "Add Account," which requires inputting account details and then updates the accounts. From the "Manage Accounts List," admin can also "View" an "Account Overview," which then allows "Editing" that leads

to "Edit details" and subsequently an update to the accounts. Ultimately, whether an account is deleted, added, or edited, the "Update Accounts" step concludes the activity, reflecting the changes made to the account list.

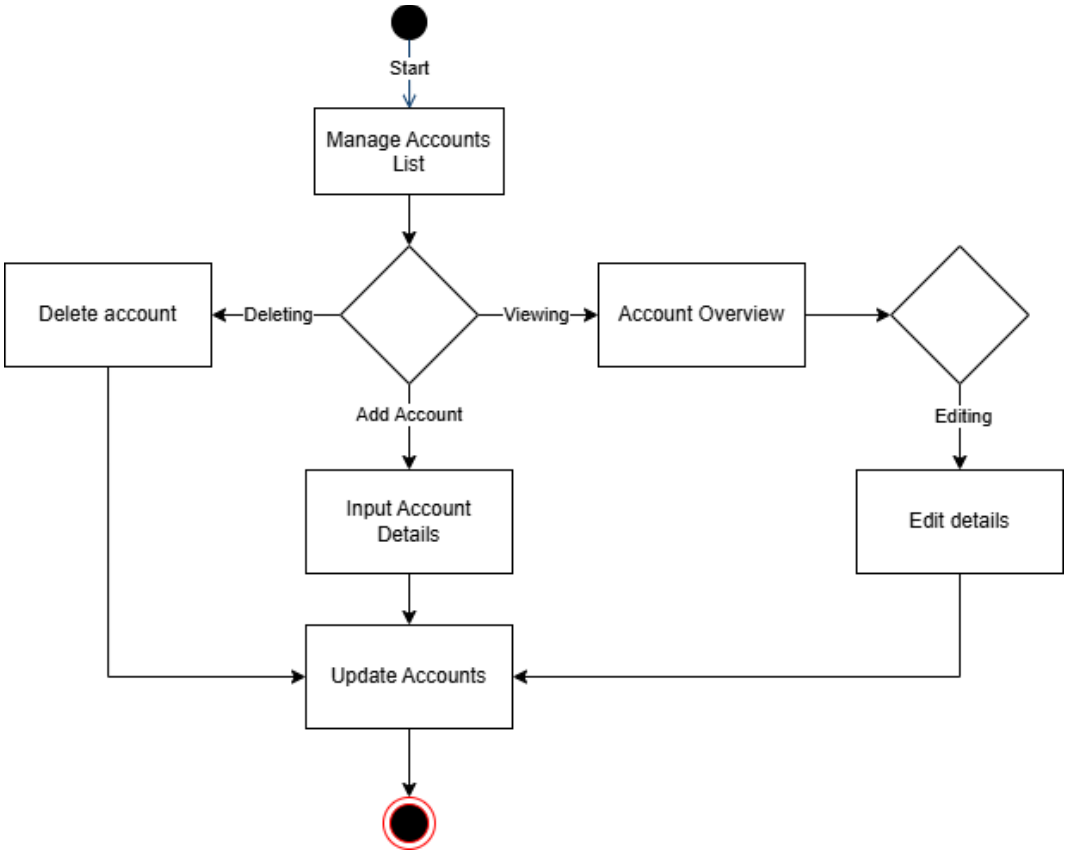


Figure 3-15. Activity Diagram of Manage User Accounts (Admin)

In Figure 3-16 below, the activity begins with the system starting to "Track Activity Logs." The admin then has the option to either "Filter" these logs or "Export" them. If the admin chooses to "Filter," they proceed to "Apply Filter," and the system then presents a decision point for "Print," which leads to "Download Activity Logs." Alternatively, if the admin chooses to "Export" from the initial decision, they directly proceed to "Download Activity Logs." Finally, after downloading the logs, the activity concludes. This flow

illustrates how admin can either refine the tracked activity logs through filtering before downloading or directly export them.

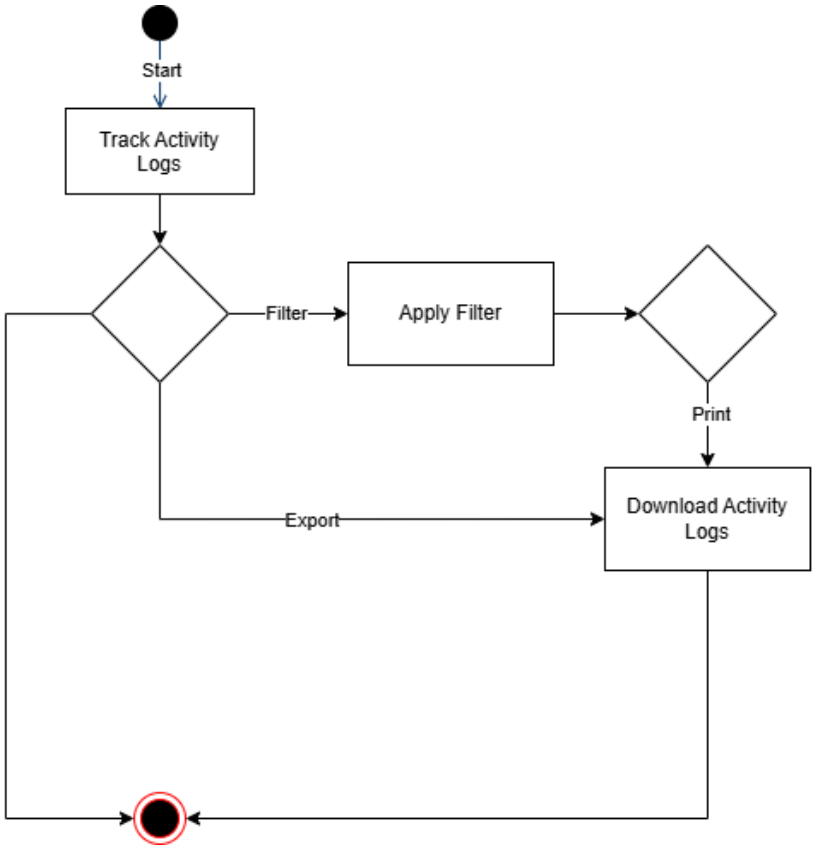


Figure 3-16. Activity Diagram of Activity Logs (Admin)

In Figure 3-17 below, the activity starts with accessing "Settings," which then presents the admin with a choice to either "Import" or "Export" data. If the admin selects "Import," they proceed to "Attach File," after which the system performs an "Update Database," and the activity concludes. If the admin chooses "Export," they are directed to "Download Selected Data," and the activity also ends. This flowchart outlines two distinct paths within the settings: one for bringing data into the system via file attachment and database update, and another for retrieving selected data through a download.

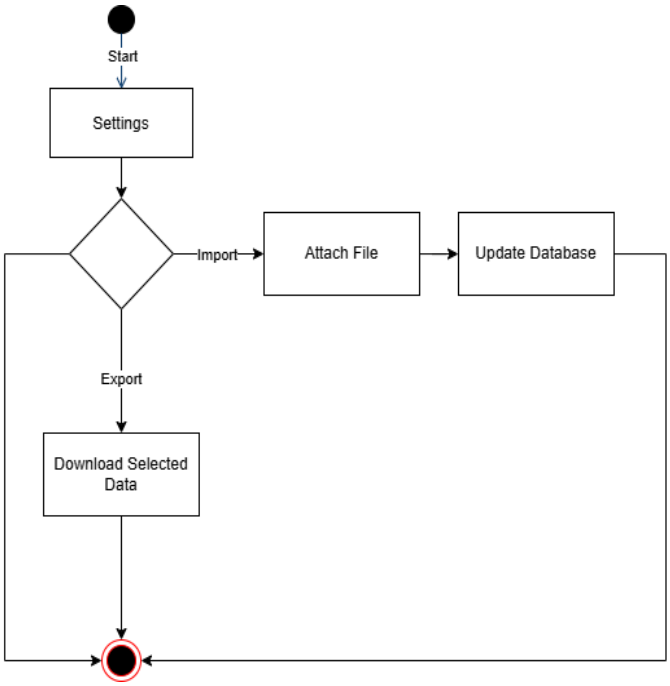


Figure 3-17. Activity Diagram of Import/Export File (Admin)

Wireframe

- **Login page**

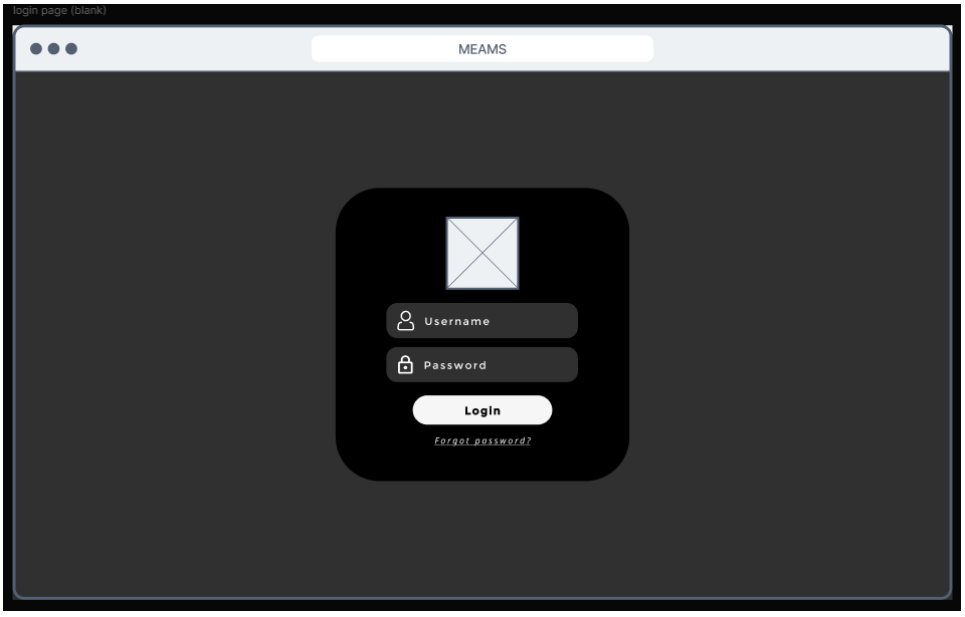


Figure 3-18. Login Page Wireframe

- **Forgot Password**

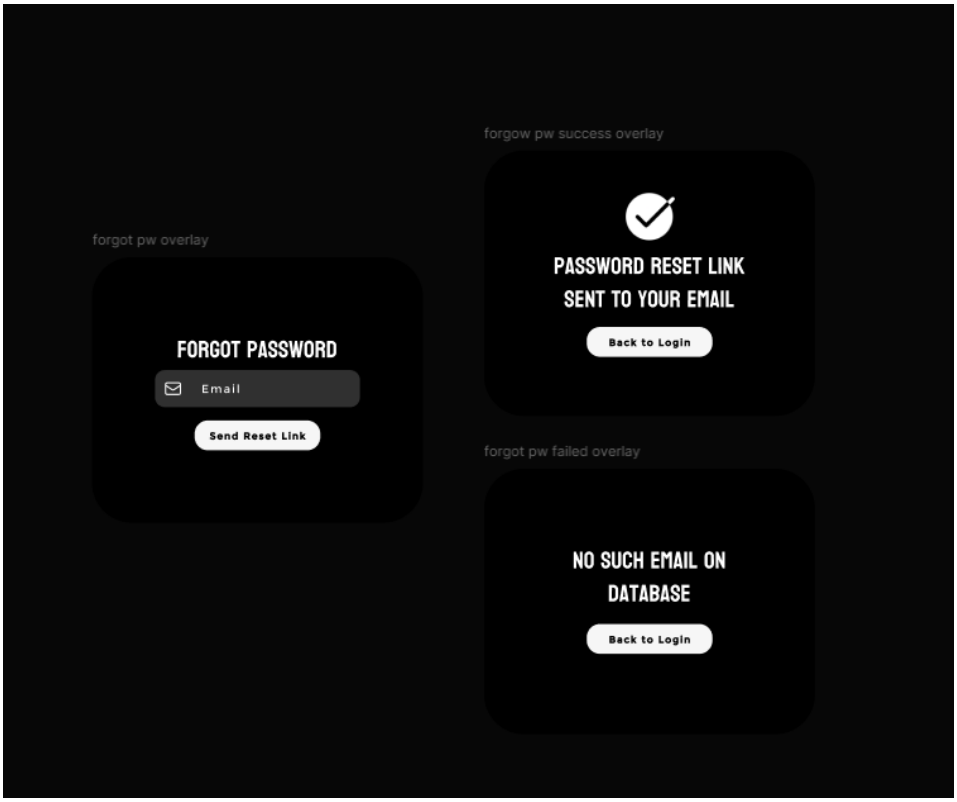


Figure 3-19. Forgot Password Wireframe

- **Dashboard (admin, staff)**



Figure 3-20. Dashboard Page Wireframe

- **Supplies Inventory (admin, staff)**

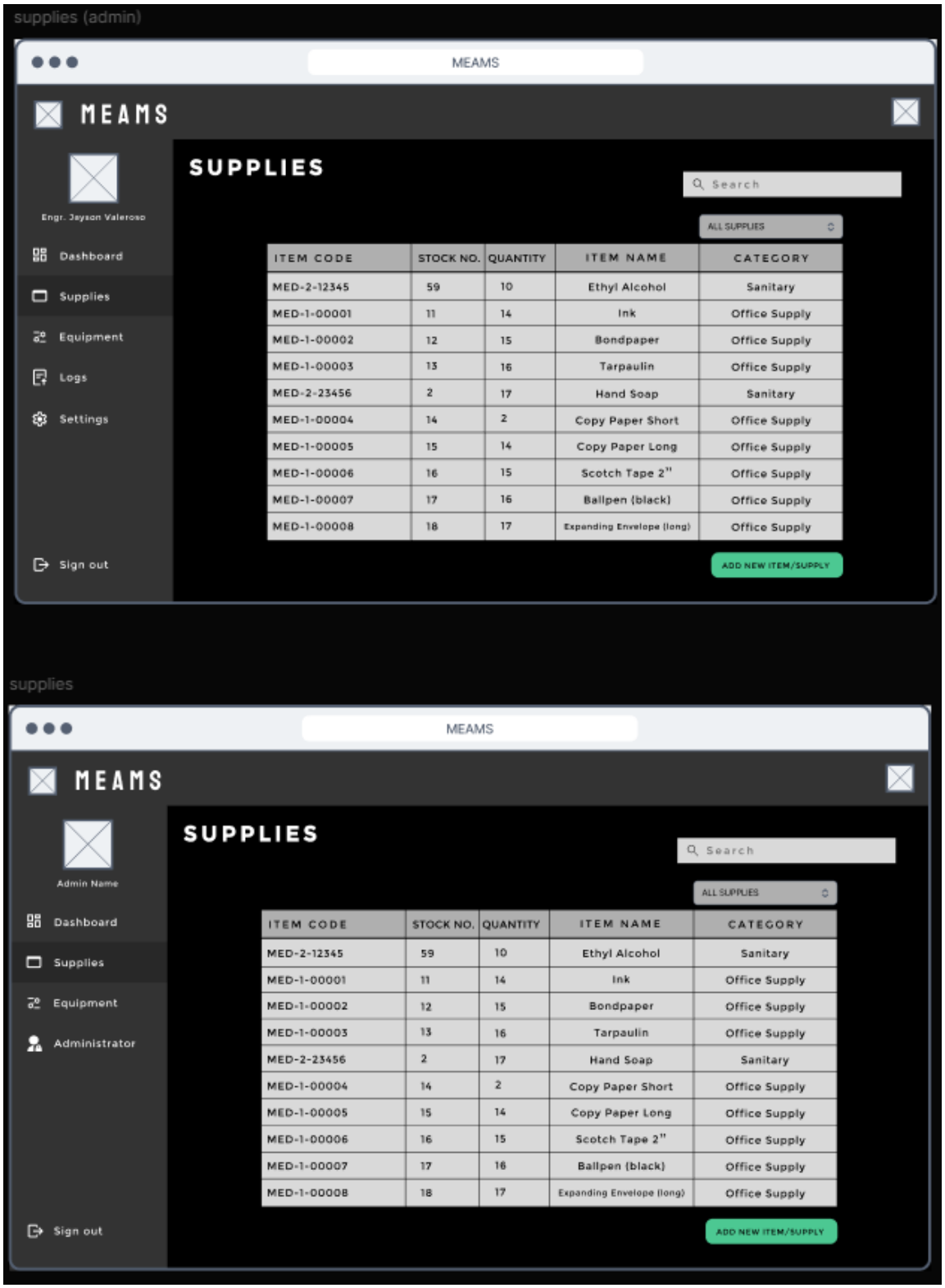


Figure 3-21. Supplies Inventory Page Wireframe

- **Item Details Overview**

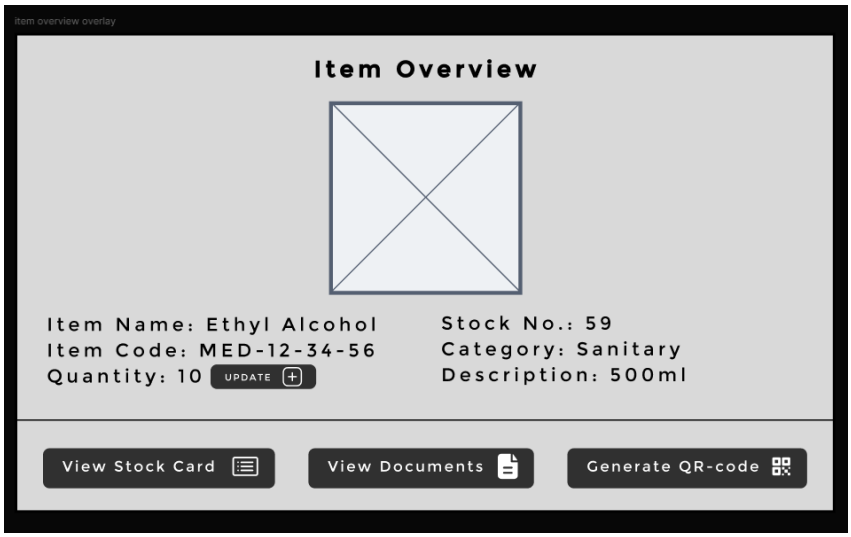


Figure 3-22. Item Overview Wireframe

- **Stock Card Overview**

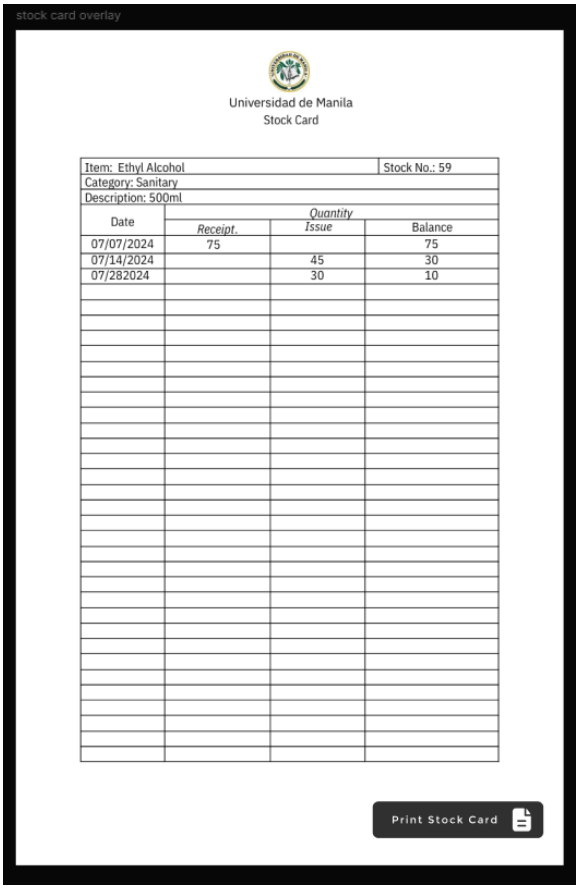


Figure 3-23. Stock Card Overview Wireframe

- **Equipment Inventory (admin, staff)**

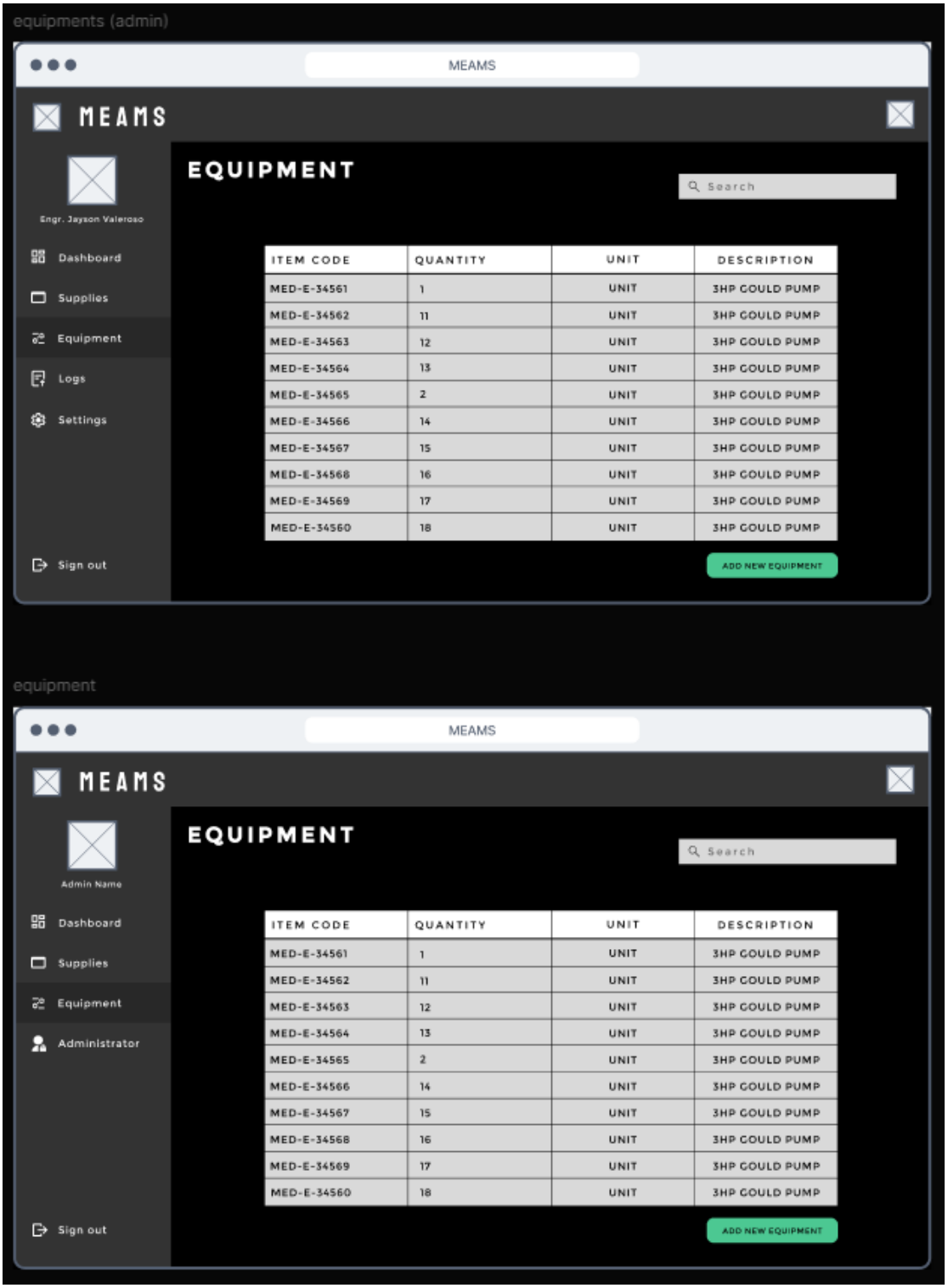


Figure 3-24. Equipment Inventory Wireframe

- **Equipment Details Overview**

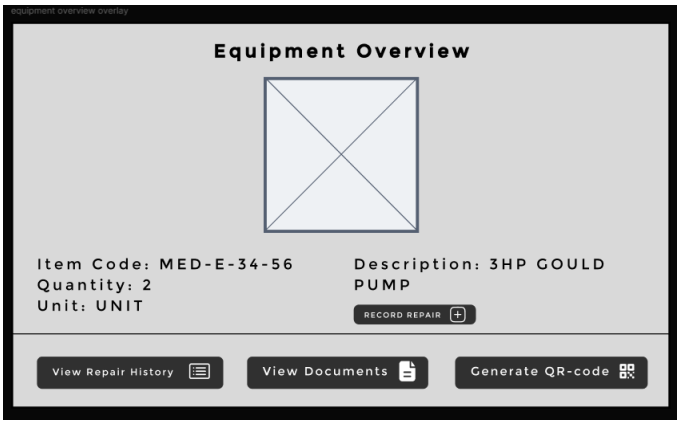


Figure 3-25. Equipment Overview Wireframe

- **Repair History Card Overview**

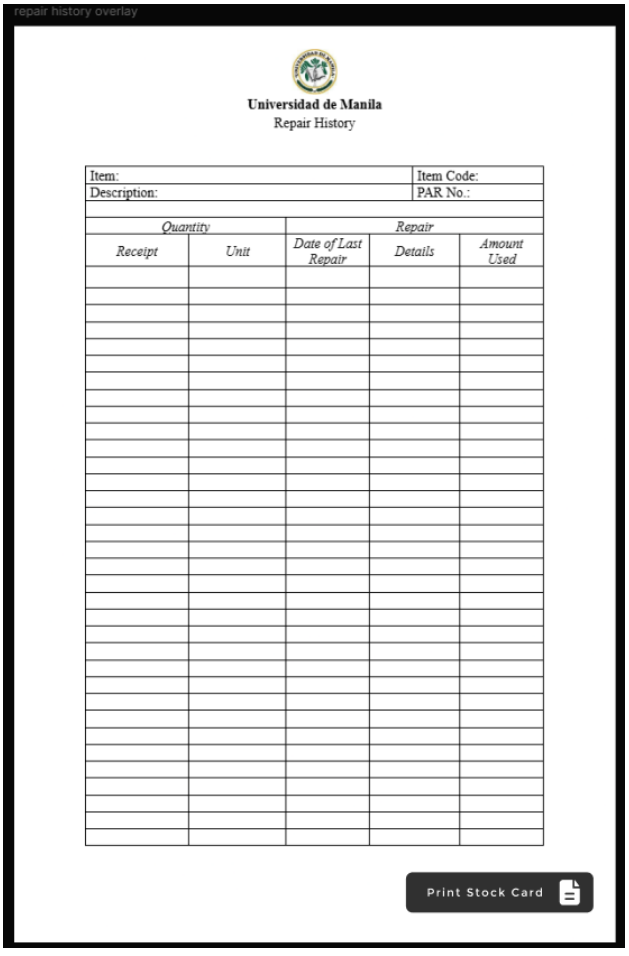


Figure 3-26. Repair History Card Overview Wireframe

- **Activity Logs**

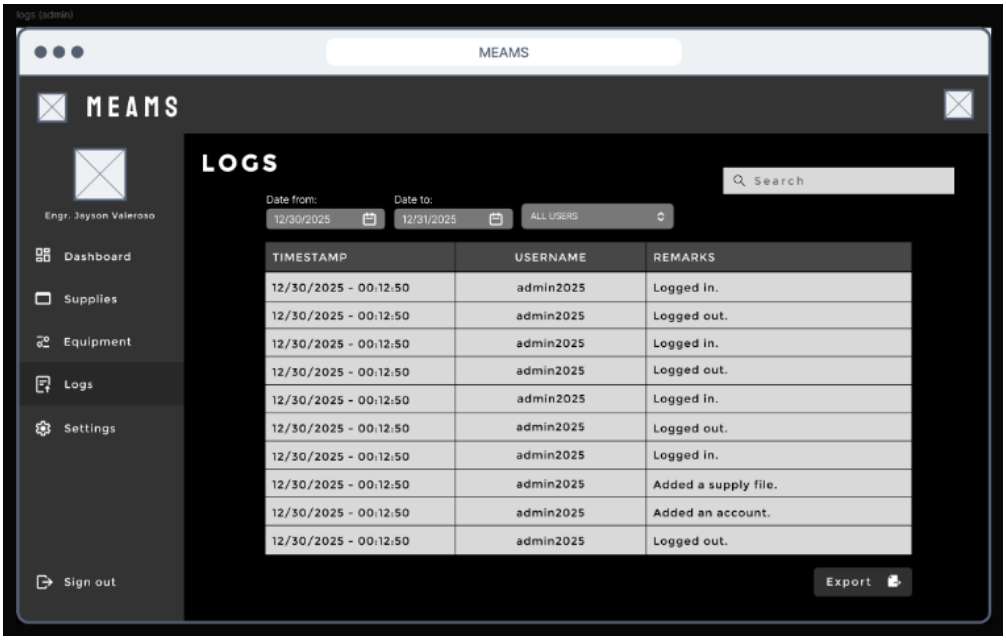


Figure 3-27. Activity Logs Page Wireframe

- **Settings**

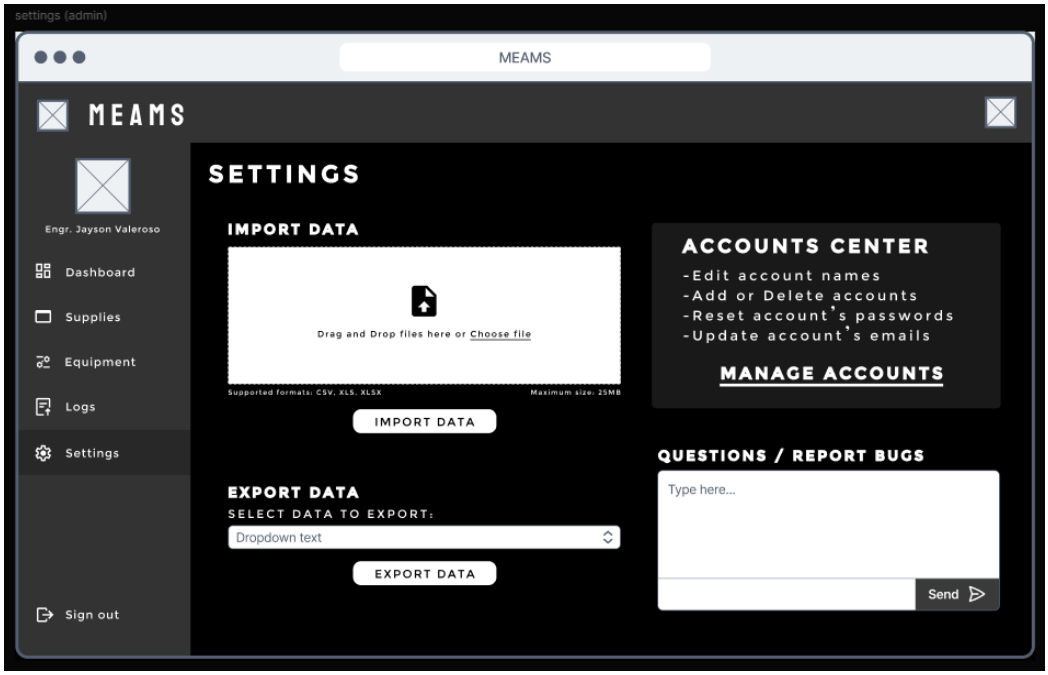


Figure 3-28. Settings Page Wireframe

- **Manage Accounts**

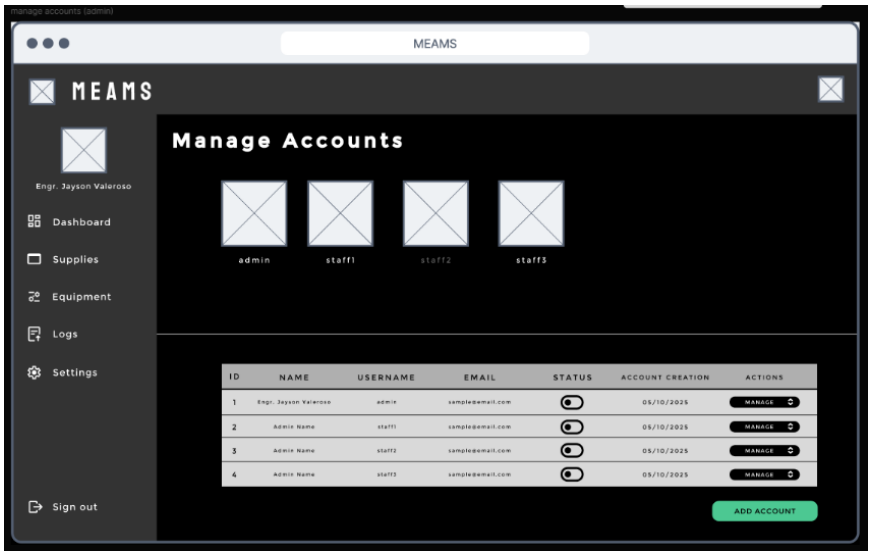


Figure 3-29. Manage Accounts Page Wireframe

Development and Testing

The development of “MEAMS: A Real-Time Tracking and Forecasting Asset Management System” followed the Agile methodology with the Scrum framework over a 32-week timeline. The project advanced through iterative sprints, each focusing on delivering prioritized features such as digitized tracking of supplies and equipment, alerts based on threshold and LCC, and interactive analytics dashboards. During sprint planning, the team refined system specifications, workflows, and database architecture while also preparing for data migration. The frontend, built with React.js was incrementally developed to provide a responsive and user-friendly interface with real-time tracking, alerts, and data visualizations powered by Chart.js. On the backend, Node.js (Express) handled APIs, user authentication, and real-time operations, while Python supported specialized tasks such as QR code generation and analytics using SARIMA time series models. Bulk uploads via CSV/XLSX were enabled through Pandas, with strict file size

limits to maintain performance, while image uploads supported detailed equipment documentation. MongoDB served as the central database, ensuring secure storage of assets, repair histories, and schedules, while maintaining data integrity. Testing was integrated into every sprint: features were immediately validated for functionality, responsiveness, and compatibility across devices. Testers also reviewed system security, identified bugs, and provided feedback that guided improvements in subsequent iterations. This iterative approach ensured MEAMS evolved into a secure, scalable, and reliable solution, meeting the MED's requirements for accurate asset tracking, repair, and efficient resource management.

Test Case

The table below is the test case for the MEAMS. These test cases guarantee that the system performs as expected, providing correct results.

The test plan for MEAMS validates that the system performs properly and achieves its objectives for UDM's MED. The system modernizes manual tracking processes, manages supply and equipment records, enables repair actions, and improves operational efficiency through data analytics. Testing identifies defects and ensures the system is reliable, secure, and user-friendly for administrators and staff with varying technical expertise.

Each test case focuses on specific system aspects. Login functionality testing ensures secure authentication with valid credentials and proper rejection of invalid attempts. QR code generation and scanning tests validate that staff can quickly access asset information on mobile devices, both online and offline—essential for field operations with intermittent connectivity. Tests verify inventory tracking accuracy, including real-time stock

updates, transaction recording, and historical repair data. Performance tests ensure system responsiveness with large datasets and multiple concurrent users during peak operations.

Test cases assess forecasting and analytics capabilities. The SARIMA model for supply forecasting is tested by comparing predictions against actual usage patterns. Equipment lifecycle management validation ensures the rule-based recommendation system correctly identifies assets approaching end-of-useful-life and suggests timely replacements. Compatibility tests verify proper function across browsers and devices. Security tests confirm role-based access controls prevent unauthorized modifications and sensitive data remains encrypted. Import/export functionality is tested with files up to 20MB to ensure smooth legacy data migration. Usability tests ensure the interface allows MED staff to perform daily tasks—recording repairs, updating inventory, generating reports—without extensive training.

Through comprehensive testing, the project team delivers a dependable platform replacing paper-based processes with digital workflows. Tests ensure the system satisfies MED operational requirements, complies with COA Circular No. 2020-006 for digital asset tracking in government institutions, and improves service delivery through data-driven decisions for resource allocation, repair actions, and budget planning. This approach aligns with ISO 25010 quality standards, validating functional suitability, performance efficiency, usability, reliability, security, and maintainability—ensuring MEAMS provides lasting value to UDM.

Table 3-2. Test Plan for MEAMS

Test Case Number	Test Step	Test Data	Expected Result	Actual Result	Status Passed/Failed
1	**Admin Authentication:** - Launch URL - Enter admin credentials - Login	URL Username Password	Admin should be able to login and redirected to the dashboard.	Admin successfully logged in and redirected to the dashboard	Passed
2	**Staff Authentication**: - Launch URL - Enter staff credentials - Login	URL Username Password	Staff should be able to login and redirected to the dashboard.	Staff successfully logged in and redirected to the dashboard	Passed
3	**Forgot Password:** - Click forgot password - Enter email - Check reset link	Test	System should send a password rest link to the registered gmail.	Password rest link successfully sent to the registered gmail	Passed
4	**User Management -** Create: -Login as Admin - Navigate to User Management - Click "Add User" - Fill in user details - Assign role - Click Save	Test	System should create a new user account and display it in the user list with “Active” status.	User account created successfully and displayed in the user list with “Active” status.	Passed
5	**User Management -** Edit: - Navigate to User Management - Select existing user - Click Edit - Modify user details - Click Save	Test	System should update user details and display the changes in the user list.	User details successfully updated and reflected in the users list.	Passed
6	**User Management -** Deactivate: - Navigate to User Management - Select active user - Click Deactivate - Confirm action	Test	System should deactivate the selected user and restrict access to the system.	User successfully deactivated and unable to log in to the system.	Passed
7	**User Management -** Reactivate: - Navigate to User Management - Select deactivated user - Click Activate/Reactivate - Confirm	Test	System should reactivate the selected user and allow login again.	User successfully reactivated and able to log in again.	Passed

Test Case Number	Test Step	Test Data	Expected Result	Actual Result	Status Passed/Failed
8	**User Management -** Delete: - Navigate to User Management - Select user - Click Delete - Confirm deletion	Test	System should permanently delete the selected user from the database and remove them from the user list.	User permanently deleted from the database and removed from the user list.	Passed
9	**Deactivated User Login Attempt:** -Deactivate user account - Attempt to login with deactivated credentials	Test	System should deny login and display message: "Your account has been deactivated. Please contact the administrator."	Login denied with message: "Your account has been deactivated. Please contact the administrator."	Passed
10	**Deleted User Login Attempt:** - Attempt login with deleted account	Test	System should deny login and display message: "Invalid credentials."	Login denied with message: "Invalid credentials.	Passed
11	**Supply Management -** Add: - Navigate to Supplies - Add new supply - Save	Test	System should successfully add new supply to the inventory list.	Supply successfully added to the inventory.	Passed
12	**Supply Management -** Update: - Select supply - Update quantity - Save	Test	System should successfully update the quantity of the selected supply.	Quantity successfully updated and reflected in the system.	Passed
13	**Supply Management -** View Stock Card: - Select supply - View stock card	Test	System should display the stock card with complete transaction history.	Stock card displayed with full transaction history.	Passed
14	**Supply Management -** Generate QR: - Select supply - Generate QR code - Download	Test	System should generate and download a QR code for the selected supply.	QR code successfully generated and downloaded.	Passed
15	**Supply Management -** Delete (Admin): - Login as admin - Select supply - Delete	Test	System should delete the selected supply record from the database.	Supply successfully deleted from the database.	Passed
16	**Equipment Management -** Add: - Navigate to Equipment - Add equipment - Save	Test	System should successfully add the equipment and automatically calculate its end-of-life.	Equipment successfully added and end-of-life automatically calculated.	Passed
17	**Equipment Management -** Record Repair: - Select equipment - Record repair details - Save	Test	System should record the repair details and save them in the repair history.	Repair details successfully recorded and saved in repair history.	Passed
18	**Equipment Management -** View Repair History: - Select equipment - View repair history	Test	System should display the complete repair history for the selected equipment.	Complete repair history successfully displayed.	Passed

Test Case Number	Test Step	Test Data	Expected Result	Actual Result	Status Passed/Failed
19	**Equipment Management -** Generate QR: - Select equipment - Generate QR code - Download	Test	System should generate a QR code for the selected equipment.	QR code successfully generated and downloaded.	Passed
20	**Dashboard -** View Real-Time Data: - Login to system - Access dashboard	Test	System should display the current stock levels and equipment status on the dashboard.	Dashboard successfully displayed current stock levels and equipment status.	Passed
21	**Dashboard -** View Forecasts: - Navigate to Dashboard - View analytics section	Test	System should display SARIMA forecasts in line graphs.	SARIMA forecasts successfully displayed in line graphs.	Passed
22	**Dashboard -** Understock Table: - Login to system - Navigate to Dashboard - View Understock Items table	Test	System should display items below the threshold with their current quantities.	Understock table displayed items below threshold with current quantities.	Passed
23	**Dashboard -** Generate Requisition Form: - View Understock table - Select understock item - Click Generate Form	Test	System should generate a requisition form automatically with pre-filled item details.	Requisition form successfully generated with pre-filled item details.	Passed
24	**Dashboard -** Repair Required Table: - Navigate to Dashboard - View Equipment Repair table	Test	System should display equipment requiring repair with their repair history.	Table successfully displayed equipment requiring repair with repair history.	Passed
25	**Dashboard -** Generate Repair Form: - View Repair table - Select equipment - Click Generate Form	Test	System should generate a repair form with equipment details and repair records.	Repair form successfully generated with equipment details and repair records.	Passed
26	**Dashboard -** Beyond Useful Life Table: - Navigate to Dashboard - View Beyond Useful Life table	Test	System should display equipment that exceeded its useful life with purchase details.	Table successfully displayed equipment that exceeded useful life with purchase details.	Passed
27	**Dashboard -** Generate Purchase Form: - View Beyond Useful Life table - Select equipment - Click Generate Form	Test	System should generate a purchase request form with equipment specifications and justification.	Purchase request form successfully generated with specifications and justification.	Passed
28	**Dashboard -** Refresh Tables: - Update inventory/equipment - Refresh dashboard - Verify table updates	Test	System should update all three tables (Understock, Repair, and Beyond Useful Life) in real-time.	All three tables successfully updated in real-time.	Passed
29	**Form Generation -** Print/Download: - Generate any form - Download or print	Test	System should allow the generated form to be downloaded as PDF or printed directly.	Generated form successfully downloaded and printed.	Passed

Test Case Number	Test Step	Test Data	Expected Result	Actual Result	Status Passed/Failed
30	**QR Scanning -** Online Mode: - Open mobile scanner - Scan supply QR code	Test	The system should display supply details on mobile.	The system displayed supply details correctly on mobile.	Passed
31	**QR Scanning -** Offline Mode: - Disable internet - Scan QR code - Reconnect	Test	The system should display data from cache and sync when online.	The system displayed cached data and synced successfully after reconnecting.	Passed
32	**Import Data:** (Admin) - Navigate to Settings - Select CSV file - Import	Test	The system should import data successfully.	System imported data successfully.	Passed
33	**Export Data:** (Admin) - Navigate to Supplies/Equipment - Click Export - Download	Test	The system should export data successfully.	System exported data successfully.	Passed
34	**Activity Logs -** View(Admin):- Navigate to Activity Logs - Apply filters	Test	The system should display filtered logs.	The system displayed filtered logs correctly.	Passed
35	**Activity Logs -** (Admin) Export: - Select date range - Export logs	Test	The system should export logs successfully.	System exported logs successfully.	Passed
36	**Role-Based Access Control:** - Login as Staff - Attempt admin function	Test	The system should display an access denied message.	System displayed access denied message as expected.	Passed
37	**Data Validation -** Invalid Input: - Enter negative quantity - Attempt save	Test	The system should display an error message: “Quantity must be positive.”	The system displayed the correct error message.	Passed
38	**Print Functionality:** - View stock card - Print document	Test	The system should format the document correctly for printing.	System formatted and printed document correctly.	Passed
39	**Performance -** Dashboard Load: - Login - Measure load time	Test	The system should load the dashboard within 3 seconds.	System loaded dashboard within 2.8 seconds.	Passed
40	**Security -** SQL Injection: - Enter SQL injection attempt - Try login	Test	The system should prevent the attack and deny login.	The system prevented SQL injection and denied login.	Passed
41	**Security -** Account Lockout: - Enter wrong password 5 times	Test	System should lock the account for 15 minutes	Account locked for 15 minutes	Passed
42	**Bug Reporting -** Staff: - Navigate to Report Bug - Submit bug report	Test	System should submit a report successfully	Bug report submitted successfully	Passed
43	**Search Functionality:** - Search for supply - View results	Test	System should display relevant items	Relevant items displayed	Passed

Implementation Plan

Table 3-3. Implementation Plan for MEAMS

STRATEGY	ACTIVITIES	PERSONNEL INVOLVED	PROPOSED SCHEDULE	DURATION
1.System Development and Testing	• Design and develop MEAMS • Finalize system modules and fix bugs • Configure server and database • Conduct system and integration testing	Development Team	May to October	6 months
2. System Deployment	• Install MEAMS on MED • Configure MongoDB and API services • Implement role-based access control and authentication	Development Team	November	1 week
3. User Training	• Conduct training for admin and staff • Hands-on practice with QR scanning and dashboards • Provide user manuals and support materials	Development Team	November	3 days
4. Pilot Testing	• Run pilot with selected MED staff • Test forecasting and QR features • Collect feedback and fix issues	MED Staff & Development Team	November	1 week
STRATEGY	**ACTIVITIES**	**PERSONNEL INVOLVED**	**PROPOSED SCHEDULE**	**DURATION**
5. System Launch	• Launch MEAMS for full MED operations • Provide on-site support and monitor performance • Address initial user concerns	Development Team & MED	November	1 week
8. Maintenance and Evaluation	• Monitor system performance and uptime • Review and fix reported bugs or errors promptly •Collect user feedback to enhance usability and efficiency • On-site technical support • Address immediate user concerns	Development Team & MED Admin	After full deployment	Continuous

The Table 3-3 presents the implementation plan for the MEAMS Asset Management System, outlining its key phases to ensure an organized and efficient deployment. The process begins with System Development and Testing (May to October), where the development team designs, develops, and tests all system modules. This is

followed by System Deployment in November, involving installation, configuration of MongoDB and API services, and setup of role-based access control. Next is User Training (3 days), where administrators and staff receive hands-on training in system use and QR functionalities. Pilot Testing then follows to assess forecasting and QR features, gather feedback, and address issues before the System Launch, which officially deploys MEAMS across all MED operations. Lastly, Maintenance and Evaluation ensure continuous monitoring, bug fixing, and user support to maintain system performance and reliability.

Network Architecture

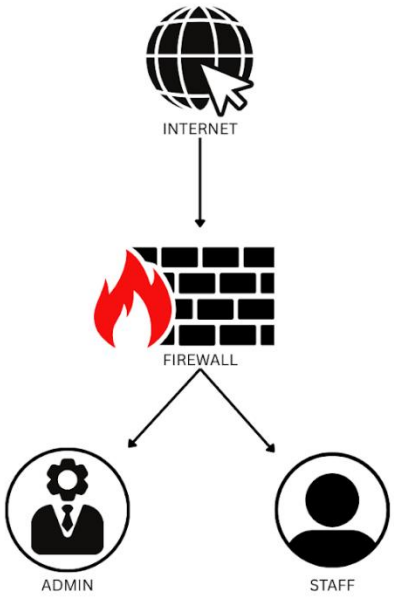


Figure 3-30. Network Architecture Diagram for MEAMS

The Figure 3-30 illustrates the structure and interaction between the components of the MEAMS. The diagram shows how the system operates securely by connecting administrators and staff to the application through the internet, with a firewall serving as a protective layer. The firewall filters and manages all incoming and outgoing traffic to ensure that only authorized users and data transactions are allowed. The web server

hosts the MEAMS application, which integrates a React.js frontend and Node.js with FastAPI backend, while the MongoDB database securely manages all equipment, supply, and repair information. This network design enables users to access the system from different internet connections while maintaining secure, efficient, and reliable communication between all components.

Network Model

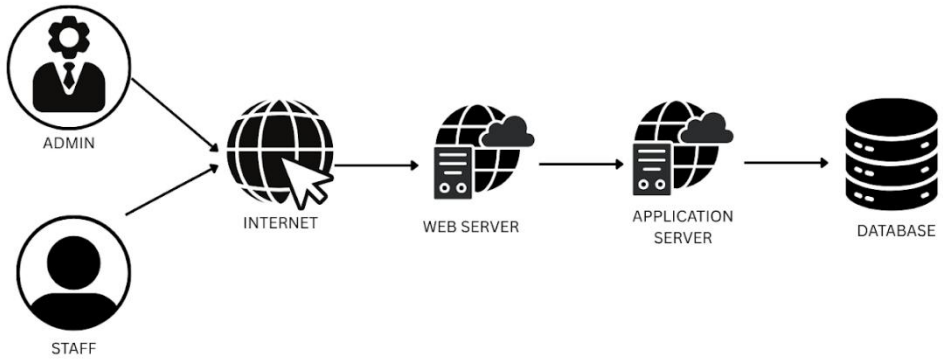


Figure 3-31. Network Model for MEAMS

The Figure 3-31 represents the flow of the MEAMS involving the administrator, staff, and the system's servers. Both the admin and staff access the system through the internet, which connects them to the web server hosting the MEAMS application. The web server manages user requests and forwards them to the application server, where the backend processes data and executes operations such as inventory tracking, QR code scanning, and forecasting. The application server then communicates with the MongoDB database, where all essential information such as equipment records, supply inventories, and accounts is securely stored and retrieved. This flow ensures seamless interaction, efficient data processing, and reliable access for all system users.

Deployment Diagram

The Deployment Diagram of the MEAMS illustrates how its components interact to ensure efficient system operation and service delivery. The system can be accessed by both the Admin and Staff using devices such as laptops, desktops, or mobile devices (for QR code scanning) connected through the internet. These devices communicate with the Web Server, which handles incoming requests and routes them to the appropriate services within the system.

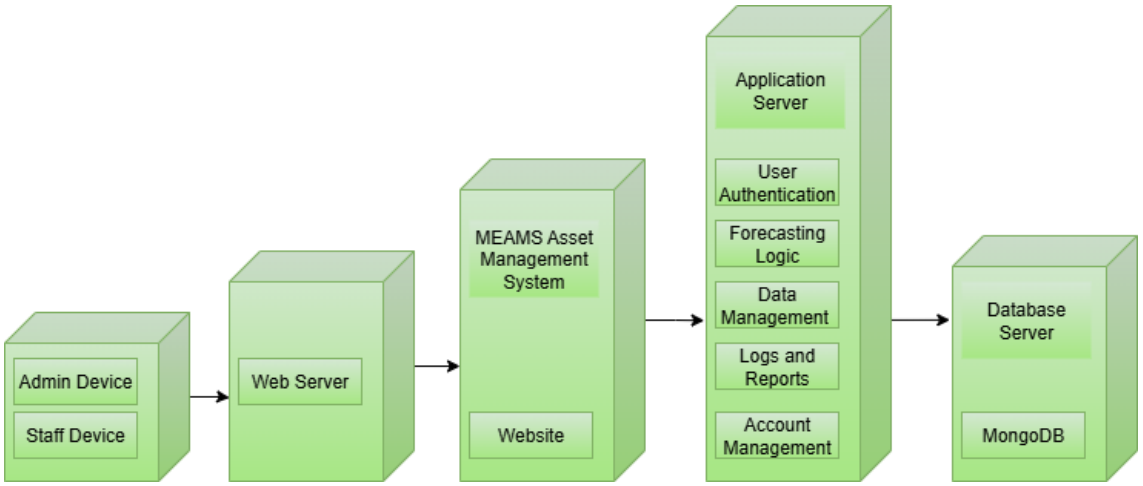


Figure 3-32. Deployment Diagram for MEAMS

At the core is the MEAMS, hosted on an Application Server, where essential processes such as user authentication, forecasting logic, data management, logs and reports, and account management are executed. The Application Server ensures that all user requests are properly processed and coordinated. All operational data, including equipment records, supply inventories, and repair histories, are securely stored in the Database Server managed by MongoDB. This architecture highlights how each component works together to provide seamless access, secure data handling, and reliable

system performance, ensuring that MEAMS effectively supports the asset management operations of the organization.

Research Design

This study utilized a descriptive-developmental research design. The descriptive component focused on collecting data to describe the existing asset management procedures and the issues encountered by the MED of the UDM in managing their assets. These include inaccurate inventory tracking, lack of repair monitoring, and the absence of real-time data updates.

Meanwhile, the developmental component involved the design and creation of MEAMS: A Real-Time Tracking and Forecasting Asset Management System, which aimed to automate the MED's manual processes and improve operational efficiency.

Data were gathered through demonstrations and survey questionnaires, which allowed the researchers to evaluate the system's performance and usability. The collected data were analyzed both quantitatively and qualitatively. Quantitative data from the survey responses were interpreted using descriptive statistics, such as weighted mean and percentage distribution, to measure the system's level of acceptability and functionality based on the ISO/IEC 25010 Software Quality Standards. Qualitative feedback was also examined to identify specific user concerns and suggestions for improvement.

This research design ensured that the development of MEAMS was guided by actual user needs and expert evaluations, leading to a reliable and user-friendly system designed for the university's MED.

Research Locale and Respondents

The study was conducted at the MED of the UDM. The MED is the university department responsible for facility, repair, and inventory management, including electrical, plumbing, air-conditioning, and carpentry services.

The respondents of the study were divided into two groups:

1. End-Users (MED Personnel) – composed of the MED Head, storekeeper, administrative staff, and repair technicians. These individuals directly use the system in performing daily operations such as asset tracking, repairs, and record management.
2. IT Specialists/Evaluators – composed of information technology professionals and academic experts who assessed the system's technical performance and compliance with ISO/IEC 25010 standards.

A purposive sampling method was used to select respondents who were directly involved in asset management and system evaluation. This ensured that feedback came from both operational users and technical experts.

Data Gathering Techniques

To ensure comprehensive and valid data collection, the researchers used a combination of demonstration-based data gathering and survey evaluation techniques:

1. System Demonstration

   The researchers conducted an on-site system demonstration at the MED. During the demo, MED personnel were shown the full functionality of the

MEAMS, including real-time tracking, forecasting analytics, and QR-based asset management. The demo served as the basis for gathering firsthand feedback from users about the system's usability and effectiveness in addressing their operational needs.

2. Survey Questionnaires

   After the demonstration, a survey questionnaire was distributed to MED personnel through printed copies and Google Forms. The questionnaire was based on the ISO/IEC 25010 Software Quality Evaluation Criteria, focusing on six attributes:

   a. Functional suitability
   b. Performance efficiency
   c. Usability
   d. Reliability
   e. Security
   f. Maintainability

3. Respondents rated each criterion using a five (5) point Likert scale, where 5 meant "Strongly Agree" and 1 meant "Strongly Disagree." Their responses were used to assess the system's overall quality and user satisfaction.

Table 3-4. Five-Point Likert Scale Interpretation

Scale	Verbal Descriptor
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

4. IT Specialist Evaluation via Demo Video and Online Survey
   For technical validation, a demo video of the MEAMS system was shared with IT specialists and academic evaluators. After viewing the video, they answered the same Google Form survey to evaluate the system's performance, design, and technical accuracy. Their expert feedback was instrumental in validating system reliability and identifying potential improvements.

Data Analysis

The collected survey data were analyzed using descriptive statistics, particularly the weighted mean, to determine the level of acceptability of MEAMS across each ISO 25010 criterion. Qualitative comments from respondents and IT specialists were also summarized and interpreted to support the quantitative results, providing a holistic evaluation of the system's effectiveness.

Chapter IV

RESULTS AND DISCUSSIONS

This chapter presents the results of the system development process and discusses the outcomes of the implementation of the MEAMS. The project involved two groups of respondents, namely the MED, who served as the primary users of the system, and the IT practitioners, who acted as validators to assess its overall functionality and performance. A total of 68 respondents participated in the evaluation, consisting of 39 from the MED and 29 IT practitioners. Their feedback provided valuable insights that contributed to assessing and improving the effectiveness of the system.

Output and User-Interface Design

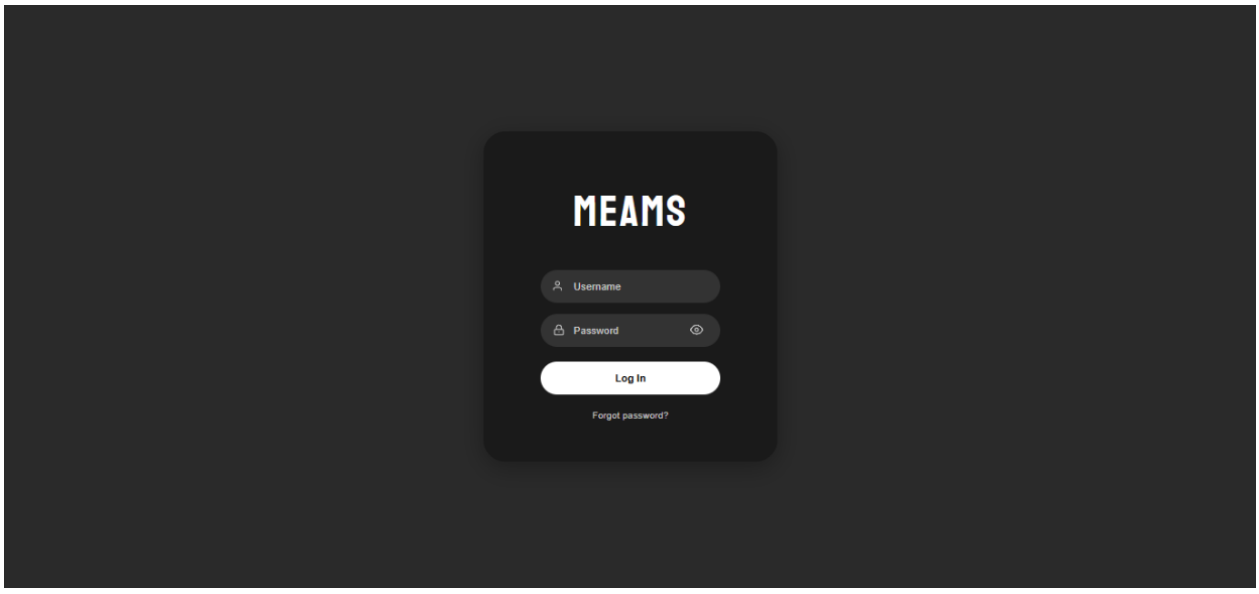


Figure 4-1. Login Page

Figure 4-1 illustrates the Login Page, which serves as the secure gateway to the MEAMS system. The interface requires users to input their Username and Password to

authenticate and access the dashboard, ensuring that sensitive asset data is restricted to authorized personnel, while also providing a "Forgot password?" link for account recovery.

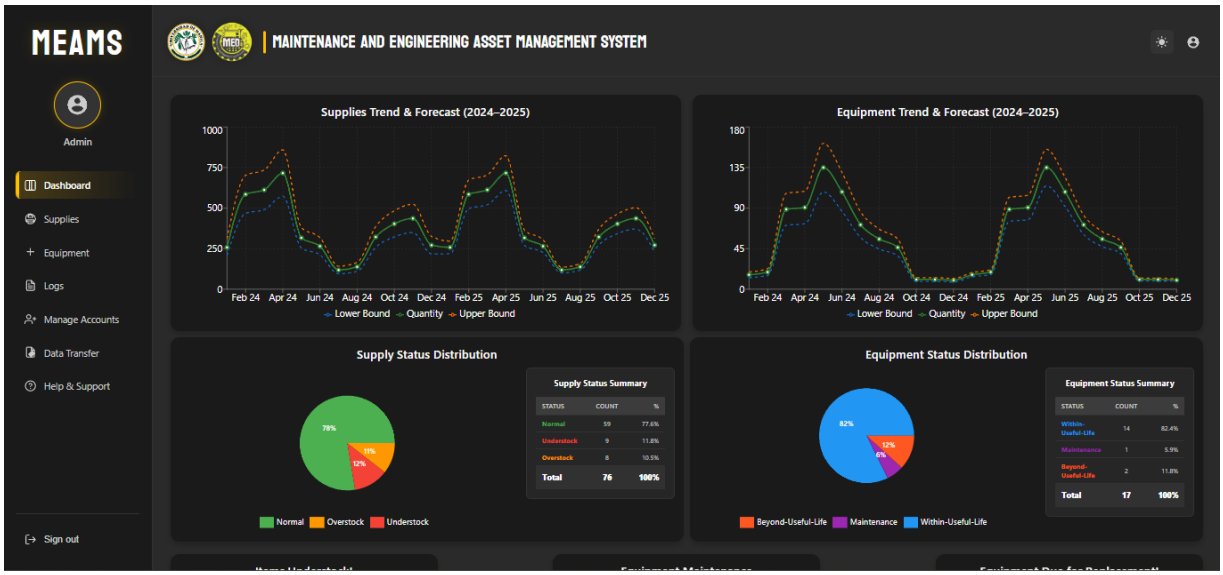


Figure 4-2. System Dashboard

Figure 4-2 displays the System Dashboard, which acts as the central command center for administrators to navigate system modules and monitor operational health. It features interactive trend forecast charts and status distribution pie charts, enabling users to instantly assess supply trends, inventory levels, and equipment lifecycles at a glance.

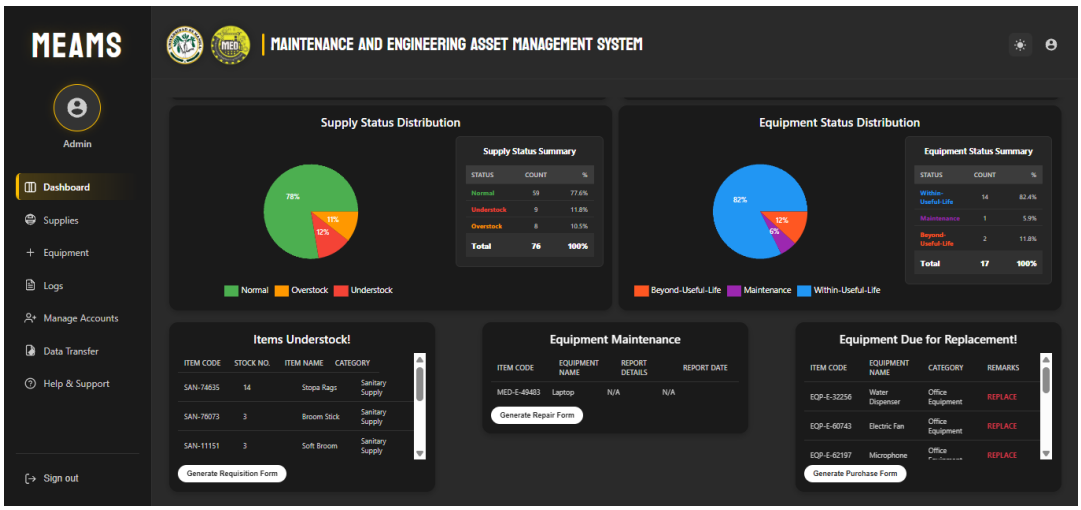


Figure 4-3. Dashboard Action Panels

Figure 4-3 illustrates the Dashboard Action Panels presented upon scrolling down the interface. This section highlights three (3) critical operational alerts: (1) Items Understock, (2) Equipment Repair, and (3) Equipment Due for Replacement. Each panel facilitates immediate action, enabling administrators to generate the necessary Requisition, Repair, or Purchase Forms directly to address flagged operational needs.

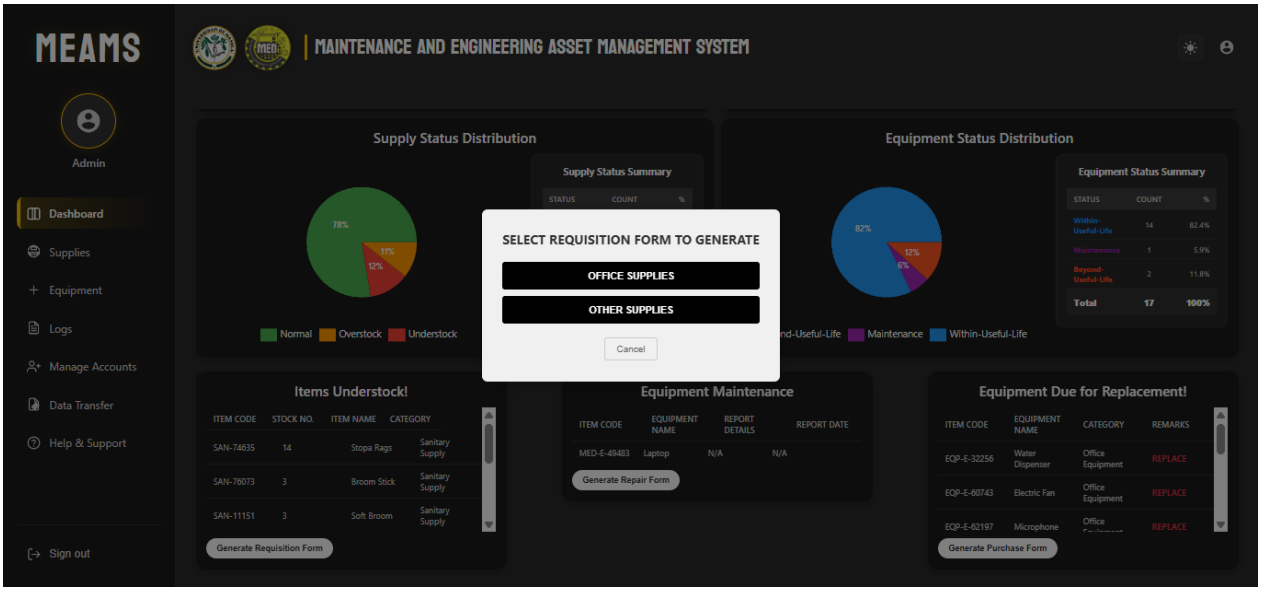


Figure 4-4. Select Requisition Form Modal

Figure 4-4 illustrates the Select Requisition Form Modal, which activates upon clicking the "Generate Requisition Form" button. This interface mandates that the administrator categorize the request as either "OFFICE SUPPLIES" or "OTHER SUPPLIES" to ensure the correct documentation is generated.

Figure 4-5 below illustrates the interface displayed when the administrator selects the "Office Supplies" category. It allows the user to customize the requisition list by adding new rows for additional items or deleting entries to ensure the request is accurate and it can also be printed.

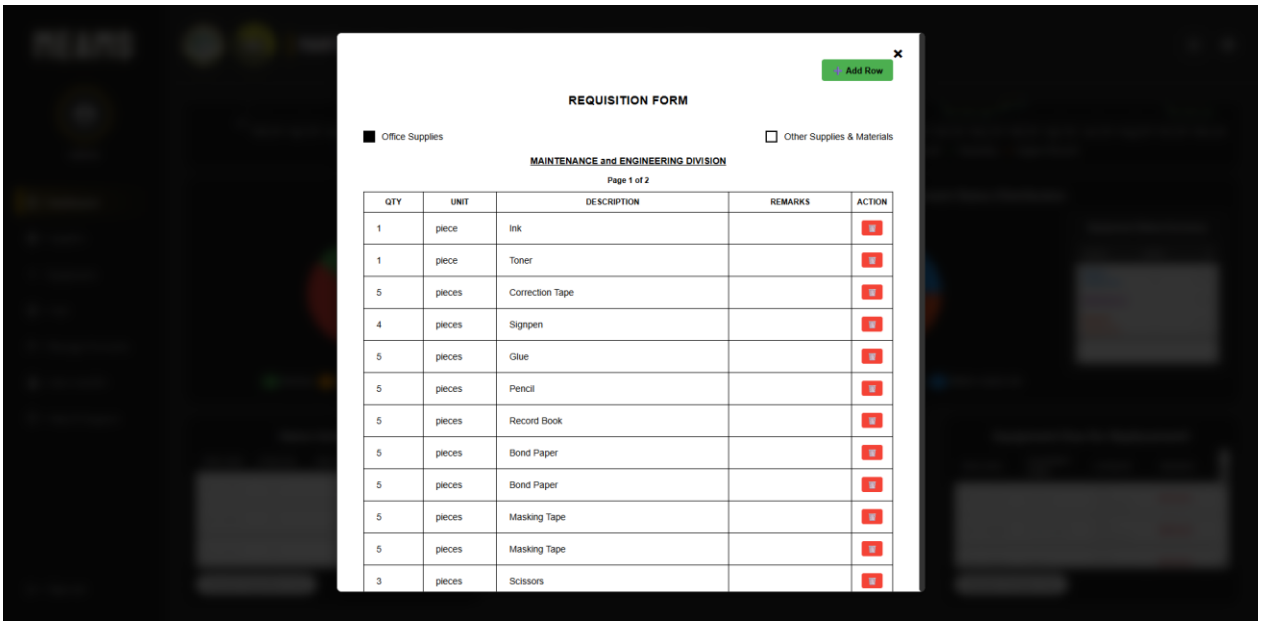


Figure 4-5. Requisition Form (Office Supplies)

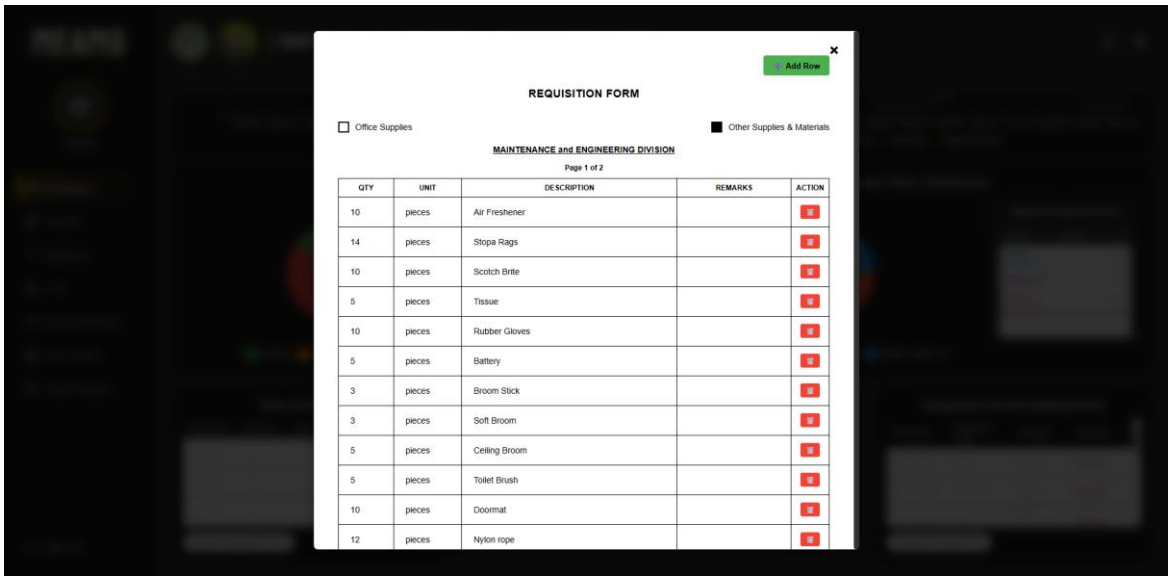


Figure 4-6. Requisition Form (Other Supplies)

Figure 4-6 illustrates the interface displayed when the administrator selects the "Other Supplies" category. It allows the user to customize the requisition list by adding new rows for maintenance materials or deleting entries to ensure the request is accurate and it can also be printed.

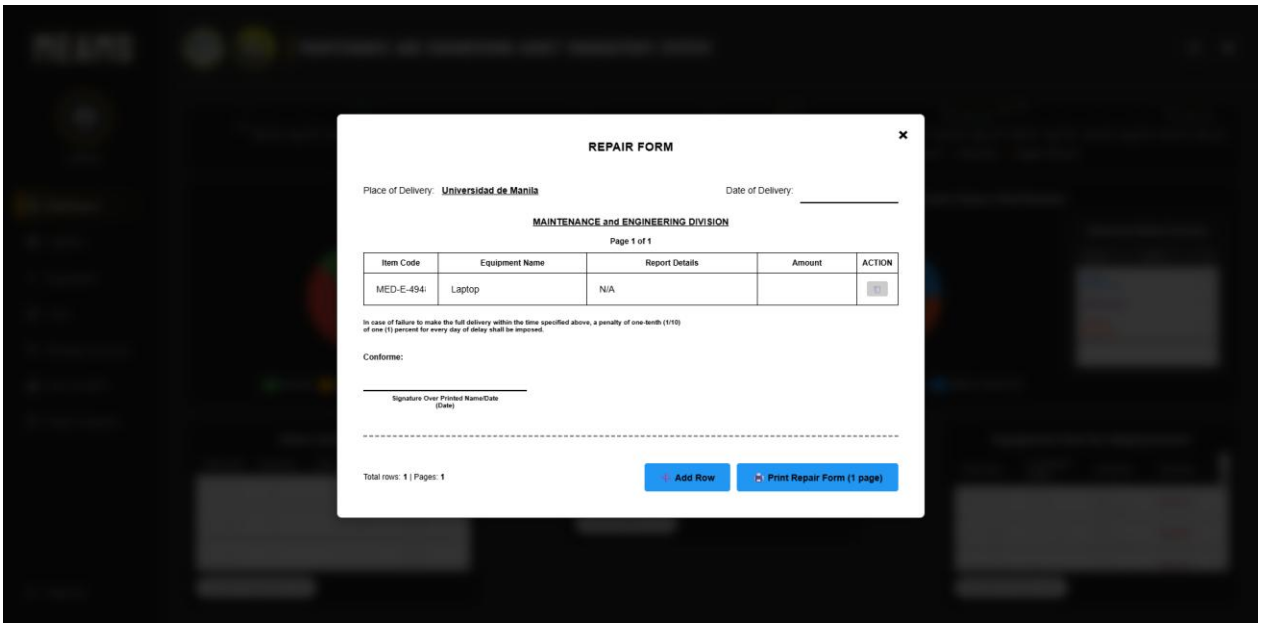


Figure 4-7. Generate Repair Form Modal

Figure 4-7 displays the Repair Form modal, which appears upon clicking the "Generate Repair Form" button. This interface allows the administrator to review and edit maintenance details, add additional items via the "+ Add Row" button, or finalize the documentation by selecting "Print Repair Form."



Figure 4-8. Generate Purchase Form Modal

Figure 4-8 displays the Purchase Form modal, which opens when the "Generate Purchase Form" button is clicked. It enables the administrator to review the equipment procurement list, remove specific items using the delete icon in the Action column, and ensure all details are correct before finalizing the purchase request and to finalize the documentation by selecting “Print Purchase Form”.

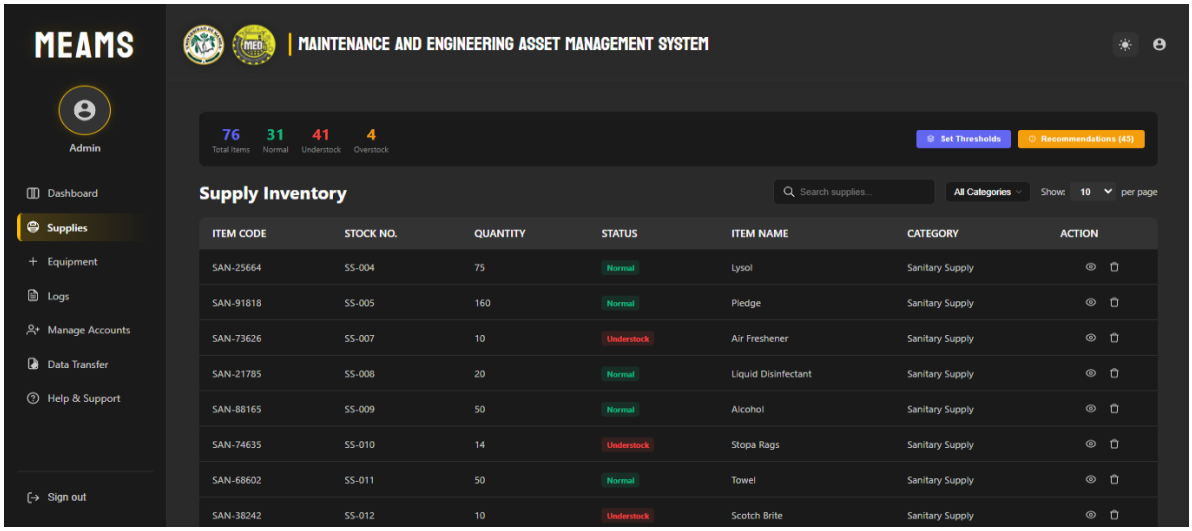


Figure 4-9. Supplies Inventory Page

Figure 4-9 depicts the Supplies Inventory Page, providing a tabular view of consumable assets alongside real-time status metrics for immediate assessment. Administrators can manage stock levels using the "Set Thresholds" and "Recommendations" buttons, or perform specific actions on items via the search bar and Action icons.

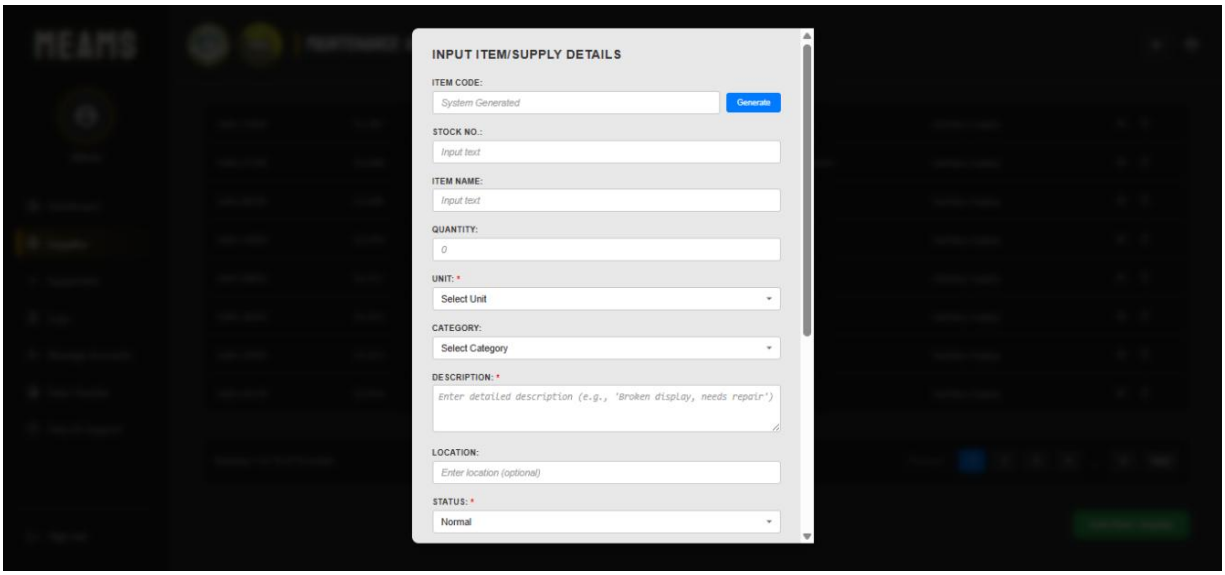


Figure 4-10. Input Item/Supply Details Modal

Figure 4-10 depicts the Input Item/Supply Details Modal, accessed by clicking the "Add Item Supply" button. It requires the administrator to input essential inventory details, such as Item Name, Stock No., and Quantity, to successfully register a new item into the database.

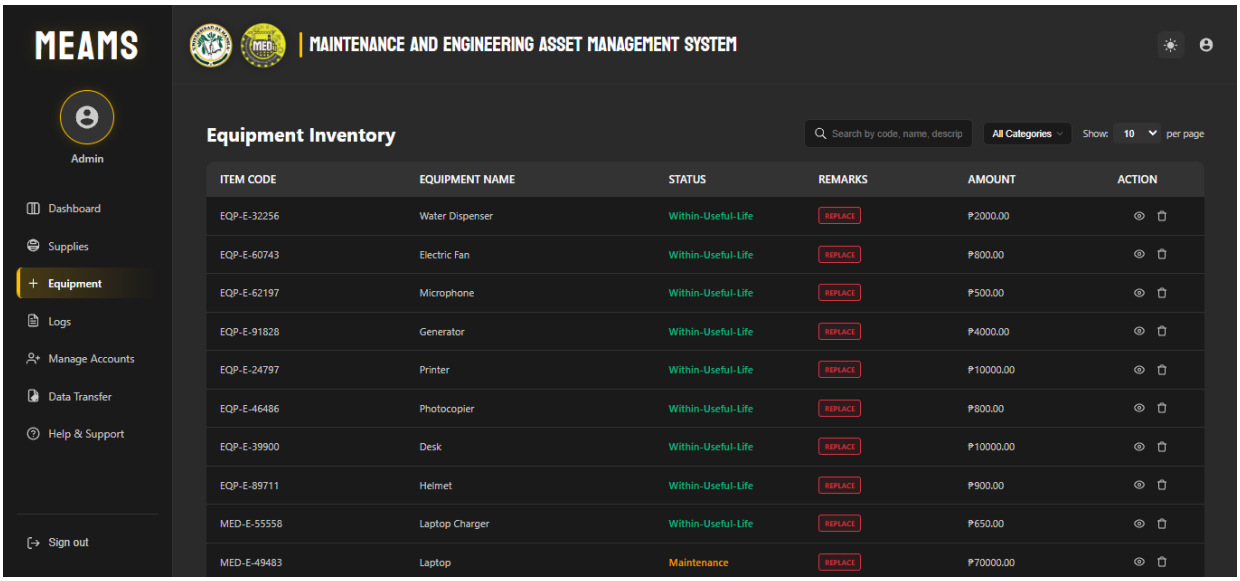


Figure 4-11. Equipment Inventory Page

Figure 4-11 illustrates the Equipment Inventory Page, presenting a searchable table of engineering assets with key details such as Item Code, Status, and Remarks. Administrators can manage these records by using filters for quick retrieval or the Action icons to view details or delete specific equipment entries.

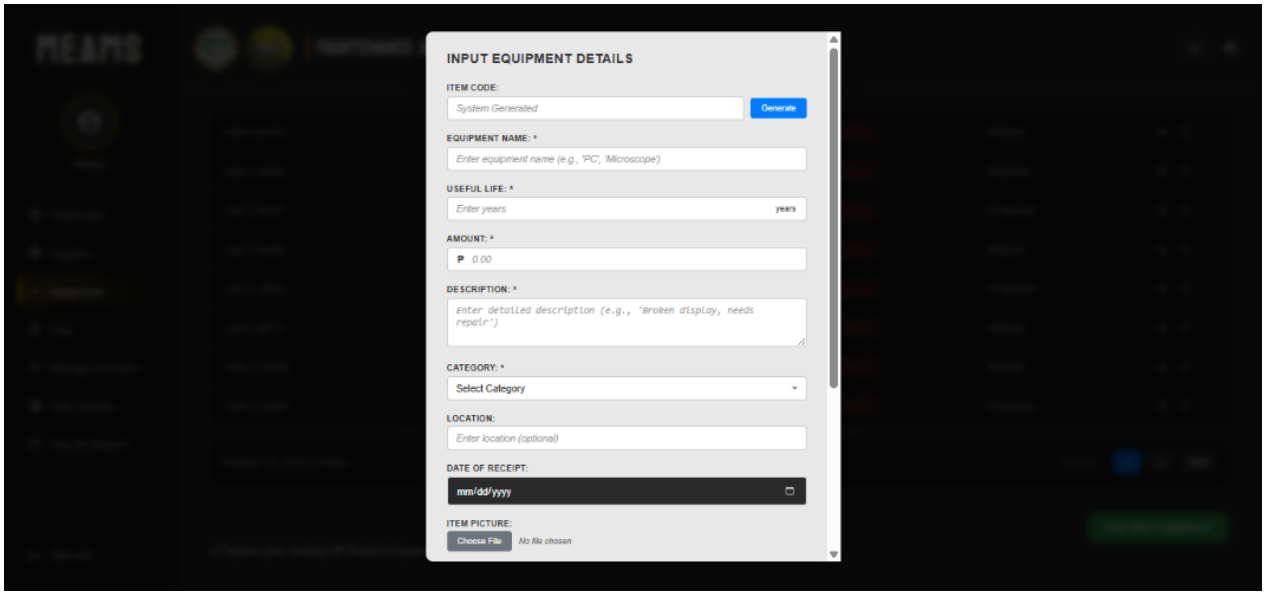


Figure 4-12. Input Equipment Details Modal

Figure 4-12 depicts the Input Equipment Details Modal, accessed by clicking the "Add New Equipment" button. It requires the administrator to input essential asset specifications, including Equipment Name, Useful Life, and Amount, to formally register the equipment in the inventory database.

Figure 4-13 below displays the System Logs Page, which maintains a comprehensive audit trail of all system activities. Administrators can monitor user actions by applying date and user filters, searching for specific events, or clicking the "Export" button to download records for external review.

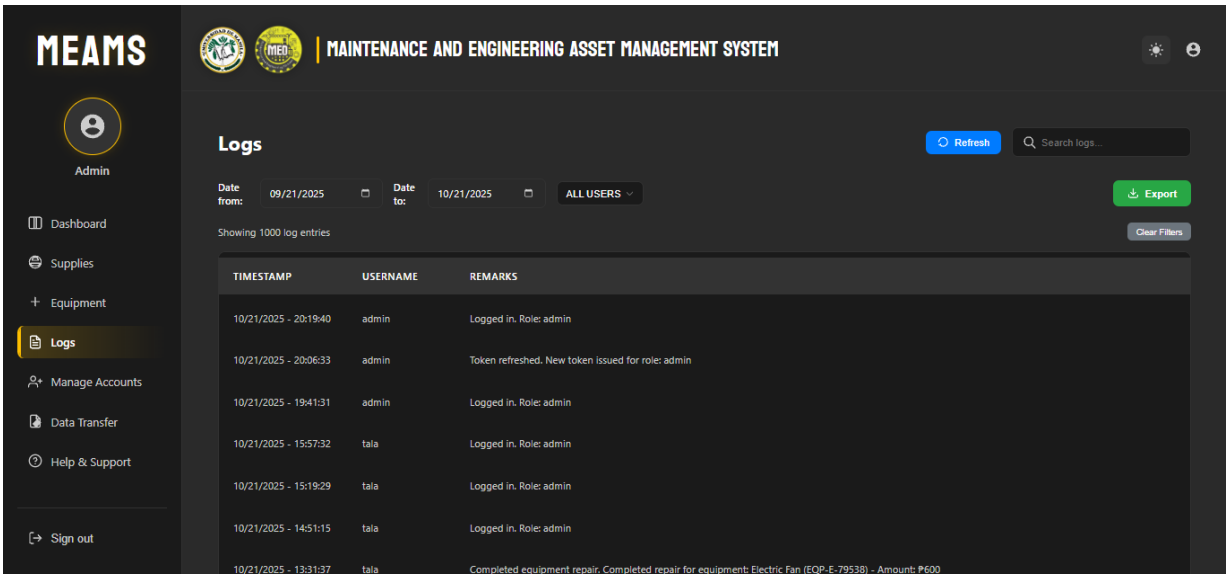


Figure 4-13. System Logs Page

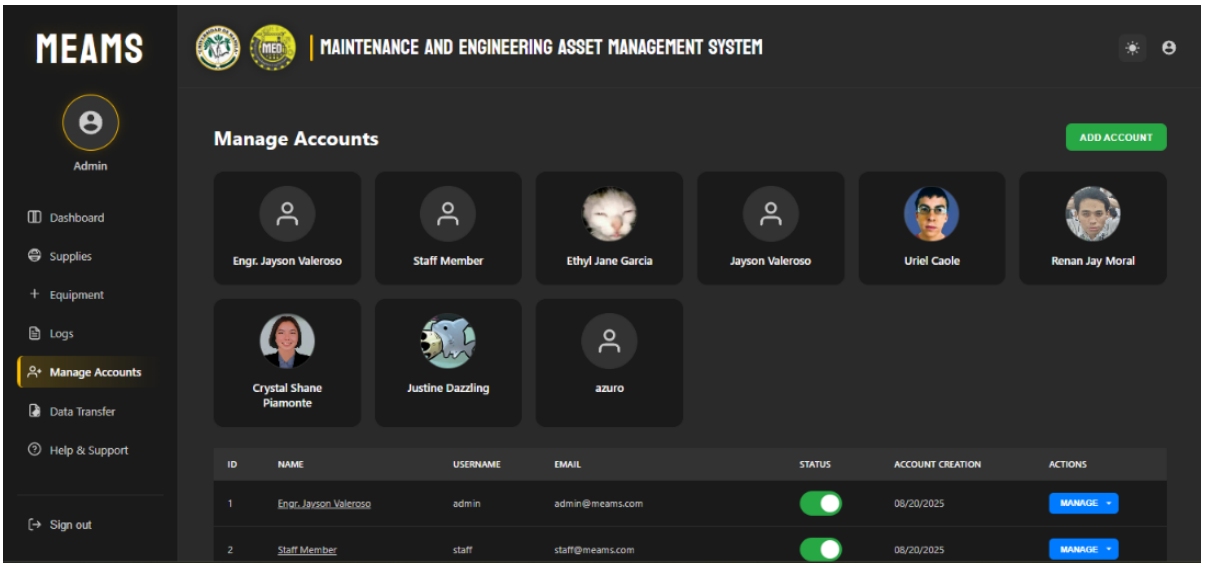


Figure 4-14. Manage Accounts Page

Figure 4-14 illustrates the Manage Accounts Page, presenting a comprehensive list of system users via interactive cards and a detailed table. Administrators can register new personnel using the "Add Account" button, click individual profiles to view details, or utilize the table actions to manage account status and permissions.

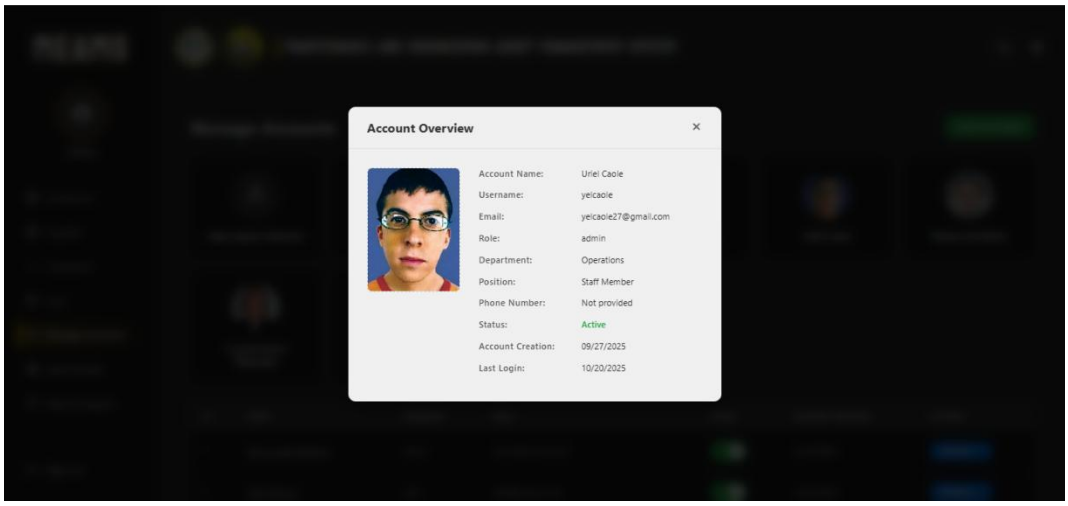


Figure 4-15. Account Overview Modal

Figure 4-15 depicts the Account Overview Modal, which is triggered by clicking a user's profile icon. This interface presents a detailed summary of the account holder, including key information such as Role, Department, Status, and login activity, enabling administrators to quickly verify user details.

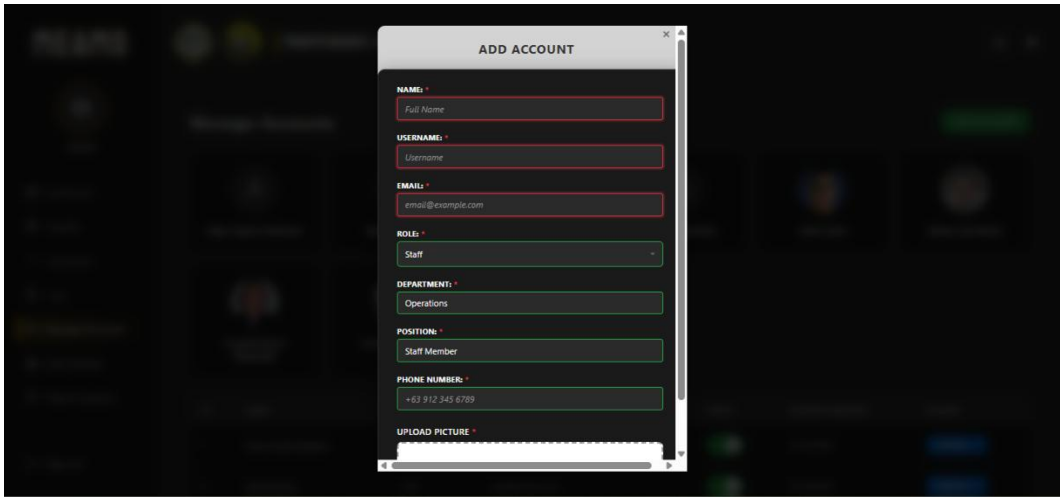


Figure 4-16. Add Account Modal Form

Figure 4-16 displays the Add Account Modal Form, which is activated by clicking the "Add Account" button. This interface enables administrators to register new users by

inputting essential details such as Full Name, Username, and Email. Upon submission, the system automatically generates a password and sends it to the provided email address, ensuring secure initial access for the new user.

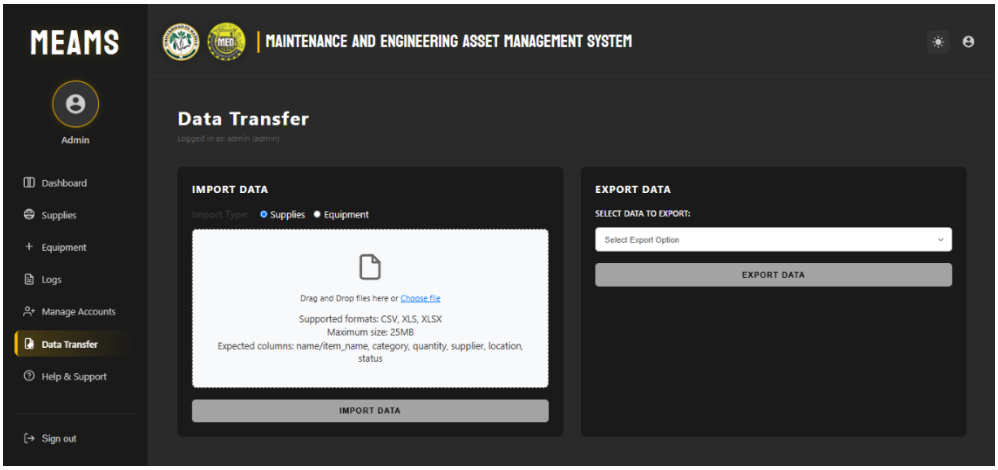


Figure 4-17. Data Transfer Page

Figure 4-17 depicts the Data Transfer Page, designed for bulk data management. Administrators can Import large datasets of supplies or equipment by uploading compatible CSV or XLSX files, or Export system records by selecting the desired data category from the dropdown menu and clicking the "EXPORT DATA" button.

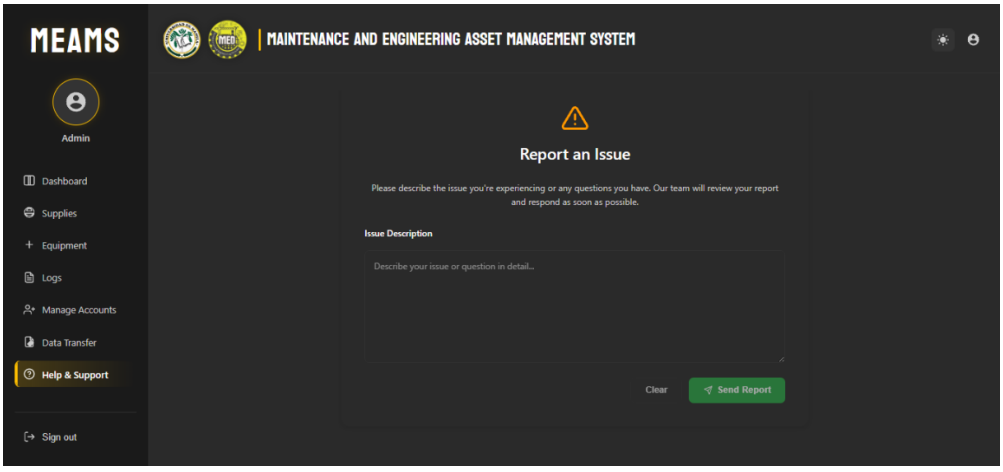


Figure 4-18. Help Page

Figure 4-18 illustrates the Help & Support Page, specifically the Report Issue interface. This section provides a dedicated text area for users to describe technical difficulties or inquiries and submit them directly to the support team for resolution by clicking the "Send Report" button.

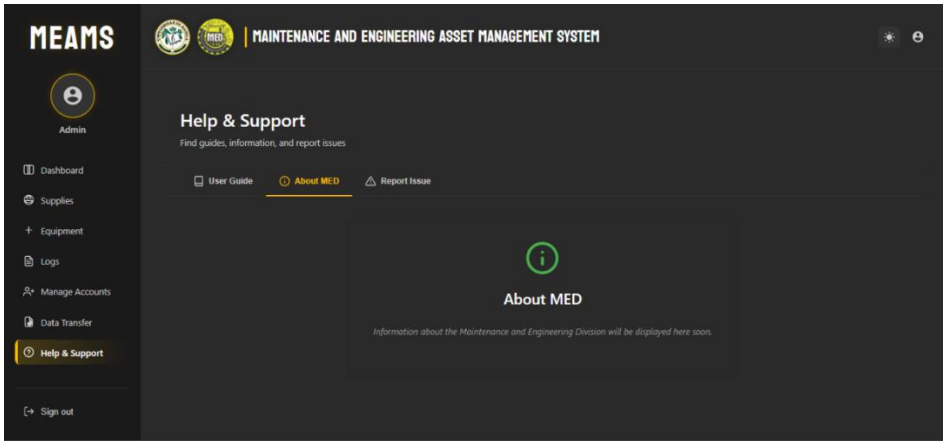


Figure 4-19. Support Page

Figure 4-19 illustrates the About MED tab within the Help & Support module. This section serves as an informational resource, displaying details about the MED to provide users with context regarding the department's role and services.

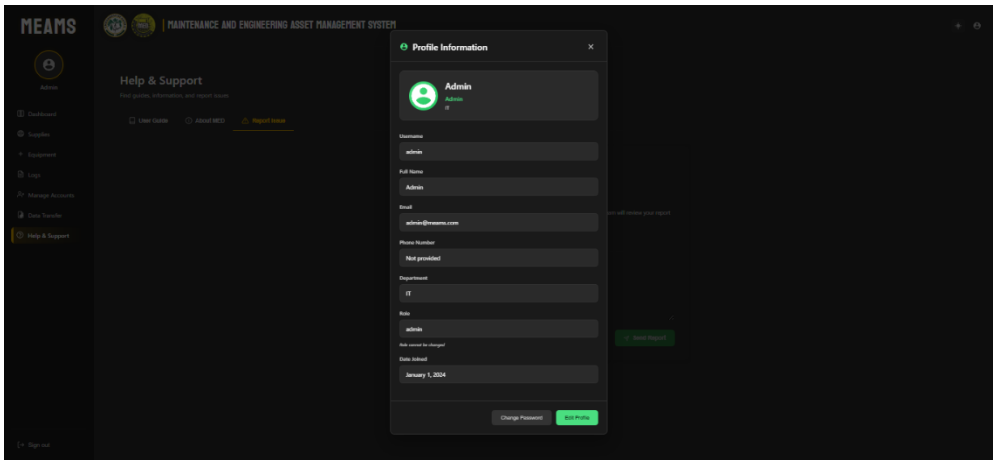


Figure 4-20. User Profile Page

Figure 4-20 illustrates the User Profile Page, which displays the logged-in user's account details such as Role and Department. This interface allows users to manage their account security by clicking "Change Password" or update their personal information via the "Edit Profile" button.

Evaluation Results of the Newly Developed System

Table 4-1. Respondents of the Study MEAMS

Respondents	No. of Respondents	Percentage
Maintenance and Engineering Division	39	57.4%
IT Practitioner	29	42.6%
Total	**68**	**100%**

The evaluation of the system follows the criteria outlined in ISO 25010, ensuring that MEAMS meets software quality standards in terms of Security, Functional Suitability, Performance Efficiency, Usability, Maintainability, and Reliability. Security ensures that the system protects sensitive asset and user data through encryption, role-based authentication, and activity logging, preventing unauthorized access and maintaining data integrity. Functional Suitability confirms that MEAMS delivers complete and accurate features such as real-time asset tracking, repair monitoring, and forecasting analytics that support the daily operations of the MED. Performance Efficiency emphasizes the system's ability to process data and render analytics swiftly, maintaining responsiveness and stability even when accessed by multiple users simultaneously. Usability highlights the intuitive and user-friendly interface that allows staff with minimal technical experience to navigate modules, record supplies, and generate reports with ease. Maintainability

reflects the system's modular architecture and clear documentation, enabling developers to efficiently apply updates, debug issues, and enhance functionality without disrupting operations. Reliability guarantees consistent performance, data accuracy, and system stability during continuous usage, ensuring that transactions and reports remain accessible without downtime. By adhering to these ISO 25010 standards, MEAMS demonstrates quality, security, and efficiency—providing a robust and sustainable solution that meets the operational needs of the MED of UDM.

Table 4-2. Verbal Interpretation Reference on Weighted Mean

Scale	Interpretation	Descriptive Equivalent
1	1.00 - 1.49	Needs Improvement (NI)
2	1.50 - 2.49	Fair (F)
3	2.50 - 3.49	Satisfactory (S)
4	3.50 - 4.49	Very Satisfactory (VS)
5	4.50 - 5.00	Excellent (E)

The Table 4-2 presents the range of values used in the survey questionnaire for evaluating the MEAMS. Each rating scale corresponds to a descriptive interpretation that represents the respondents' overall perception of the system. This structure helps ensure a clearer understanding of the gathered feedback across different user groups. The Weighted Mean formula was used to determine the respondents' assessment of the system, going beyond a simple average by assigning specific weights to each response. This approach provides a more accurate reflection of how respondents evaluated the MEAMS across various ISO 25010 criteria.

Formula:

$$w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

Where:

W = *weighted average*

n = *number of terms to be average*

w_i = *weighted applied to x values*

x_i = *data values to be averaged*

Table 4-3. Evaluation of MEAMS in terms of Security

Security	MED		IT Practitioner		OVERALL	
	WM	VI	WM	VI	WM	VI
The system ensured that only authorized personnel could access asset and inventory records.	4.97	E	4.97	E	4.84	E
The system protected sensitive data (e.g., asset details, repair history) from unauthorized access.	4.82	E	4.59	E	4.72	E
The system provided secure login and authentication features.	4.82	E	4.59	E	4.72	E
The system-maintained confidentiality and accountability of user actions.	4.85	E	4.55	E	4.72	E
GENERAL WEIGHTED MEAN					**4.75**	**E**

Legend: E = Excellent, VS = Very Satisfactory, S = Satisfactory, F = Fair, NI = Needs Improvement, WM= Weighted Mean, VI= Verbal Interpretation

The Table 4-3 presents the evaluation results of MEAMS in terms of Security, which recorded an overall general weighted mean of 4.75, interpreted as Excellent (E). Respondents highly rated the system's role-based authentication, activity log tracking,

and database encryption, ensuring the protection of sensitive asset data. For MED, the item “The system ensured that only authorized personnel could access asset and inventory records” received the highest rating with a weighted mean of 4.97, emphasizing their strong confidence in access restriction and control. Similarly, IT experts also gave their highest rating 4.97 to the same indicator, reflecting their recognition of MEAMS’s robust implementation of permission handling and account-level restrictions.

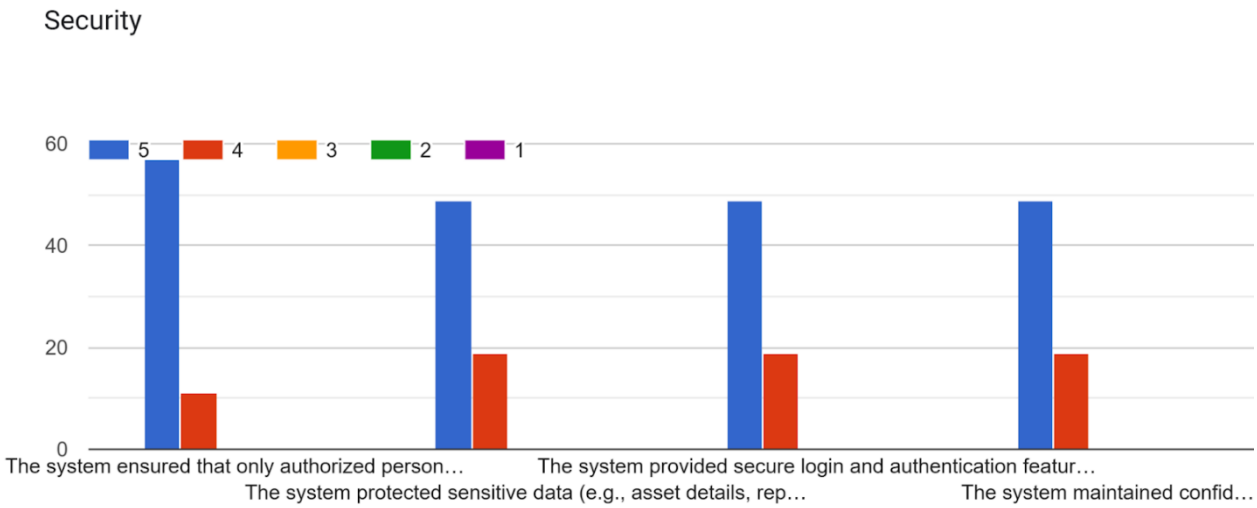


Figure 4-21. Qualitative Results of MEAMS in terms of Security

Figure 4-21 illustrates that 57 out of 68 respondents strongly agreed that their data remained secure and confidential during system use, showing trust in the system’s built-in protection features. These findings indicate that MEAMS fulfills ISO 25010’s requirements for maintaining confidentiality, integrity, and accountability, thus reinforcing the system’s credibility in secure asset management.

The findings of this study parallel the research conducted by Llabres (2023) on an Auto Parts Inventory Management System, which similarly utilized the ISO 25010 standard. In that study, the system achieved a weighted mean of 4.60 for Confidentiality, interpreted as Excellent. Llabres attributed this high rating to the system's ability to restrict

data access to authorized personnel only. This corroborates the MEAMS evaluation results (4.75), confirming that implementing strict role-based authentication and database encryption are effective industry-standard methods for achieving 'Excellent' security in asset management applications.

Table 4-4. Evaluation of MEAMS in terms of Functional Suitability

Functional Suitability	MED		IT Practitioner		OVERALL	
	WM	VI	WM	VI	WM	VI
The system provided all the necessary features to support asset tracking and management.	4.85	E	4.66	E	4.76	E
The system performed functions as intended without missing important requirements.	4.79	E	4.38	VS	4.62	E
The system accurately generated reports and forecasting outputs.	4.90	E	4.45	VS	4.71	E
The system met the operational needs of the Maintenance and Engineering Division.	4.87	E	4.55	E	4.74	E
GENERAL WEIGHTED MEAN					**4.71**	**E**

Legend: E = Excellent, VS = Very Satisfactory, S = Satisfactory, F = Fair, NI = Needs Improvement, WM= Weighted Mean, VI= Verbal Interpretation

The Table 4-4 presents the evaluation of MEAMS in terms of Functional Suitability, which obtained an overall general weighted mean of 4.71, interpreted as Excellent (E). Respondents affirmed that the system effectively supports core asset management functions, including real-time asset tracking, operational monitoring, and forecasting analytics. Among the MED, the item “The system accurately generated reports and forecasting outputs” received the highest weighted mean of 4.90, reflecting their confidence in the system’s ability to produce reliable and data-driven insights for asset evaluation and decision-making. Meanwhile, IT experts gave the highest rating of 4.66 to

“The system provided all the necessary features to support asset tracking and management,” emphasizing MEAMS’s completeness in meeting functional requirements from a technical perspective. These results indicate that MEAMS successfully meets the operational needs of the MED while maintaining high functional effectiveness, ensuring both performance reliability and practical usability across user groups.

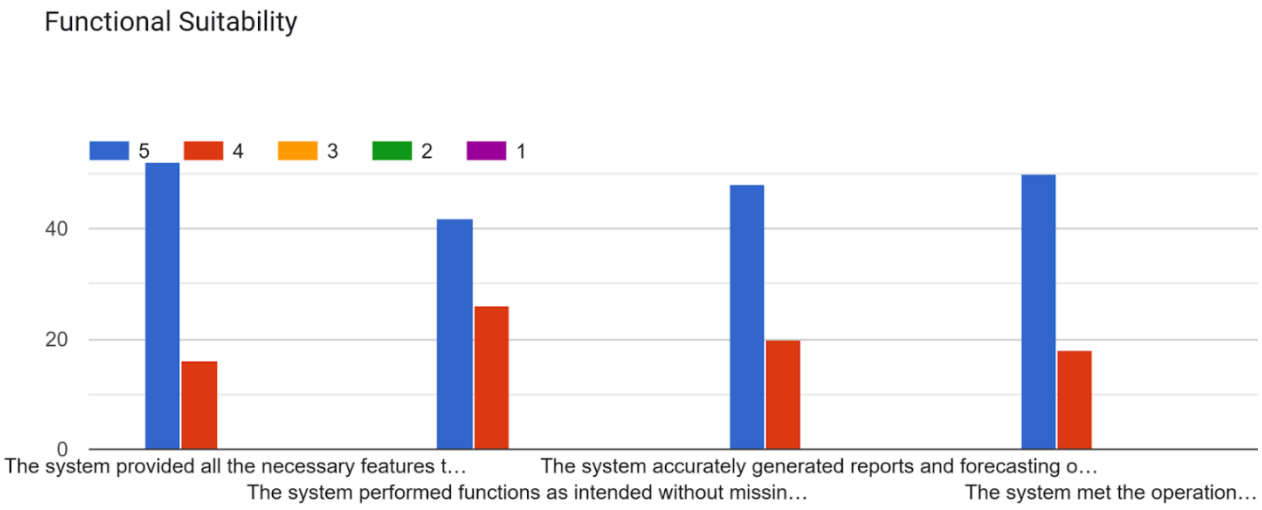


Figure 4-22. Qualitative Result of MEAMS in terms of Functional Suitability

In Figure 4-22 further illustrates that most respondents strongly agreed that the system’s features are functional and reliable, with 52 out of 68 participants confirming that MEAMS provides the necessary capabilities to manage assets effectively. These findings affirm that MEAMS is functionally suitable for digitizing and streamlining the division’s asset management processes.

The high Functional Suitability rating of MEAMS (4.71) is corroborated by the findings of a local study on a Records Tracking Management System (RTMS) (2023), which similarly achieved exceptional ratings under the ISO 25010 standard. In that study, the system's ability to provide real-time transparency on item status and maintain a

detailed audit trail was directly linked to meeting organizational goals and stakeholder needs. This mirrors the feedback from the MED regarding MEAMS, where the real-time tracking of supplies and repair history monitoring were identified as the key features driving their 'Excellent' (4.89) satisfaction rating. Both studies confirm that in management systems, the capability to perform efficient, real-time monitoring is the primary determinant of functional suitability.

Table 4-5. Evaluation of MEAMS in terms of Performance Efficiency

Performance Efficiency	MED		IT Practitioner		OVERALL	
	WM	VI	WM	VI	WM	VI
The system responded quickly during asset searches, updates, and report generation.	4.85	E	4.76	E	4.81	E
The system handled multiple transactions without slowing down.	4.79	E	4.31	VS	4.59	E
The system efficiently managed data retrieval through QR code scanning.	4.87	E	4.72	E	4.81	E
The system operated smoothly even during peak usage.	4.87	E	4.55	E	4.74	E
GENERAL WEIGHTED MEAN					**4.74**	**E**

Legend: E = Excellent, VS = Very Satisfactory, S = Satisfactory, F = Fair, NI = Needs Improvement, WM= Weighted Mean, VI= Verbal Interpretation

The Table 4-5 presents the evaluation of MEAMS in terms of Performance Efficiency, which obtained an overall general weighted mean of 4.74, interpreted as Excellent (E). Respondents commended the system's responsiveness, stability, and ability to process real-time transactions and dashboard analytics without performance degradation. Among MED, the highest rating of 4.87 was given to both "The system efficiently managed data retrieval through QR code scanning" and "The system operated

smoothly even during peak usage," highlighting their confidence in MEAMS's capability to handle intensive scanning and data operations efficiently. Meanwhile, IT experts awarded the highest score of 4.76 to "The system responded quickly during asset searches, updates, and report generation," emphasizing the system's optimized response time and backend processing. These results indicate that MEAMS consistently executed tasks within an acceptable time frame, even during concurrent operations, reinforcing its robustness and efficiency in real-world application.

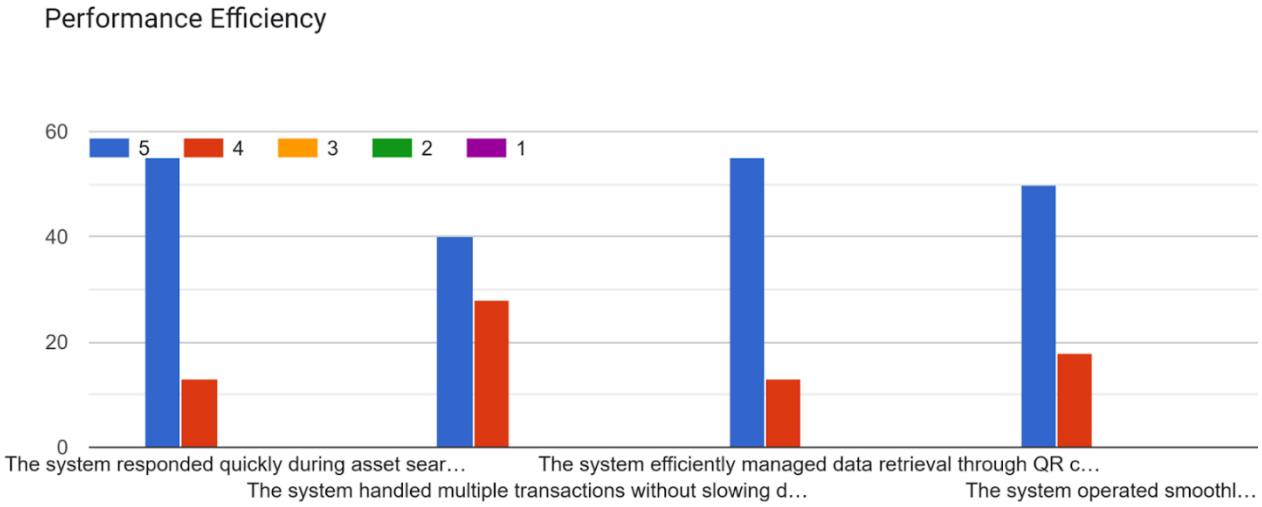


Figure 4-23. Qualitative Result of MEAMS in terms of Performance Efficiency

The Figure 4-23 shows that 55 out of 68 respondents strongly agreed that the system runs smoothly without frequent crashes or performance lags, demonstrating that MEAMS efficiently utilizes system resources while maintaining stability. These results validate that the system meets the ISO 25010 standard for performance efficiency in real-world use.

The evaluation results of MEAMS (4.74) are strongly supported by the local study of Salita et al. (2023) on a Web-based Barangay Profiling and Management System. In

their ISO 25010 evaluation, the system achieved a weighted mean of 4.75 for Time Behavior, interpreted as 'Excellent.' The researchers attributed this high score to the system's capability to handle data retrieval and dashboard updates with minimal latency, directly ensuring that officials could access essential records promptly. This finding mirrors the MEAMS evaluation, where the MED reported a similar satisfaction level (4.87) regarding the system's fast loading times and real-time dashboard updates. Both studies confirm that optimizing response time and resource utilization is critical for maintaining 'Excellent' performance in management systems handling real-time operational data.

Table 4-6. Evaluation of MEAMS in terms of Usability

Usability	MED		IT Practitioner		OVERALL	
	WM	VI	WM	VI	WM	VI
The system was easy to learn and use.	4.90	E	4.86	E	4.88	E
The interface was user-friendly and intuitive.	4.90	E	4.62	E	4.78	E
The dashboard visualizations were clear and easy to understand.	4.87	E	4.55	E	4.74	E
The system helped reduce manual work and simplified staff tasks.	4.82	E	4.59	E	4.72	E
GENERAL WEIGHTED MEAN					**4.78**	**E**

Legend: E = Excellent, VS = Very Satisfactory, S = Satisfactory, F = Fair, NI = Needs Improvement, WM= Weighted Mean, VI= Verbal Interpretation

The Table 4-6 presents the evaluation of MEAMS in terms of Usability, which obtained an overall general weighted mean of 4.78, interpreted as Excellent (E). Respondents affirmed that MEAMS is highly user-friendly and intuitive, noting its clear interface layout, ease of feature navigation, and logically structured dashboard components. Among MED, the highest rating of 4.90 was given to both “The system was

easy to learn and use" and "The interface was user-friendly and intuitive," indicating that staff were able to perform operations efficiently without the need for extensive technical training. Meanwhile, IT practitioners gave the highest rating of 4.86 to "The system was easy to learn and use," recognizing MEAMS's accessibility and seamless workflow integration even from a technical standpoint. These results imply that the system's interface design effectively enhances user productivity, minimizes operational errors, and promotes smooth adoption among both administrative and technical users.

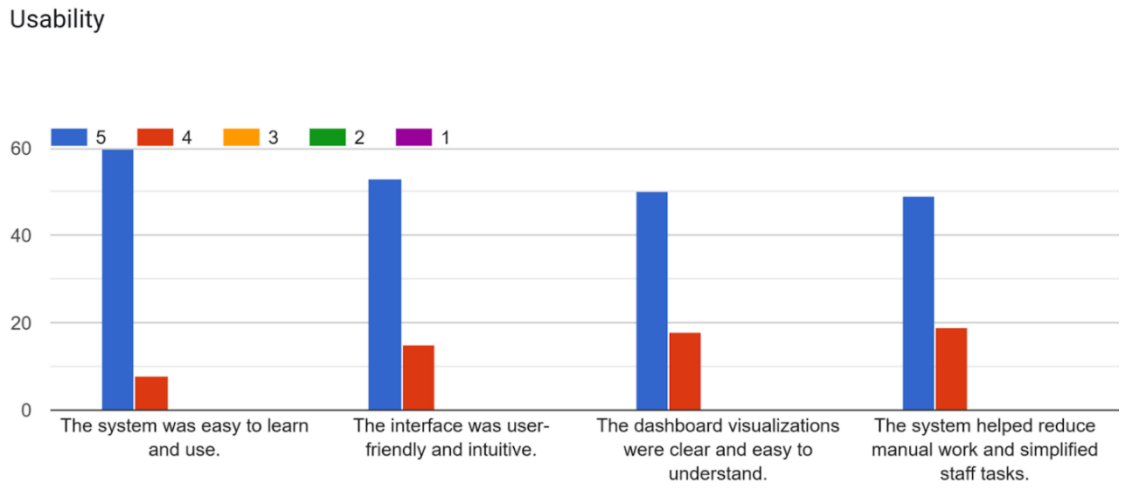


Figure 4-24. Qualitative Result of MEAMS in terms of Usability

Figure 4-24 illustrates the qualitative results of MEAMS in terms of usability, showing that 60 out of 68 respondents strongly agreed that the interface and visual indicators contribute to ease of use and improved work accuracy. This finding aligns with ISO 25010 standards for usability, confirming that MEAMS provides a positive, efficient, and engaging user experience across its modules.

The 'Excellent' Usability rating of MEAMS (4.78) is supported by the study of Encarnacion (2025) on the Development and Evaluation of an Advanced Property Management System (PMS). In their ISO 25010 assessment, the system achieved a high

mean rating of 4.62 for Usability, with researchers highlighting that a user-friendly interface was crucial for ensuring operational efficiency among property owners and managers. This aligns with the feedback from the MED regarding MEAMS, where the intuitive design and visually organized dashboard (4.90) were key factors in their ability to navigate the system without extensive training. Both studies confirm that prioritizing clear layouts and intuitive navigation directly correlates with 'Excellent' usability scores in asset management applications.

Table 4-7. Evaluation of MEAMS in terms of Maintainability

Maintainability	MED		IT Practitioner		OVERALL	
	WM	VI	WM	VI	WM	VI
The system was easy to update and maintain without major disruptions.	4.87	E	4.66	E	4.78	E
The system's modular design allowed efficient troubleshooting and modifications.	4.82	E	4.55	E	4.71	E
The system documentation supported maintenance activities.	4.82	E	4.59	E	4.72	E
The system could be enhanced or expanded when necessary.	4.90	E	4.62	E	4.78	E
GENERAL WEIGHTED MEAN					**4.75**	**E**

Legend: E = Excellent, VS = Very Satisfactory, S = Satisfactory, F = Fair, NI = Needs Improvement, WM= Weighted Mean, VI= Verbal Interpretation

The Table 4-7 presents the evaluation of MEAMS in terms of Maintainability, which obtained an overall general weighted mean of 4.75, interpreted as Excellent (E). Respondents commended the system's modular structure and organized coding framework, which allow updates and troubleshooting to be performed efficiently without causing major system disruptions. Among MED, the highest rating of 4.90 was given to

"The system could be enhanced or expanded when necessary," highlighting their confidence that MEAMS is scalable and adaptable to future organizational needs. Meanwhile, IT practitioners gave their highest score of 4.66 to "The system was easy to update and maintain without major disruptions," emphasizing the system's stable architecture and ease of maintenance from a technical perspective. These findings indicate that MEAMS supports long-term sustainability through streamlined update processes, error-handling capabilities, and well-structured documentation.

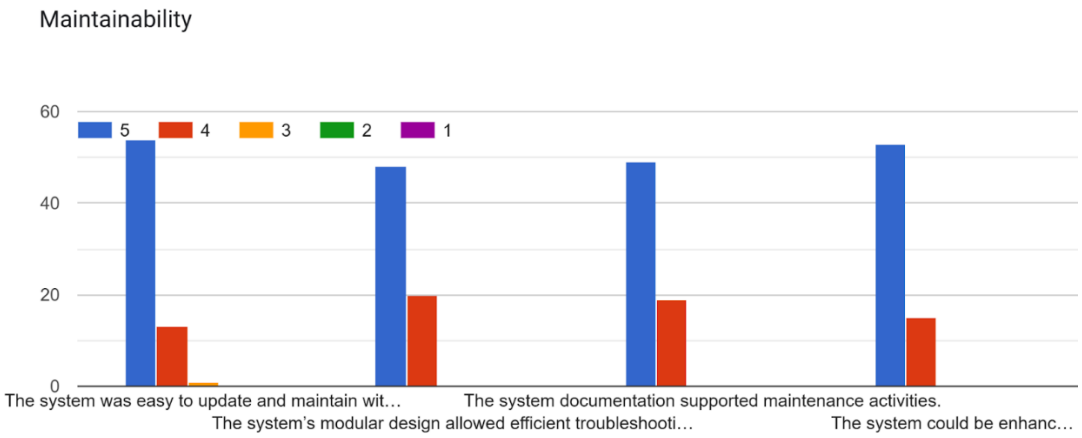


Figure 4-25. Qualitative Result of MEAMS in terms of Maintainability

Figure 4-25 illustrates the qualitative results of MEAMS in terms of maintainability, showing that 54 out of 68 respondents strongly agreed that the system can be easily maintained and enhanced to adapt to future technological advancements. This finding confirms that MEAMS aligns with ISO 25010 maintainability standards by supporting efficient modification, testing, and continuous improvement over time.

The 'Excellent' Maintainability rating of MEAMS (4.75) is supported by the local study on an Online Order and Inventory Management System (2025). In their ISO 25010 evaluation, the system achieved a weighted mean of 4.60 for Maintainability, with IT

experts rating it as 'Completely Acceptable' (equivalent to Excellent). The study highlighted that systems designed with high modularity and modifiability allow for easier updates and error corrections without disrupting operations. This mirrors the findings for MEAMS, where the MED gave a high rating (4.90) specifically acknowledging the system's structured coding and modular components. Both studies confirm that a modular architecture is the primary factor in achieving high maintainability scores in asset and inventory management systems.

Table 4-8 Evaluation of MEAMS in terms of Reliability

Reliability	MED		IT Practitioner		OVERALL	
	WM	VI	WM	VI	WM	VI
The system consistently operated without crashing or errors.	4.74	E	4.48	E	4.63	E
The system was available whenever needed.	4.90	E	4.62	E	4.78	E
The system could recover quickly from errors or failures.	4.85	E	4.34	VS	4.63	E
The system maintained data accuracy and integrity over time.	4.74	E	4.62	E	4.69	E
GENERAL WEIGHTED MEAN					**4.68**	**E**

Legend: E = Excellent, VS = Very Satisfactory, S = Satisfactory, F = Fair, NI = Needs Improvement, WM= Weighted Mean, VI= Verbal Interpretation

The Table 4-8 presents the evaluation of MEAMS in terms of Reliability, which achieved an overall general weighted mean of 4.68, interpreted as Excellent (E). Respondents commended the system for maintaining consistent performance during extended operations, with no observable instances of crash, data loss, or prolonged downtime during evaluation. Among MED personnel, the highest rating of 4.90 was given

to “The system was available whenever needed,” demonstrating their strong confidence in MEAMS’s operational stability and responsiveness during critical asset management activities. Meanwhile, IT practitioners gave their highest score of 4.62 to both “The system was available whenever needed” and “The system maintained data accuracy and integrity over time,” highlighting the system’s ability to reliably handle data transactions and preserve information quality throughout usage. These results indicate that MEAMS ensures dependable performance in saving, retrieving, and updating asset records, reinforcing its suitability for mission-critical applications within the MED.

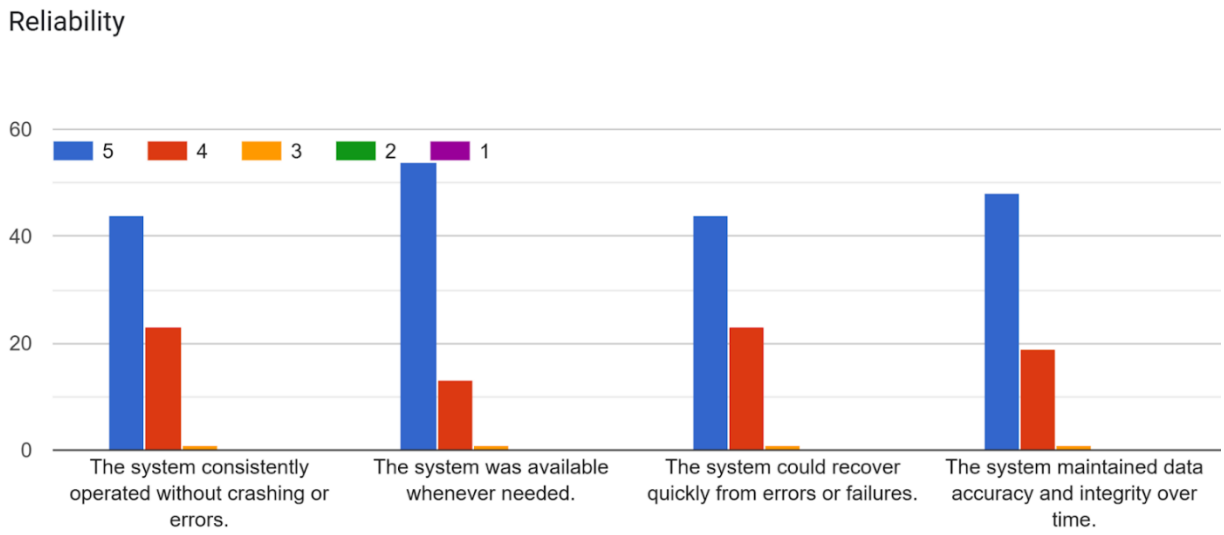


Figure 4-26. Qualitative Result of MEAMS in terms of Reliability

The Figure 4-26 further supports these findings, where 54 out of 68 respondents strongly agreed that the system consistently performs as intended without errors, system crashes, or corrupted files. This proves that MEAMS has achieved operational stability and dependability, satisfying the ISO 25010 criteria for reliability and recoverability.

The evaluation results for MEAMS’ Reliability, with a general weighted mean of 4.68 interpreted as Excellent, are supported by a related local study on an Online Equipment Management System for a Property Management Department. This study also

applied the ISO 25010 standard for system quality assessment and reported a similarly high Reliability rating, with a weighted mean of 4.56 for recoverability and fault handling. The authors emphasized that the system's automated backup and recovery mechanisms ensured consistent performance even after failures, aligning with the MEAMS respondents' commendation of dependable operations with no significant downtime or data loss. This parallel confirms that in asset and equipment management systems, reliable data saving, retrieving, and updating capabilities are fundamental to achieving high user confidence and excellent reliability ratings, as reflected in MEAMS's robust performance during continuous operations and positive feedback from the MED. This local evidence reinforces the validity of MEAMS's Reliability findings within the ISO 25010 framework.

Chapter V

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter provides a summary of the results and conclusions outlined from the research questions and the project's outcomes. Furthermore, it provides recommendations based on the findings and the purpose of the project.

Summary of Findings

The MEAMS: A Real-Time Tracking and Forecasting Asset Management System for the MED of UDM, was evaluated using ISO 25010 software quality standards, specifically focusing on Security, Functional Suitability, Performance Efficiency, Usability, Maintainability, and Reliability. A survey was conducted among respondents from the MED staff, administrators, and IT experts to assess the system’s effectiveness, resulting in high overall ratings across all categories.

Based on the data gathered, the following findings were determined:

a. The survey result for Security showed an Excellent performance, with a general weighted mean of 4.75, indicating that respondents were confident in the system’s capability to protect sensitive data through encryption, role-based authentication, and activity log monitoring.
b. The survey result for Functional Suitability achieved an Excellent rating, with a general weighted mean of 4.71, demonstrating that MEAMS effectively meets operational needs through accurate asset tracking, repair monitoring, and forecasting features.

c. The survey result for Performance Efficiency showed an Excellent rating, with a general weighted mean of 4.74, emphasizing the system's responsiveness and stability during simultaneous user operations.
d. The survey result for Usability earned an Excellent rating, with a general weighted mean of 4.78, confirming that users found the interface intuitive, easy to navigate, and visually organized.
e. The survey result for Maintainability obtained an Excellent rating, with a general weighted mean of 4.75, verifying that the modular architecture and clear documentation make it simple to update, debug, and enhance the system.
f. The survey result for Reliability reflected an Excellent performance, with a general weighted mean of 4.68, showing that MEAMS consistently performs as intended with minimal downtime and accurate data handling.

Conclusions

Based on the findings of the study, the following conclusions were drawn:

1. The MEAMS effectively modernized the asset tracking and management of the MED by replacing manual, paper-based methods with a digital platform that enhanced operational efficiency and accuracy.

2. The integration of QR code-based labeling and digitized stock cards allowed for real-time asset updates and inventory monitoring, significantly reducing discrepancies and improving resource allocation within the MED.

3. The system's forecast analytics feature utilized the SARIMA model to forecast demand trends for supplies and equipment based on historical requisition data.

Through threshold detection and life cycle costing, MEAMS effectively identifies understocked supplies and equipment nearing the end of their useful life, supporting timely, repair, and replacement decisions.

4. The system adhered to ISO 25010 quality standards, ensuring functional suitability, security, performance efficiency, usability, reliability, and maintainability, which contributed to overall user satisfaction and system dependability.

5. Feedback from both MED and IT practitioners demonstrated strong acceptance of the system's usability, reliability, and security, confirming that MEAMS effectively supports the division's asset management needs.

Recommendations

Based on the work accomplished during this project and the conclusions drawn, the following recommendations are proposed:

1. **Enhance Functional Suitability:** Regularly update the MEAMS to incorporate new features such as real-time notifications, comprehensive reporting tools, and mobile applications to further increase user engagement and operational functionality.

2. **Optimize Compatibility:** Conduct regular testing across various devices and browsers to maintain seamless cross-platform performance and adapt to emerging technologies.

3. **Improve Usability:** Introduce onboarding aids such as video tutorials, FAQs, and intelligent chatbots to support new users. Implement regular usability

testing with feedback mechanisms to refine the user interface and simplify complex workflows.

4. **Ensure Reliability:** Establish routine system, performance monitoring, stress testing, and an effective disaster recovery plan to maintain high system availability and quick recovery from failures.

5. **Strengthen Security:** Adopt multi-factor authentication, conduct regular penetration testing, and maintain comprehensive audit logs to enhance protection against evolving security threats.

6. **Enhance Portability:** Develop offline access capabilities for critical functions and create lightweight mobile applications to improve asset management in environments with limited or intermittent internet connectivity.

7. **Guide Future Researchers:** Future studies should explore advanced technologies such as artificial intelligence, machine learning, and IoT integration to further improve repair actions and asset management. Engaging end-users throughout the development lifecycle is recommended to ensure systems meet evolving operational needs efficiently and inclusively.

8. **Establish Data Governance Policies:** Since the system utilizes the SARIMA model for forecasting, the accuracy of predictions depends heavily on the quality of historical data. It is recommended to enforce strict data entry standards and conduct regular data audits to ensure consistency, thereby preventing data irregularities that could compromise the reliability of future trend analysis and decision-making.

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APPENDICES

APPENDIX A: CONSENT FORM

August 14, 2025

Engr. Jayson L. Valeroso
Maintenance and Engineering Division
Universidad de Manila
Cecilia Munoz-Palma cor. Antonio J. Villegas Streets, Manila, Philippines

Subject: Request for Client Partnership in a Capstone Project and Research

Dear Engr. Jayson Valeroso,

Greetings!

We are the BSIT-DS42 students of Universidad de Manila. We would like to formally request you to be our client for our project in Capstone Project and Research.

As part of our academic requirements, we are tasked to develop a software solution that addresses real-world challenges faced by businesses or organizations. We believe that your company/organization would be an excellent partner for this project, as we aim to analyze your workflow, identify key challenges, and propose a tailored software solution to improve efficiency and productivity.

Our commitment is to work closely with your team, ensuring that the proposed system aligns with your needs. We also guarantee that all collected data will be kept confidential and used solely for academic purposes.

We would appreciate the opportunity to discuss this collaboration further at your convenience. Please let us know a suitable time for a meeting. We look forward to your positive response and hope to work with you on this exciting project.

Thank you for your time and consideration.

Sincerely,

Caole, Christoper Uriel
Dazo, Justine
Garcia, Ethyl Jane
Moral, Renan Jay
Piamonte, Crystal Shane
BSIT-DS42
Universidad De Manila

Approved by:

Prof. Christina P. Atal
Professor, Capstone Project and Research

Engr. Jayson L. Valeroso
Head, Maintenance and Engineering Division

APPENDIX B: SURVEY QUESTIONNAIRE

Evaluation of MEAMS: A Real-Time Tracking and Forecasting Asset Management System

This survey aims to evaluate the MEAMS system based on ISO 25010 standards. Your responses will help assess the system's quality in terms of security, functional suitability, performance efficiency, usability, maintainability, and reliability.
No personal identifiers (such as your name, ID number, or contact information) will be included or published in the results of this study. By continuing to answer this form, you are giving your consent for your responses to be used for the stated purpose.

Please rate each statement using the following scale:

5 – Strongly Agree
4 – Agree
3 – Neutral
2 – Disagree
1 – Strongly Disagree

ethylgarcia07@gmail.com Switch account

Not shared

* Indicates required question

Admin Demonstration Video
https://drive.google.com/file/d/1Hjv3hBZE5HFFMibSnjQ-0qfYqHjBGxOp/view?usp=sharing

Staff Demonstration Video
https://drive.google.com/file/d/1IQoo0nHaImEmFdZVrP0UHbZmtksfRpi6/view?usp=sharing

Name (Optional):

Your answer

Type of Respondent *

- ◯ Maintenance Engineering Division
- ◯ IT Practitioner

Security *

	5	4	3	2	1
The system ensured that only authorized personnel could access asset and inventory records.	○	○	○	○	○
The system protected sensitive data (e.g., asset details, repair history) from unauthorized access.	○	○	○	○	○
The system provided secure login and authentication features.	○	○	○	○	○
The system maintained confidentiality and accountability of user actions.	○	○	○	○	○

Functional Suitability *

	5	4	3	2	1
The system provided all the necessary features to support asset tracking and management.	○	○	○	○	○
The system performed functions as intended without missing important requirements.	○	○	○	○	○
The system accurately generated reports and forecasting outputs.	○	○	○	○	○
The system met the operational needs of the Maintenance and Engineering Division.	○	○	○	○	○

Performance Efficiency *

	5	4	3	2	1
The system responded quickly during asset searches, updates, and report generation.	○	○	○	○	○
The system handled multiple transactions without slowing down.	○	○	○	○	○
The system efficiently managed data retrieval through QR code scanning.	○	○	○	○	○
The system operated smoothly even during peak usage.	○	○	○	○	○

Usability *

	5	4	3	2	1
The system was easy to learn and use.	○	○	○	○	○
The interface was user-friendly and intuitive.	○	○	○	○	○
The dashboard visualizations were clear and easy to understand.	○	○	○	○	○
The system helped reduce manual work and simplified staff tasks.	○	○	○	○	○

Maintainability *

	5	4	3	2	1
The system was easy to update and maintain without major disruptions.	○	○	○	○	○
The system's modular design allowed efficient troubleshooting and modifications.	○	○	○	○	○
The system documentation supported maintenance activities.	○	○	○	○	○
The system could be enhanced or expanded when necessary.	○	○	○	○	○

Reliability *

	5	4	3	2	1
The system consistently operated without crashing or errors.	○	○	○	○	○
The system was available whenever needed.	○	○	○	○	○
The system could recover quickly from errors or failures.	○	○	○	○	○
The system maintained data accuracy and integrity over time.	○	○	○	○	○

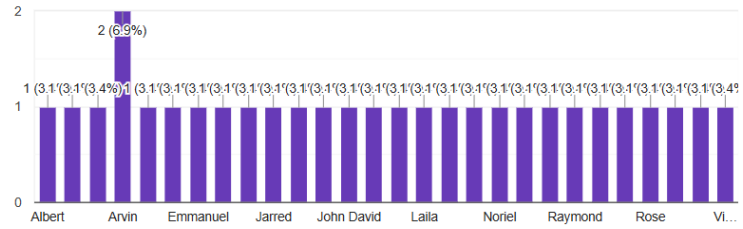
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APPENDIX C: SURVEY RESULTS

Name (Optional):

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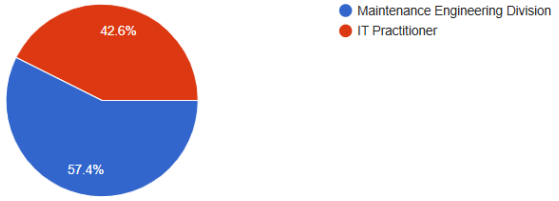
29 responses



Type of Respondent

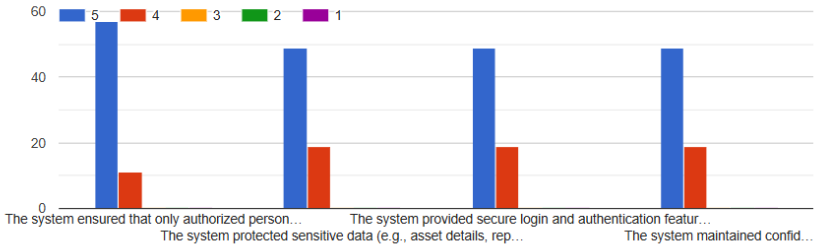
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68 responses



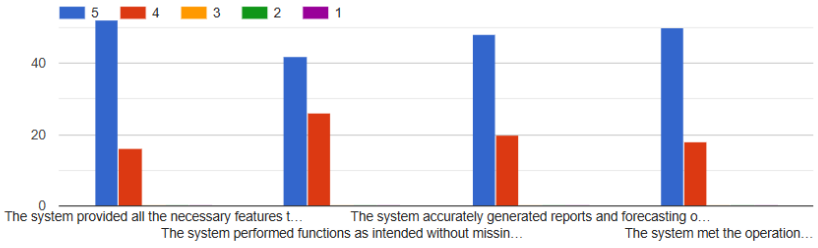
Security

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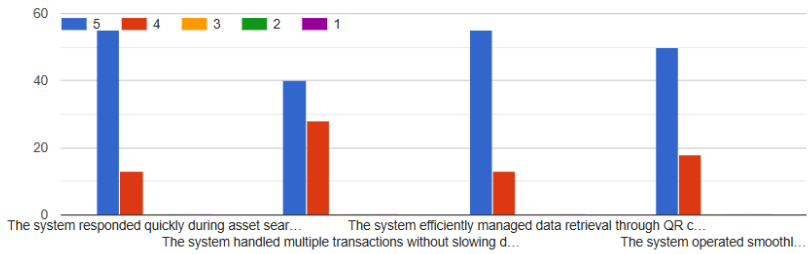
Functional Suitability

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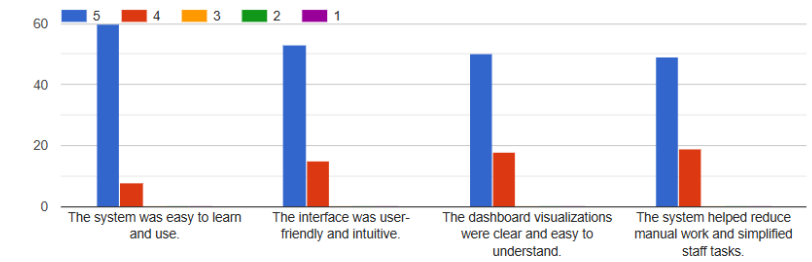
Performance Efficiency

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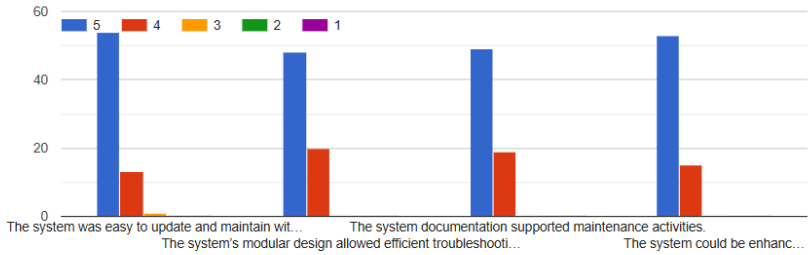
Usability

Copy chart



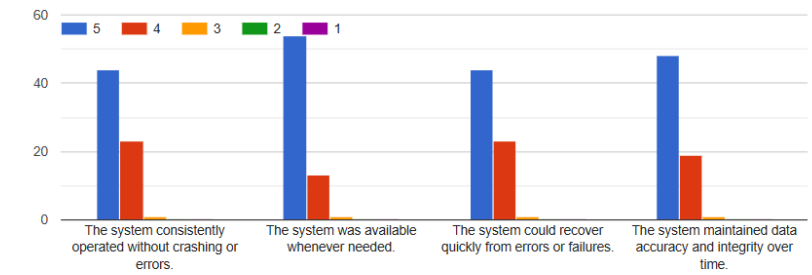
Maintainability

Copy chart



Reliability

Copy chart



APPENDIX D: SOURCE CODE

```
52      // Function to get auth token from AuthContext
53      const getAuthToken = () => {
54        return adminToken || authToken;
55      };
56
57      // Function to download file from blob
58      const downloadFile = (blob, filename) => {
59        const url = URL.createObjectURL(blob);
60        const link = document.createElement('a');
61        link.href = url;
62        link.download = filename;
63        link.style.visibility = 'hidden';
64        document.body.appendChild(link);
65        link.click();
66        document.body.removeChild(link);
67        URL.revokeObjectURL(url);
68      };
69
70      // Function to download file from base64 data (for ZIP files)
71      const downloadBase64File = (base64Data, filename) => {
72        try {
73          const binaryString = atob(base64Data);
74          const bytes = new Uint8Array(binaryString.length);
75          for (let i = 0; i < binaryString.length; i++) {
76            bytes[i] = binaryString.charCodeAt(i);
77          }
78          const blob = new Blob([bytes], { type: 'application/zip' });
79          downloadFile(blob, filename);
80        } catch (error) {
81          console.error('Error downloading base64 file:', error);
82          alert('Error downloading file. Please try again.');
83        }
84      };
```

```
187
188     const getAuthToken = () => localStorage.getItem('authToken');
189
190     const handleSignOut = async () => {
191       try {
192         const token = getAuthToken();
193         if (token) {
194           try {
195             await fetch(`${process.env.REACT_APP_API_URL}/logout`, {
196               method: 'POST',
197               headers: {
198                 Authorization: `Bearer ${token}`,
199                 'Content-Type': 'application/json',
200               },
201             });
202           } catch {}
203         }
204         localStorage.removeItem('authToken');
205         navigate('/login');
206       } catch {
207         localStorage.removeItem('authToken');
208         navigate('/login');
209       }
210     };
211
212     const toggleStaffMenu = () => setShowStaffMenu(!showStaffMenu);
213
214     const handleProfileClick = (e) => {
215       e.preventDefault();
216       setShowProfileModal(true);
217       setShowStaffMenu(false);
218     };
219
220     const isActive = (path) => location.pathname === path;
221
222     return (
223       <div className="staff-layout-container">
224         <aside className="staff-sidebar">
225           <div className="logo">
226           <Link to="/dashboard" style={{ textDecoration: 'none' }}>
```

```
90      const handlePasswordChange = async (e) => {
118       setIsLoading(true);
119
120       try {
121         const response = await fetch(`${process.env.REACT_APP_API_URL}/api/change-password`, {
122           method: 'POST',
123           headers: {
124             'Content-Type': 'application/json',
125             'Authorization': `Bearer ${authToken}`
126           },
127           body: JSON.stringify({
128             current_password: currentPassword,
129             new_password: newPassword,
130           }),
131         });
132
133         const data = await response.json();
134
135         if (response.ok) {
136           setPasswordChangeMessage('Password changed successfully! Redirecting...');
137
138           // Store the token and redirect after successful password change
139           if (userRole === 'admin') {
140             localStorage.setItem('adminToken', authToken);
141             localStorage.setItem('authToken', authToken);
142             setTimeout(() => {
143               window.location.href = '/administrator';
144             }, 2000);
145           } else if (userRole === 'staff') {
146             localStorage.setItem('authToken', authToken);
147             setTimeout(() => {
148               window.location.href = '/dashboard';
149             }, 2000);
150           }
151         } else {
152           setPasswordChangeMessage(data.detail || 'Failed to change password');
153         }
154       } catch (error) {
155         setPasswordChangeMessage('Connection error. Please try again.');
156         console.error('Password change error:', error);
```

```
5     function ManageAccountsPage() {
96      const fetchAccounts = async () => {
104       }
105
106       const response = await fetch(`${process.env.REACT_APP_API_URL}/api/accounts`, {
107         headers: {
108           'Authorization': `Bearer ${token}`,
109           'Content-Type': 'application/json'
110         }
111       });
112
113       if (!response.ok) {
114         if (response.status === 401) {
115           setError('Authentication failed. Please log in again.');
116           return;
117         }
118         if (response.status === 403) {
119           setError('Access denied. Admin privileges required to manage accounts.');
120           return;
121         }
122         throw new Error(`HTTP error! status: ${response.status}`);
123       }
124
125       const data = await response.json();
126       if (data.success) {
127         console.log('Accounts data received:', data.data);
128         setAccounts(data.data || []);
129         setError('');
130       } else {
131         setError('Failed to fetch accounts');
132       }
133     } catch (error) {
134       console.error('Error fetching accounts:', error);
135       setError('Failed to load accounts. Please try again.');
136     } finally {
137       setLoading(false);
138     }
139   };
140
```

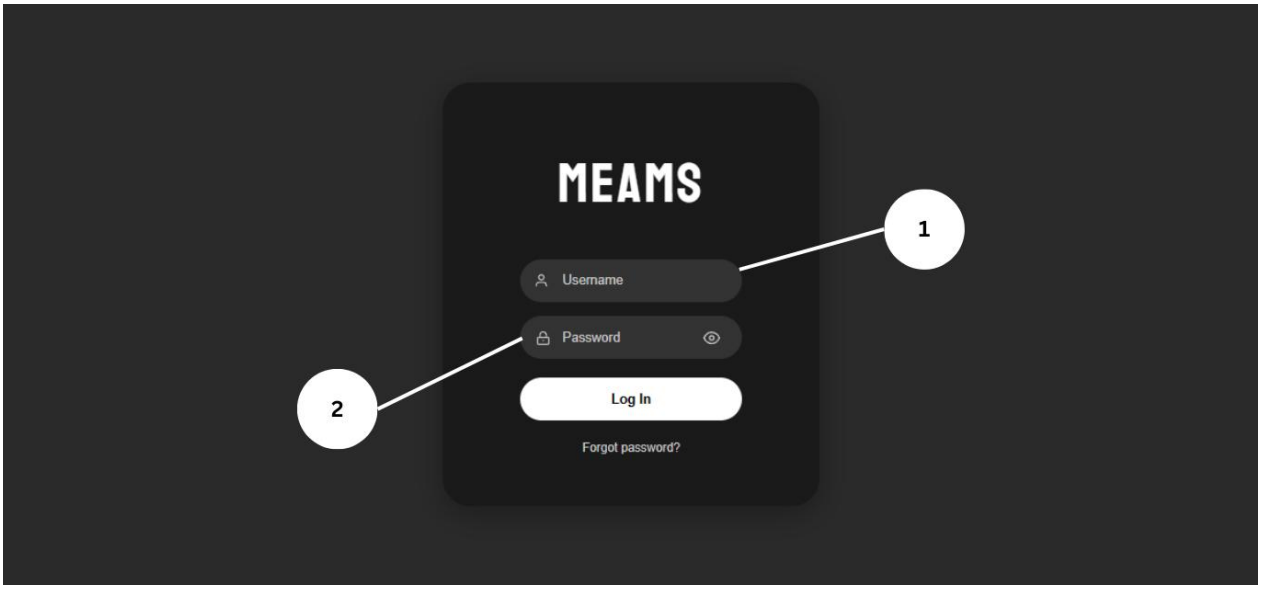
```
220     const handleDeleteSupply = async (supplyId, supplyName) => {
263     const paginatedSupplies = filteredSupplies.slice(startIndex, endIndex);
264
265     // Pagination handlers
266     const handlePageChange = (page) => {
267       setCurrentPage(page);
268     };
269
270     const handleItemsPerPageChange = (e) => {
271       setItemsPerPage(parseInt(e.target.value));
272       setCurrentPage(1);
273     };
274
275     const generatePageNumbers = () => {
276       const pageNumbers = [];
277       const maxVisiblePages = 5;
278
279       if (totalPages <= maxVisiblePages) {
280         for (let i = 1; i <= totalPages; i++) {
281           pageNumbers.push(i);
282         }
283       } else {
284         if (currentPage <= 3) {
285           for (let i = 1; i <= 4; i++) {
286             pageNumbers.push(i);
287           }
288           pageNumbers.push('...');
289           pageNumbers.push(totalPages);
290         } else if (currentPage >= totalPages - 2) {
291           pageNumbers.push(1);
292           pageNumbers.push('...');
293           for (let i = totalPages - 3; i <= totalPages; i++) {
294             pageNumbers.push(i);
295           }
296         } else {
297           pageNumbers.push(1);
298           pageNumbers.push('...');
299           for (let i = currentPage - 1; i <= currentPage + 1; i++) {
300             pageNumbers.push(i);
```

```
4   function ResetPasswordPage() {
48        setIsValidToken(false);
49        setMessage('Error validating reset token. Please try again.');
50      } finally {
51        setIsLoading(false);
52      }
53    };
54
55    const handleResetPassword = async (e) => {
56      e.preventDefault();
57
58      // Validation
59      if (!password || !confirmPassword) {
60        setMessage('Please fill in all fields');
61        return;
62      }
63
64      if (password.length < 8) {
65        setMessage('Password must be at least 8 characters long');
66        return;
67      }
68
69      if (password !== confirmPassword) {
70        setMessage('Passwords do not match');
71        return;
72      }
73
74      setIsLoading(true);
75      setMessage('');
76
77      try {
78        const response = await fetch(`${process.env.REACT_APP_API_URL}/api/auth/reset-password`, {
79          method: 'POST',
80          headers: {
81            'Content-Type': 'application/json',
82          },
83          body: JSON.stringify({
84            token: token,
85            new_password: password,
86          }),
```

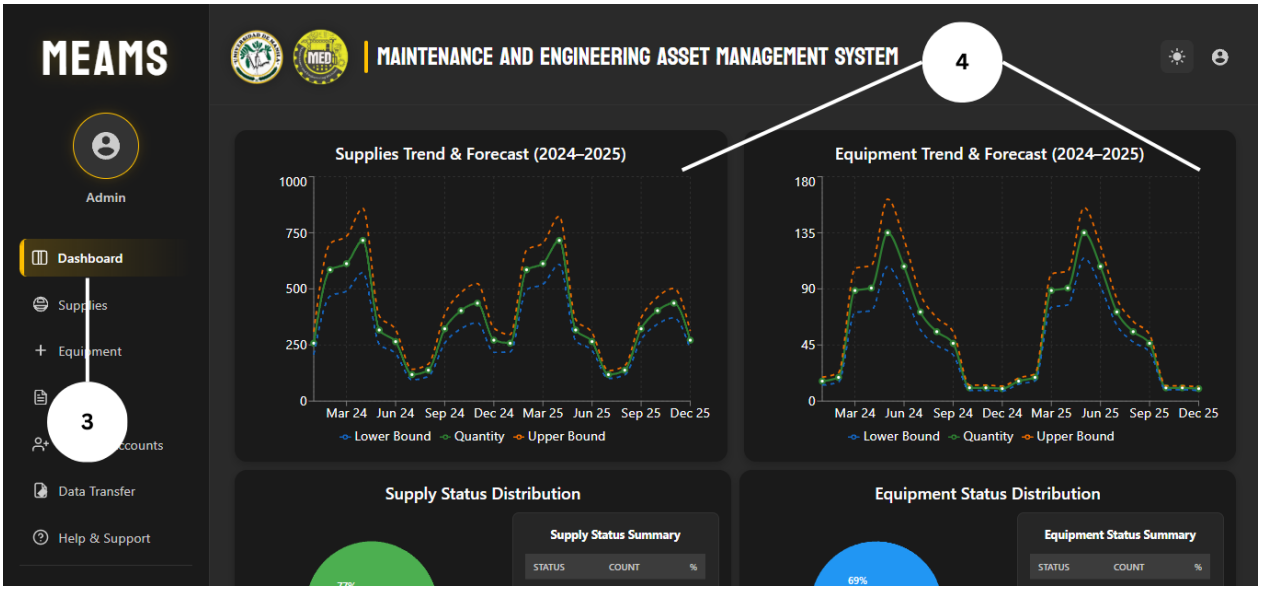
```
82        category: '',
83        itemId: '',
84        understock: '',
85        overstock: ''
86      });
87      const [statusStats, setStatusStats] = useState({
88        total: 0,
89        understock: 0,
90        normal: 0,
91        overstock: 0
92      });
93      const [recommendations, setRecommendations] = useState([]);
94      const [showRecommendations, setShowRecommendations] = useState(false);
95
96      const categories = ['Sanitary Supply', 'Office Supply', 'Construction Supply', 'Electrical Supply'];
97
98      // Define unit options
99      const unitOptions = ['piece', 'pack', 'box', 'bottle', 'gallon', 'set', 'roll', 'bag', 'meter', 'ream'];
100
101     // Status options
102     const statusOptions = ['Understock', 'Normal', 'Overstock'];
103
104     useEffect(() => {
105       loadSupplies();
106     }, []);
107
108     // Reset to first page when search term or category changes
109     useEffect(() => {
110       setCurrentPage(1);
111     }, [searchTerm, selectedCategory]);
112
113     // Listen for status change events with better notifications
114     useEffect(() => {
115       const handleStatusChange = (event) => {
116         const { item, oldStatus, newStatus, quantity, thresholds } = event.detail;
117         console.log('📊 Status changed:', event.detail);
118
119         const urgency = supplyThresholdManager.calculateUrgency(item);
120         const urgencyEmoji = urgency === 'critical' ? '🚨' : urgency === 'high' ? '⚠️' : '📋';
121
```

APPENDIX E: USERS MANUAL

Admin's Manual

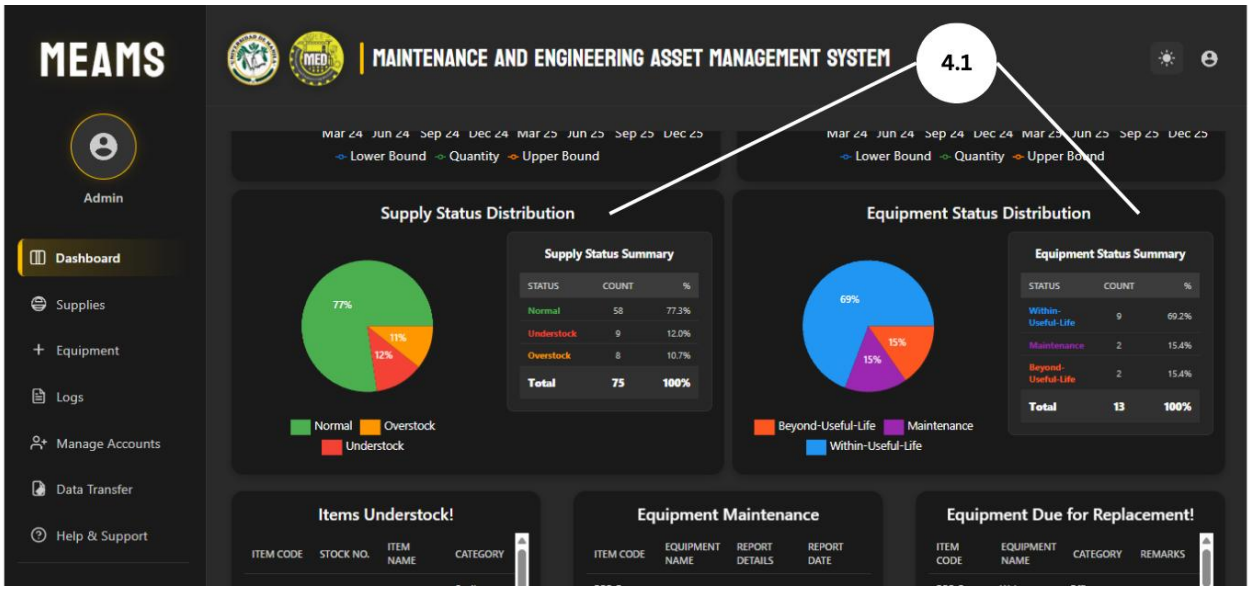


1. Input your assigned username.
2. Input your unique password.

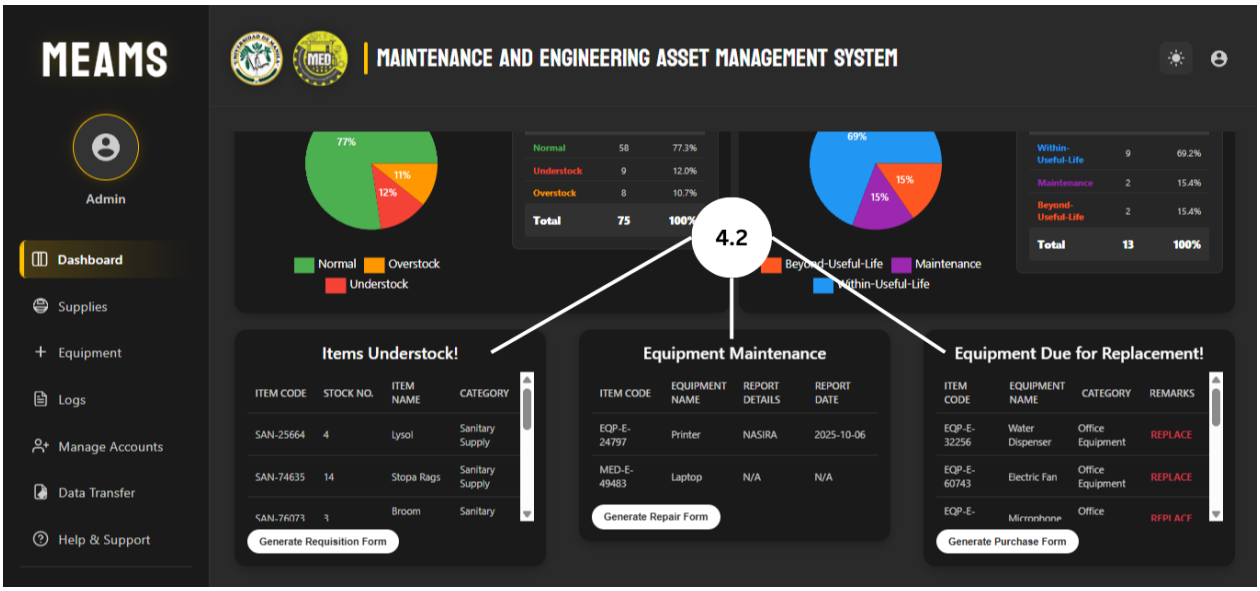


3. Dashboard Button - Displays a quick overview of the system's asset health through visual distribution summaries and forecasting charts for both supplies and equipment.

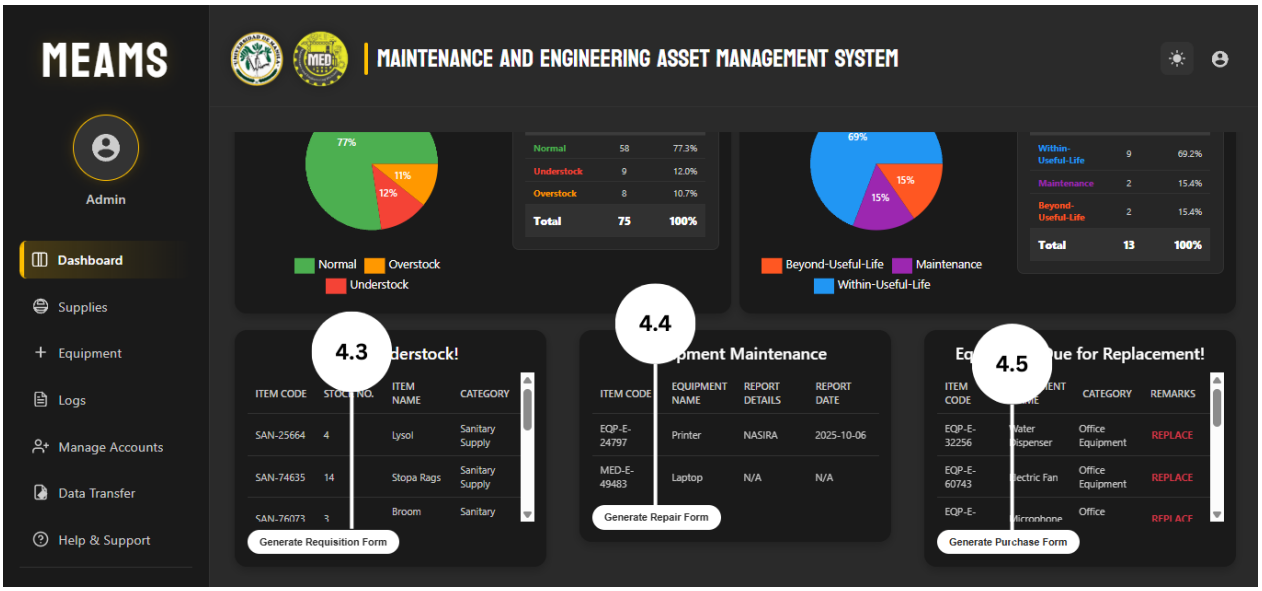
4. Displays key system metrics and statistics using Supplies Trend & Forecast and Equipment Trend & Forecast charts.



4.1. Provides a concise visual summary of Supply Status Distribution and Equipment Status Distribution via pie charts and summary tables.



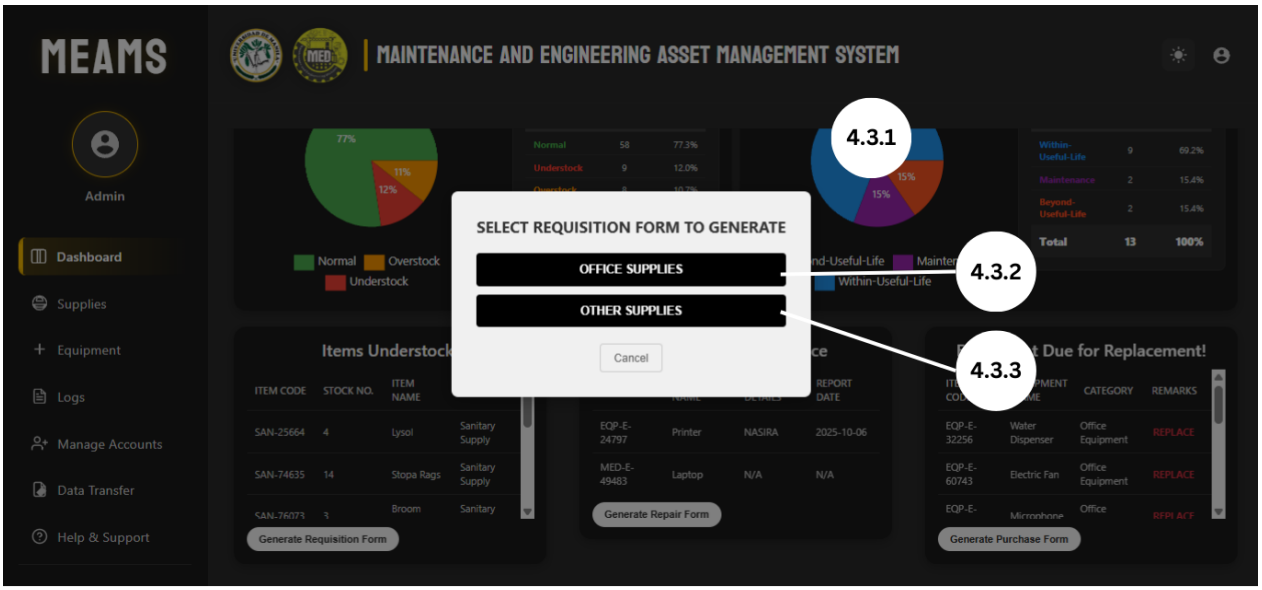
4.2. Features three actionable panels: Items Understock!, Equipment Maintenance, and Equipment Due for Replacement! for quick form generation.



4.3. Where you can Generate a Requisition Form.

4.4. Where you can Generate a Repair Form.

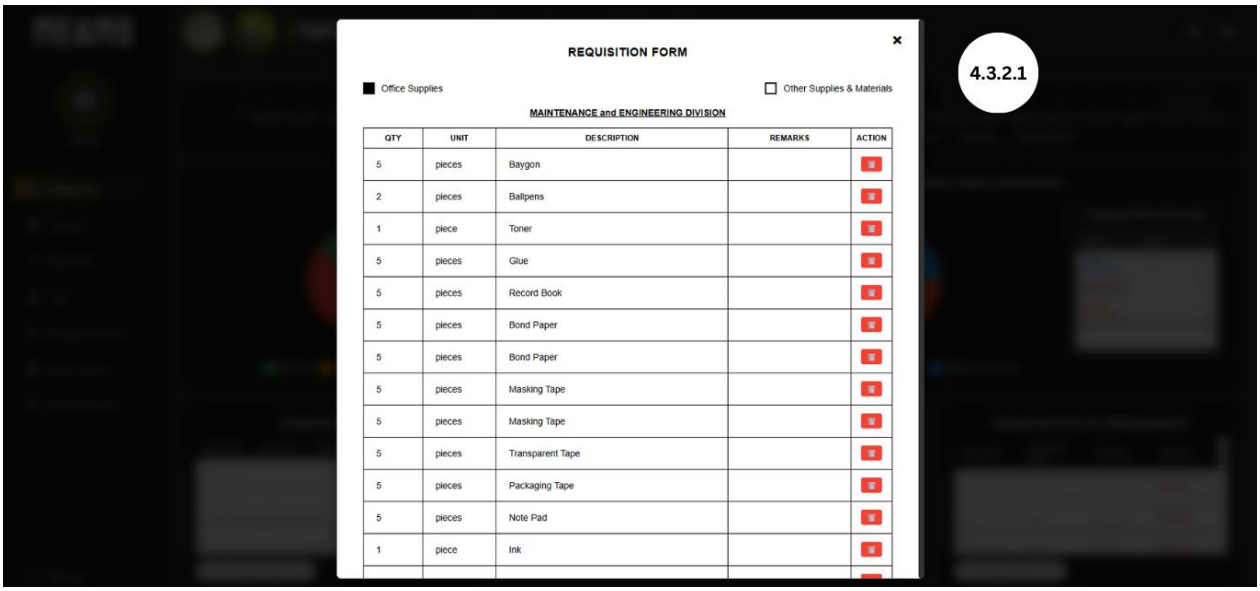
4.5. Where you can Generate a Purchase Form.



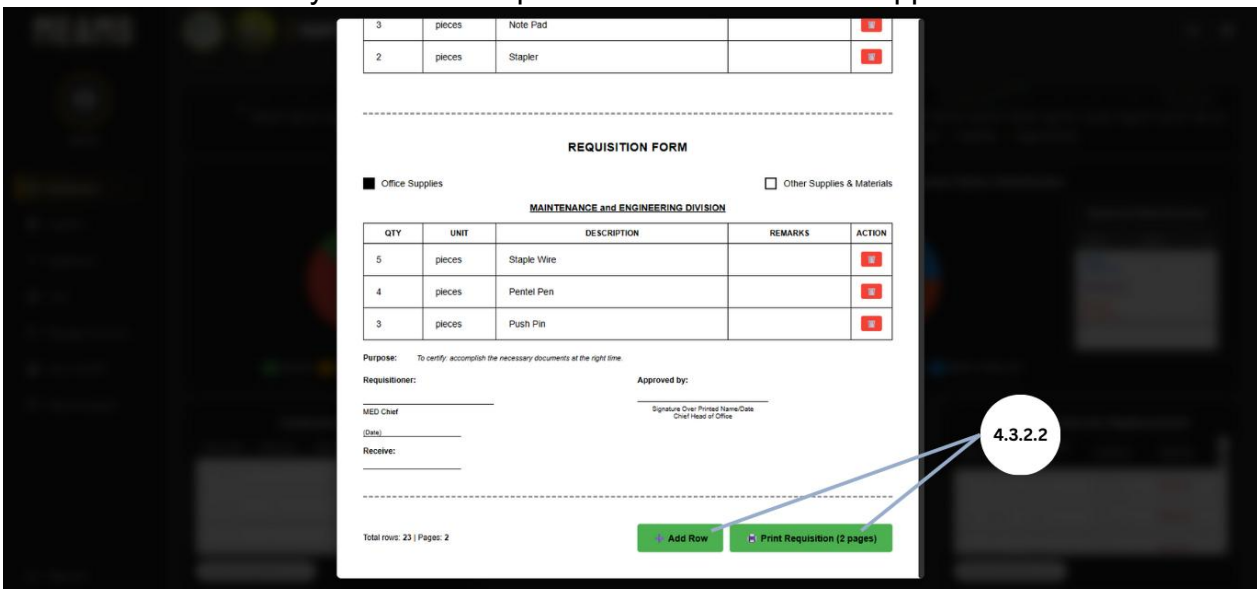
4.3.1. Overlay of “Generate Requisition Form.

4.3.2. Generate a Requisition Form for Office Supplies.

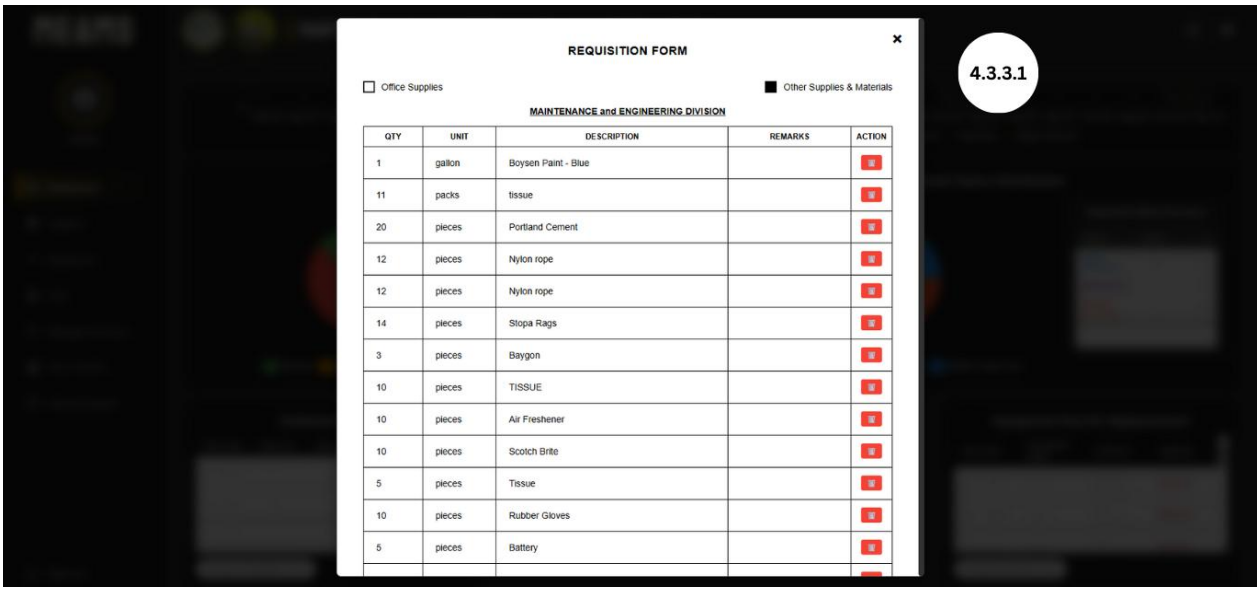
4.3.3. Generate a Requisition Form for Other Supplies.



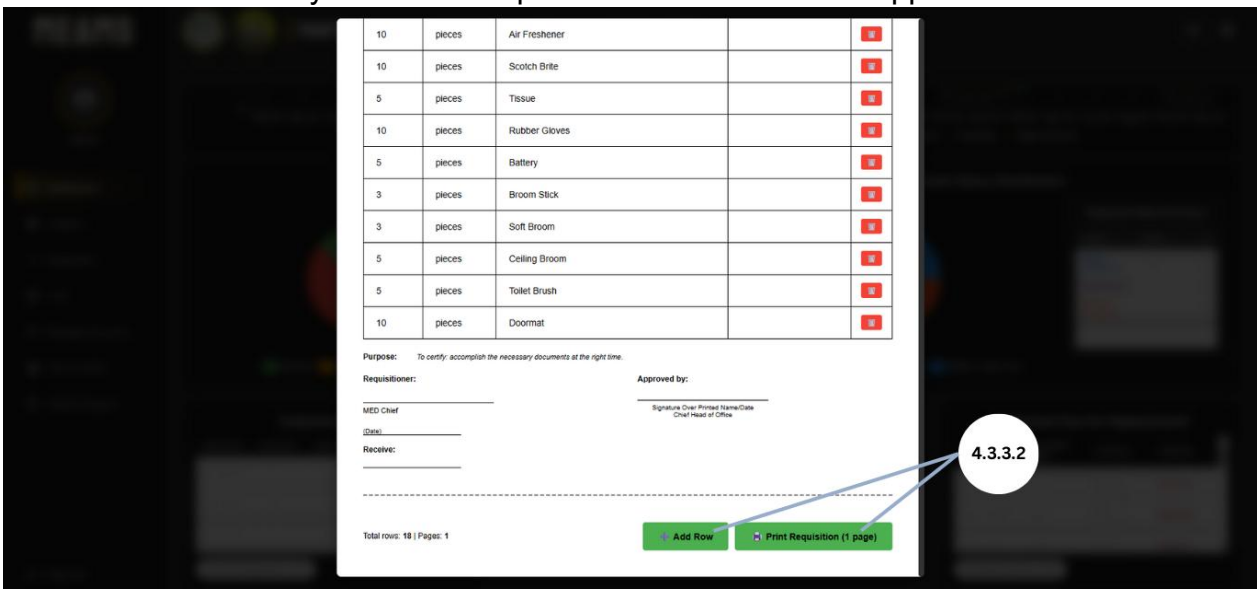
4.3.2.1. Overlay of Modal Requisition Form for Office Supplies.



4.3.2.2. Add Row and Print the Requisition Form for Office Supplies.



4.3.3.1. Overlay of Modal Requisition Form for Other Supplies.

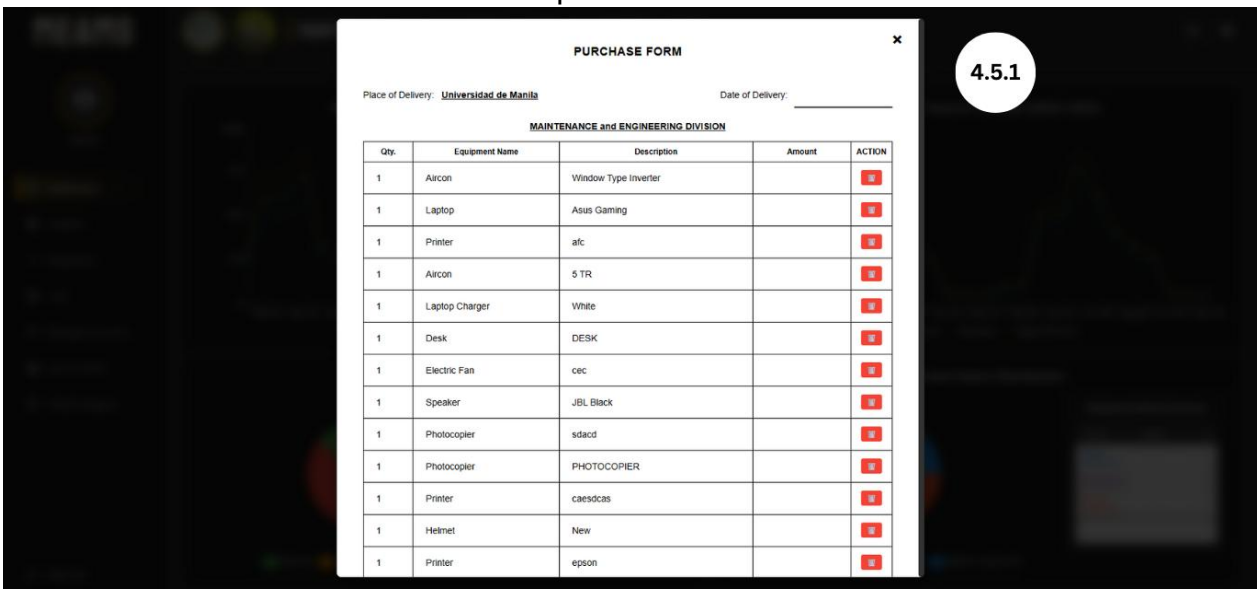


4.3.3.2. Add Row and Print the Requisition Form for Other Supplies.



4.4.1. Overlay of Modal Repair Form.

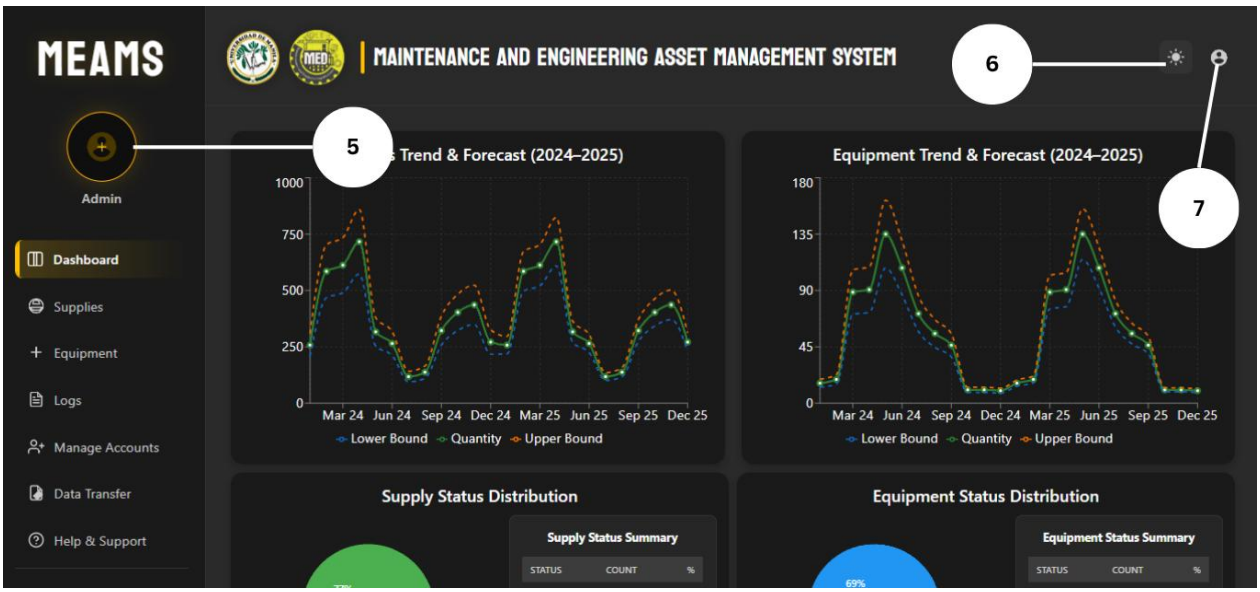
4.4.2. Add Row and Print the Repair Form.



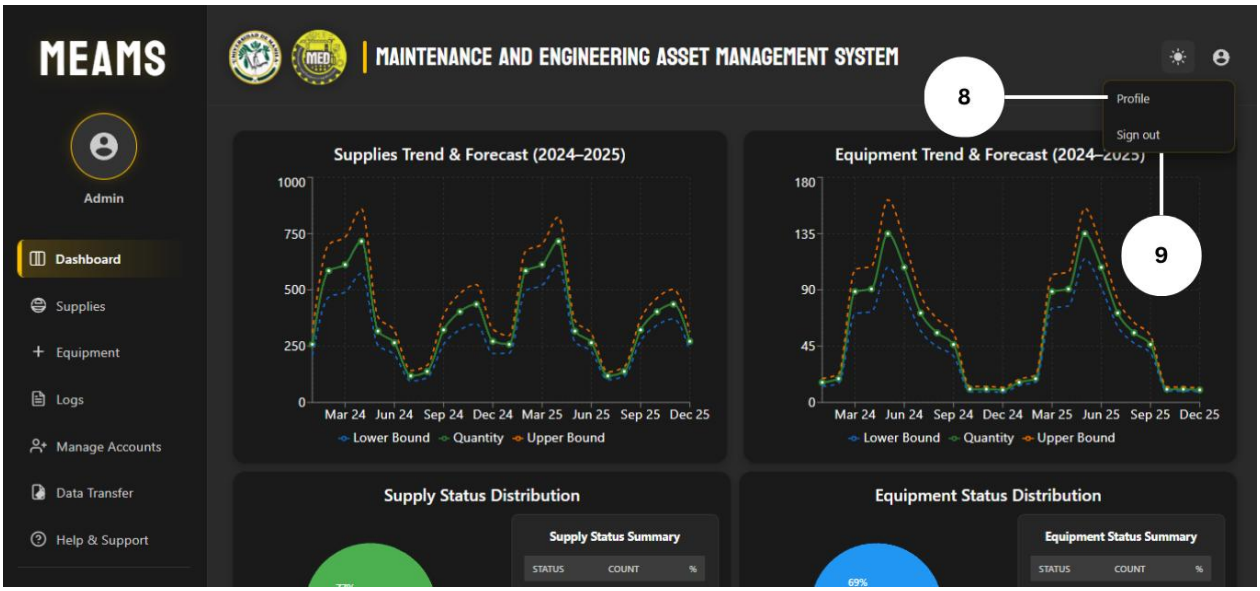
4.5.1. Overlay of Modal Purchase Form.



4.5.2. Add Row and Print the Modal Purchase Form.

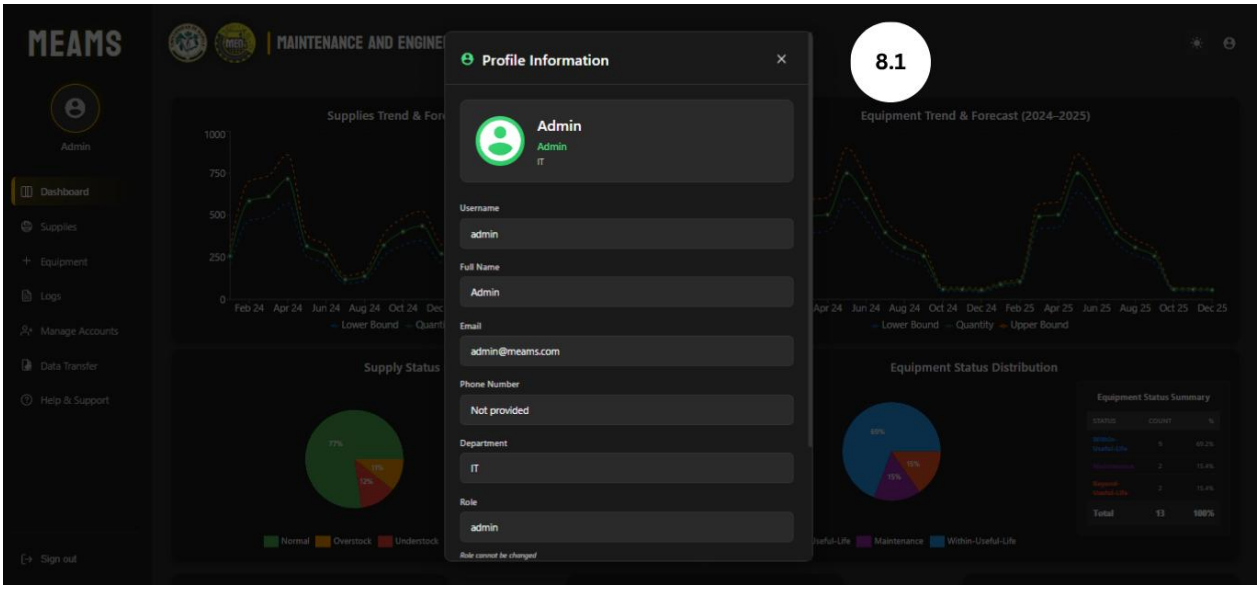


5. Where you can change the Profile Picture of the User.
6. Switch to Light Mode/Dark Mode.
7. Clickable Dropdown.

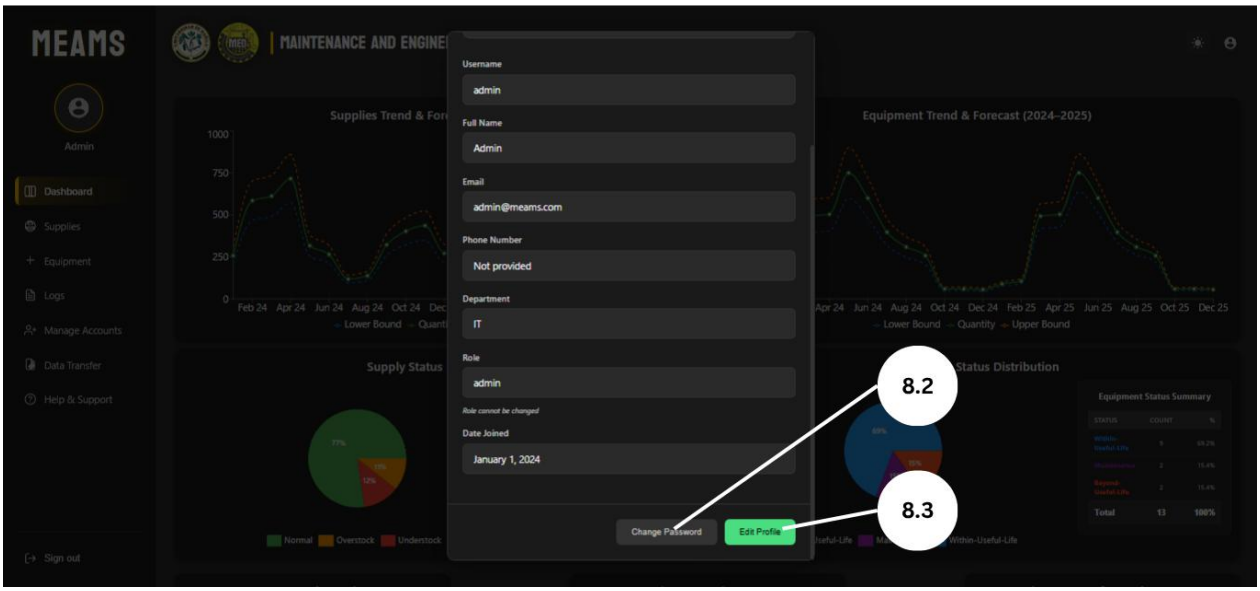


8. You can see your Profile Information.

9. Sign out from the System.

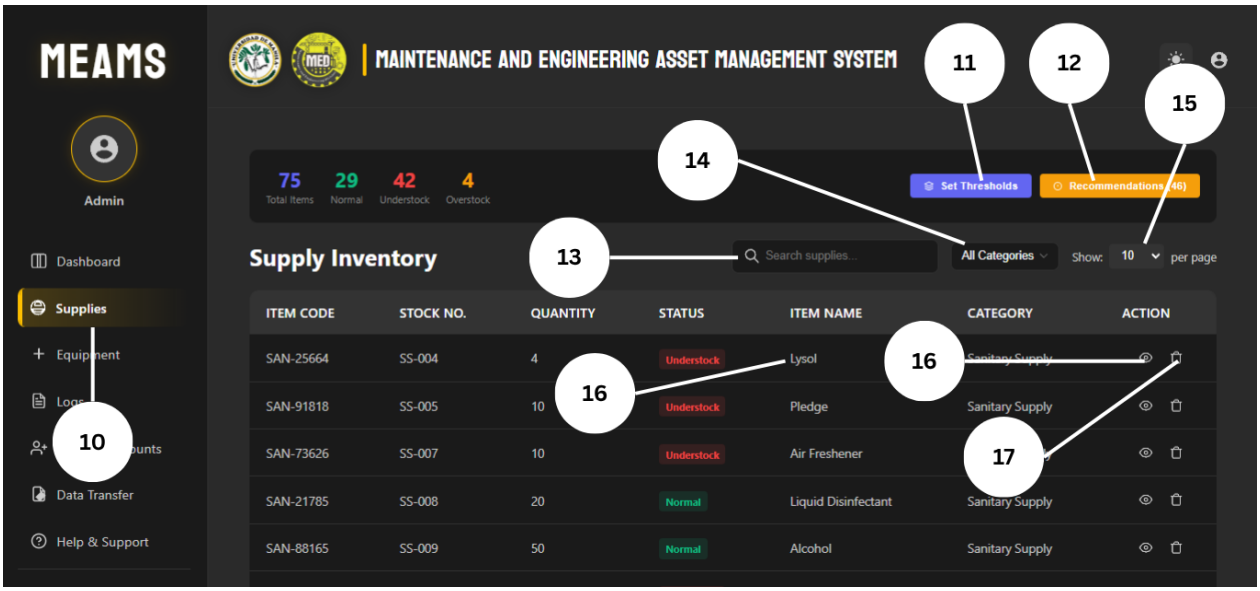


8.1. Overlay of Profile Information.



8.2 If you want to change Password.

8.3 If you want to edit your Profile Information.



10. Supplies Button - Displays Supplies Inventory.

11. Set a Thresholds for specific supply per category.

12. It displays recommendations for critical or low condition for supplies.

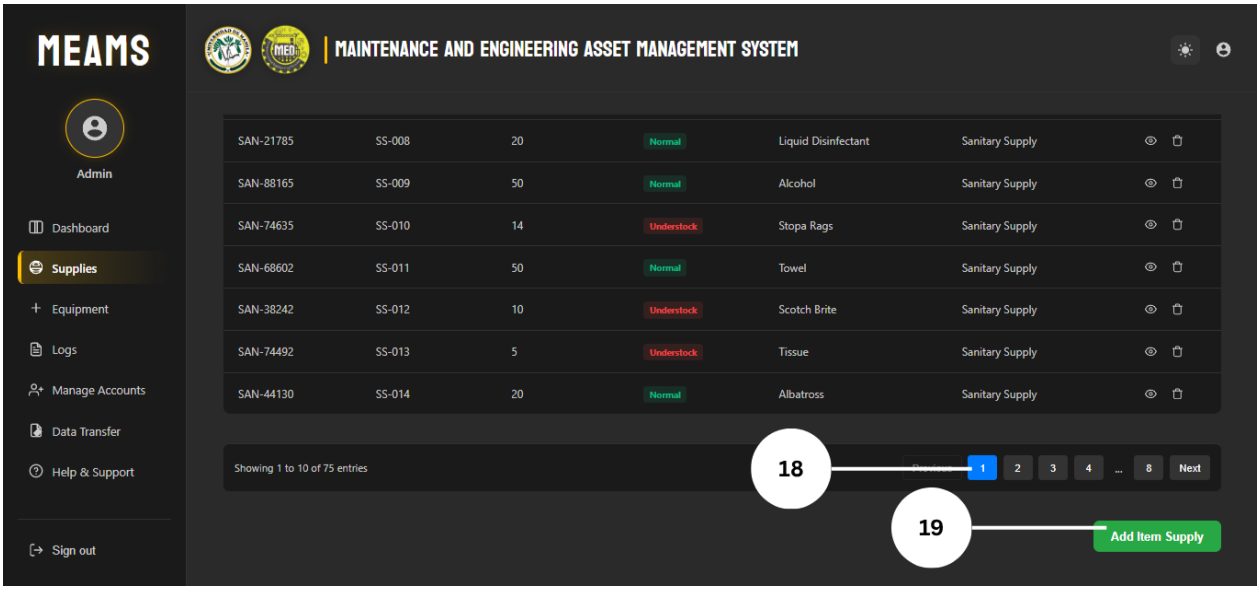
13. If you want to search a specific supply.

14. You can choose a specific category of supplies.

15. A dropdown menu to select the number of inventory entries shown per page.

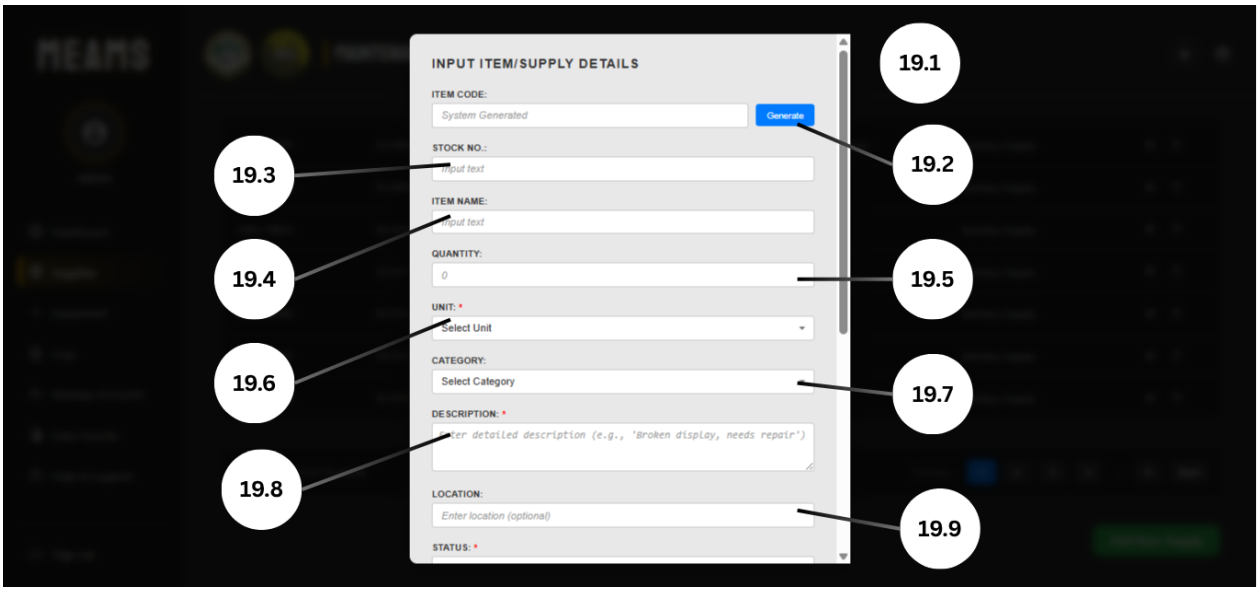
16. Can be click for Supply Details.

17. Delete a Supply.



18. Shows the current range of displayed entries and for navigating the full inventory list.

19. Action button for Adding an Item Supply



19.1. Modal form for Adding Supply.

19.2. Generate a Unique Item Code.

19.3. Input a Stock No.

19.4. Input a Item Name.

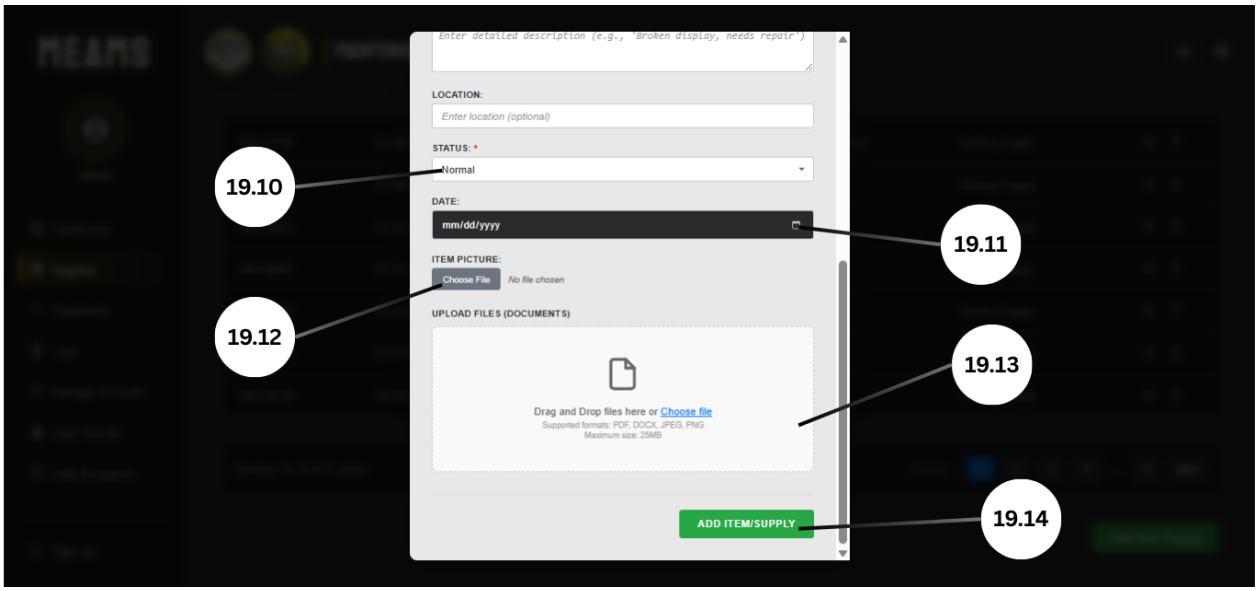
19.5. Input the Quantity of the Supply.

19.6. Select a Unit for the Supply.

19.7. Select a Category for the Supply.

19.8. Description of the Supply.

19.9. Location of the Supply (Optional)



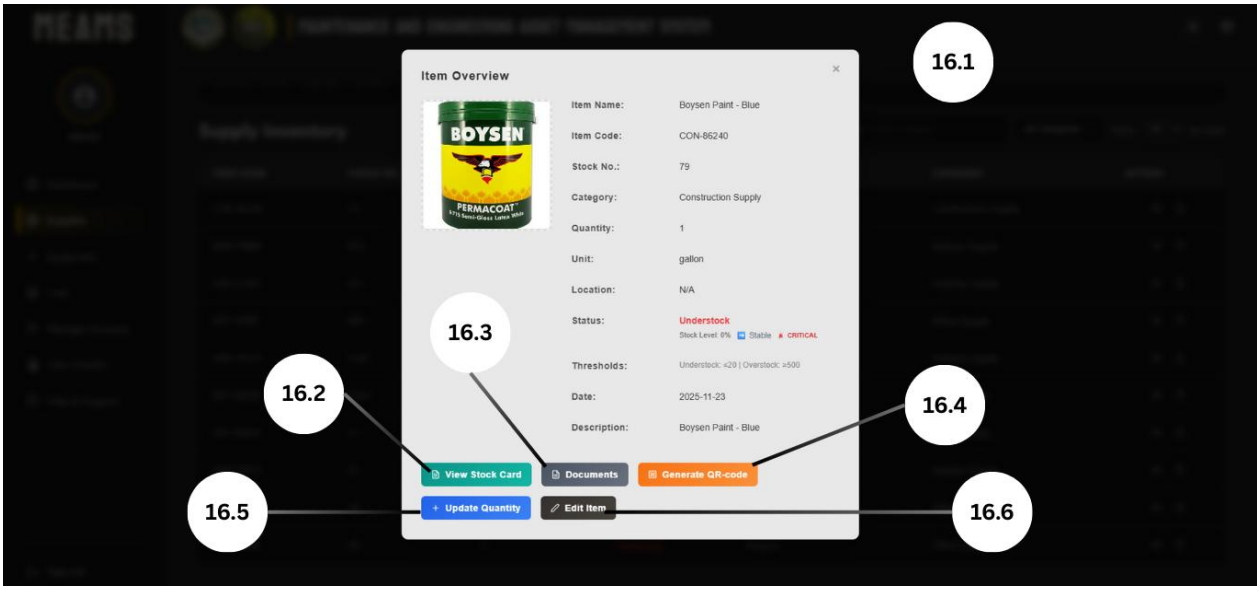
19.10. Select a Status for the Supply.

19.11. Input a Date when the Supply bought.

19.12. Import a Picture for the Supply.

19.13. Import a Document/Picture for the Supply.

19.14. Add the Supply.



16.1. Overlay of the Supply Details.

16.2. When clicked, the Stock Card will appear.

16.3. Import necessary documents for the Supply.

16.4. When clicked, the generated QR code will appear.

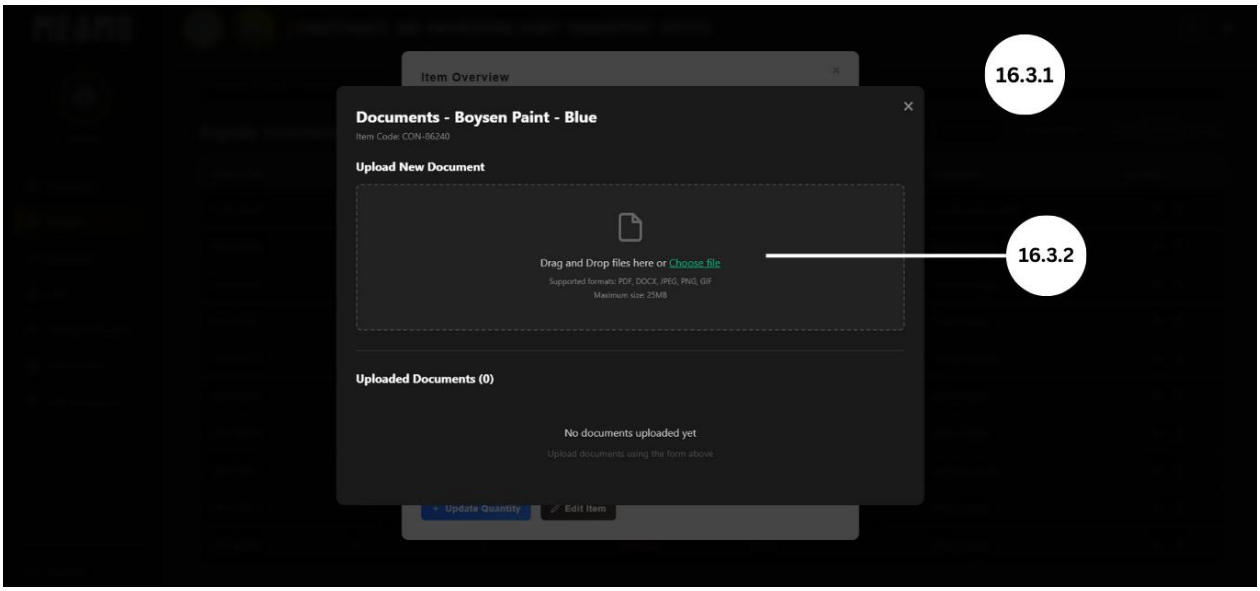
16.5. Update the Quantity of the Supply.

16.6. Edit the Details of the Supply.



16.2.1. Overlay of the Stock Card form.

16.2.2. Can be Printed.



16.3.1. Overlay of the Import Documents for Supply.

16.3.2. Drag and Drop or Import a Document for the Supply.

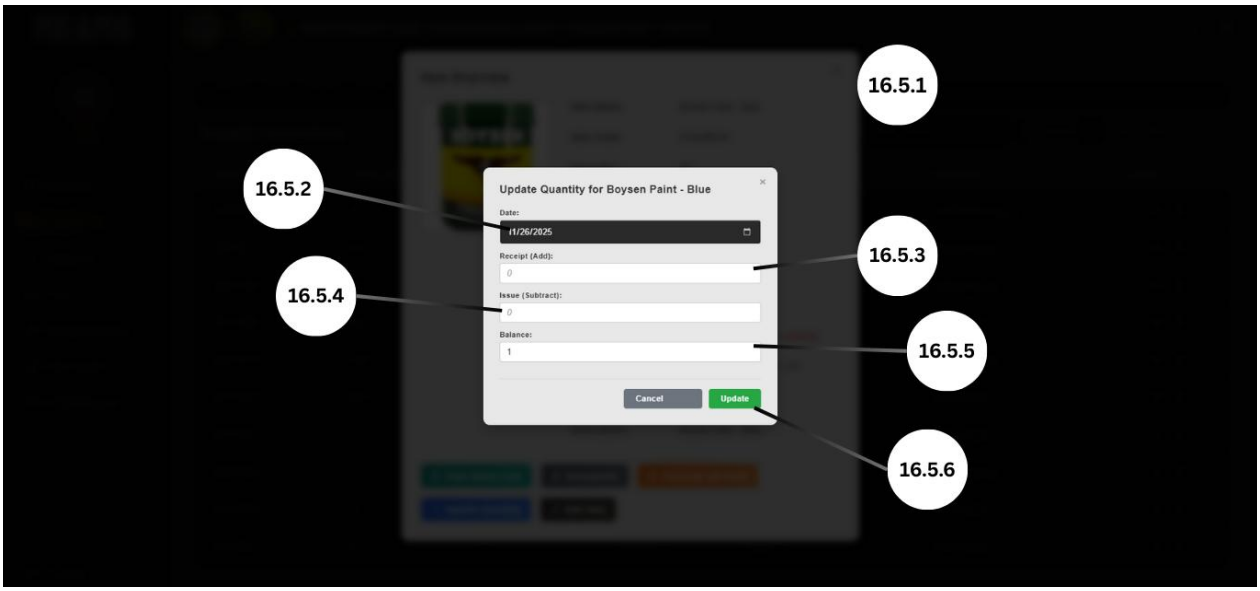


16.4.1. Overlay of the QR Code Details.

16.4.2. QR Code can be downloaded.

16.4.3. QR Code can be Print.

16.4.4. Data can be copied also.



16.5.1. Overlay of the Update Quantity.

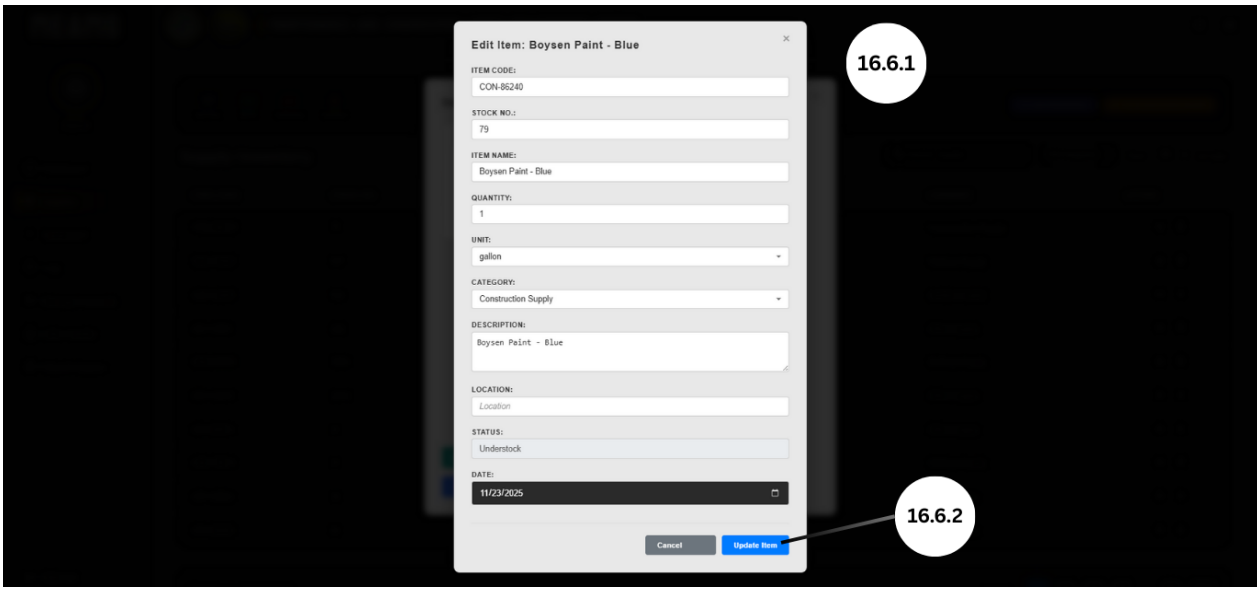
16.5.2. Input when the supply stored or distributed.

16.5.3. If the Supply quantity is added.

16.5.4. If the Supply quantity is subtracted.

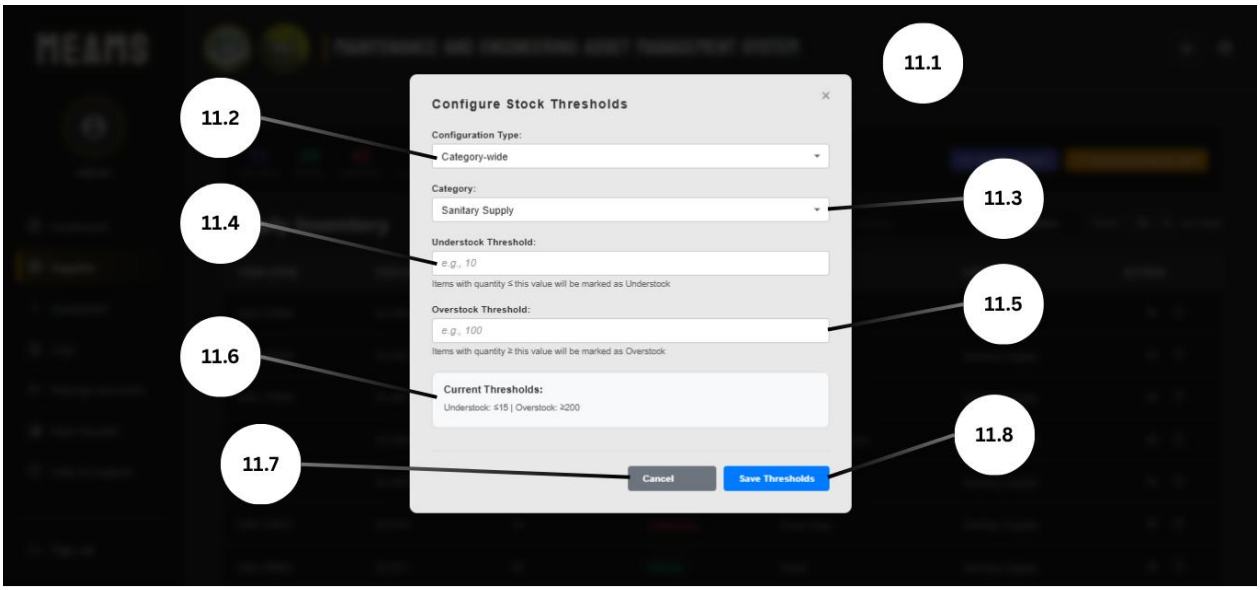
16.5.5. Total of the Quantity.

16.5.6. Update the edited quantity



16.6.1. Overlay of the Edit Item.

16.6.2. Update the edited item.



11.1. Overlay of the Set Thresholds.

11.2. Choose if per Item or per Category.

11.3. Choose a specific Category.

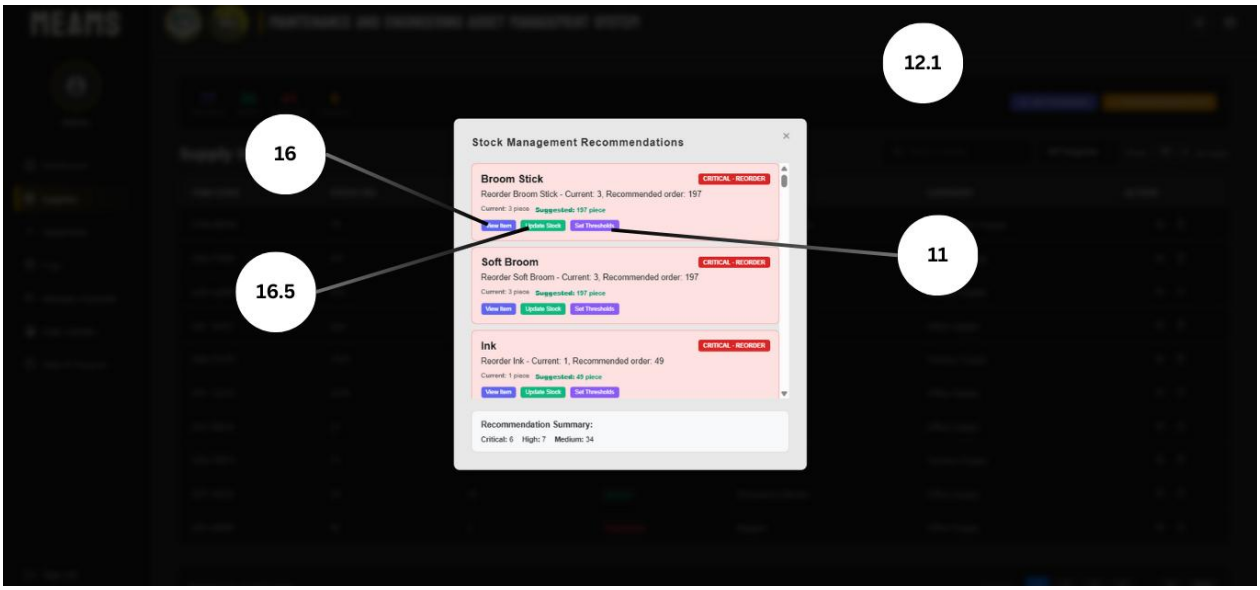
11.4. Set an Understock Threshold for the Item.

11.5. Set an Overstock Threshold for the Item.

11.6. Shows the Current Thresholds.

11.7. Cancel.

11.8. Save Thresholds.

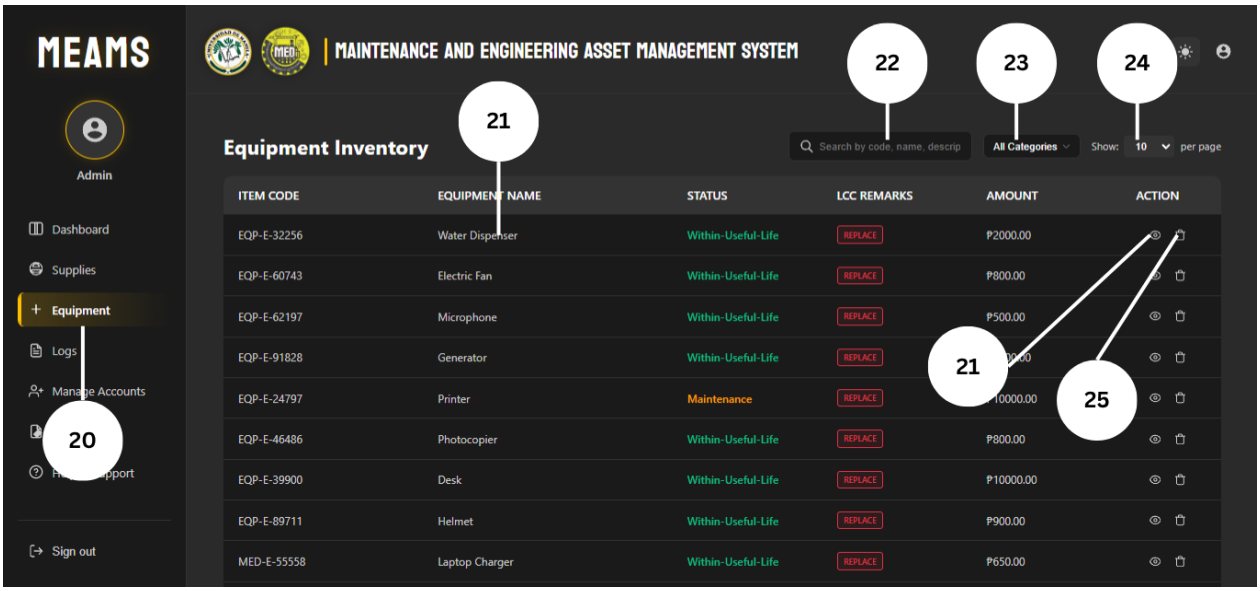


12.1. Overlay of the Recommendations.

16. View the Supply Details.

16.5. Update Quantity.

11. Set Thresholds for the Supply.



20. Equipment Button - Displays Equipment Inventory.

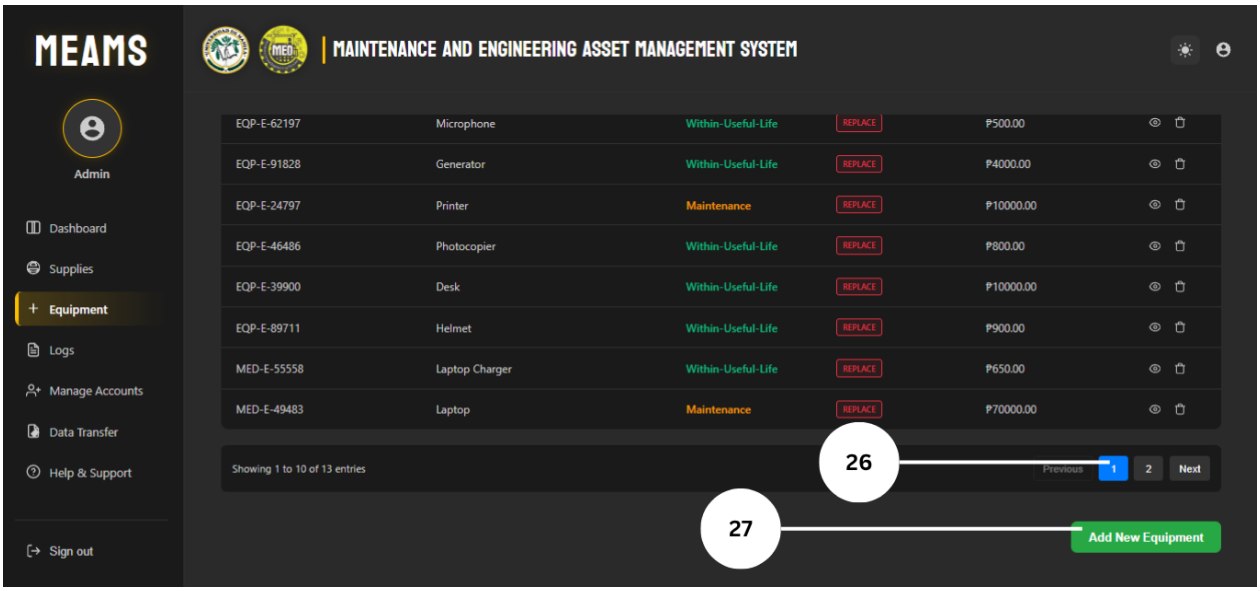
21. Can be clicked for Equipment Details.

22. If you want to search a specific Equipment.

23. You can choose a specific Category.

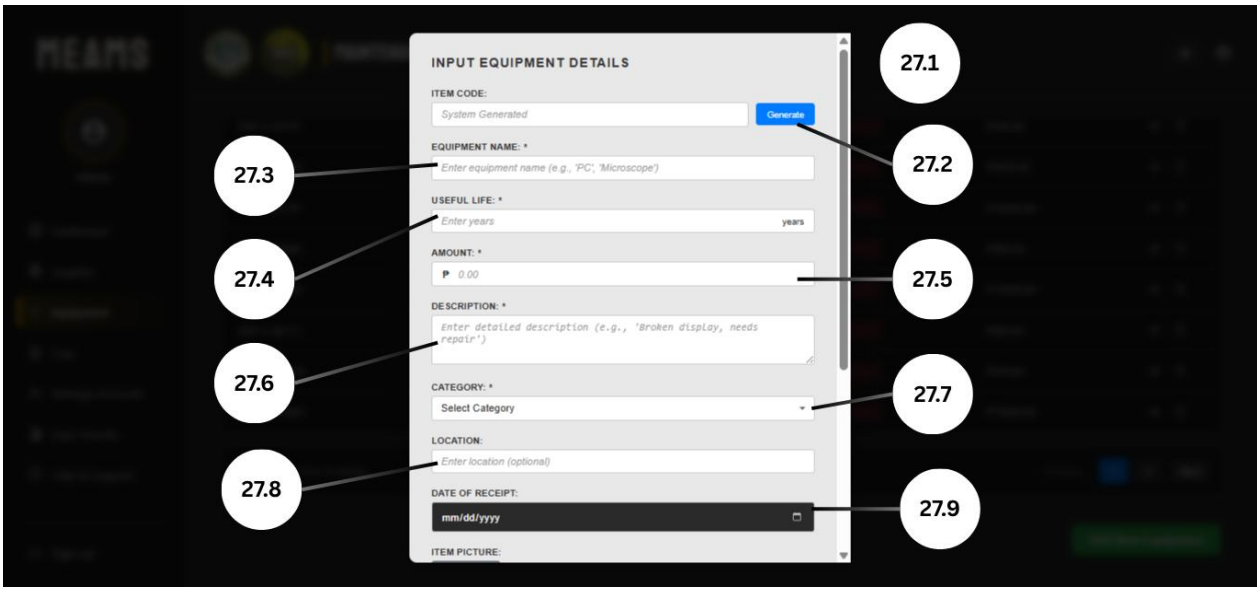
24. A dropdown menu to select the number of inventory entries shown per page.

25. Delete an Equipment.



26. Shows the current range of displayed entries and for navigating the full inventory list.

27. Action Button for adding an Equipment.



27.1. Modal form for Adding Equipment.

27.2. Generate a Unique Item Code for Equipment.

27.3. Input an Equipment Name.

27.4. Input the Useful Life of the Equipment.

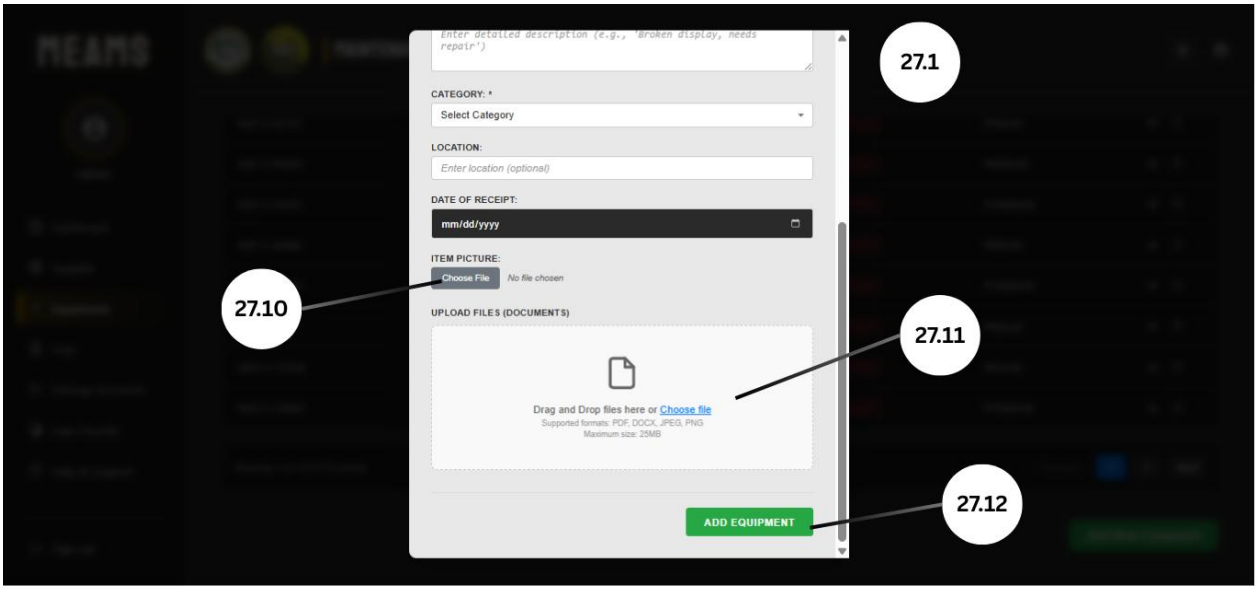
27.5. Input the Amount of the Equipment.

27.6. Description of the Equipment.

27.7. Select a category of the Equipment.

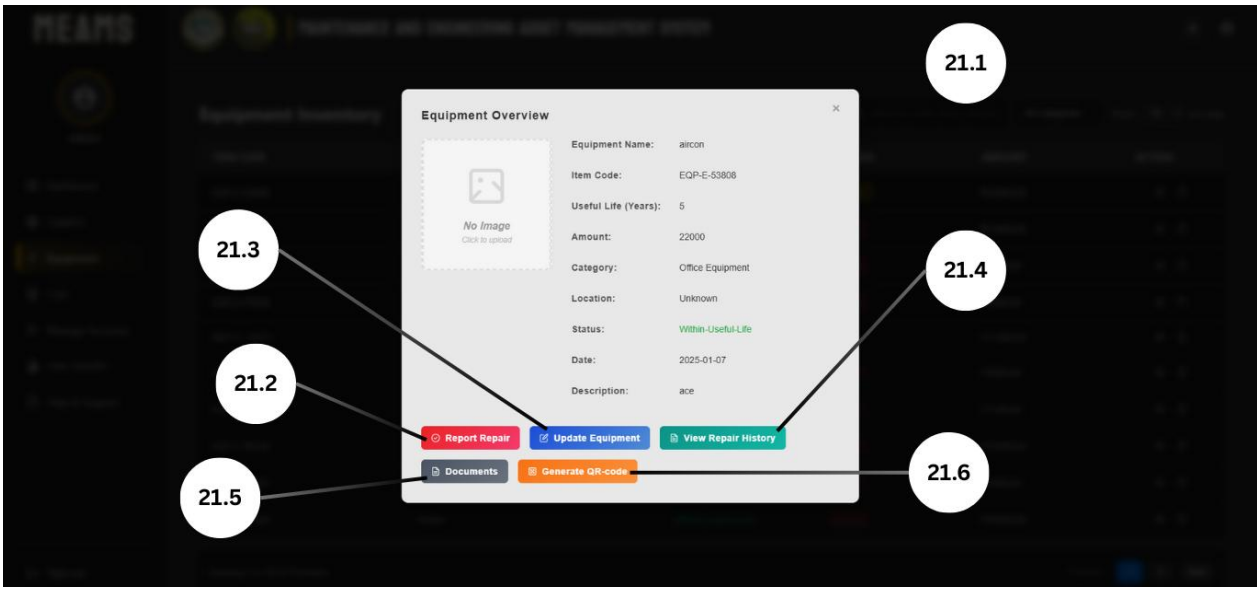
27.8. Location of the Equipment (Optional)

27.9. Date of Receipt of the Equipment.



27.10. Import a Picture of the Equipment.

27.11. Import/Upload a Documents for the Equipment.



21.1. Overlay of the Equipment Details

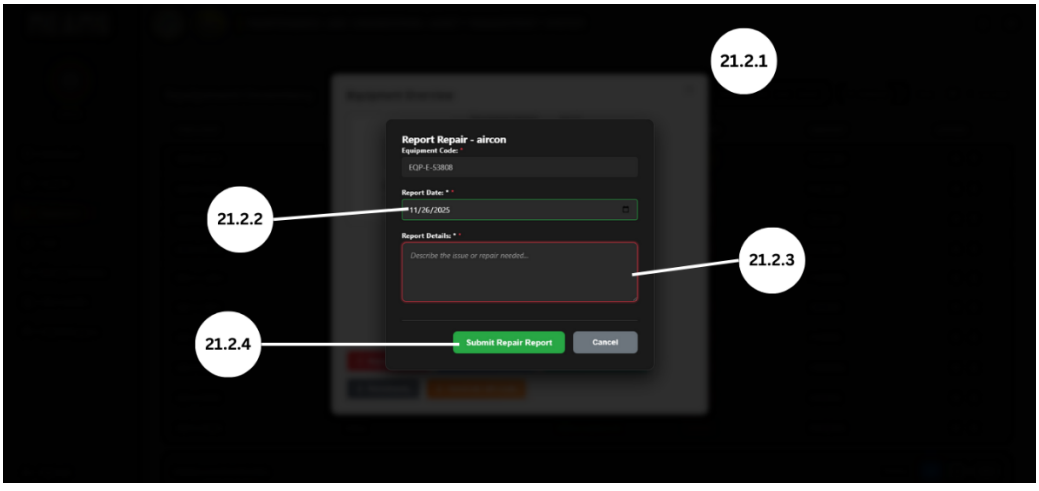
21.2. Report Repair If Equipment has issue.

21.3. Update the Equipment when the equipment is repaired.

21.4. View the Repair History.

21.5. Import necessary documents for the Equipment.

21.6. When clicked, the generated QR code will appear.

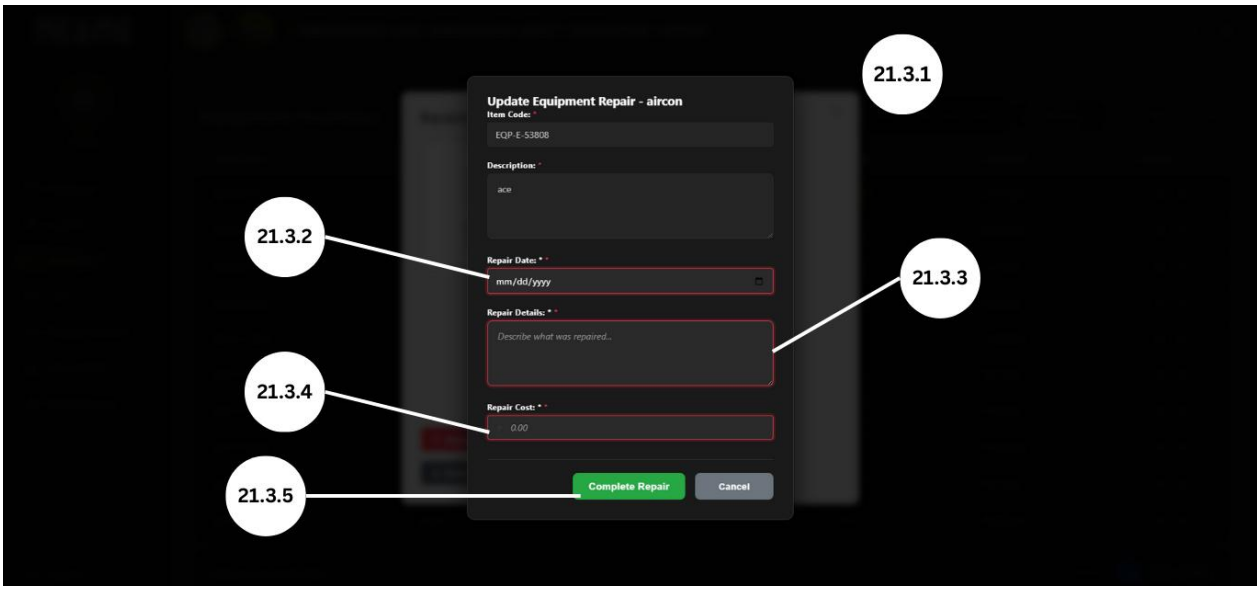


21.2.1. Overlay of the Report Repair.

21.2.2. When is the Report been issued.

21.2.3. Description of the Issue and the repair needed.

21.2.4. Submit Report Repair.



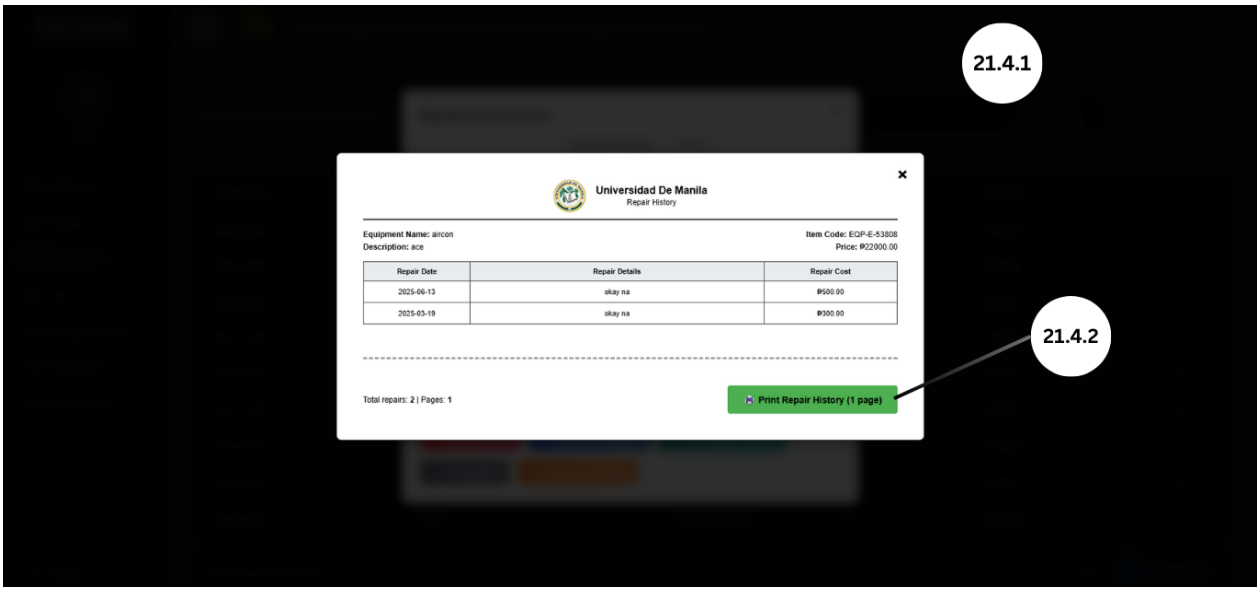
21.3.1. Overlay of the Update Equipment Repair.

21.3.2. When the Equipment Repaired.

21.3.3. Description what has been repaired.

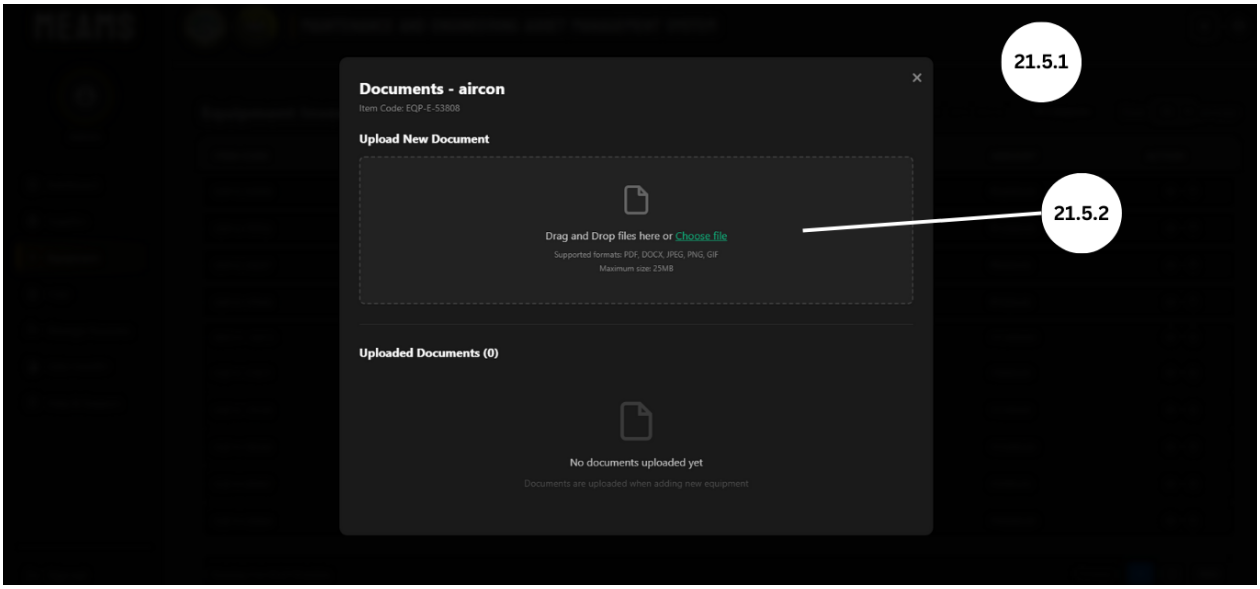
21.3.4. Amount Used for the respiration.

21.3.5. Complete Repair.



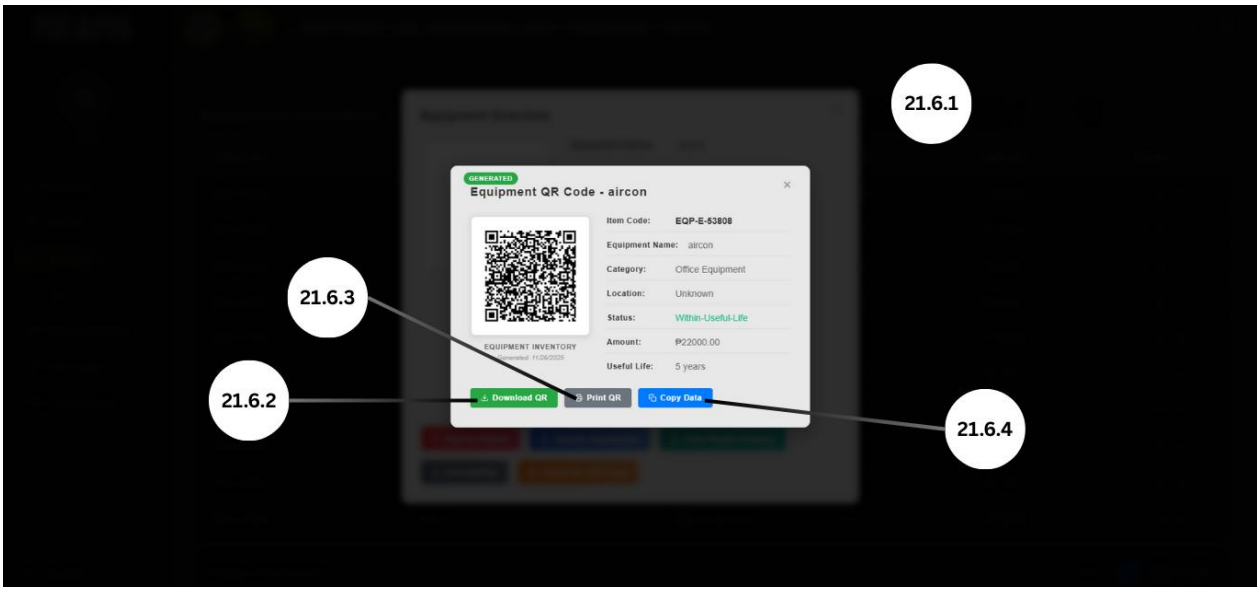
21.4.1. Overlay of the Repair History.

21.4.2. Print the Repair History.



21.5.1. Overlay of the Import Documents for the Equipment.

21.5.2. Drag and Drop or Import a Document for the Supply.

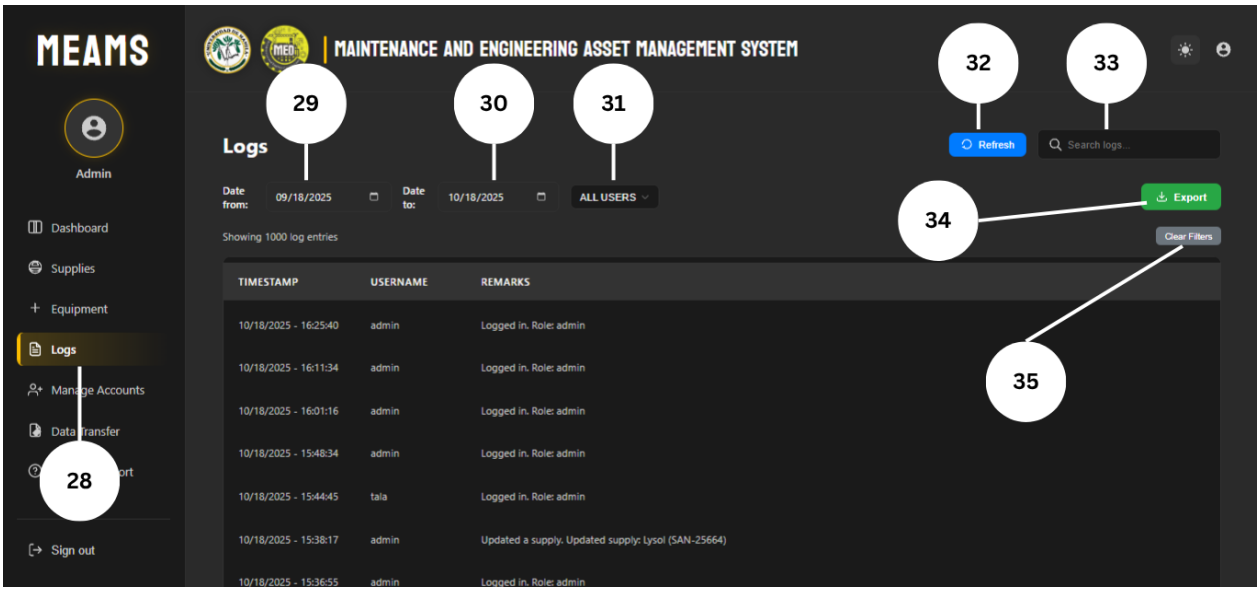


21.6.1. Overlay of the QR Code Details.

21.6.2. QR Code can be downloaded.

21.6.3. QR Code can be printed.

21.6.4. Data can also be copied.



28. Logs Button - Displays Logs Activity.

29. A date picker used to specify the starting date for filtering the system's log entries.

30. A date picker used to specify the ending date for filtering the system's log entries.

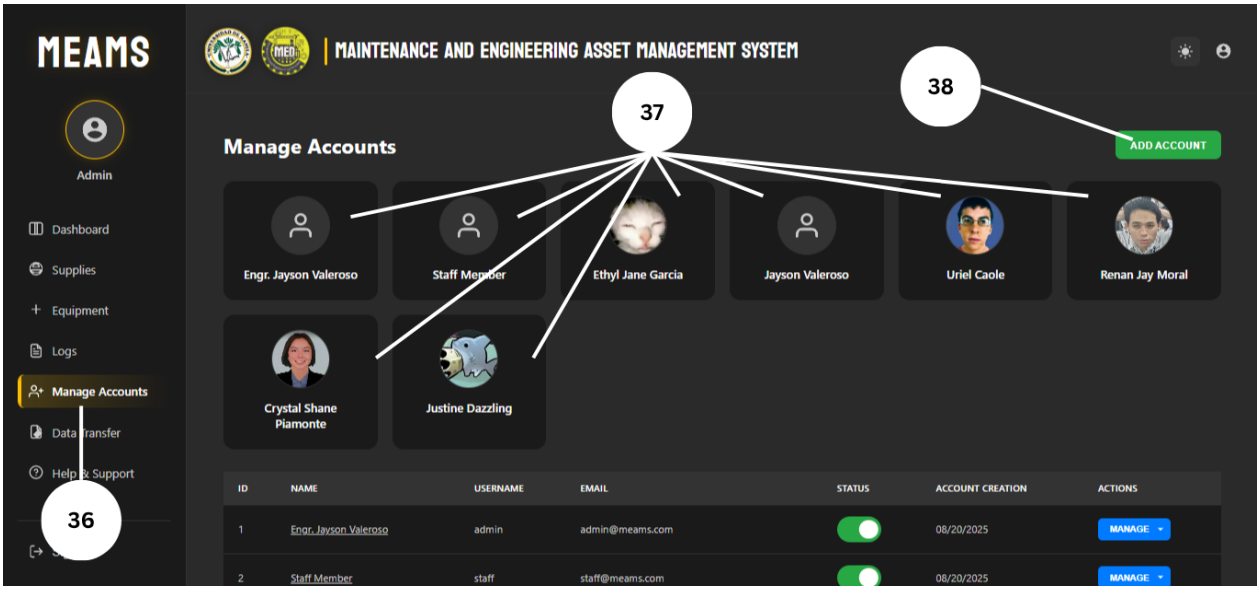
31. A dropdown menu that allows the administrator to filter log entries by the actions of ALL USERS or a specific individual user.

32. Manually reload the log entries.

33. If you want to search a specific Username or remarks.

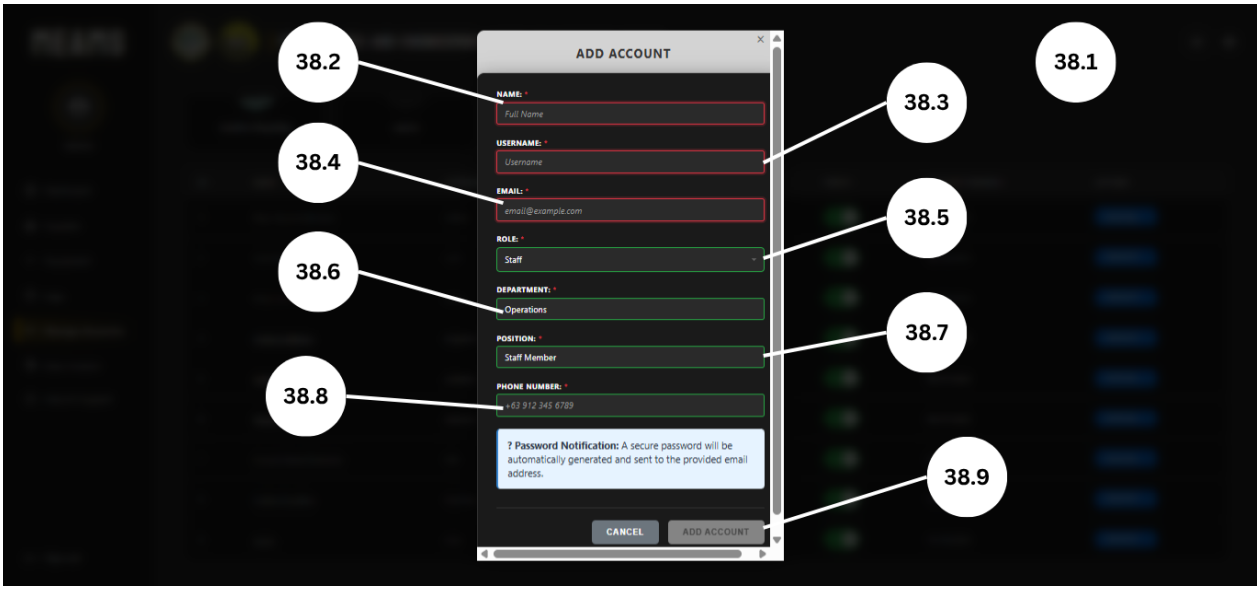
34. Used to download the currently displayed log data.

35. Resets the date range and user selections to display all log entries without any active filters.



36. Manage Accounts Button - Displays Manage Accounts.

37. A clickable card displaying the user's picture and name, used to view or edit individual account details.
38. Action button used to open the form for creating a new user account in the system.



38.1. Overlay of the Add Account.

38.2. Name of the new user.

38.3. Unique Username for the new user’s login credentials.

38.4. A required field to input the user's Email address, used for system communication and password generation.

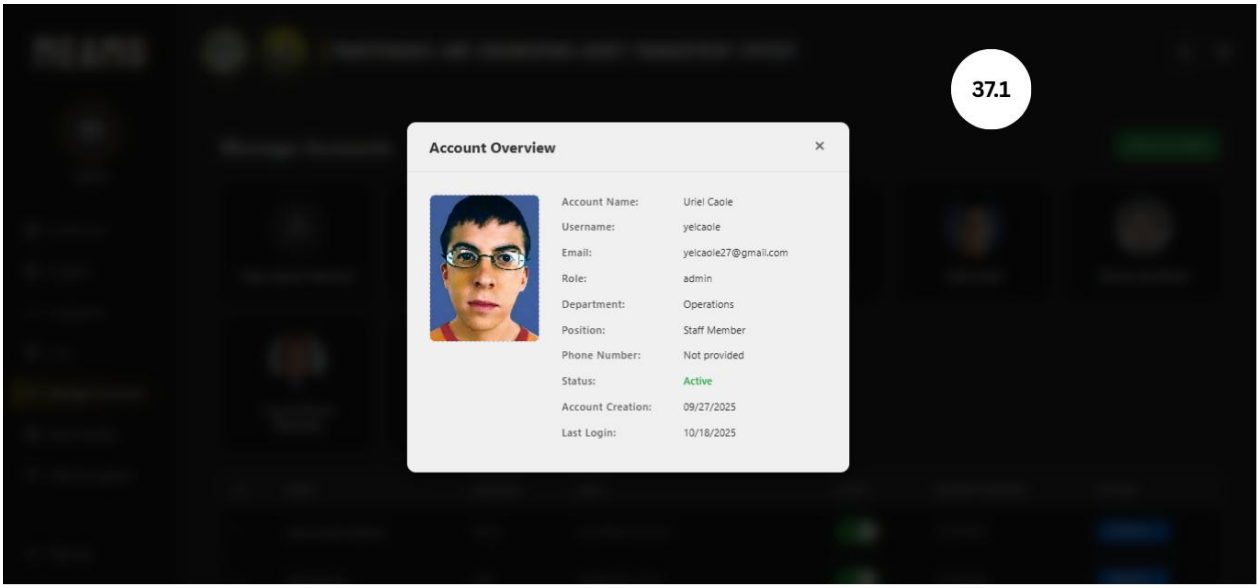
38.5. A dropdown menu used to assign the user's system ROLE (e.g., Staff, Admin), defining their access.

38.6. A required dropdown menu used to assign the user to a specific DEPARTMENT.

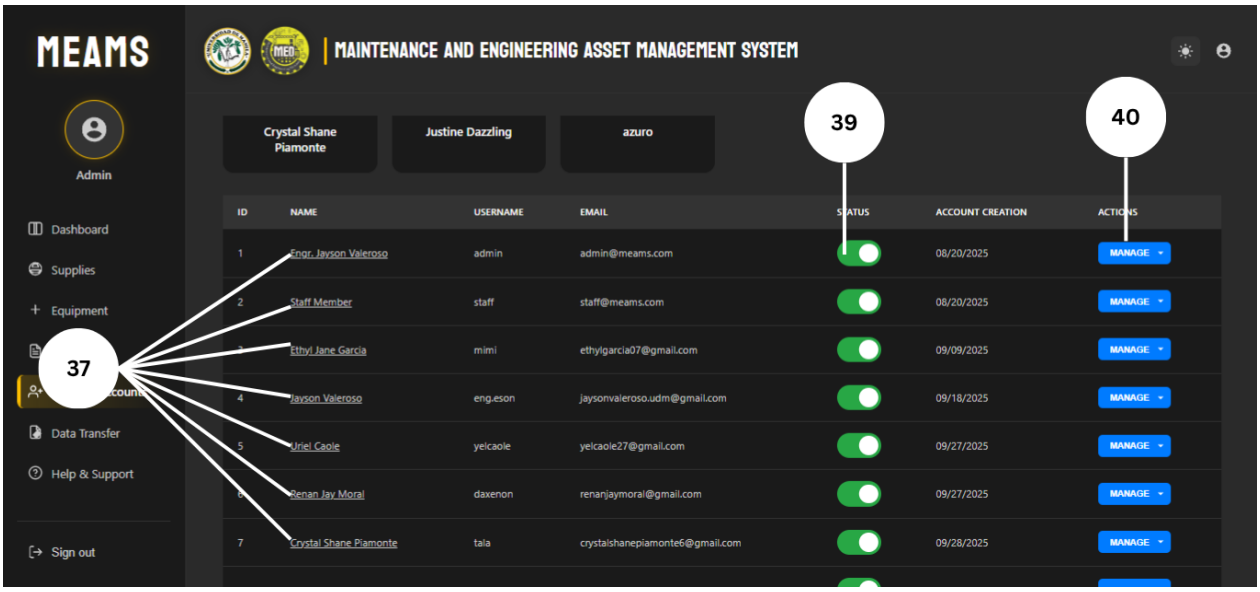
38.7. A required field used to enter the user's official job Position within the organization.

38.8. An optional field for inputting the user's contact PHONE NUMBER.

38.9. Add the newly registered account.



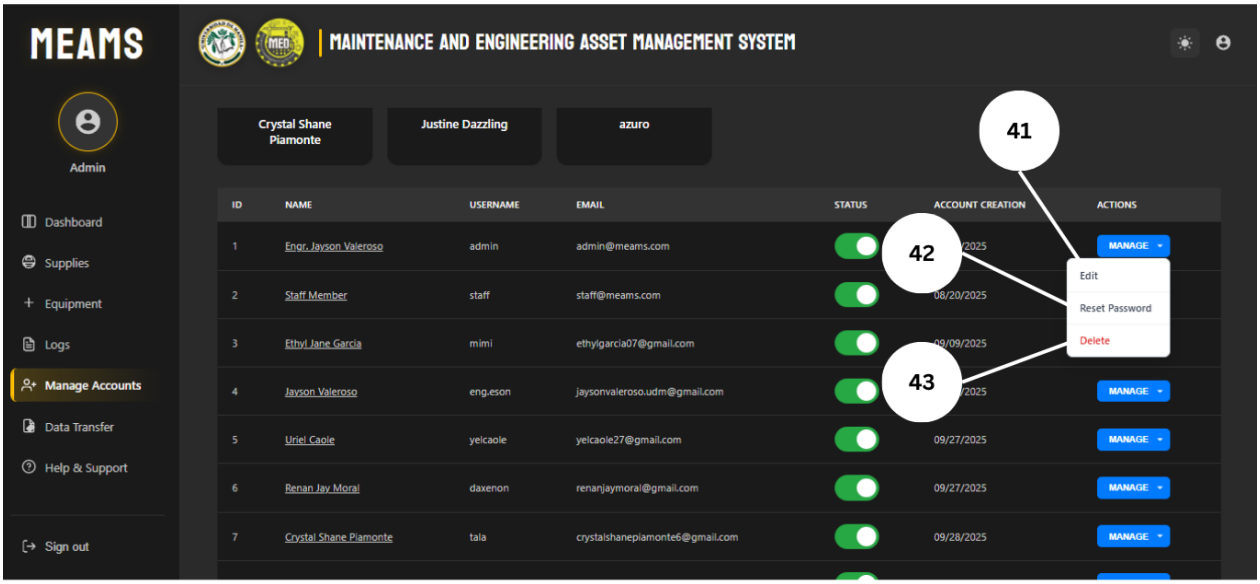
37.1. Overlay of the Account Details.



37. When clicked, Account Details will appear.

39. A switch used by the administrator to quickly activate or deactivate a user account.

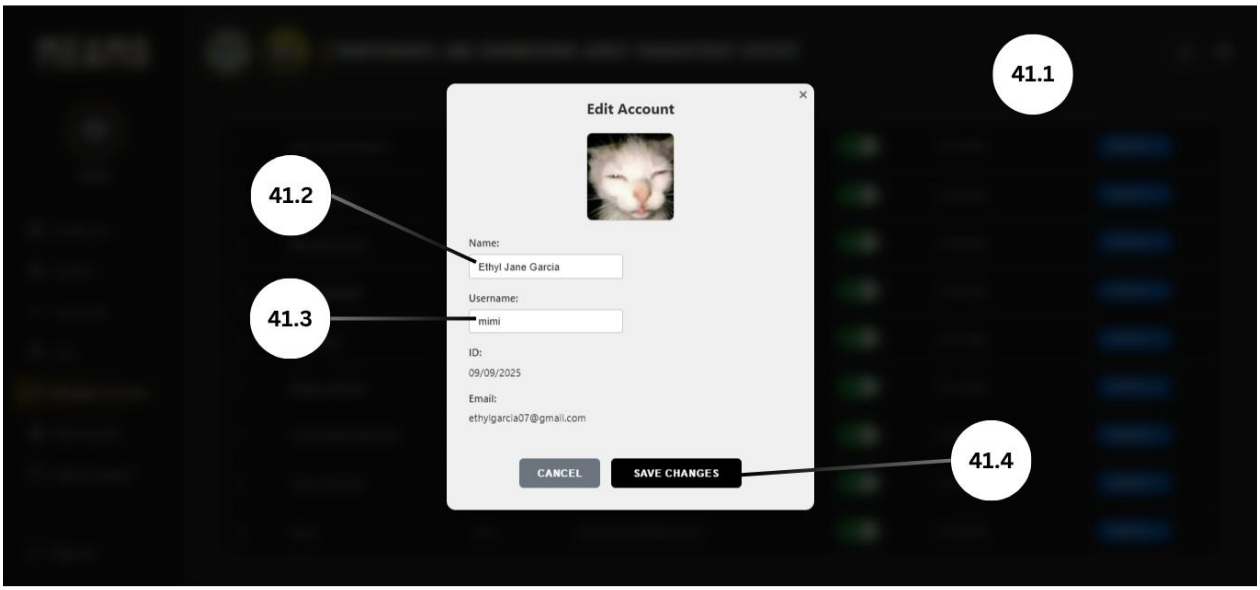
40. Clicking this button opens a modal or new view for detailed account management options.



41. Used to open a form where the administrator can modify the user's profile information.

42. Reset Password of that specific account.

43. Permanently remove the account.

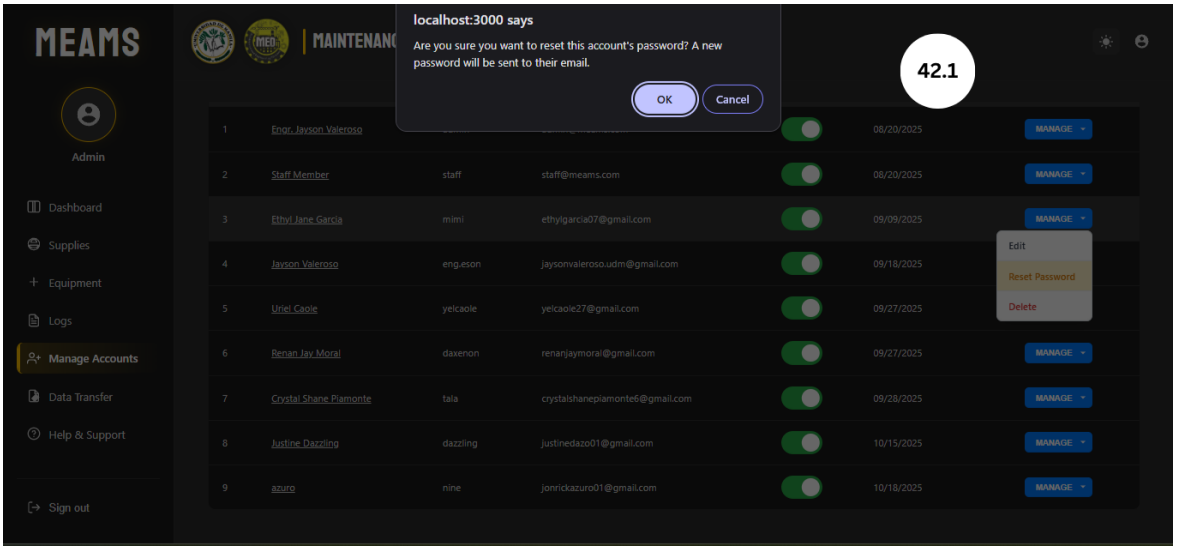


41.1. Overlay of the Account Edit.

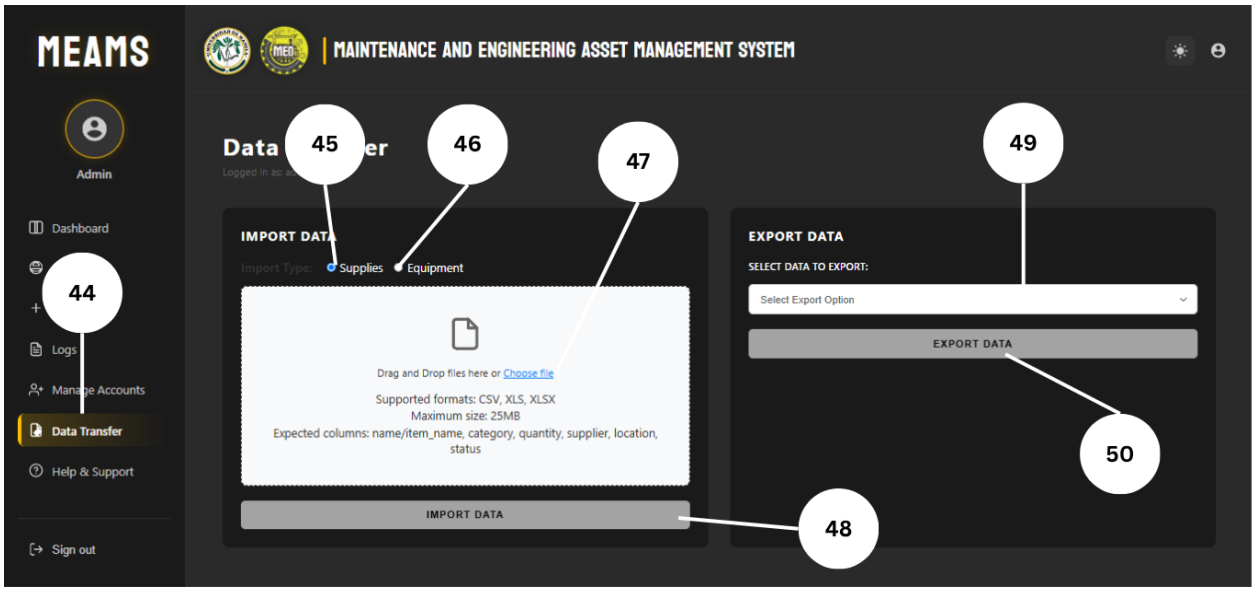
41.2. The field that allows the administrator to modify the user's name in the system.

41.3. The field that allows the administrator to modify the user's username for login.

41.4. Save the changes.



42.1. confirmation dialog that appears when the "Reset Password" action is selected.



44. Data Transfer Button - Displays Data Transfer for importing and exporting.

45. Used to select Supplies as the target asset type for data import.

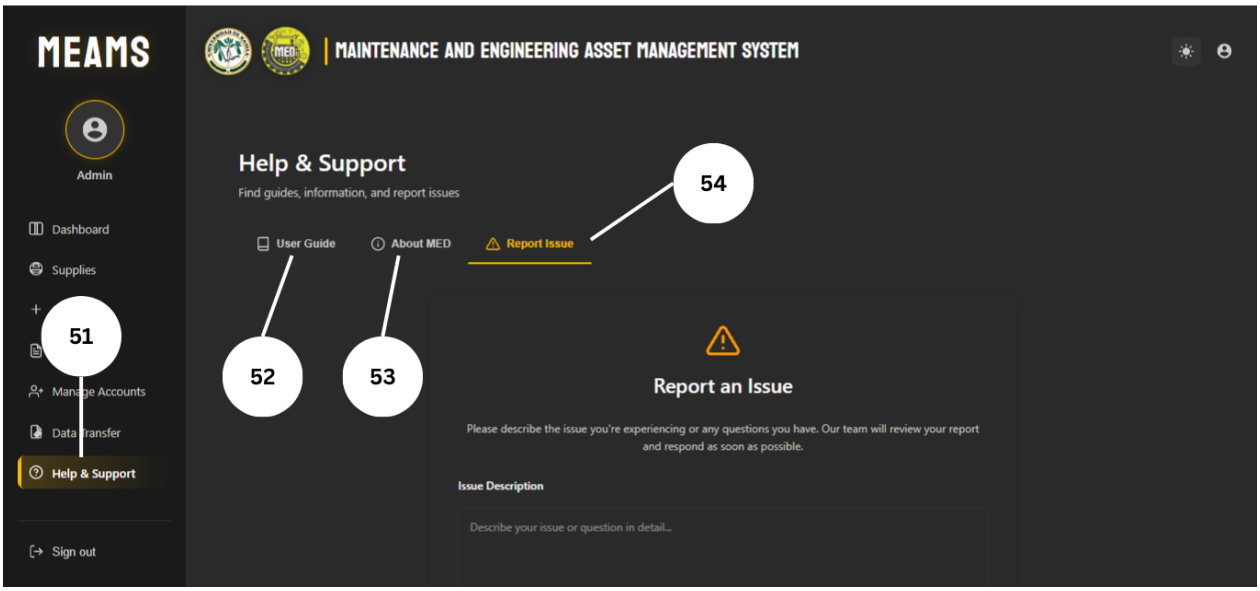
46. Used to select Equipment as the target asset type for data import.

47. The drag-and-drop zone used to upload the CSV or Excel file for data importation.

48. The action button used to begin processing and inserting the data from the uploaded file into the system.

49. A dropdown menu used to choose which system data (e.g., Supplies, Equipment, Logs) to generate for download.

50. The action button used to generate and download the selected system data in a spreadsheet format.

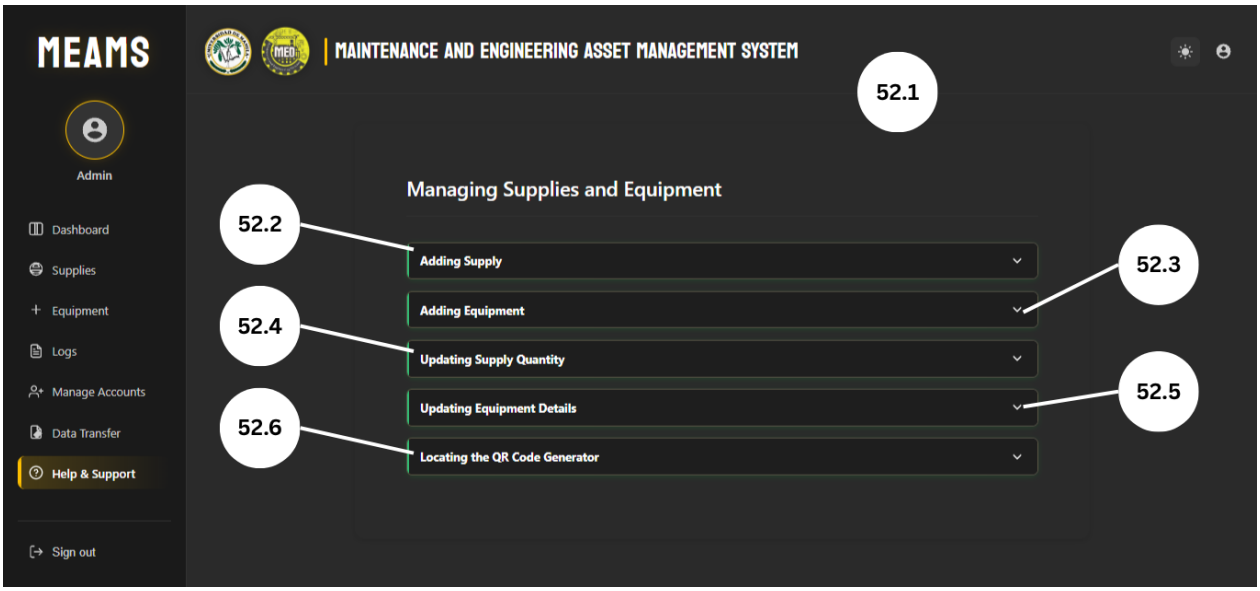


51. Help & Support Button - Displays Help & Support for guides, information and Issue reporting.

52. Selecting this tab displays documentation and tutorials on how to use the system features.

53. Selecting this tab displays information and details regarding the Maintenance and Engineering Division (MED).

54. Selecting this tab displays the form used to submit detailed descriptions of any system problems or questions to the support team.



52.1. Overlay of the User Guide Section.

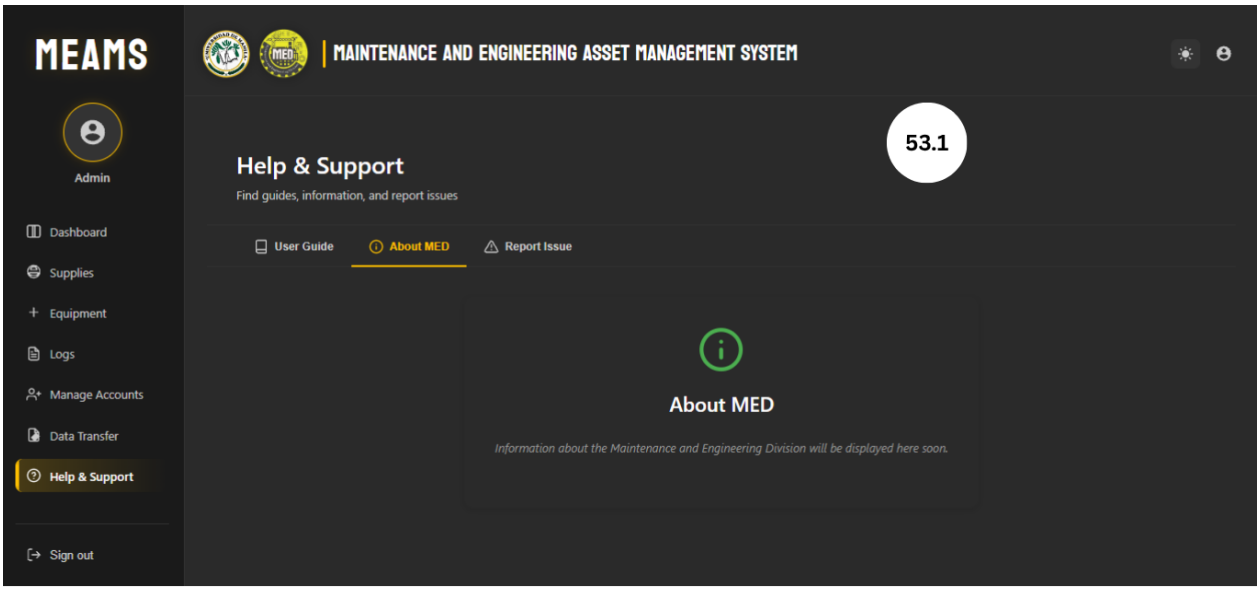
52.2. A collapsible guide that provides step-by-step instructions on how to add new supplies to the inventory.

52.3. A collapsible guide that provides step-by-step instructions on how to add new equipment assets.

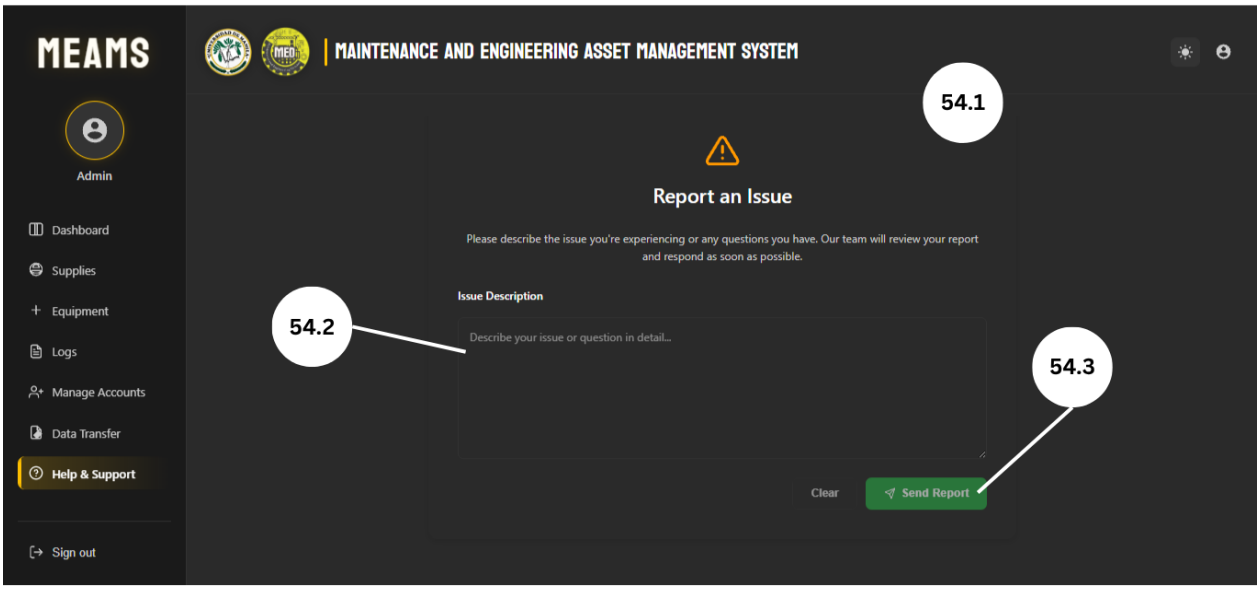
52.4. A collapsible guide detailing the process for modifying the stock quantity of an existing supply item.

52.5. A collapsible guide detailing the process for editing the information of an existing equipment asset.

52.6. A collapsible guide that provides instructions on where to find and use the QR code generation tool.



53.1. Overlay of the About Med Section.

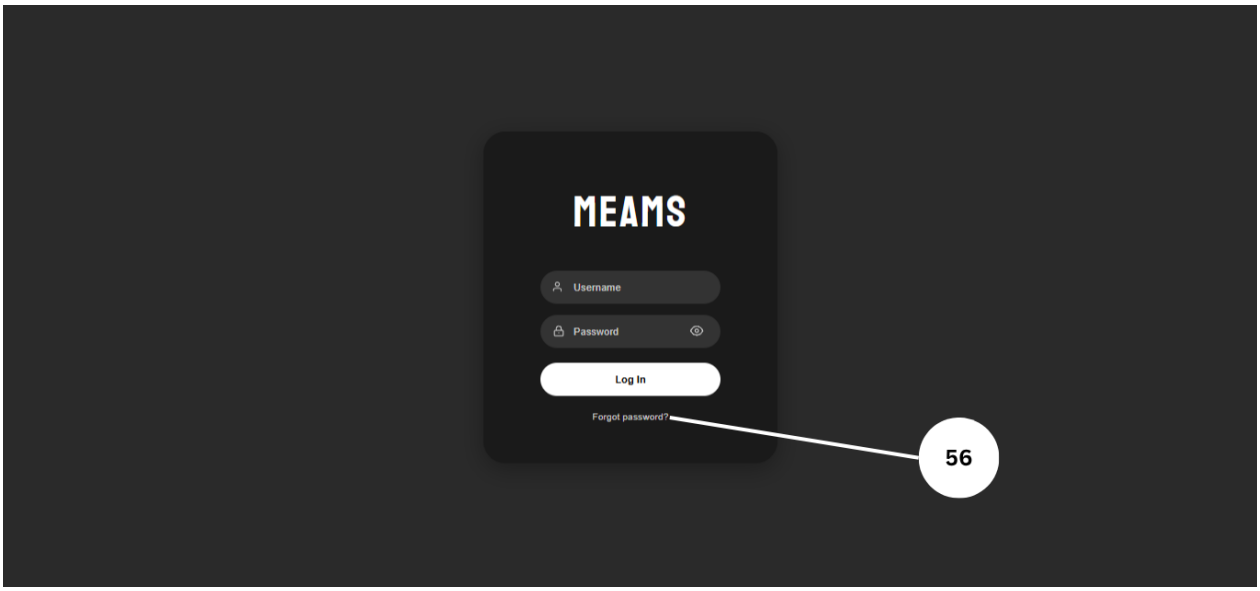


54.1. Overlay of the Report Issue Section.

54.2. A large text area where the user must detail the problem or question they are experiencing.

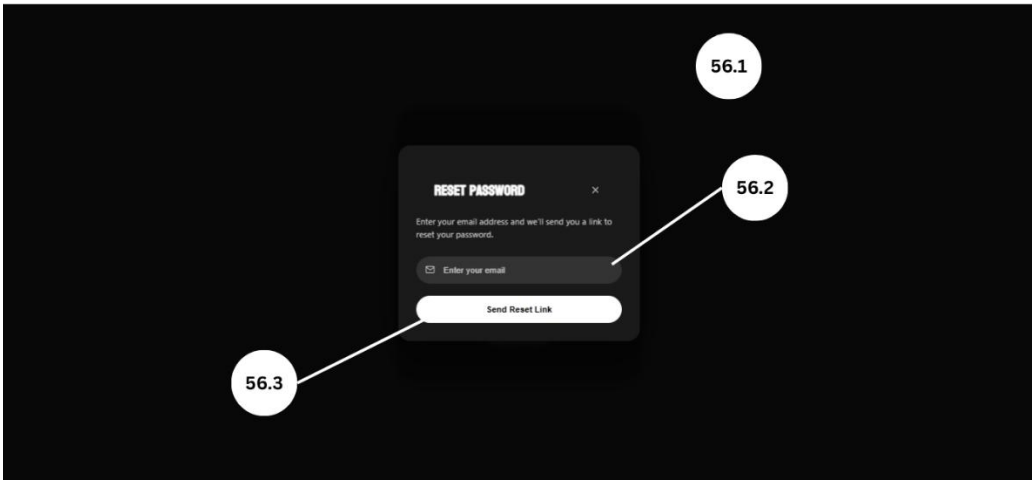
54.3. Submit the completed issue description to the support team for review and response.

55. Sign Out.



56. Clicking this initiates the process for users to recover or reset their forgotten account password.

57. forgotten account password.

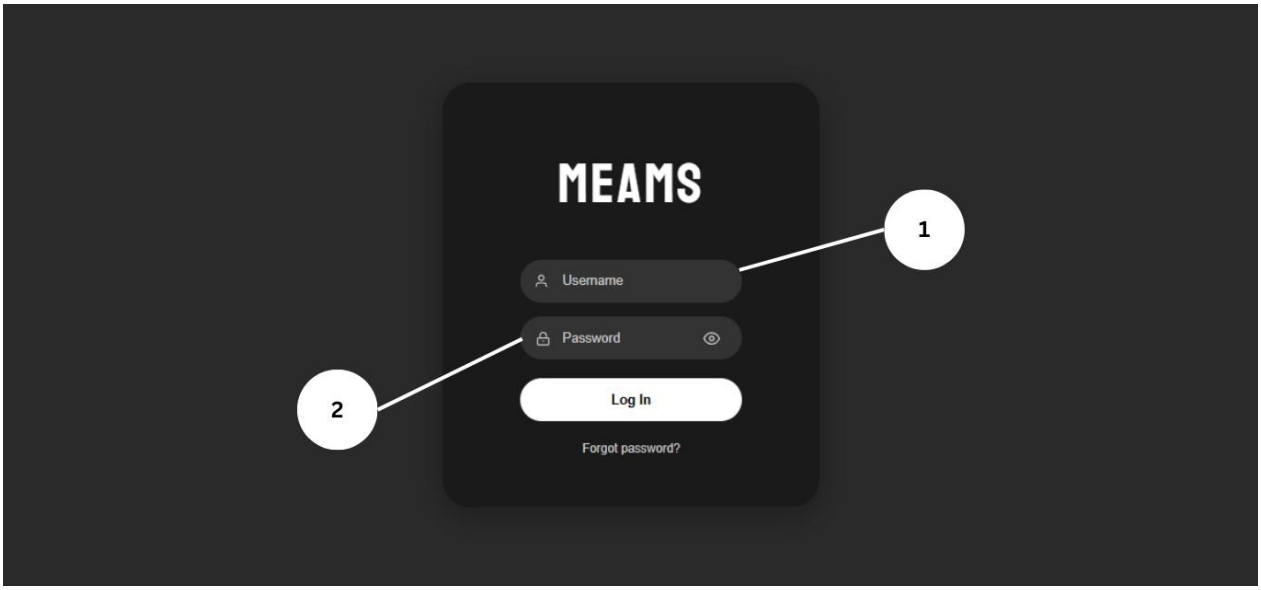


56.1. Overlay of the Reset Password.

56.2. The input field where the user must enter their registered email address to receive the reset link.

56.3. The action button used to initiate the password reset process by sending the link to the provided email.

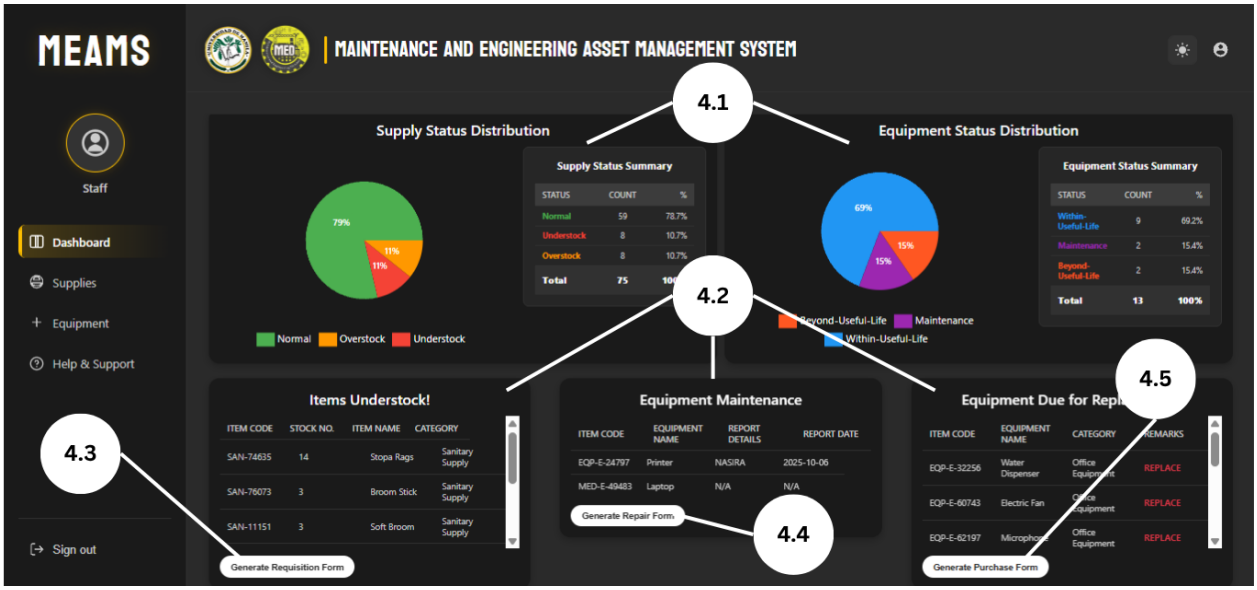
STAFF MANUAL



1. Input your assigned username.
2. Input your unique password.



3. Dashboard Button - Displays a quick overview of the system's asset health through visual distribution summaries and forecasting charts for both supplies and equipment.
4. Displays key system metrics and statistics using Supplies Trend & Forecast and Equipment Trend & Forecast charts.



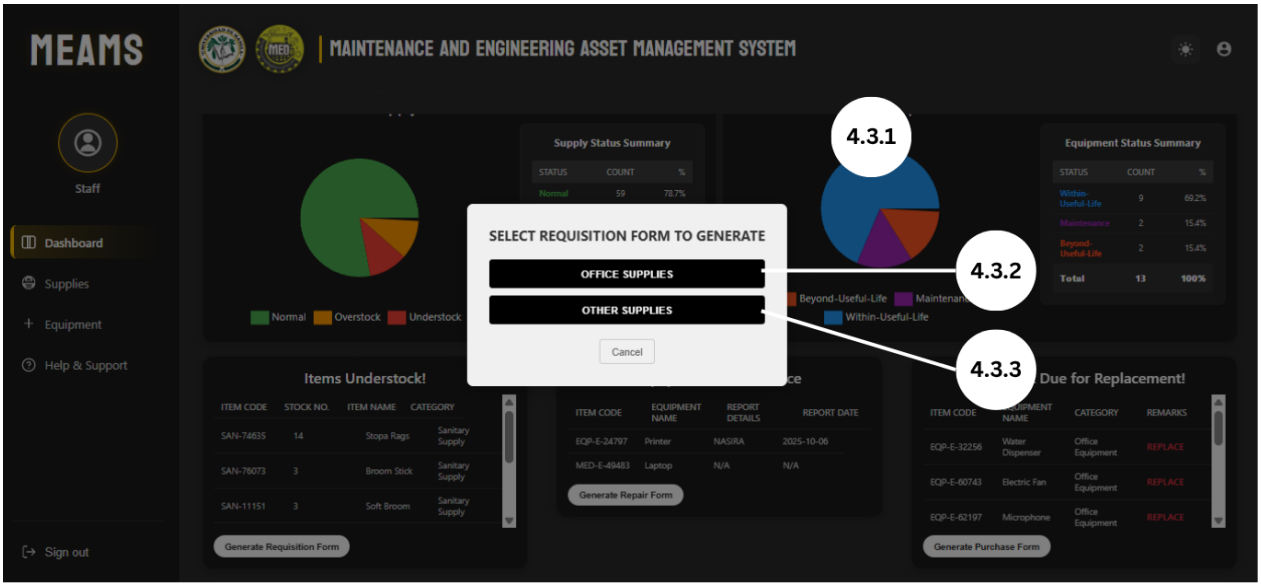
4.1. Provides a concise visual summary of Supply Status Distribution and Equipment Status Distribution via pie charts and summary tables.

4.2. Features three actionable panels: Items Understock!, Equipment Maintenance, and Equipment Due for Replacement! for quick form generation.

4.3. Where you can Generate a Requisition Form.

4.4. Where you can Generate a Repair Form.

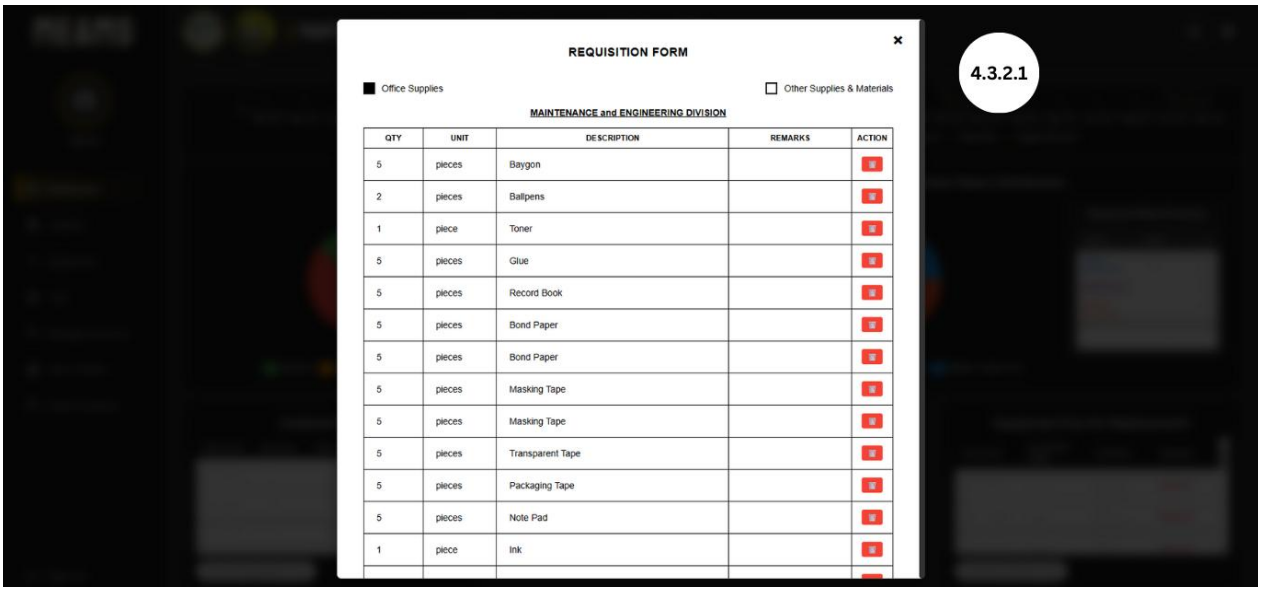
4.5. Where you can Generate a Purchase Form.



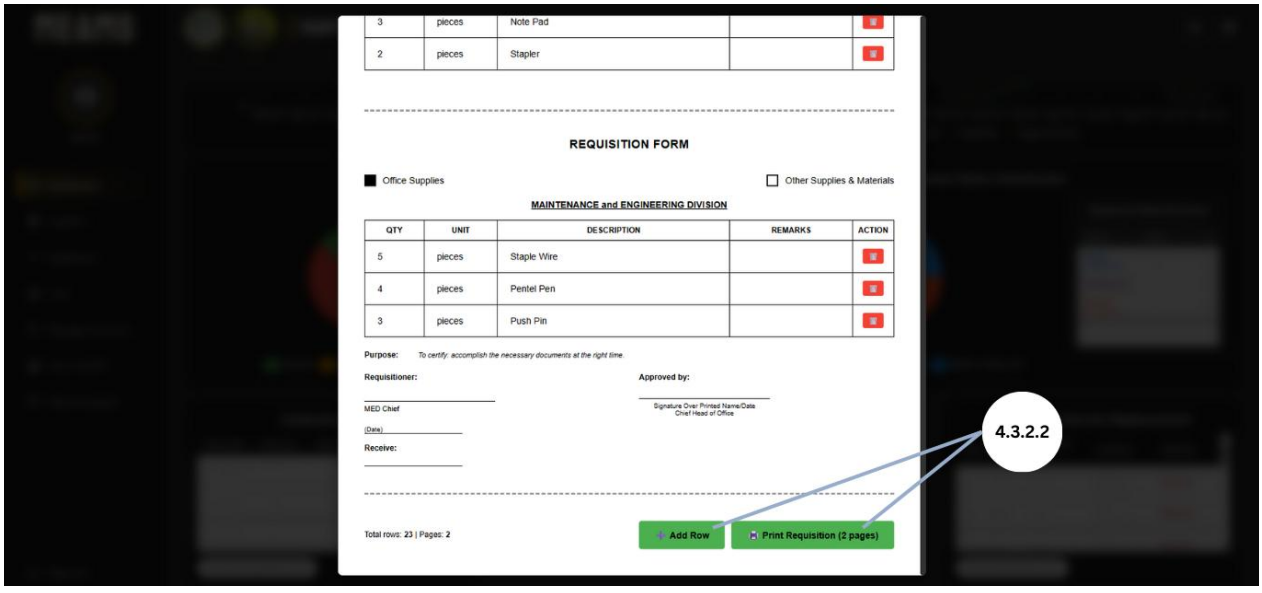
4.3.1. Overlay of “Generate Requisition Form.

4.3.2. Generate a Requisition Form for Office Supplies.

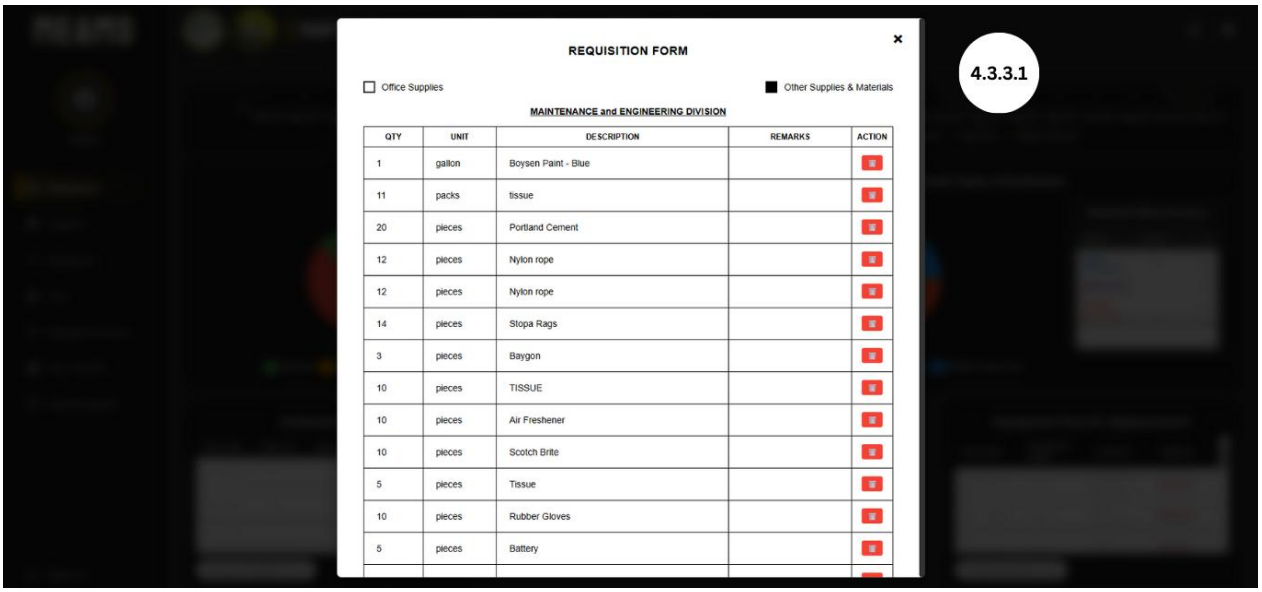
4.3.3. Generate a Requisition Form for Other Supplies.



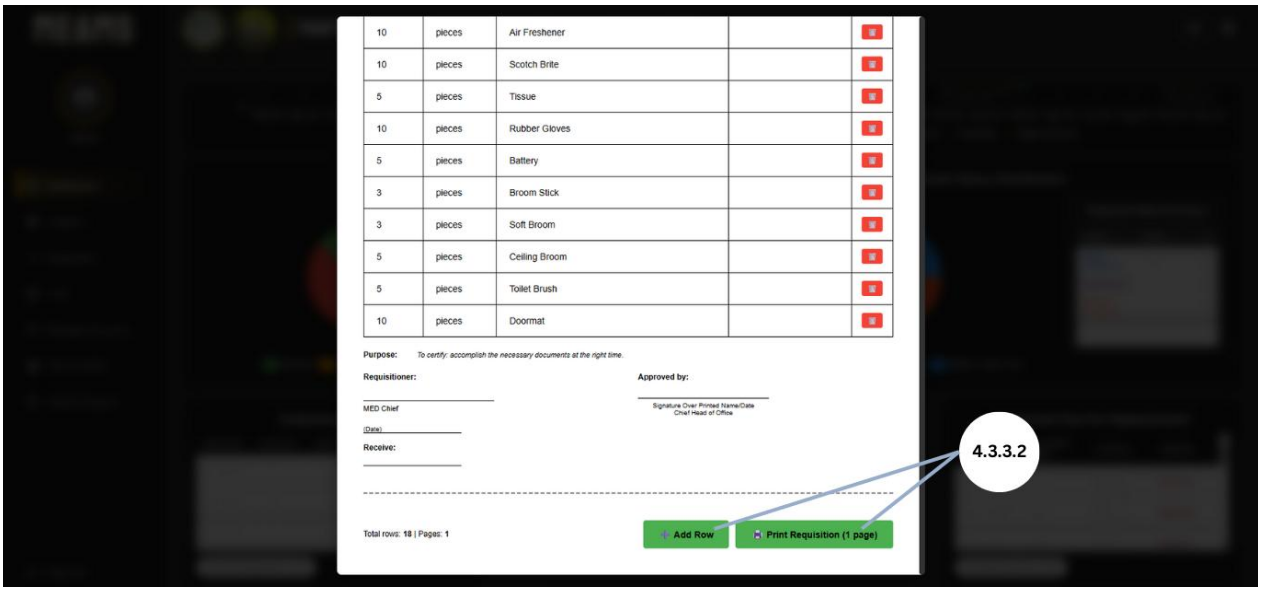
4.3.2.1. Overlay of Modal Requisition Form for Office Supplies.



4.3.2.2. Add Row and Print the Requisition Form for Office Supplies.



4.3.3.1. Overlay of Modal Requisition Form for Other Supplies.

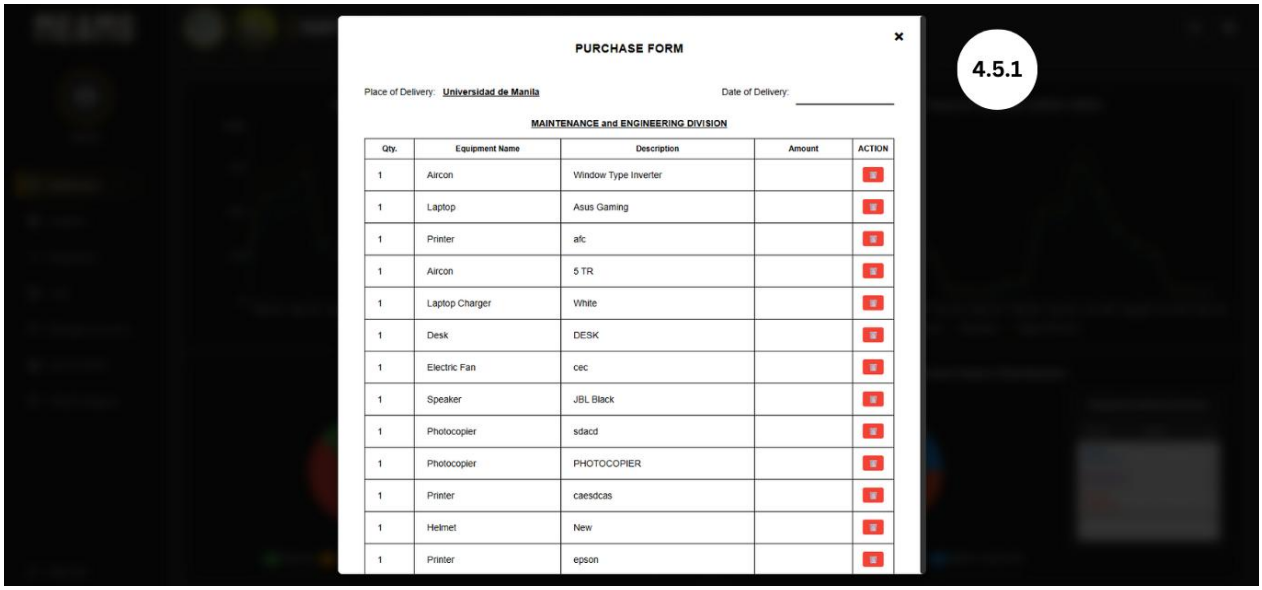


4.3.3.2. Add Row and Print the Requisition Form for Other Supplies.

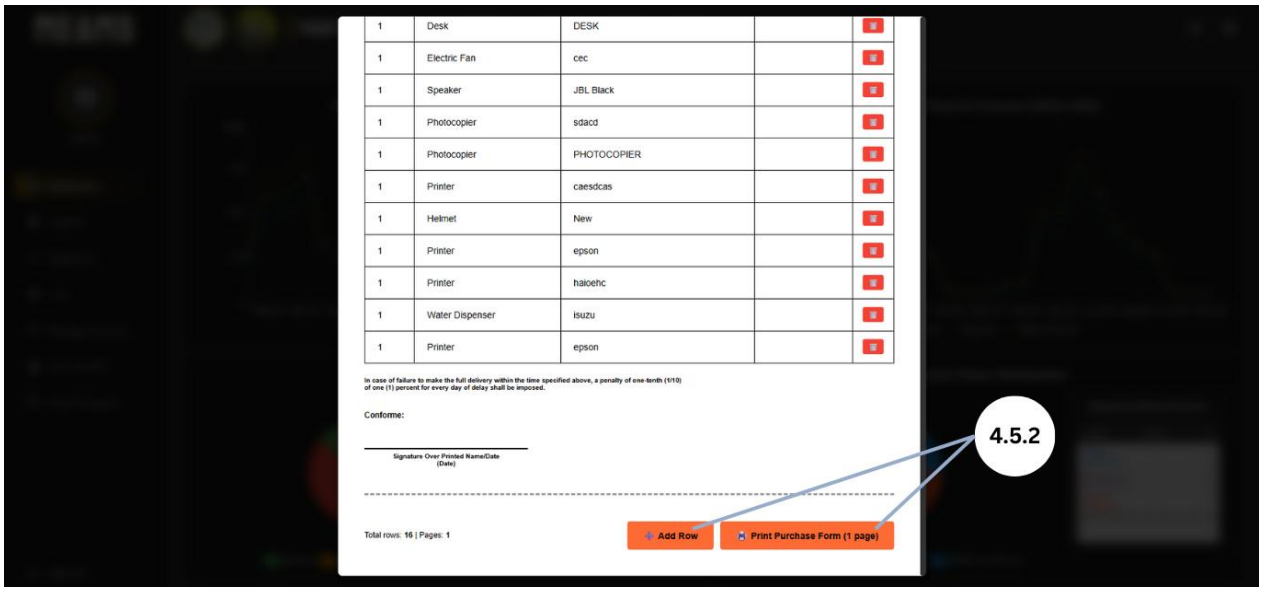


4.4.1. Overlay of Modal Repair Form.

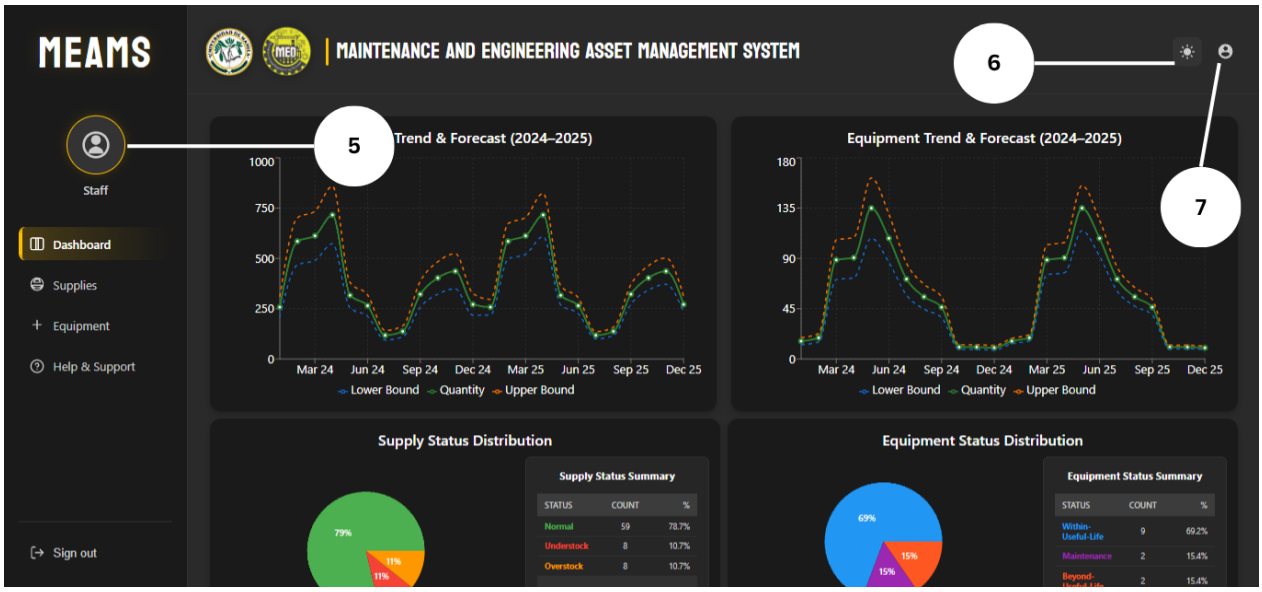
4.4.2. Print the Repair Form.



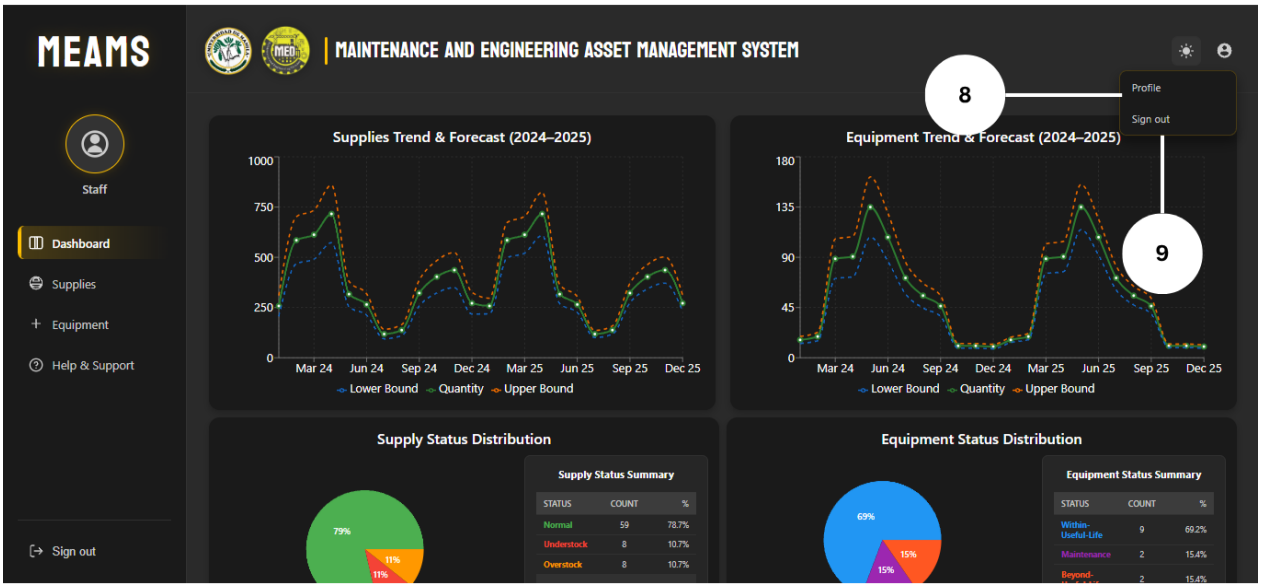
4.5.1. Overlay of Modal Purchase Form.



4.5.2. Print the Modal Purchase Form.

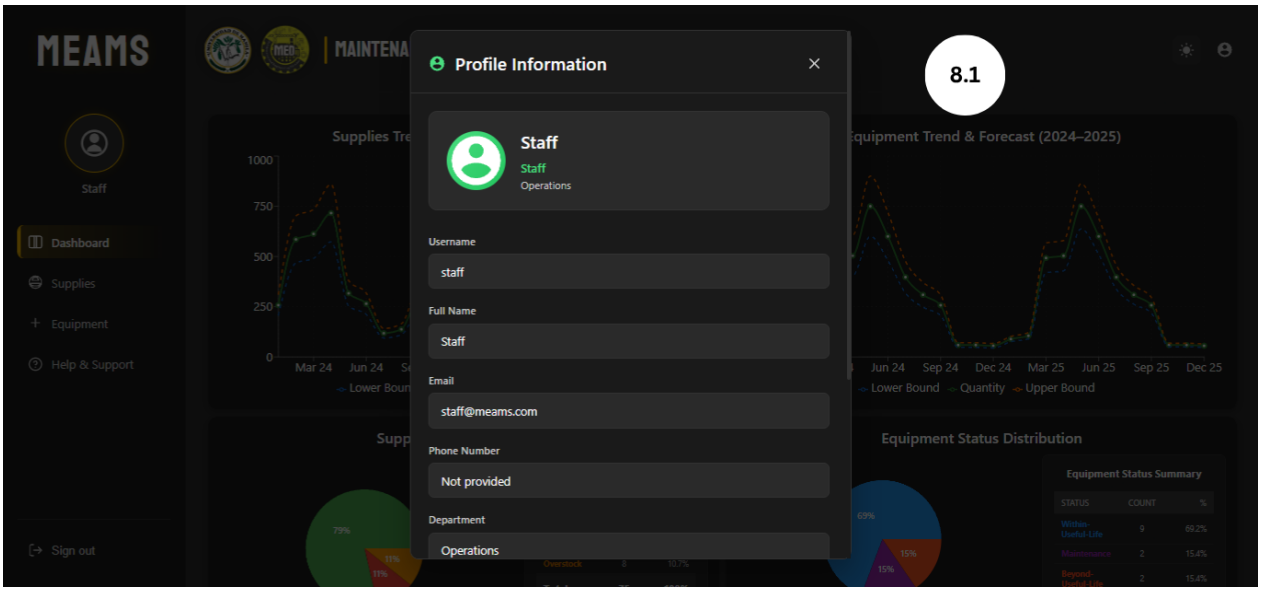


5. Where you can change the Profile Picture of the User.
6. Switch to Light Mode/Dark Mode.
7. Clickable Dropdown.

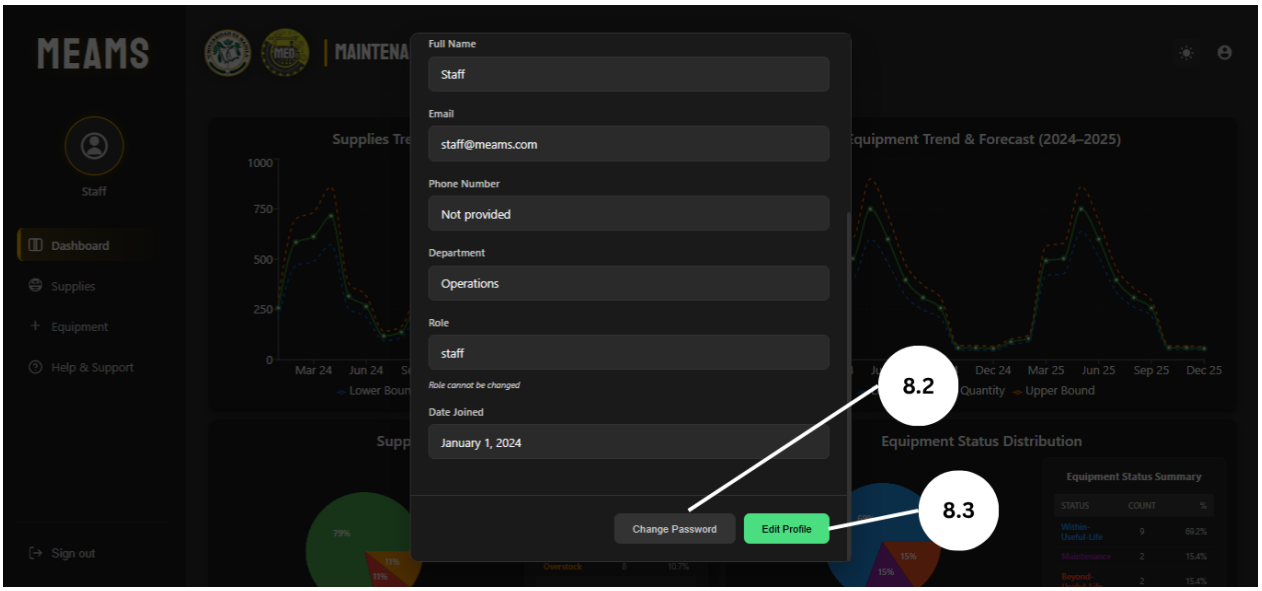


8. You can see your Profile Information.

9. Sign out from the System.

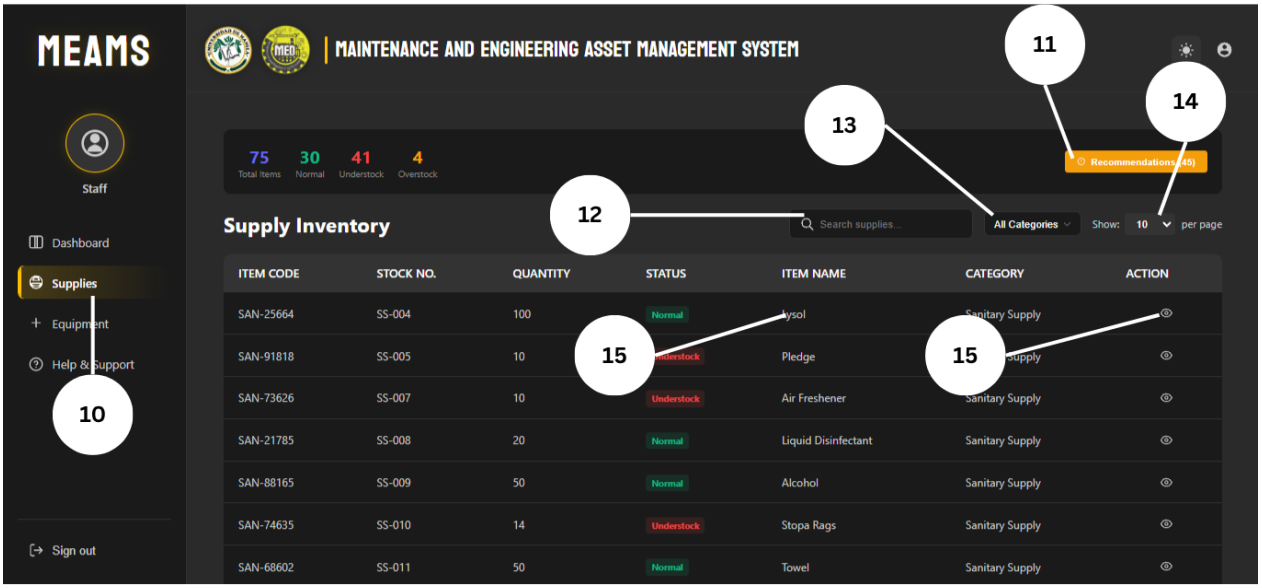


8.1. Overlay of Profile Information.



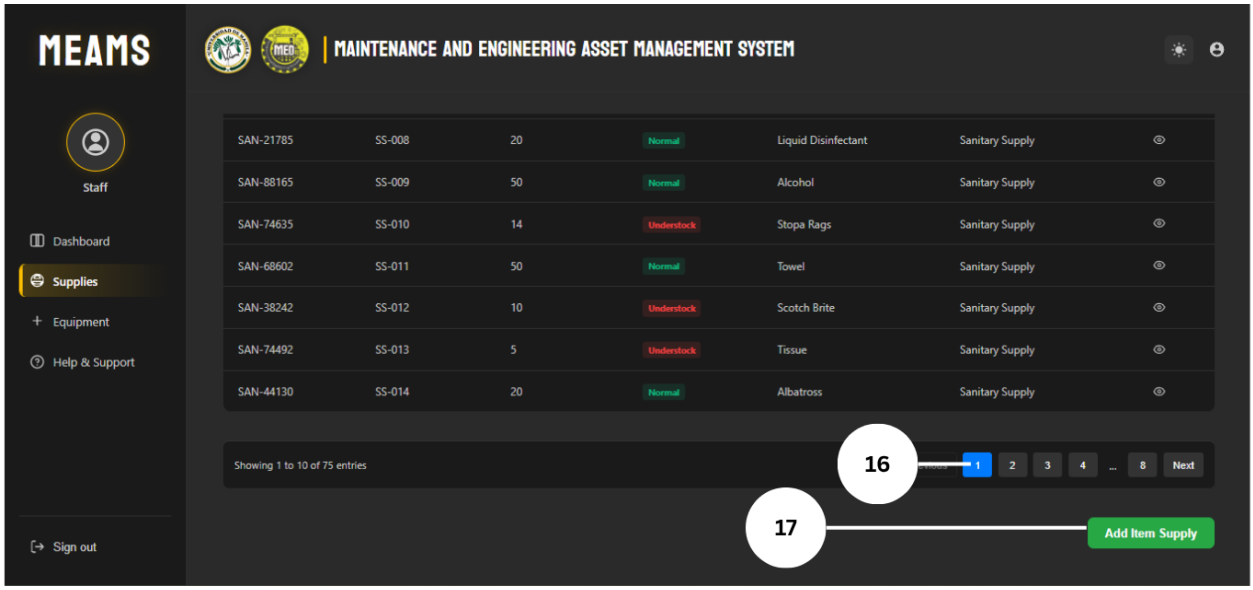
8.2 If you want to change Password.

8.3 If you want to edit your Profile Information.

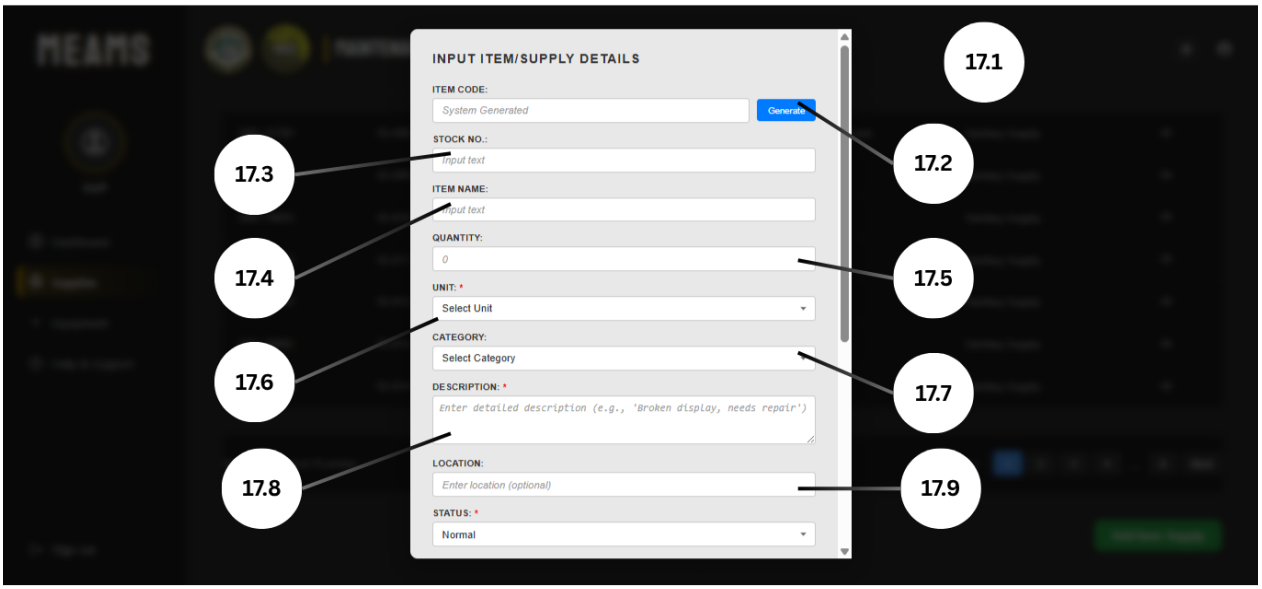


10. Supplies Button - Displays Supplies Inventory.
11. It displays recommendations for critical or low conditions for supplies.
12. If you want to search for a specific supply.

13. You can choose a specific category of supplies.
14. A dropdown menu to select the number of inventory entries shown per page.
15. Can be clicked for Supply Details



16. Shows the current range of displayed entries and for navigating the full inventory list.
17. Action button for Adding an Item Supply.



17.1. Modal form for Adding Supply.

17.2. Generate a Unique Item Code.

17.3. Input a Stock No.

17.4. Input a Item Name.

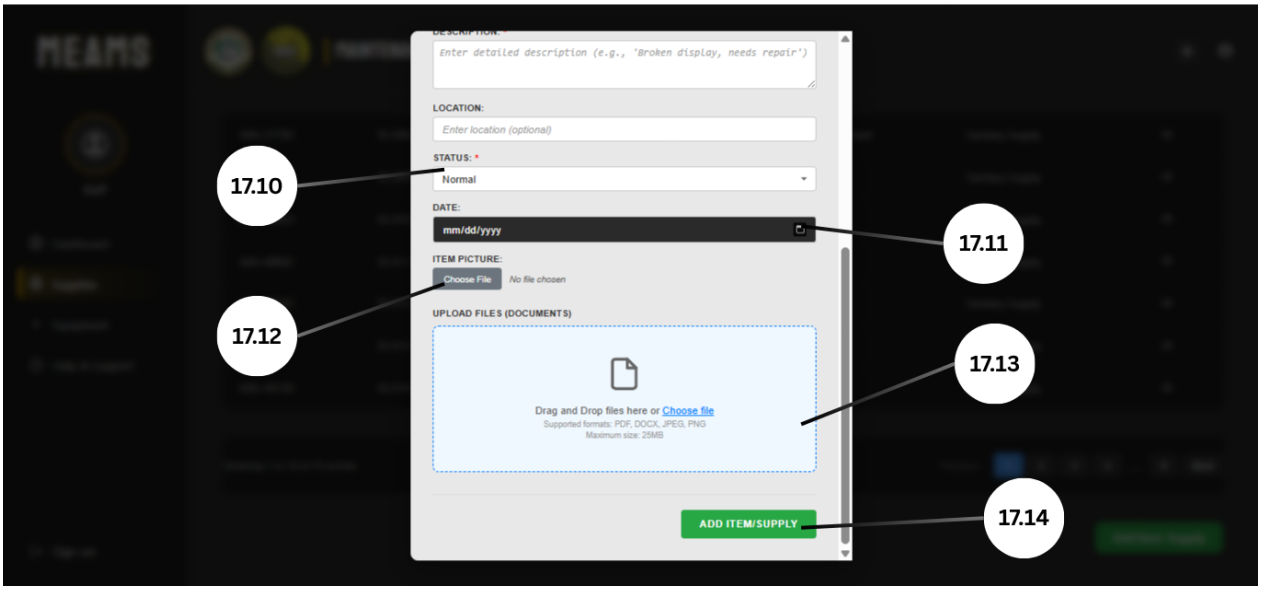
17.5. Input the Quantity of the Supply.

17.6. Select a Unit for the Supply.

17.7. Select a Category for the Supply.

17.8. Description of the Supply.

17.9. Location of the Supply (Optional)



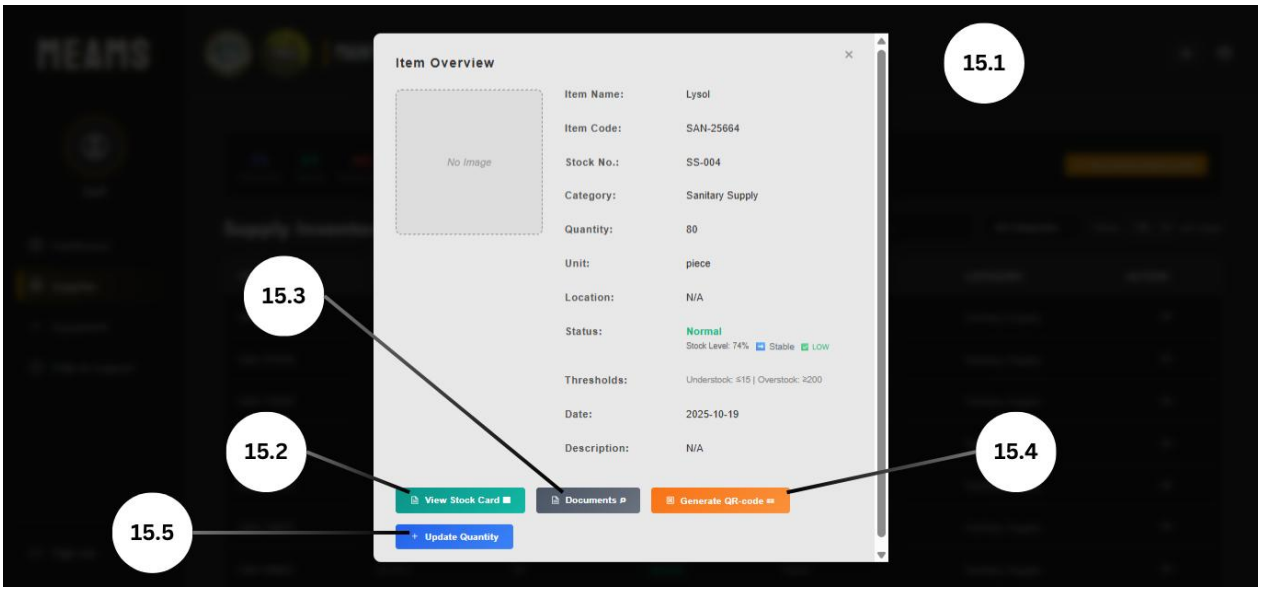
17.10. Select a Status for the Supply.

17.11. Input a Date when the Supply bought.

17.12. Import a Picture for the Supply.

17.13. Import a Document/Picture for the Supply.

17.14. Add the Supply.



15.1. Overlay of the Supply Details.

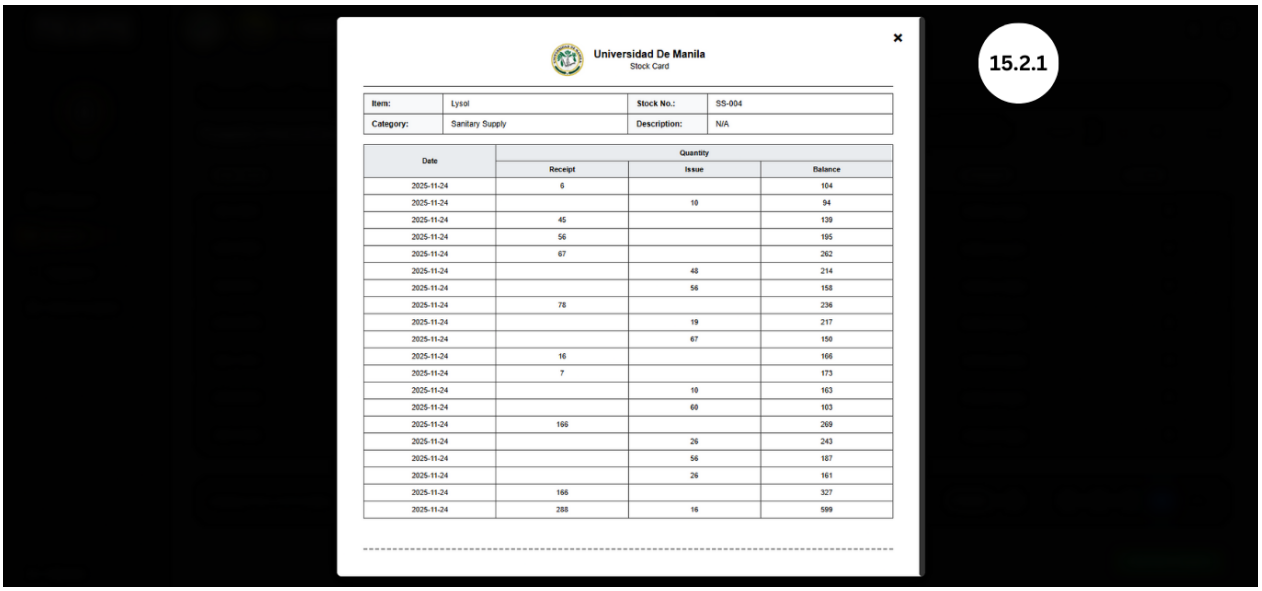
15.2. When clicked, the Stock Card will appear.

15.3. Import necessary documents for the Supply.

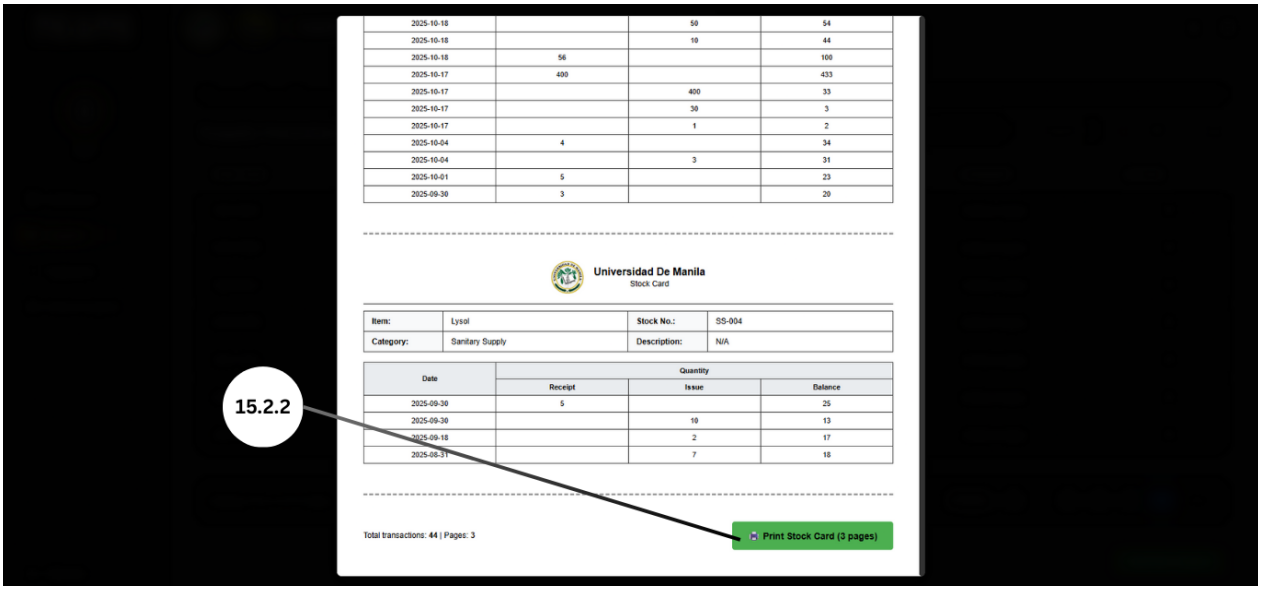
15.4. When clicked, the generated QR code will appear.

15.5. Update the Quantity of the Supply.

15.6. Edit the Details of the Supply.

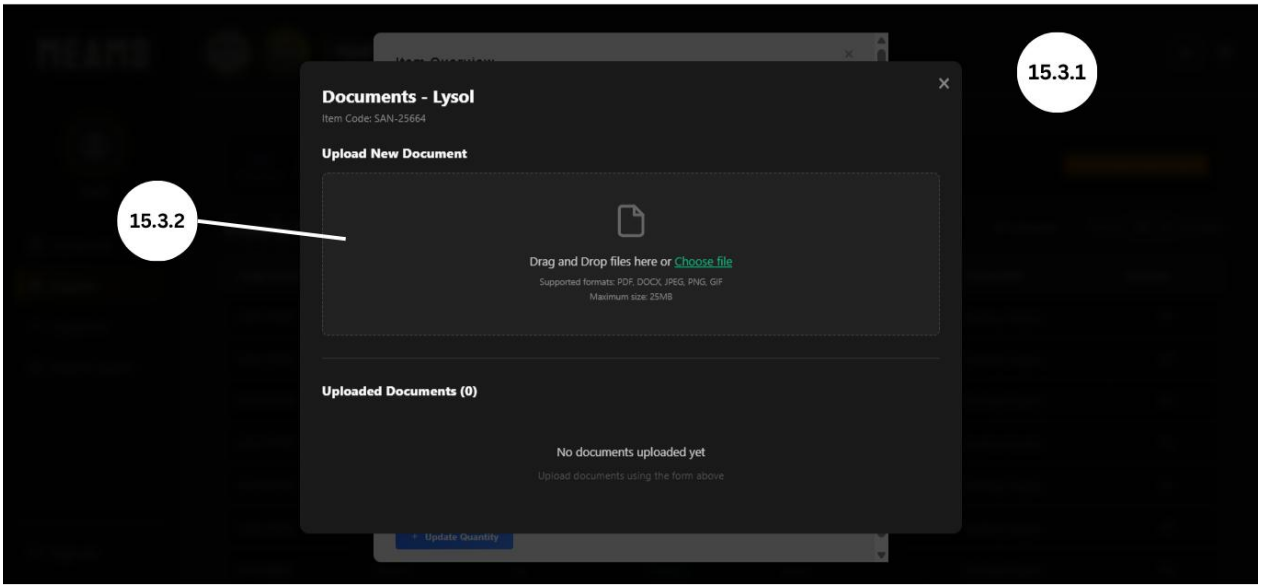


15.2.1. Overlay of the Stock Card form.



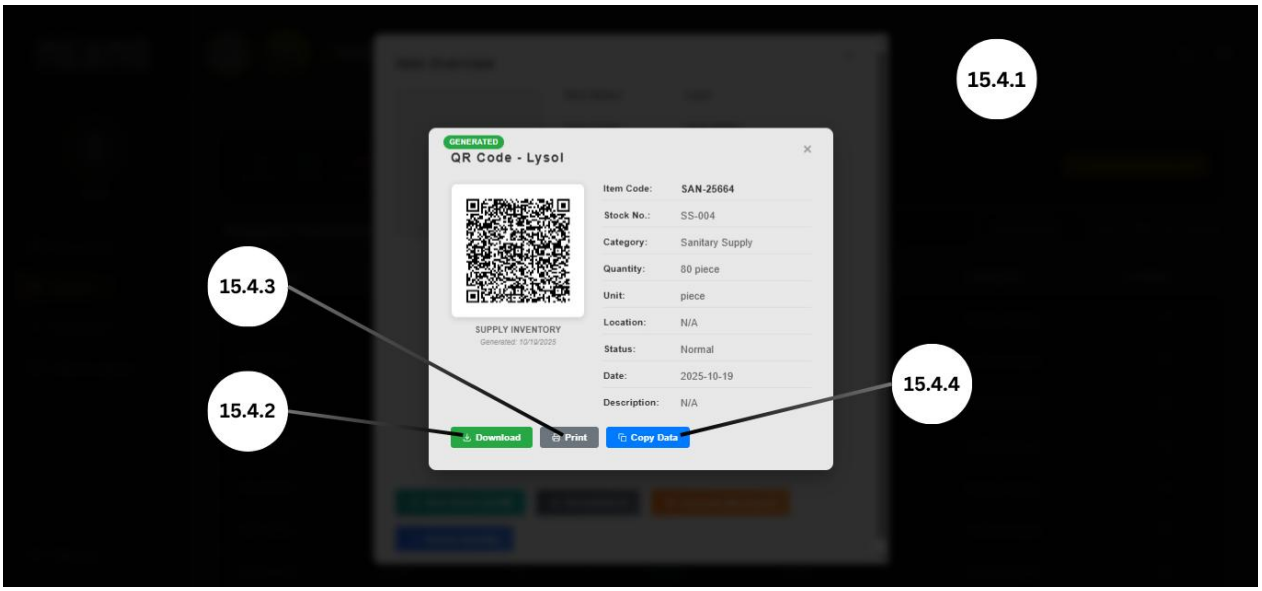
15.2.2. Can be downloaded as PDF.

15.2.3. Can be Printed also.



15.3.1. Overlay of the Import Documents for Supply.

15.3.2. Drag and Drop or Import a Document for the Supply.



15.4.1. Overlay of the QR Code Details.

15.4.2. QR Code can be downloaded.

15.4.3. QR Code can be Print.

15.4.4. Data can be copied also.



15.5.1. Overlay of the Update Quantity.

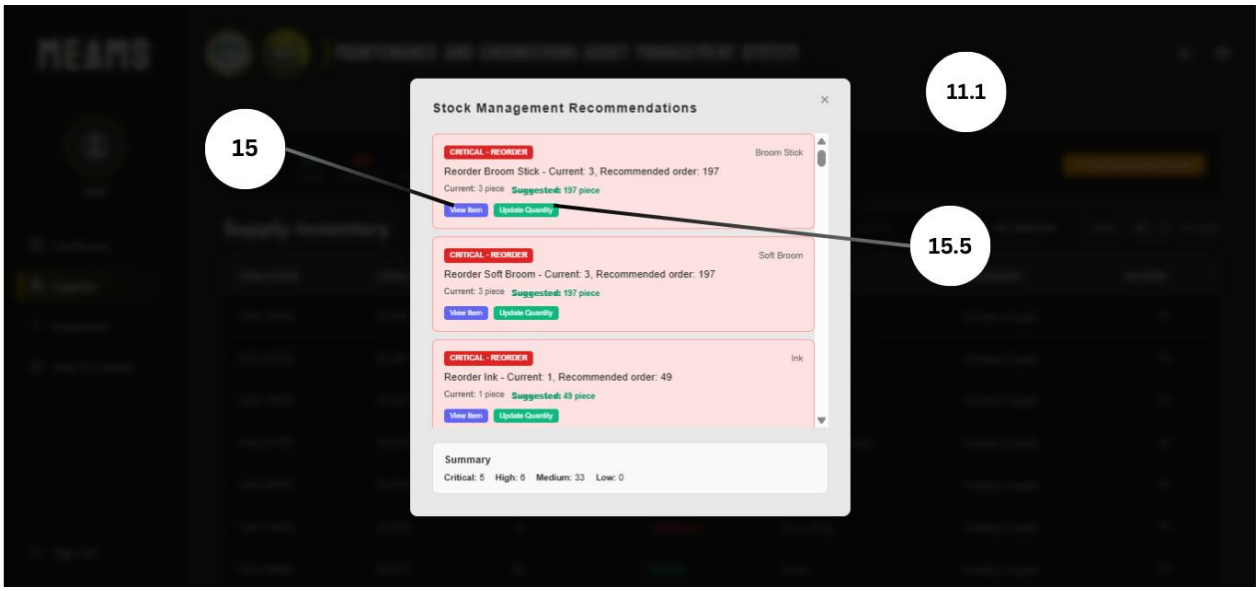
15.5.2. Input when the supply stored or distributed.

15.5.3. If the Supply quantity is added.

15.5.4. If the Supply quantity is subtracted.

15.5.5. Total of the Quantity.

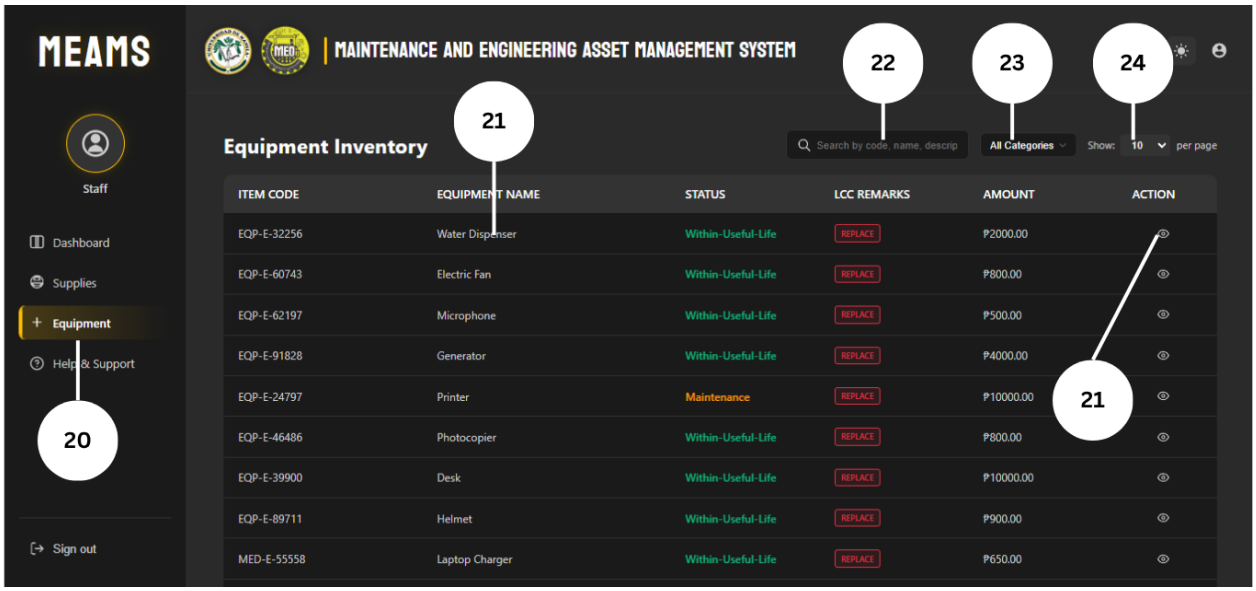
15.5.6. Update the edited quantity.



11.1. Overlay of the Recommendations.

15. View the Supply Details.

15.5. Update Quantity.



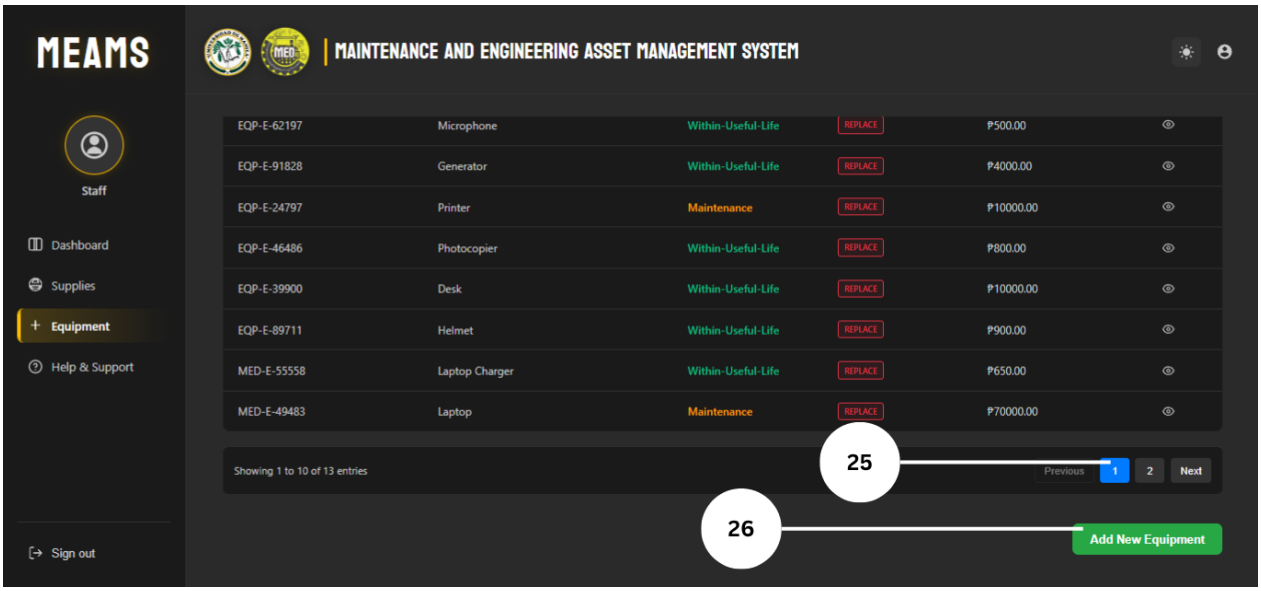
20. Equipment Button - Displays Equipment Inventory.

21. Can be clicked for Equipment Details.

22. If you want to search a specific Equipment.

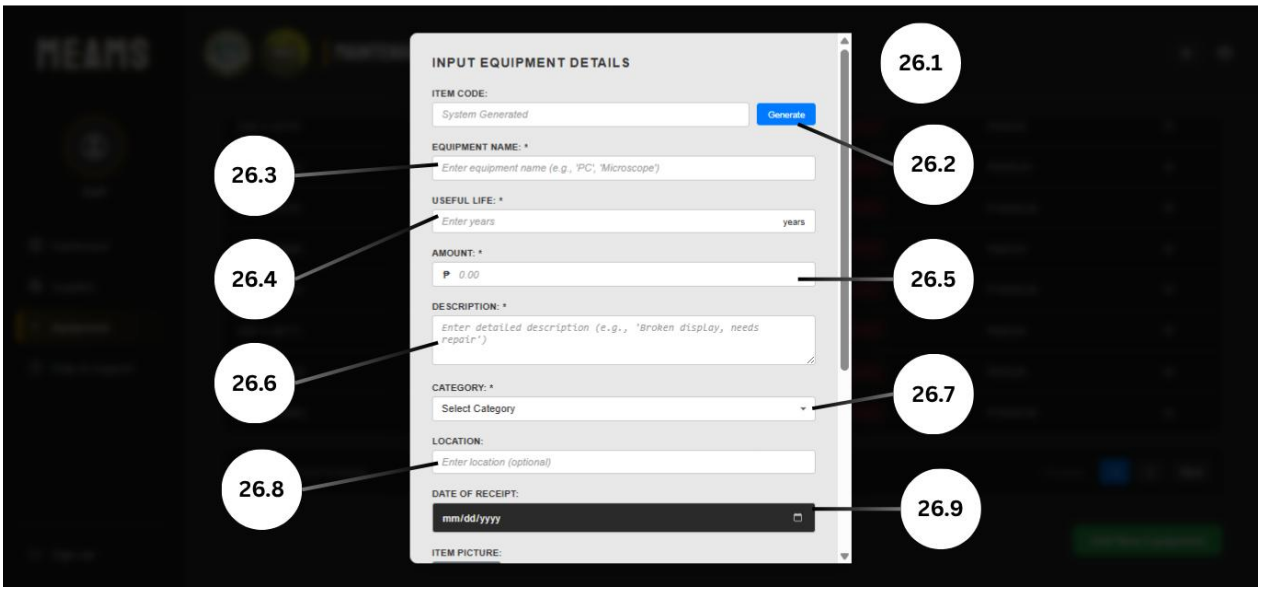
23. You can choose a specific Category.

24. A dropdown menu to select the number of inventory entries shown per page.



25. Shows the current range of displayed entries and for navigating the full inventory list.

26. Action Button for adding an Equipment.



26.1. Modal form for Adding Equipment.

26.2. Generate a Unique Item Code for Equipment.

26.3. Input an Equipment Name.

26.4. Input the Useful Life of the Equipment.

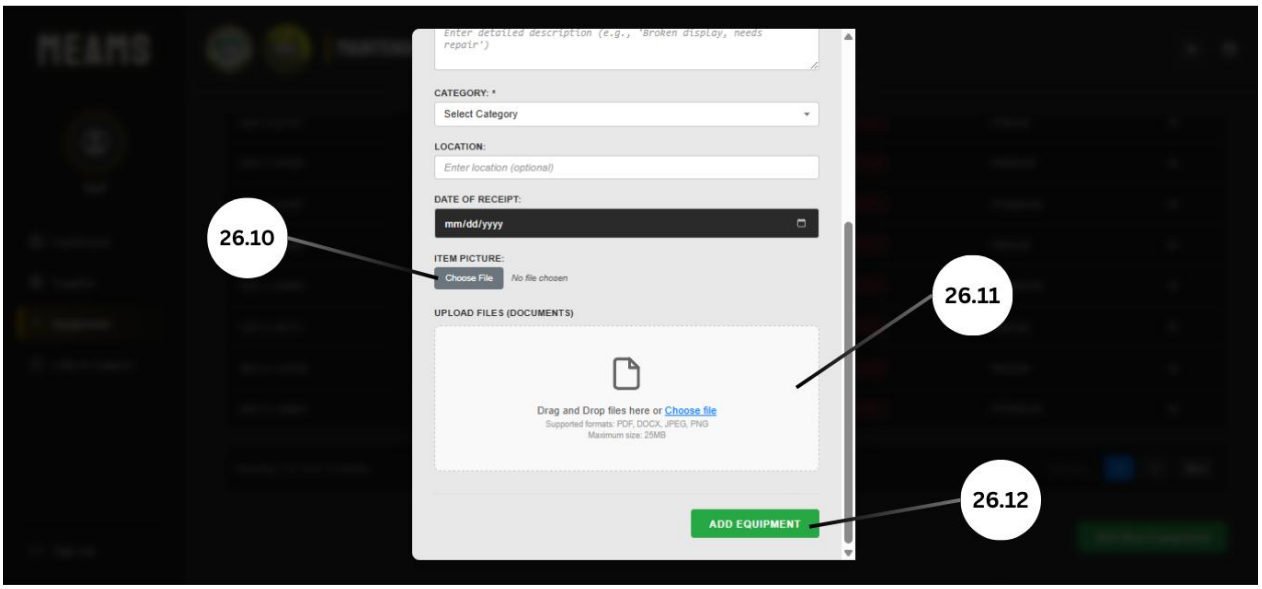
26.5. Input the Amount of the Equipment.

26.6. Description of the Equipment.

26.7. Select a category of the Equipment.

26.8. Location of the Equipment (Optional)

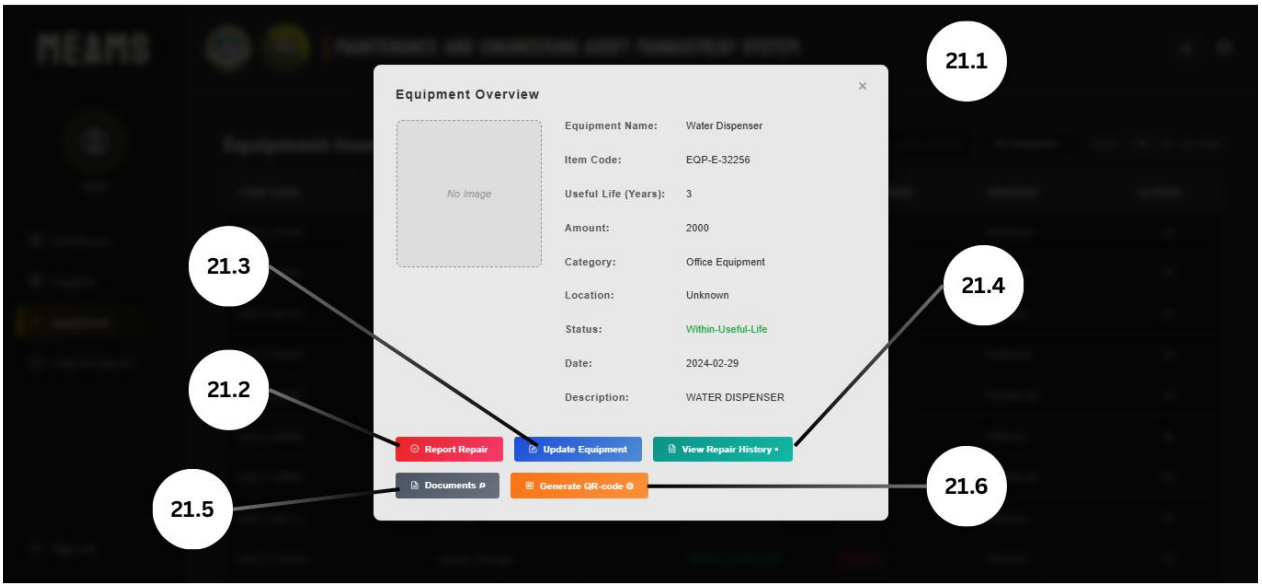
26.9. Date of Receipt of the Equipment.



26.10. Import a Picture of the Equipment.

26.11. Import/Upload a Documents for the Equipment.

26.12. Add newly registered equipment.



21.1. Overlay of the Equipment Details

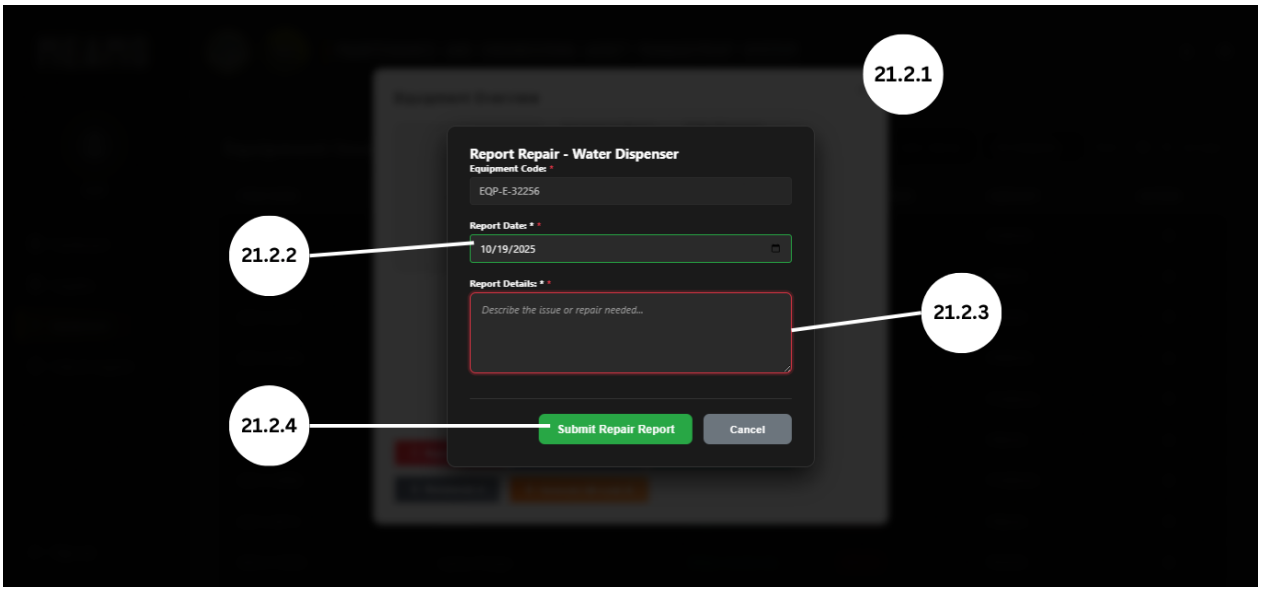
21.2. Report Repair If Equipment has issue.

21.3. Update the Equipment when the equipment is repaired.

21.4. View the Repair History.

21.5. Import necessary documents for the Equipment.

21.6. When clicked, the generated QR code will appear.

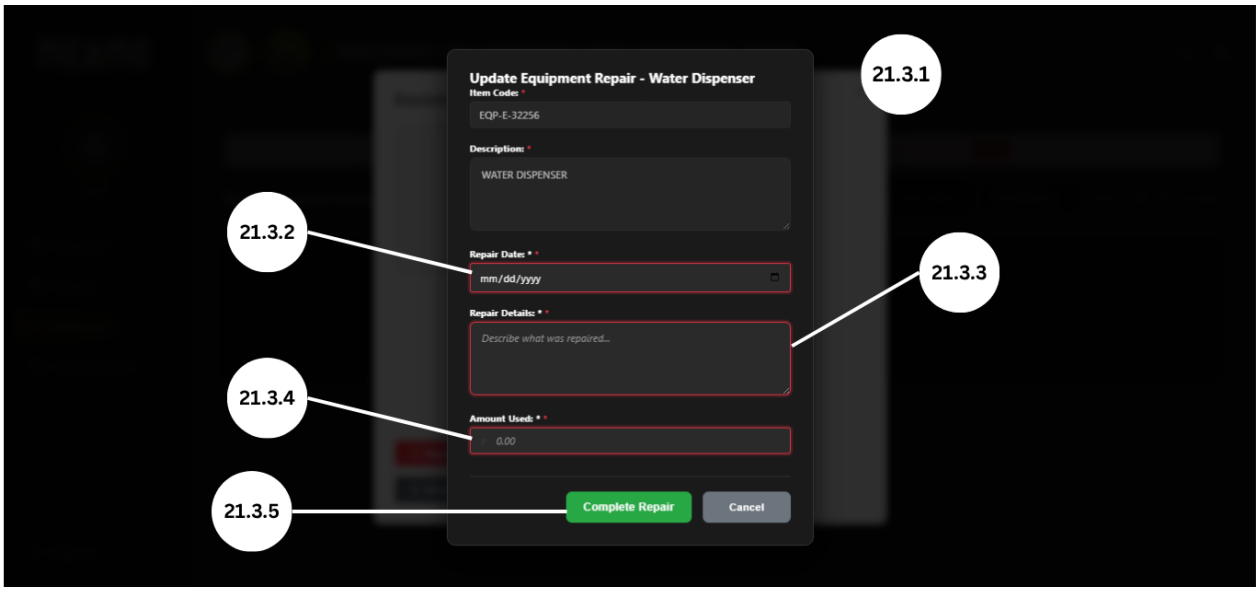


21.2.1. Overlay of the Report Repair.

21.2.2. When is the Report issued.

21.2.3. Description of the Issue and the repair needed.

21.2.4. Submit Report Repair.



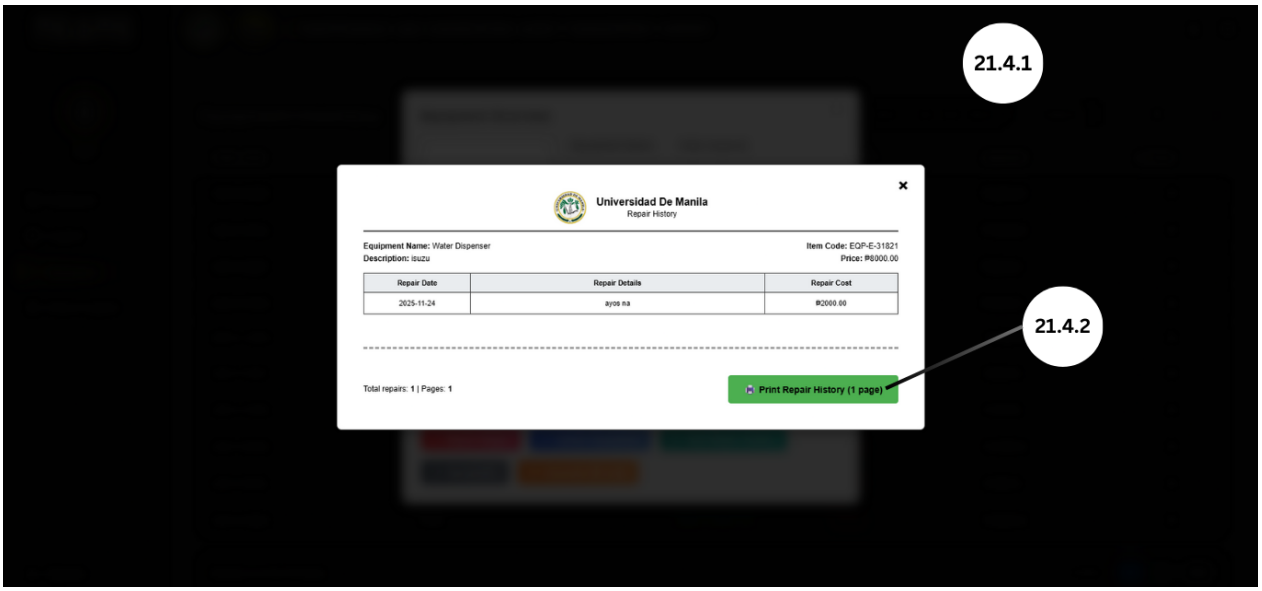
21.3.1. Overlay of the Update Equipment Repair.

21.3.2. When the Equipment Repaired.

21.3.3. Description what has been repaired.

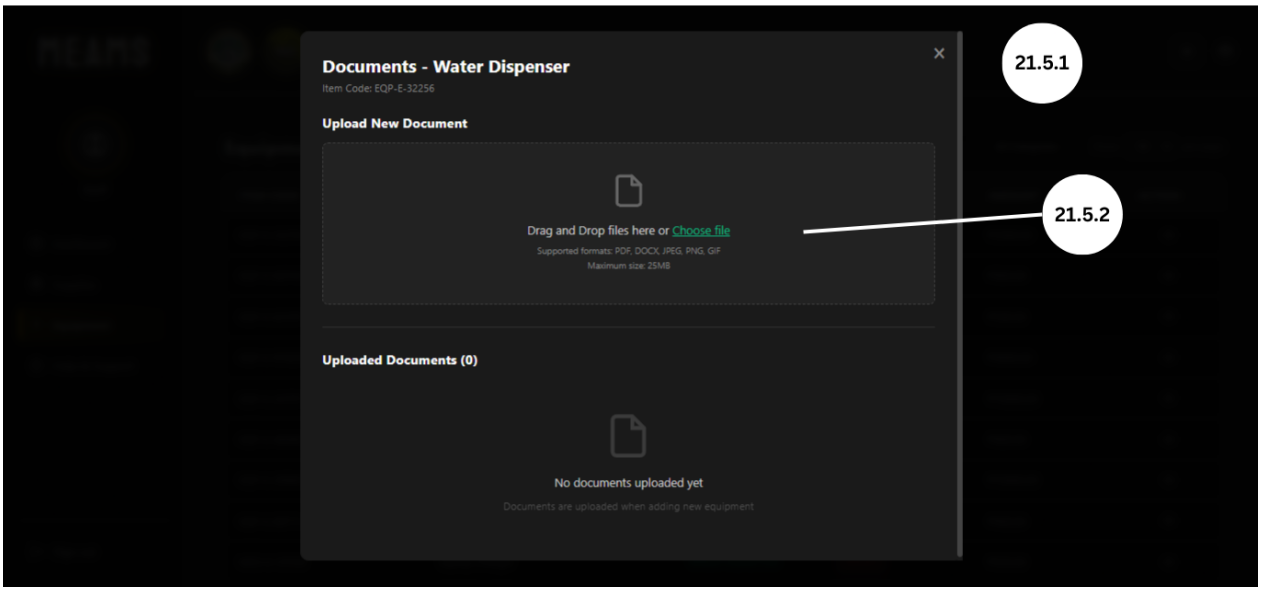
21.3.4. Amount Used for the respiration.

21.3.5. Complete Repair.



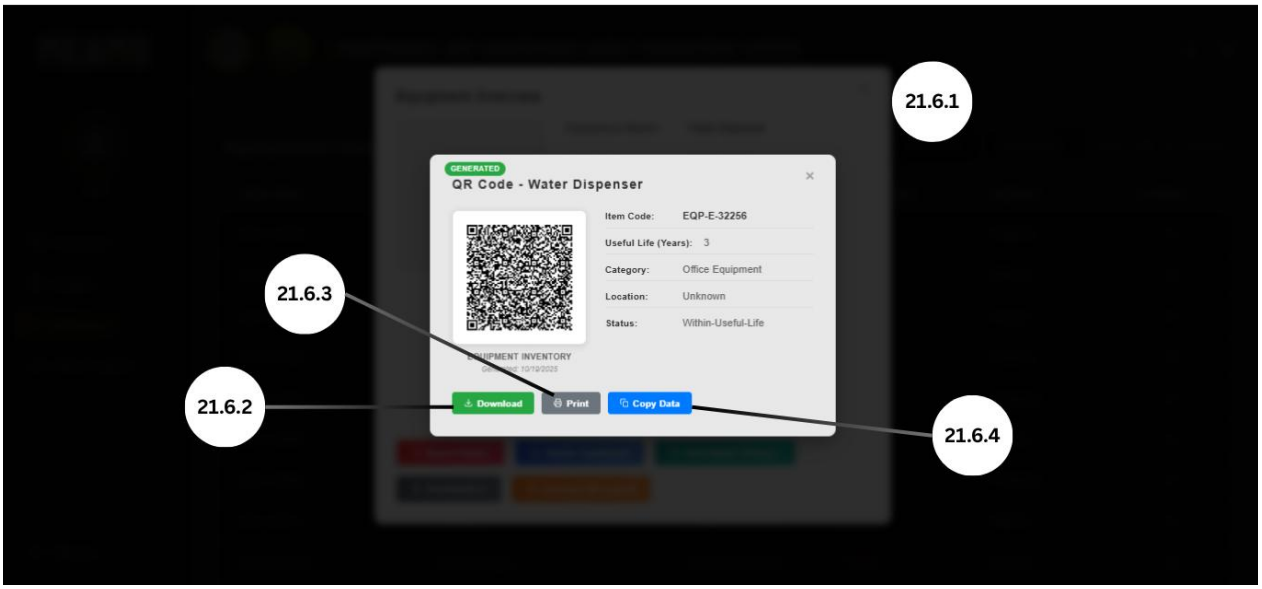
21.4.1. Overlay of the Repair History.

21.4.2. Print the Repair History.



21.5.1. Overlay of the Import Documents for the Equipment.

21.5.2. Drag and Drop or Import a Document for the Supply.

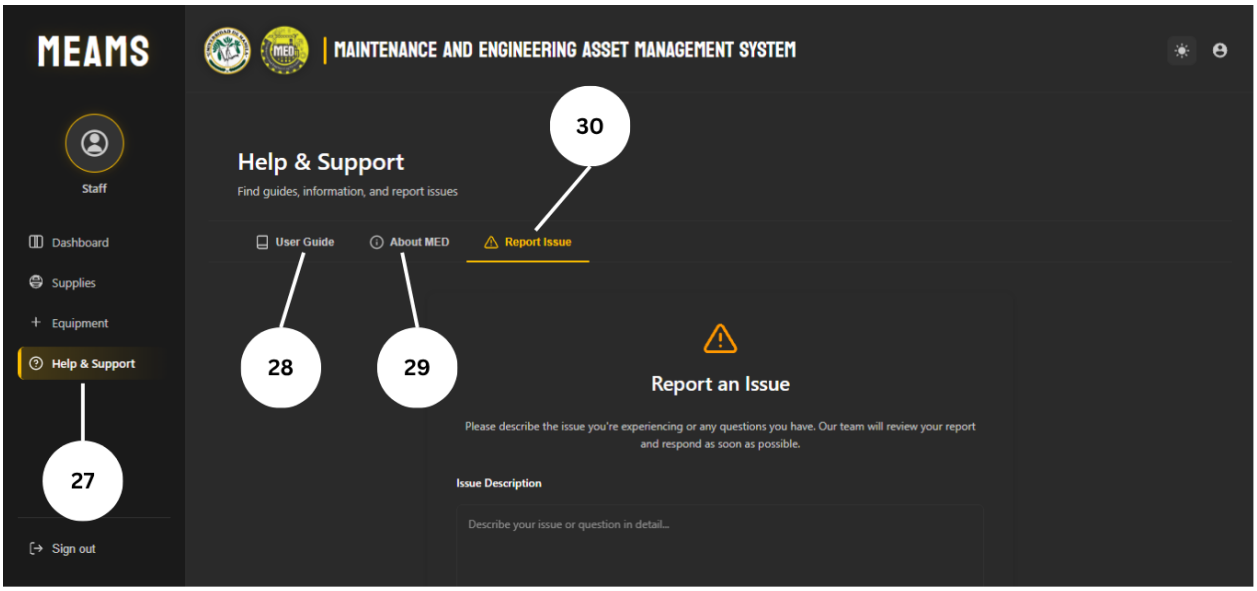


21.6.1. Overlay of the QR Code Details.

21.6.2. QR Code can be downloaded.

21.6.3. QR Code can be printed.

21.6.4. Data can also be copied.

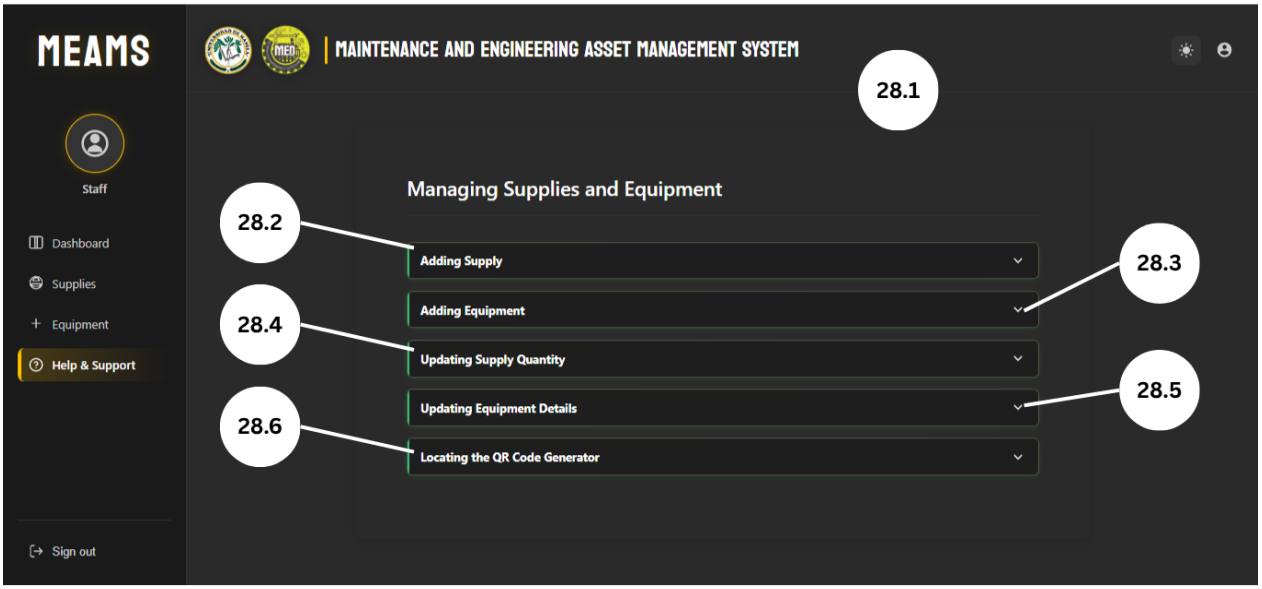


27. Help & Support Button - Displays Help & Support for guides, information and Issue reporting.

28. Selecting this tab displays documentation and tutorials on how to use the system features.

29. Selecting this tab displays information and details regarding the Maintenance and Engineering Division (MED).

30. Selecting this tab displays the form used to submit detailed descriptions of any system problems or questions to the support team.



28.1. Overlay of the User Guide Section.

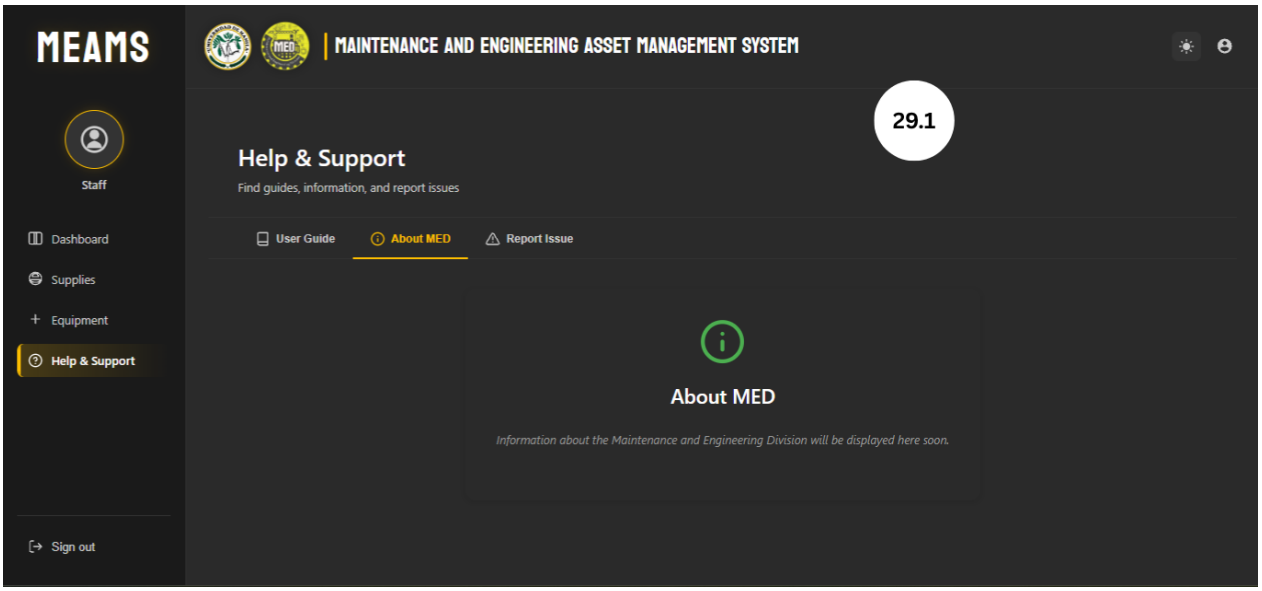
28.2. A collapsible guide that provides step-by-step instructions on how to add new supplies to the inventory.

28.3. A collapsible guide that provides step-by-step instructions on how to add new equipment assets.

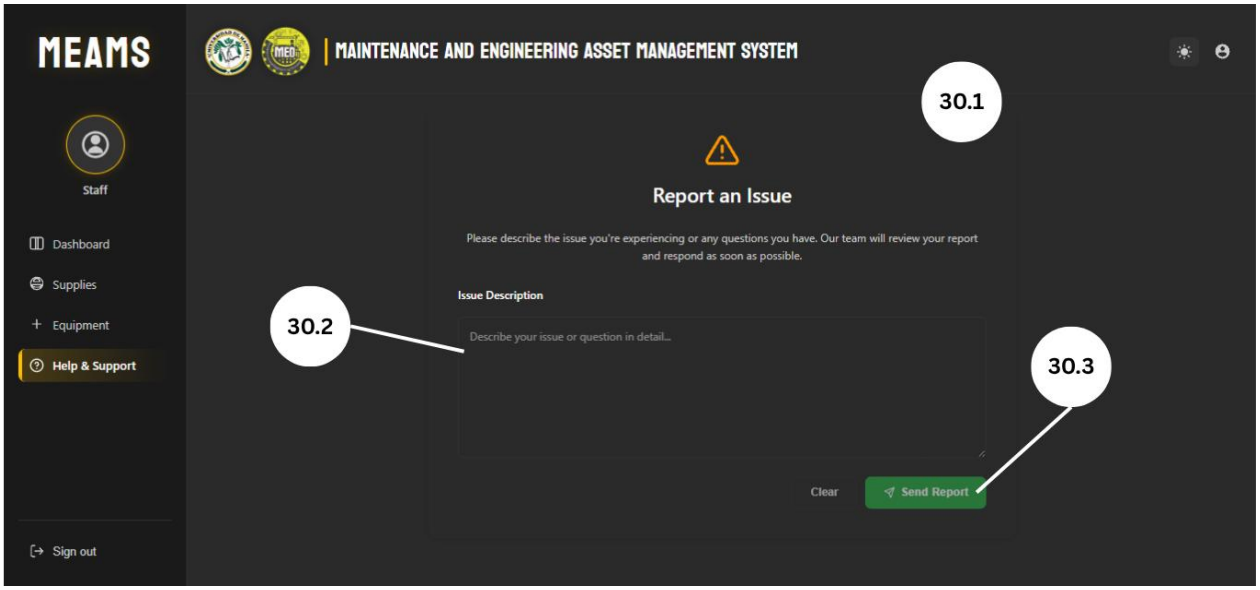
28.4. A collapsible guide detailing the process for modifying the stock quantity of an existing supply item.

28.5. A collapsible guide detailing the process for editing the information of an existing equipment asset.

28.6. A collapsible guide that provides instructions on where to find and use the QR code generation tool.



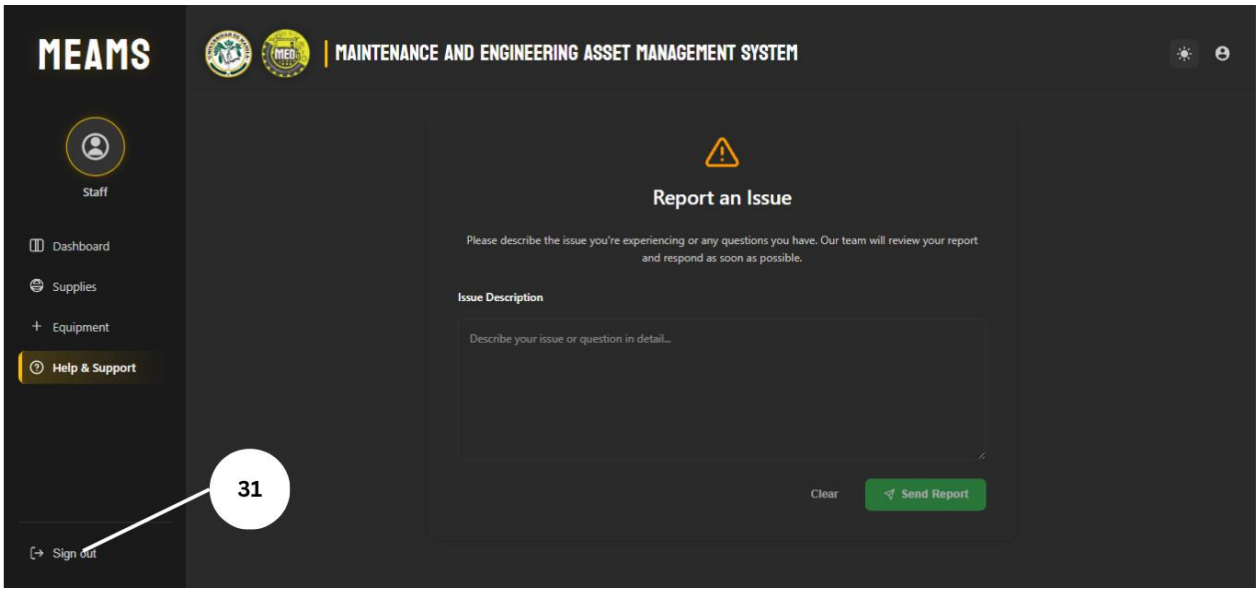
29.1. Overlay of the About Med Section.



30.1. Overlay of the Report Issue Section.

30.2. A large text area where the user must detail the problem or question they are experiencing.

30.3. Submit the completed issue description to the support team for review and response.



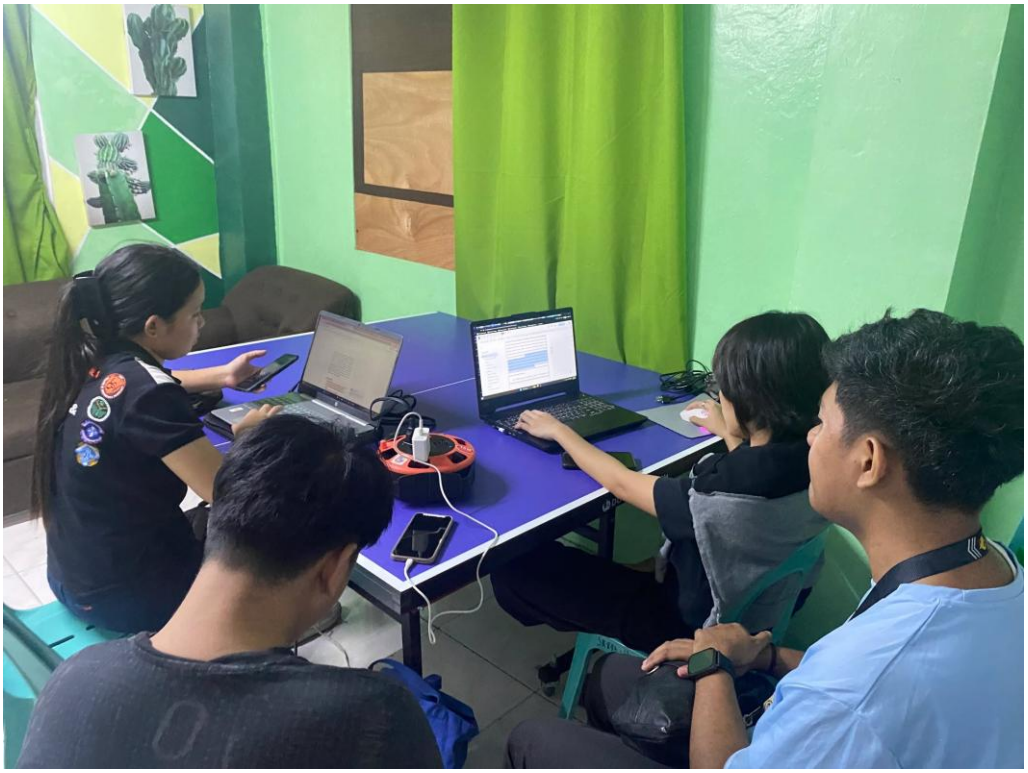
31. Sign out

APPENDIX F: DOCUMENTATION












Maintenance and Engineering Division
MEAMS: A REAL-TIME TRACKING AND FORECASTING ASSET MANAGEMENT SYSTEM
NATURE'S SPRING




M 305
COMP LAB 1
AMB. FRANCIS C. CHUA WING
PIAMONTE
CAOLE
GARCIA
MORAL
DAZO

APPENDIX G: CURRICULUM VITAE



CHRISTOPER URIEL B. CAOLE

Data Science Student

I am a motivated IT student with a passion for technology and innovation, eager to learn and expand my knowledge and skills. I strive to embrace challenges, gain practical experience, and grow to contribute meaningfully in a professional environment.

Contact

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https://github.com/yelcaole27

1901 K Int 61 Zamora St. Pandacan Manila

Education

2020-2022

ICT Computer Programming
Philippine Christian University
Taft Ave. Malate, Manila

2022-Present

BS Data Science
Universidad De Manila
Cecilia Muñoz St, Ermita, Manila

Programming Language

- Python
- Java
- SQL
- HTML/CSS
- JavaScript

Projects

MEAMS: A Real-Time Tracking and Forecasting Asset Management System — 2025

Team Member

An automatic Asset Management System that tracks the supply and demand of assets in real time and provides analytical presentations to give users insights that support decision-making. This was created using React, JavaScript, Python, and MongoDB.

UDM REGISTRAR'S AUTOMATED ENROLLMENT FORM SYSTEM — 2024

Team Member

An automatic enrollment form system that enroll the students to their subjects if they pass the prerequisites. After enrolling them it automatically make an enrollment form in which the user can print. This was created using Python, MySQL, Flask, PyPDF2, jinja2.

Inventory Management System — 2023

Team Member

An inventory management system that tracks the inventory stocks and allows the user to manage the inventory efficiently. This was created using MySQL, Java, Java Swing.

Skills

- Statistics
- Mathematics
- Analytics
- Hacking
- Creativity
- Presentation

Seminars

- No Cap, Just Clarity: The Real Deal on Explainable AI (XAI) — September 27, 2024
- Empowering Innovation with Azure OpenAI Assistant — September 28, 2024
- Build a Canvas App with Copilot in Power Apps — September 30, 2024
- Artificial Intelligence & Machine Learning: Shaping the Future of IT — October 4, 2024



JUSTINE DAZO

Data Science Student

Fututе Data Scintis with a strong sense of responsibility, rapid flexibility, and a thirst for lifelong learning. Known for being reliable, quick-thinking, and willing to take on new tasks.

Contact

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Education

2020-2022

ICT Computer Programming

Jose Abad Santos High School
Numancia St., Binondo, Manila

2022-Present

BS Data Science

Universidad De Manila
Cecilia Muñoz St, Ermita, Manila

Programming Language

- Python
- Java
- SQL
- HTML/CSS
- JavaScript
- PHP

Projects

MEAMS: A Real-Time Tracking and Forecasting Asset Management System — **2025**

Front-end Dev, QA Testing

An automatic Asset Management System that tracks the supply and demand of assets in real time and provides analytical presentations to give users insights that support decision-making. This was created using React, JavaScript, Python, and MongoDB.

UDM REGISTRAR'S AUTOMATED ENROLLMENT FORM SYSTEM — **Jun 2024 - Jun 2024**

Member

An automatic enrollment form system that enroll the students to their subjects if they pass the prerequisites. After enrolling them it automatically make an enrollment form in which the user can print. This was created using Python, MySQL, Flask, PyPDF2, jinja2.

Inventory Management System — **Mar 2023 - Mar 2023**

Member

An inventory management system that tracks the inventory stocks and allows the user to manage the inventory efficiently. This was created using MySQL, Java, Java Swing.

Skills

- Statistics
- Mathematics
- Analytics
- Hacking
- Creativity
- Presentation

Seminars

- No Cap, Just Clarity: The Real Deal on Explainable AI (XAI) — September 27, 2024
- Empowering Innovation with Azure OpenAI Assistant — September 28, 2024
- Build a Canvas App with Copilot in Power Apps — September 30, 2024
- Artificial Intelligence & Machine Learning: Shaping the Future of IT — October 4, 2024



ETHYL JANE GARCIA

Data Science Student

I am an aspiring Data Analyst with a strong interest in data analysis and data-driven problem-solving. A fast learner who thrives in collaborative environments, I bring experience in leadership roles and adaptability under pressure. I am eager to develop my expertise further and apply analytical skills to support effective, data-informed decision-making in the workplace.

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Education

2020-2022

Information Technology

AMA Computer Learning Center
Juan Luna St, Tondo, Manila

2022-Present

BS Data Science

Universidad De Manila
Cecilia Muñoz St, Ermita, Manila

Programming Language

- Python
- Java
- SQL
- HTML/CSS
- JavaScript
- PHP

Projects

MEAMS: A REAL-TIME TRACKING AND FORECASTING ASSET MANAGEMENT SYSTEM — 2025

Project Manager

An automatic Asset Management System that tracks the supply and demand of assets in real time and provides analytical presentations to give users insights that support decision-making. This was created using React, JavaScript, Python, and MongoDB.

UDM REGISTRAR'S AUTOMATED ENROLLMENT FORM SYSTEM — 2024

Database Manager

An automatic enrollment form system that enroll the students to their subjects if they pass the prerequisites. After enrolling them it automatically make an enrollment form in which the user can print. This was created using Python, MySQL, Flask, PyPDF2, jinja2.

INVENTORY MANAGEMENT SYSTEM — 2023

Front-End Developer

An inventory management system that tracks the inventory stocks and allows the user to manage the inventory efficiently. This was created using MySQL, Java, Java Swing.

Skills

- Statistics
- Mathematics
- Analytics
- Hacking
- Creativity
- Presentation

Seminars

- No Cap, Just Clarity: The Real Deal on Explainable AI (XAI) — September 27, 2024
- Empowering Innovation with Azure OpenAI Assistant — September 28, 2024
- Artificial Intelligence & Machine Learning: Shaping the Future of IT — October 4, 2024
- Crafting Custom Copilots With Azure AI Studio — November 12, 2024



RENAN JAY MORAL

Junior Data Scientist

Motivated Junior Data Scientist with growing expertise in machine learning, predictive modeling, and data analysis. Skilled in handling academic projects that translate theory into practical solutions. Dedicated to applying data-driven approaches to solve real-world challenges and support informed decision-making.

Contact

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Blk.12 Lot.17 Lemonada St, Model Community, Tondo Manila

Education

2022-Present

BS Data Science

Universidad De Manila
Cecilia Muñoz St, Ermita, Manila

2020-2022

Information Technology

Arellano University
Legarda St, Sampaloc, Manila

Programming Language

- Python
- Java
- SQL
- HTML/CSS

Projects

MEAMS: A REAL-TIME TRACKING AND FORECASTING ASSET MANAGEMENT SYSTEM — 2025

Machine Learning & Documentation

An automatic Asset Management System that tracks the supply and demand of assets in real time and provides analytical presentations to give users insights that support decision-making.

ASSET MANAGEMENT SYSTEM FOR FIRST ESTAN TRADING CORPORATION — 2024

Front-End Developer

An inventory management system that tracks the inventory stocks and allows the user to manage the inventory efficiently.

UDM'S AUTOMATED ENROLLMENT FORM SYSTEM — 2023

Documentation

An automatic enrollment form system that enroll the students to their subjects if they pass the prerequisites. After enrolling them it automatically make an enrollment form in which the user can print.

Skills

- Statistics
- Mathematics
- Analytics
- Hacking
- Creativity
- Presentation

Seminars

- No Cap, Just Clarity: The Real Deal on Explainable AI (XAI) — September 27, 2024
- Empowering Innovation with Azure OpenAI Assistant — September 28, 2024
- Artificial Intelligence & Machine Learning: Shaping the Future of IT — October 4, 2024
- Crafting Custom Copilots With Azure AI Studio — November 12, 2024



CRYSTAL SHANE PIAMONTE

Data Science Student

I am a dedicated and motivated individual with a strong interest in data science. I embrace new challenges and focus on developing practical skills that add value to the workplace. My goal is to apply my current knowledge while refining my expertise through hands-on experience, collaboration, and innovative problem-solving.

Contact

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2231 Simeon de jesus St. Balut
Tondo Manila

Education

2020-2022

Information and Communications Technology

Timoteo Paez Integrated School

Younger St. Tondo, Tondo VII, Metro Manila

2022-Present

BS Data Science

Universidad De Manila

Cecilia Muñoz St, Ermita, Manila

Programming Language

- Python
- Java
- SQL
- HTML/CSS
- JavaScript
- PHP

Projects

MEAMS: A REAL-TIME TRACKING AND FORECASTING ASSET MANAGEMENT SYSTEM — 2025

Member

An automatic Asset Management System that tracks the supply and demand of assets in real time and provides analytical presentations to give users insights that support decision-making. This was created using React, JavaScript, Python, and MongoDB.

FACULTY CLASSROOM SCHEDULING SYSTEM — 2024

Member

A system that functions to show the visibility of available rooms, facilitating, room reservations, and tracking room changes for the professors. This was created using Java JFrame and MySQL Workbench.

UDM VISITORS & SECURITY TIME LOG SYSTEM — 2023

Member

A system that functions to allow the security admin to streamline and automate their visitor management process by using Time log System to monitor, track and record UDM's visitors' informations. This was created using Java JFrame and MySQL Workbench.

Skills

- Statistics
- Mathematics
- Analytics
- Hacking
- Creativity
- Presentation

Seminars

- Unlocking AI with Teachable Machine - A No Code Approach — December 20, 2024
- Exploring Artificial Intelligence in Mechanical Engineering — December 10, 2024
- Artificial Intelligence and Machine Learning: Shaping the future of IT — October 4, 2024
- LinkedIT: Charting IT Pojects success with agile scrum methodologies — October 1, 2024